Billix

The Billix

Household Utility Savings Handbook

Understand every bill. Cut what you can.
Prove what you save.



Your utility bills, covered

- Electricity
- Heating fuel
- Water
- Sewer
- Trash
- Internet
- Phone

2026 Edition



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Understand every bill. Cut what you can. Prove what you save.

Electricity, heating fuel, water, sewer, trash, internet and phone

The Billix Household Utility Savings Handbook, 2026 Edition

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This book is education, not professional advice. It explains how household bills and services generally work in the United States. It is not financial, legal, tax, engineering or medical advice, and it does not replace advice from a professional who knows your situation.

Rules and prices vary and change. Utility rates, tenant rights, assistance programs, tax rules and consumer protections differ by state, city, provider and lease, and they change often. State rules in this book are examples only. The information was verified through October 2026. Check current rules with your provider, your state agencies and the official sources cited in each chapter before you act.

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Check your lease and local rules before changing anything in a rented home, and follow local codes and permit rules for any work on a home you own.

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Contents

Sections are numbered by chapter: 2.11 is section 11 of Chapter 2.

Introduction: Why understanding your bills matters 6

How to use this book 7 · The opening assessment 9 · The Annual Household Utility Spending Tracker 10

PART I: FOUNDATIONS

Chapter 1. Understanding household utility bills 13

1.1 What a utility is 13 · 1.2 How a bill is made 14 · 1.3 Fixed versus variable charges 16 · 1.4 Billing periods 17 · 1.5 Actual, estimated and smart-meter readings 18 · 1.6 Comparing bills correctly 18 · 1.7 Warning signs, and what to do about them 20 · 1.8 How to measure a saving honestly 21

The Billix 15-Minute Bill Audit, after 1.8 (blank record sheet: B2) 23

PART II: ENERGY

Chapter 2. Electricity 28

2.1 From power plant to outlet 28 · 2.2 Watts, kilowatts and kilowatt-hours 29 · 2.3 Anatomy of an electric bill 30 · 2.4 Why electric bills rise 31 · 2.5 Appliance by appliance 32 · 2.6 Lighting 33 · 2.7 Standby power 34 · 2.8 Refrigerators and freezers 35 · 2.9 Laundry and dishwashing 36 · 2.10 Time-of-use and other rate plans 37 · 2.11 Choosing an electricity supplier, where allowed 39 · 2.12 Electricity-saving myths 41

Tool 2.1: the Home Electricity Consumption Audit 44

Chapter 3. Natural gas, propane, heating oil and other fuels 48

3.1 The common heating fuels 49 · 3.2 Reading a gas bill 49 · 3.3 Why gas bills rise 52 · 3.4 The real cost of heat 53 · 3.5 Lowering gas use 54 · 3.6 Propane and heating oil 55 · 3.7 Safety 57

Tool 3.1: the Fuel Cost Comparison 59

Chapter 4. Heating, cooling, insulation and home efficiency 63

4.1 How heating and cooling systems work 63 · 4.2 What drives heating and cooling costs 65 · 4.3 How heat moves through a house 66 · 4.4 Thermostat management 67 · 4.5 Air leaks and insulation 68 · 4.6 Windows and sunlight 70 · 4.7 HVAC maintenance 71 · 4.8 Ceiling fans 72 · 4.9 Smart thermostats and utility thermostat programs 73 · 4.10 Repair or replace 74

Tool 4.1: the Heating and Cooling Checklist 77

PART III: WATER

Chapter 5. Water 81

5.1 Where household water comes from 81 · 5.2 Reading a water bill 82 · 5.3 Finding unexpected use 84 · 5.4 Toilets 86 · 5.5 Faucets 87 · 5.6 Showers: one shower, two bills 87 · 5.7 Laundry 89 · 5.8 Dishwashing 89 · 5.9 Outdoor water 90 · 5.10 Leaks 91 · 5.11 Water-heating costs and safe temperatures 92 · 5.12 Putting the savings math together 93

Tool 5.1: the Household Water Audit 96 · *Tool 5.2: the Water Leak Inspection Checklist* 97 · *Tool 5.3: the Water Savings Calculator* 98

Chapter 6. Sewer, wastewater and stormwater 100

6.1 How wastewater systems work 100 · 6.2 How sewer is charged 102 · 6.3 Why sewer costs rise 103 · 6.4 Outdoor water and sewer 104 · 6.5 Septic systems 105 · 6.6 Stormwater fees and credits 106

PART IV: CONNECTIONS AND WASTE

Chapter 7. Home internet and TV 111

7.1 How home internet works 112 · 7.2 Speed explained 112 · 7.3 Reading an internet bill 113 · 7.4 Broadband labels 114 · 7.5 How much speed you need 116 · 7.6 Lowering your internet bill in ten steps 118 · 7.7 Asking for a better rate 118 · 7.8 Modems and routers: rent or buy 120 · 7.9 TV, streaming and bundles 121

Tool 7.1: the Internet Plan Comparison Worksheet 123 · *the Provider Negotiation Script (A1)*

Chapter 8. Mobile phone service 126

8.1 How cell plans work 126 · 8.2 Plan terms decoded 127 · 8.3 Reading a cell phone bill 128 · 8.4 Lowering your mobile costs 129 · 8.5 The real cost of a "free" phone 130 · 8.6 Switching carriers safely 132

Tool 8.1: the Mobile Plan Comparison Worksheet 135

Chapter 9. Trash and recycling 138

9.1 How waste collection is billed 139 · 9.2 Common charges 140 · 9.3 Lowering waste costs, including duplicate charges (paying twice) 141 · 9.4 What not to do 143

Tool 9.1: the Waste Service Audit 144

PART V: PROTECT AND DECIDE

Chapter 10. Billing disputes, negotiation and consumer protection **151**

10.1 Reviewing bills systematically **152** · 10.2 Billing errors **152** · 10.3 Contacting customer service **153** · 10.4 What can and cannot be negotiated **155** · 10.5 Budget billing **156** · 10.6 Autopay **157** · 10.7 Scams and high-pressure offers **158** · 10.8 Escalation: who handles what **160** · 10.9 Deposits, late fees, reconnection fees, collections and credit reports **161**

Tool 10.1: the Billix Electric Bill Analyzer **152** · *the Provider Call Log (A2)* · *the billing dispute letter (A3)* · *the fee explanation request (A4)* · *the rate or plan review request (A5)* · *the service cancellation checklist (A6)*

Chapter 11. Renters, homeowners and moving **168**

11.1 Why housing status matters **168** · 11.2 Renters **169** · 11.3 Homeowners **173** · 11.4 Shared housing **174** · 11.5 Starting, moving and stopping service **175**

Tool 11.1: the Housing Action Matrix **177** · *Tool 11.2: the Move-In and Move-Out Utility Checklist* **179**

Chapter 12. Financial assistance, discounts and hardship programs **183**

Shutoff notice? Do this first (the chapter's opening box) · 12.1 Why assistance exists **184** · 12.2 LIHEAP **185** · 12.3 The Weatherization Assistance Program **186** · 12.4 Phone and internet help: Lifeline **186** · 12.5 Utility company programs **188** · 12.6 Water assistance **190** · 12.7 Finding help safely **191** · 12.8 If you are facing a shutoff **193**

Tool 12.1: the Billix Utility Assistance Finder Checklist **197**

Chapter 13. Smart upgrades and return on investment **202**

13.1 Upfront price versus running cost **202** · 13.2 Simple payback **203** · 13.3 Total cost of ownership **204** · 13.4 Comparing nine upgrades **206** · 13.5 Rebates and incentives **206** · 13.6 Solar, batteries and community solar **208** · 13.7 Avoiding costly mistakes **210**

Tool 13.1: Should I Buy It? **212**

PART VI: YOUR PLAN

Chapter 14. The Billix 30-day and 90-day savings plans **218**

14.1 Before you start: the three labels **219** · 14.2 The 30-day plan **219** · 14.3 The 90-day plan **222** · 14.4 Measuring and reporting your savings **223**

Tool 14.1: the 30-Day Action Plan **226** · *Tool 14.2: the 12-Month Bill Calendar* **226** · *the Household Savings Priority Matrix (B3)* · *the 90-Day Household Savings Scorecard (B4)*

APPENDIX 229

A. Scripts and letters **229**

A1 The Provider Negotiation Script **229** · A2 The Provider Call Log **230** · A3 The billing dispute letter **231** · A4 The fee explanation request **232** · A5 The rate or plan review request **232** · A6 The service cancellation checklist **232**

B. Core worksheets **233**

B1 The Annual Household Utility Spending Tracker **233** · B2 The Billix 15-Minute Bill Audit **235** · B3 The Household Savings Priority Matrix **237** · B4 The 90-Day Household Savings Scorecard **238**

C. Glossary **239**

D. Reference index **242**

D1 Topics by chapter **242** · D2 Tools **244** · D3 Where to get help **244** · D4 State rules for propane and heating-oil deliveries (examples) **245**

About Billix **245**

BEFORE YOU START

Introduction: Why understanding your bills matters

Most households know what they pay in rent and roughly what they spend on groceries. Fewer know what they spent on electricity, water, internet and phones last year. These bills arrive every month, often paid automatically, and they rarely get a second look.

That inattention costs money. Utility bills carry recurring charges that are easy to forget, prices that rise quietly when promotions end, fees that can be avoided, and use that nobody notices until a bill jumps. Add up a year of them, and many households find they spend more than they guessed.

The good news is that saving on utilities does not always mean living with less. Some of the biggest savings in this book come from a phone call, a corrected bill, a plan that better fits the household or a discount nobody had asked for.

Eight ways a bill can shrink

Way	What it means	Example	Chapters
Reduce use	Use fewer units: kWh, therms, gallons or gigabytes	Fixing a running toilet	2 to 6
Lower the unit price	Pay less for each unit	A better rate plan, supplier or fuel contract	2, 3
Remove avoidable fees	Stop paying charges you do not need	Returning a rented router; avoiding late fees	7 to 10
Correct billing errors	Get back money you were wrongly charged	A duplicate fee or an estimated reading	1, 10
Obtain eligible discounts	Use programs you qualify for	An income-based rate or Lifeline	12
Replace inefficient equipment	Buy something that costs less to run	A heat pump water heater when the old tank fails	13
Change providers or plans	Pay less for the same service	A cheaper internet or phone plan	7, 8
Budget more effectively	Make bills predictable and plan for peaks	Budget billing; a seasonal savings cushion	10, 14

Budgeting is not the same as spending less. Budget billing and other ways of smoothing payments make bills predictable. They help a household plan, but they do not lower the total cost of what it uses. This book keeps the two ideas separate.

Why the difference matters

Each way of saving is proven differently. A cheaper plan shows up as a lower price. A repaired leak shows up as lower use. And lower use rarely lowers a bill by the same percentage, because fixed charges stay.

Worked example: using 10% less electricity (Illustrative)

- **Before:** 600 kWh × $0.18 + a $12 customer charge = **$120.00.**
- **After using 10% less:** 540 kWh × $0.18 + $12 = **$109.20.**
- **The result:** use fell 10%, but the bill fell $10.80, or 9%. The $12 customer charge did not change.

Section 1.8 builds on this. Its Rule 2 is the idea at the center of the book: value every saving at your marginal price, the price of one more unit, not at your bill's average.

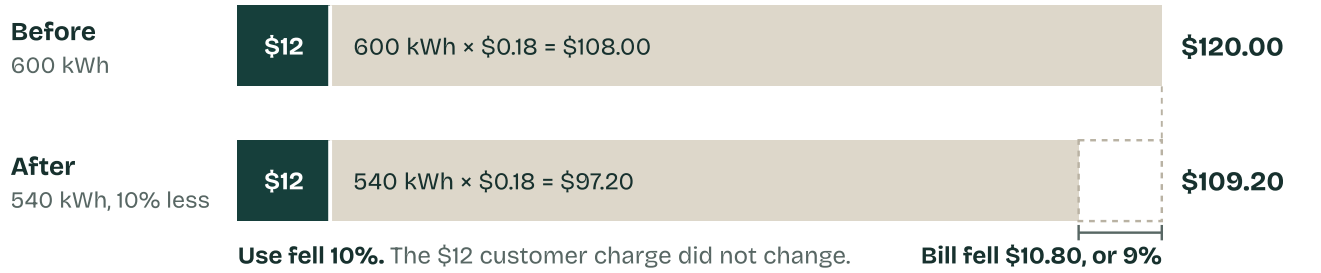


Figure 0.1. The 10% example: use fell 10%, but the bill fell 9%, because the $12 customer charge did not change. Illustrative.

What this book will not do

- **Sell you anything.** It does not ask you to buy a product, sign up with a supplier or make an upgrade.
- **Recommend providers.** Real companies appear only as examples. You compare offers using your own numbers.
- **Promise savings.** It shows you how to estimate a saving and how to prove it with your own bills.

This handbook covers household utilities: electricity, heating fuels, water, sewer, trash, internet and phone. Billix also covers other recurring household costs, from housing and insurance to subscriptions and local fees, which are outside the scope of this book.

How to use this book

Start where your money is

Bill just jumped, or a shutoff notice? Use the table below now, and do the assessment later.

Otherwise, begin with the opening assessment below. It shows which bills matter most in your home. Then follow the path that fits. You do not need to read the chapters in order, and each chapter opens with a **Start here** box that sends common problems to the right section.

If...	Start with
A bill just jumped	The 15-Minute Bill Audit after 1.8, including "Split the change" (blank sheet: B2), then the section its results point to
You are behind on a bill or facing a shutoff	The "Shutoff notice? Do this first" box at the start of Chapter 12, then 12.8
A rate plan, supplier or fuel offer arrived	2.10 (rate plans), 2.11 (electricity suppliers) or 3.6 (propane and heating oil)
A call or text about a bill seems suspicious	10.7
You rent, or share a home	11.2 if you rent; 11.4 if you share
You are about to move	11.5 and Tool 11.2
Internet and phone feel too expensive	7.6 (internet) and 8.4 (phone)
You are thinking of buying equipment or solar	13.3 and Tool 13.1 for equipment; 13.6 for solar
You want a step-by-step plan	14.2, the 30-day plan

Put opportunities in order

As a rule, work from the cheapest and most certain savings to the most expensive and least certain:

1. **Errors and avoidable fees.** No cost, and the money is already yours.
2. **Plans, providers and discounts.** Usually no cost, often a phone call.
3. **Habits and settings.** No cost, but they need follow-through.
4. **Low-cost fixes,** such as bulbs, showerheads and weatherstripping.
5. **Major upgrades,** timed to when equipment is near the end of its life, and only after the math.

The Household Savings Priority Matrix (B3) turns this order into a score for each idea.

How the book labels things

Savings labels. Every saving carries one of three labels:

- **Illustrative:** a made-up example that shows a method.
- **Estimated:** a projection from your own data, not yet confirmed by your bills.
- **Verified:** confirmed by your own bills over several billing periods, after adjusting for billing days, weather and price, with no other big change in the home. For a flat-price service, one full bill at the new price is enough.

Cost tiers. No-cost; Low-cost, under $50; Moderate, about $50 to $500; Major investment, over $500.

Callouts. Six kinds of box appear throughout:

- **Safety warning:** a risk to people or property.
- **Call a professional:** work that needs a licensed expert.
- **Check your lease:** something a renter should confirm first.
- **Last Verified:** when a fast-changing fact was last checked.
- **Start here:** at the start of each chapter, where to go first for a common problem.
- **Find it on your bill:** the line to look for on a bill, notice, quote, delivery ticket or rent statement, what it means and where to record it.

The Billix rate card

Example prices for the worked examples, not averages. Worked examples use one set of illustrative prices, so the math is consistent from chapter to chapter. They are not national averages, not benchmarks and not your prices. To find your own marginal price, see 1.8, Rule 2.

For real averages, see the sources the chapters cite. The US Energy Information Administration (EIA) publishes the average electricity bill and price per kWh (Chapter 2's opening) and a monthly residential gas price (3.2). It also publishes weekly propane and heating-oil prices for the US and some regions (3.6). For water and sewer, 5.2 gives a survey's average combined bill. Use any average only as a rough check, never in place of your marginal price.

Service	Low example	Middle, used in worked examples	High example
Electricity, per extra kWh	$0.13	$0.18	$0.26
Electricity, monthly customer charge	$6	$12	$24
Natural gas, per therm	$1.00	$1.30	$1.60
Propane, per gallon	$2.10	$2.60	$3.30
Heating oil, per gallon	$4.25	$5.25	$5.75
Water, per 1,000 gallons	$4.00	$6.00	$9.00
Sewer, per 1,000 gallons	$5.00	$8.00	$12.00

Some chapters rerun an example at the low and high prices to show how much the answer depends on your rates. Worked examples also use a $15 monthly gas customer charge. The low and high values are illustrative too: they test an example, not whether your own price is high.

Meet the households

Seven fictional households return throughout the book, each with the same home, equipment and bills every time. Each has a case letter, so Case B always means the Okafors. Their figures are illustrative.

Case	Household	Who they are	Main chapter	Also appears in
A	Sam	A first-time renter in an all-electric one-bedroom apartment	11	1
B	The Okafors	A family of four in a 1962 house with a gas furnace and an electric water heater	2	1, 3, 4, 5, 13
C	Dana	A remote worker in a townhouse with a heat pump, sharing it with a partner	4, 7	14
D	The Harrises	Two retired adults on a fixed income, behind on their electric bill	12	1, 9, 10
E	The Lindqvists	A rural family heating with propane, with a well and septic system	3	6.5
F	The Brooks household	A suburban family whose water bill jumped by $160	5	Step 3 of the opening assessment, 6.3, 10
G	Jordan and Casey	Roommates comparing phone plans and splitting shared bills	8	11.4

Providers named "Anytown" are fictional and marked SAMPLE.

The opening assessment

Before you change anything, find out what you pay. This assessment is the first part of the Annual Household Utility Spending Tracker. It takes about an hour, most of it spent finding bills.

Step 1: Gather up to 12 months of bills

One month rarely tells the truth. Heating and cooling swing with the seasons, and a single bill can include a spike, a credit or an estimated reading. Gather up to 12 months for every service:

- **Online accounts:** most providers let you download past bills and usage history.
- **Paper bills and email:** search your inbox for each provider's name.
- **Fuel deliveries:** collect every propane, oil or wood delivery receipt from the past year.
- **Bank and card statements:** these catch services you forgot you pay for, and bills paid automatically.
- **Rent, HOA or tax statements:** note any utility included in them.

If you have fewer than 12 months, use what you have and note it.

Step 2: Fill in the assessment

Use the blank assessment, which is Part 1 of B1 in Appendix B and is also in the companion worksheet pack. It has one row for each service, from electricity and heating fuel to trash and other recurring household services, and seven columns: utility, provider, monthly cost, annualized cost, billing frequency, billing type and notes. A completed sample follows the tracker description below.

Billing frequency: monthly, every two months, quarterly, per delivery or yearly.

Billing type: for example, metered use, flat fee, tiered rates, time-of-use, budget billing, per delivery, ratio-billed through rent, or included in rent, taxes or HOA dues.

Step 3: Annualize each bill correctly

The annualized cost is what each service costs over a full year. Work it out first, then divide by 12 for the monthly cost, so every service is compared on the same basis.

If the bill comes…	Annualized cost =	Watch out for
Monthly	The total of the last 12 bills	Multiplying one month by 12. A July electric bill × 12 can overstate the year; an April one can understate it.
Every two months	The total of the last 6 bills	Multiplying one bill by 6, if any bill was unusual
Quarterly	The total of the last 4 bills	The same: use actual bills where you can
Per delivery (propane, oil, wood)	The total of all deliveries in the last 12 months	A large delivery just before or after the 12-month window
Yearly	The yearly charge	Tank rentals, service plans and memberships billed once a year
Included in rent, taxes or HOA dues	The amount, if the statement shows it; otherwise write "included"	Ratio-billed shares, which can change monthly (11.2)

If you have only one or two bills, estimate the year and mark it **Estimated** in the Notes column. Replace it as more bills arrive.

Worked example: a water bill that arrives every two months (Illustrative)

- **The Brooks household's last six bills:** five at $181.20 and one, after a leak, at $340.80.
- **Annualized correctly:** (5 × $181.20) + $340.80 = **$1,246.80,** or about $103.90 a month.
- **Annualized from one bill:** $181.20 × 6 = $1,087.20, which misses the leak. $340.80 × 6 = $2,044.80, which treats the leak as normal.

Step 4: Read your results

- **Rank the services by annualized cost.** The top two or three usually deserve your first attention.
- **Mark any bill that changed sharply** during the year. That is where the 15-Minute Bill Audit (after 1.8; blank sheet: B2) starts.
- **Note every promotion, contract or price-lock end date** you find. The 12-Month Bill Calendar (Tool 14.2) keeps track of them.
- **Keep your completed assessment.** In a year, fill in a fresh copy (B1) and compare.

The Annual Household Utility Spending Tracker

Purpose: to see a full year of household bills in one place, so you know which bills matter most, when they peak, and what to work on first.

The tracker has two parts, and Appendix B1 prints a blank copy of each:

1. **The opening assessment** (Steps 1 to 4 above): one line per service, annualized.
2. **The month-by-month sheet:** each bill's total in the month it arrived, with a provider list. Fill it in as bills arrive through the year.

Both are in the companion worksheet pack as a fillable PDF and a spreadsheet. They also work on paper.

A completed sample (Illustrative)

A fictional household of three in a house with gas heat. All figures are illustrative.

Utility	Provider	Monthly cost	Annualized cost	Billing frequency	Billing type	Notes
Electricity	Anytown Electric (SAMPLE)	$135.00	$1,620	Monthly	Metered, with a $12 customer charge	Peaks in July and August
Natural gas	Anytown Gas (SAMPLE)	$65.00	$780	Monthly	Metered, with a customer charge	Heating season November to March
Water	Anytown Water & Sewer (SAMPLE)	$45.00	$540	Every two months	Metered, tiered	Total of six bills
Sewer	Same bill as water	$60.00	$720	Every two months	Based on water use	Check whether winter averaging applies (6.2)
Internet	Anytown Broadband (SAMPLE)	$65.00	$780	Monthly	Flat price	Promotion ends in March
Mobile phone	Anytown Wireless (SAMPLE)	$90.00	$1,080	Monthly	Flat price, two lines	Includes device insurance
Trash	City	$25.00	$300	Quarterly	Flat fee, 96-gallon cart	Cart size never checked (9.3)
Streaming services	Two services	$25.00	$300	Monthly	Subscriptions	One rarely used
Total		**$510.00**	**$6,120**			

Reading the sample

Rank	Service	Share of the year	First question to ask
1	Electricity	26.5%	Why do July and August peak? (2.4 and 4.2)
2	Water and sewer together	20.6%	Does sewer use winter averaging, and are there leaks? (6.2 and 5.10)
3	Mobile phone	17.6%	Is device insurance worth it, and is there a cheaper plan? (8.4)
4	Internet	12.7%	What happens when the promotion ends in March? (7.3 and 7.7)

Water and sewer are shown together here because one meter drives both. Natural gas ties with internet for fourth place, at $780 a year each.

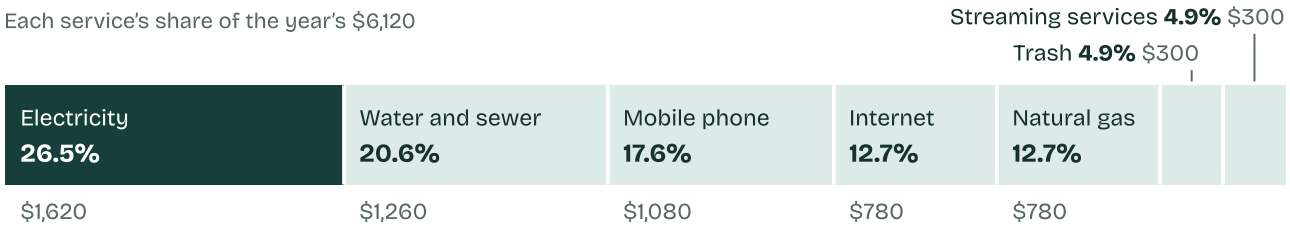


Figure 0.2. The sample tracker's share of the year, by service, of $6,120 in all. Water and sewer are shown together because one meter drives both. Illustrative.

This household's quickest wins are probably in the Notes column: a promotion about to end, device insurance, a rarely used subscription and a trash cart that was never sized. Each is a phone call or a few minutes online. The electricity peak needs more work, starting with the 15-Minute Bill Audit (after 1.8).

Calculations: monthly cost = annualized cost ÷ 12. Share of the year = the service's annualized cost ÷ the household total × 100.

Next step: turn to Chapter 1, or to the chapter for your largest bill.

PART I

Foundations

1 Understanding Household Utility Bills

PART I · FOUNDATIONS

1 Understanding Household Utility Bills

Every utility bill, whatever the service, is built the same way: something is measured, a price is applied, fixed charges and taxes are added, and credits or past balances are carried over. Once you can see those parts, you can tell a high bill caused by your home from one caused by the price, the calendar or a mistake.

That difference matters. You cannot fix a rate increase by taking shorter showers, and you cannot fix a running toilet by calling customer service. Most wasted effort on bills comes from treating one kind of problem as another.

By the end of this chapter you will be able to:

- Name the kind of provider behind each of your bills, and who oversees it
- Find the fixed and variable charges on any bill
- Compare two bills fairly, even when the billing periods differ
- Tell an actual meter reading from an estimate
- Spot the warning signs of a billing error, and know what to do next
- Work out what a saving is really worth, without fooling yourself

Keep your most recent bill nearby. Every section ends with something to find on it.

START HERE:

- → **Your bill jumped?** Run the 15-Minute Bill Audit after 1.8, and split the change into dollars.
- → **Not sure the reading is right?** Go to 1.5.
- → **Comparing two bills with different dates?** Go to 1.4, then 1.6.
- → **A charge or fee you do not recognize?** Go to 1.7.
- → **Wondering whether a saving is real?** Go to 1.8.

1.1 What a utility is

A **utility** is a provider of an essential service that reaches your home through a network of wires, pipes or cables. Who owns that provider decides two things that matter to you: who sets the price, and where you go when something is wrong.

Your bills come from five kinds of provider:

Provider	Typical services	Who sets the price	Where to take a problem
Investor-owned utility, owned by shareholders	Many electric and gas companies, some water companies	The state utility commission, through a public process called a rate case	The utility first, then the state commission
Public or municipal utility, owned by a city, county or district	Water and sewer for most Americans, some electric systems	A city council or utility board	The utility, then its board or council
Cooperative, owned by its members	Many rural electric systems	A board the members elect	The co-op, then its board
Competitive supplier	Electricity or gas supply, in states that allow choice	The contract you agreed to	The supplier, then the state commission or attorney general
Subscription service	Internet, mobile, TV, private trash haulers	The company's plan or contract	The company, then the FCC (phone, internet, TV) or a state consumer office

Water shows how much ownership varies. About 8 in 10 Americans get their tap water from a system owned by a local government, and more than 1 in 10 from a privately owned one. Electricity choice is narrower than many people think: about 14.6 million homes, in 15 states and Washington, D.C., bought their electricity supply from a company other than their utility in 2025. Most of Texas runs its own competitive market.

Provider structures differ because they grew up locally. A town may have built its own water system a century ago, while a private company built the power lines. In most places you cannot choose who delivers your water, gas or electricity. The pipes and wires are the same whoever you pay.

Ownership also decides which protections apply. State rules on bills, deposits and shutoffs usually cover investor-owned utilities. City-owned utilities and co-ops often set their own rules, so their customers should look up the provider's own policies.

Find it on your bill: the provider's name, and any phone number for a state utility commission. Some states require that number on the bill. Ohio, for example, requires it on every electric bill.

1.2 How a bill is made

Every metered bill goes through the same seven steps, and each step is a place where your bill can change or go wrong.

1. **Service is provided.** Electricity, gas or water flows to your home.
2. **Use is recorded or estimated.** A meter is read, by a person or remotely, or the provider estimates it.
3. **A rate is applied.** The provider multiplies your use by the prices in its approved rate schedule (often called a **tariff**) or in your contract.
4. **Fixed and variable charges are added.** Some charges are a flat amount each month; others grow with every unit you use.
5. **Taxes and approved surcharges are added.** These are set by governments and regulators, not by your habits.
6. **Credits, discounts, deposits and past balances are applied.**
7. **The amount due is produced.**

Here is what those steps produce on a sample electric bill. The numbers are illustrative, built from US averages. The 874 kWh of use is the average US home's monthly use in 2025.

Illustrative example: Anytown Electric Co. sample bill (SAMPLE, not a real company)

Line	Section	Rate	Amount
Customer charge	Delivery	$12.00 per month	$12.00
Distribution charge	Delivery	$0.046/kWh	$40.20
Transmission charge	Delivery	$0.024/kWh	$20.98
Storm recovery rider	Delivery	$0.005/kWh	$4.37
Energy-efficiency program charge	Delivery	$0.0015/kWh	$1.31
Low-income assistance charge	Delivery	$1.50 per month	$1.50
Generation (supply) charge	Supply	$0.073/kWh	$63.80
Fuel adjustment	Supply	$0.004/kWh	$3.50
Capacity charge	Supply	$0.003/kWh	$2.62
Clean-energy rider	Supply	$0.003/kWh	$2.62
Prior-period credit	Supply	-$0.0004/kWh	-$0.35

Line	Section	Rate	Amount
Charges before taxes			**$152.55**
State gross receipts tax and regulatory fee	Taxes and fees	2.6%	$3.97
Local franchise fee	Taxes and fees	1.7%	$2.59
Total amount due			**$159.11**

Three things stand out. Delivery, the cost of the wires that bring power to you, is $80.36, slightly more than supply at $72.19. Nine lines change with every kWh. And two lines, the customer charge and the low-income charge, would appear even if you used nothing.

Anytown Electric Co. SAMPLE

Account 0000-000-000 · Example Street, Anytown

Electricity used this period			**874 kWh**
CHARGE		2 RATE	AMOUNT
DELIVERY			
1 Customer charge		$12.00 per month	$12.00
Distribution charge		$0.046/kWh	$40.20
Transmission charge		$0.024/kWh	$20.98
Storm recovery rider		$0.005/kWh	$4.37
Energy-efficiency program charge		$0.0015/kWh	$1.31
Low-income assistance charge		$1.50 per month	$1.50
SUPPLY			
Generation (supply) charge		$0.073/kWh	$63.80
Fuel adjustment		$0.004/kWh	$3.50
Capacity charge		$0.003/kWh	$2.62
Clean-energy rider		$0.003/kWh	$2.62
Prior-period credit		-$0.0004/kWh	-$0.35
Charges before taxes			**$152.55**
TAXES AND FEES			
5 State gross receipts tax and regulatory fee	2.6%		$3.97
Local franchise fee	1.7%		$2.59
6 **Total amount due**			**$159.11**

3 **Delivery $80.36**

4 **Supply $72.19**

1 **Customer charge**
$12.00 a month, even at zero use. With the $1.50 low-income charge, $13.50 is fixed.

2 **Per-kWh lines (shaded)**
Nine lines change with every kWh. These are the ones your habits can change.

3 **Delivery**
$80.36 for the wires that bring power to you, including both fixed charges.

4 **Supply**
$72.19 for the electricity itself. Switching supplier, where allowed, changes only this part.

5 **Taxes and fees**
2.6% and 1.7% of the $152.55 before taxes: $6.56, set by governments and regulators.

6 **Total amount due, $159.11**
Delivery $80.36 + supply $72.19 + taxes and fees $6.56.

Figure 1.1. Annotated sample bill: the Anytown Electric Co. bill from 1.2 at 874 kWh, with the delivery and supply sections bracketed and the per-kWh lines shaded. SAMPLE: a fictional provider, with illustrative figures built from US averages.



Figure 1.2. Where the $159.11 goes on the Anytown Electric Co. sample bill. Illustrative.

Not every bill looks like this. Nationally, generating electricity is the largest part of the average price, about 56% in 2025. This sample's split is closer to homes that buy supply from a separate company. Some bills show one combined energy charge instead of many lines. Every bill still follows the same seven steps.

Why two similar homes pay different amounts. Two neighbors who use the same 874 kWh can still get different bills. They may be on different rate plans or pay different city franchise fees. They may also buy supply from different companies, have billing periods of different lengths, or carry different credits and past balances. Section 2.3 explains each of these electricity charges in detail.

Find it on your bill: the subtotal before taxes, and every line whose rate is "per kWh", "per therm" or "per 1,000 gallons". Those lines are the ones your habits can change.

1.3 Fixed versus variable charges

Every charge on a bill is either **fixed**, the same each month whatever you use, or **variable**, growing with each unit you use. This split decides how much any saving can do for you, because cutting use only shrinks the variable part.

Charge	Fixed or variable	Still charged at zero use?	What it usually pays for
Customer charge (also called base, basic service or meter charge)	Fixed	Yes	The meter, billing and the connection to your home
Minimum bill	Fixed floor	Yes	Ensures every customer pays a set amount. One Florida utility, for example, sets a $30 minimum base bill
Water base charge	Fixed, often by meter size	Yes	Pipes, treatment and debt that cost the same whatever you use
Usage (energy, consumption, volumetric) charge	Variable	No	The electricity, gas or water itself
Delivery charges	Usually variable, sometimes fixed	Sometimes	Poles, wires, pipes and their upkeep
Riders and adjustments	Usually variable	Rarely	Fuel costs, storm repairs, efficiency programs
Equipment rental	Fixed	Yes	A rented modem, router or water meter
Taxes and fees	A percentage of the bill or per unit	Yes, on the fixed part	State and local government

The Anytown sample bill shows how the split works. At zero use it still costs $14.08: the $12.00 customer charge, the $1.50 low-income charge and $0.58 of taxes. Every kWh after that adds about $0.166, including taxes.

Illustrative example: the same Anytown bill at three levels of use

Monthly use	Variable charges	Fixed charges	Taxes and fees	Total	Fixed share of charges before taxes
0 kWh	$0.00	$13.50	$0.58	$14.08	100%
500 kWh	$79.55	$13.50	$4.00	$97.05	14.5%
1,000 kWh	$159.10	$13.50	$7.42	$180.02	7.8%

The smaller your bill, the larger the share that is fixed, and the less a cut in use will move it. For a small apartment, fixed charges can be a large part of the bill. For a large home in July, they are a small slice.

Fixed charges are rising in many places. A Billix analysis of current residential rates in a federal database found a typical electric customer charge of about $12 a month, with the middle half between $9 and $17. That is our estimate, not an official figure. In the third quarter of 2025 alone, the NC Clean Energy

Technology Center counted 43 actions in 25 states and Puerto Rico on raising residential fixed charges or minimum bills by 10% or more. Some changes are paired with lower per-unit prices. California's three large utilities now charge a monthly base services charge of $6.00, $12.08 or $24.15 depending on income, with lower prices per kWh.

What you can control. You control the variable charges, through how much you use and, on some plans, when. You rarely control fixed charges. The exceptions are choosing a different rate plan where one is offered, owning your own equipment instead of renting it (see 7.8), and, occasionally, a water meter larger than the home needs.

Find it on your bill: every charge that does not say "per kWh", "per therm", "per CCF" or "per 1,000 gallons". Add them up. That is roughly what you would pay at zero use.

1.4 Billing periods

A bill covers a **billing period**, the days between two meter readings, and that period is rarely the same length twice. Before you compare two bills, divide each one by its number of days.

Most electric and gas bills cover about a month, but meter-reading dates move with weekends, holidays and reading routes, so one bill may cover 28 days and the next 33. Many water and sewer bills come every other month, and some water, sewer and trash bills come quarterly. There is no national rule on how long a period must be.

The length alone can move a bill a lot.

Illustrative example: same habits, different months

- A home uses 35 kWh a day.
- 35 kWh per day × 33 days = 1,155 kWh
- 35 kWh per day × 28 days = 980 kWh
- 1,155 kWh ÷ 980 kWh = 1.18, so the longer bill shows 18% more use with no change in habits.

The fix is to compare **use per day**:

- Bill 1: 900 kWh ÷ 30 days = 30 kWh per day
- Bill 2: 930 kWh ÷ 31 days = 30 kWh per day

Bill 2 shows more electricity, but the home used exactly the same amount each day. The extra 30 kWh is simply the extra day.

Do the same with money. **Cost per day** is the total due divided by the number of days. It is the fairest single number for comparing two bills, because it folds in every charge, fixed ones included.

For bills that arrive less often than monthly, convert to a yearly figure before comparing them with monthly bills. Add up the last 6 bills for an every-other-month service, or the last 4 for a quarterly one. For irregular bills, add up the last 12 months. Step 3 of the opening assessment in the Introduction shows how, and why multiplying one bill can mislead.

Find it on your bill: the service period dates and the number of days. Many bills print the day count next to the dates. If yours does not, count it from the dates.

1.5 Actual, estimated and smart-meter readings

Your bill is only as good as the reading behind it. An **actual reading** comes from your meter. An **estimated reading** is the provider's guess, usually based on your past use, made when the meter could not be read.

Smart meters send readings to the utility automatically, with no meter reader. About 84% of US electric meters were smart meters in 2024, up from about 43% in 2015. That still leaves about 23.5 million homes whose electric meters are read another way.

Regulator-ordered tests have generally found smart meters accurate. In 2010, independent tests of more than 750 smart meters for California's utility commission found them working correctly. The commission said the high bills that had prompted the tests coincided with a heat wave and rate increases. A smart meter can still produce an estimated bill when it stops reporting.

Why estimates matter. An estimate is not wrong in itself, and rules allow it when a meter cannot be reached. The trouble starts when estimates run low for several months. When the next actual reading comes, the bill catches up all at once. New York's regulator warns that this can bring "an extremely large bill", and its rules require the utility to make a special effort to get an actual reading after six months of estimates. Texas requires an actual reading at least every third month, and the word "Estimated" must appear prominently on the bill.

How far back a utility may bill you for under-billed use depends on your state, from 3 months to 4 years. In California, PG&E may back-bill a residential customer for only 3 months of undercharges caused by its own billing error. Florida's limit for a utility's mistake is 12 months, and Pennsylvania's is 4 years. New York generally bars charges for gas and electric service billed more than 3 months late. Ask your state commission for your rule.

How to check a reading yourself

1. Find the meter. Electric meters are usually outside; gas meters outside or in a basement; water meters in a basement, a utility closet or a box near the street.
2. Photograph the display with the date visible. Most newer meters show a number on a digital screen. If you have an older dial meter, your utility's website usually explains how to read it.
3. Find the "current reading" and "previous reading" on your bill. Current minus previous is the use you were billed for. If the bill lists a meter multiplier, multiply by it.
4. Compare your photo with the bill's current reading, allowing for the days between the two.
5. If the bill says the reading was estimated, or your reading is far off, call the provider and ask:
   - Was this reading actual or estimated?
   - When is the next actual reading?
   - Can you correct this bill using my reading or photo?

SAFETY WARNING: Read meters from the outside only. Never open a meter cover, break a seal or touch wiring or gas fittings. Tampering with a meter is dangerous and illegal.

Find it on your bill: the letter E, the abbreviation Est. or the word Estimated beside the reading, or a "reading type" marked actual or estimated.

1.6 Comparing bills correctly

The most common mistake is comparing this month's bill with last month's. Compare it with the **same month last year**, per day, and then check the price, because season, calendar and price can each move a bill more than your habits do.

Many bills print a 12- or 13-month usage chart for exactly this reason. Some states require it: Ohio requires 12 months of usage history on electric bills, and New York requires 13 months.

What changed	How it moves the bill	Where to check
Weather	Heating and cooling are the largest seasonal uses, so a colder January or hotter July raises use	Heating or cooling degree days, if your bill shows them, or local weather records
Number of days	More days, more use	The service period (see 1.4)
Price per unit	A rate change raises every unit you use	The rate column, or usage charges divided by units used
Fixed charges	A higher customer charge or a new rider adds the same amount each month	The fixed lines (see 1.3)
People at home	More people, or more time at home, means more use	Your own records
Equipment	A new appliance, a failing one, or a new load such as an electric vehicle	Your own records
Plan or provider	A promotion ending, a supplier contract renewing, a switch to time-of-use pricing	Messages on the bill, and your contract

EIA notes that the share of home electricity going to each use changes from year to year with the weather. It measures that weather in **degree days**, a count of how far each day's average temperature falls below or rises above 65 degrees F. More heating degree days means a colder period, and more cooling degree days a hotter one.

Case A: Sam's two March bills

Illustrative example. Sam is a fictional first-time renter in an all-electric one-bedroom apartment. Taxes are left out to keep the math simple.

Sam's March bill went up, so Sam assumed the effort to cut electricity use had failed. Here are the two bills side by side.

	Last March	This March
Days in the period	30	31
Electricity used	450 kWh	434 kWh
Use per day	15.0 kWh	14.0 kWh
Price per kWh	$0.17	$0.18
Customer charge	$11.00	$12.00
Total	**$87.50**	**$90.12**
Cost per day	$2.92	$2.91

The math for this March:

- 434 kWh × $0.18 per kWh = $78.12
- $78.12 + $12.00 customer charge = $90.12
- Use per day: 434 kWh ÷ 31 days = 14.0 kWh, down from 15.0, a 6.7% cut

Sam did use less: about 1 kWh less every day. The bill still rose $2.62 because the price per kWh went up a cent, the customer charge went up a dollar, and the period had one more day. Per day, Sam paid almost exactly what Sam paid a year earlier.

What was the cut worth? If Sam had kept last year's 15.0 kWh a day:

- 15.0 kWh × 31 days = 465 kWh
- 465 kWh × $0.18 = $83.70
- $83.70 + $12.00 = $95.70

- $95.70 - $90.12 = $5.58

Sam's lower use saved about $5.58 this month, even though the bill went up. Section 1.8 explains why that figure is labeled Estimated rather than Verified.

Find it on your bill: the usage chart, and last year's bill for the same month. Write down the days, the use and the total for each.

1.7 Warning signs, and what to do about them

Most odd bills have an ordinary cause, but eight signs are worth checking every time. Each points to a different first step.

Warning sign	What it can mean	Check first
A sudden spike with no change in habits	A catch-up after estimates, a longer period, a rate change, a leak or failing equipment	Reading type, days, price per unit, then the home itself
The same charge twice	A billing error, or a payment not applied	Last bill's payments and credits
A late fee you did not expect	A payment posted late or to the wrong account, or a moved due date	Your payment confirmation and the due date
A service you do not recognize	An add-on, a protection plan or a third-party charge	Your account's list of services (see 10.2)
A price that jumped	A rate case, a new rider or a supplier contract renewing	Bill messages, inserts and your contract
The wrong name, address or meter number	You may be billed for someone else's meter	The meter number on the bill against the one on your meter
A reading that does not match your meter	A misread or an estimate	Your dated meter photo (see 1.5)
Service dates that overlap or skip	A problem from a move or account change	Your move-in or move-out date (see 11.5)

Building your evidence

Before you call, gather what supports your case: the bills in question and the same months last year, a dated photo of the meter, payment receipts, and any letters or inserts. During the call, write down the date, the representative's name and any reference number. The Provider Call Log (A2) has space for all of it.

Where to take it

1. **The provider first.** Ask for an explanation of the charge, and for a correction if it is wrong.
2. **A supervisor or the provider's complaints team,** if the first answer does not resolve it.
3. **The right outside body.** For an investor-owned utility, that is your state utility commission. For phone, internet or TV, the FCC accepts complaints. For a city utility or co-op, it is usually the board or council. Section 10.8 has the full map.

Keep paying the part of the bill you do not dispute. In some states, such as Illinois, late fees apply only to amounts that are not in dispute.

Case D: the Harris household finds the real problem

Illustrative example. The Harrises are a fictional retired couple on a fixed income. One of them uses powered medical equipment at home.

The Harrises' electric bills had seemed high for months, and they assumed the rates were the problem. Running through the eight signs took ten minutes. Their usage was in line with the same months last year, and every reading was actual. What had grown was a past-due balance of $420, with a new late fee every month.

Late-fee rules vary by state. Pennsylvania, Illinois and New York, for example, allow up to 1.5% a month on a past-due balance:

- $420 × 0.015 = $6.30 a month
- $6.30 × 12 months = $75.60 a year, if the balance never shrank

The problem was not the price of electricity but the unpaid balance and the fees it generates. That changes what to do: ask the utility about a payment arrangement (10.4, 12.5), ask whether medical-needs protections apply to their account (12.8), and check which assistance programs they might qualify for (12.1 lists them). Their case continues in Chapter 12, after 12.8.

SAFETY WARNING: In most states, a shutoff must be preceded by a separate written notice, not just a line on your bill. If you receive one, call the provider the same day. If anyone in the home depends on powered medical equipment, say so on that call. Many states offer extra protection, but often only if you ask.

Find it on your bill: the past-due balance, any late fee, and the meter number. Check that the meter number matches the one on your meter.

1.8 How to measure a saving honestly

A saving is real only if you measure it the right way. Use the price that actually changes, leave fixed charges out, count each reduction once, adjust for weather and days, and say how sure you are. The rest of this book follows these five rules, and you should hold any savings claim, including ours, to the same standard.

Rule 1: Using less is not the same as paying less

Cutting use by 10% does not cut the bill by 10%, because fixed charges stay.

Illustrative example, using the Billix example rate card: $0.18 per kWh plus a $12.00 customer charge, taxes left out.

- Before: 600 kWh × $0.18 = $108.00, plus $12.00 = $120.00
- After a 10% cut: 540 kWh × $0.18 = $97.20, plus $12.00 = $109.20
- Saving: $120.00 - $109.20 = $10.80, which is 9.0% of the bill

A 10% cut in use produced a 9% cut in the bill. The more of your bill is fixed, the bigger that gap.

Rule 2: Use the marginal price, not the average

The **marginal price** is what one more unit costs you, or what one fewer saves. It is usually not your total bill divided by your use, because that average includes the fixed charges.

Bill saving = units saved × marginal price per unit

On the Anytown sample bill, the average is $159.11 ÷ 874 kWh = $0.182 per kWh. But each kWh actually adds about $0.166, all per-kWh charges plus taxes. Using 100 kWh less saves about $16.59, not $18.20.

On a **tiered** plan, the marginal price can be higher than the average.

Illustrative example: the first 500 kWh cost $0.16 each, every kWh above that costs $0.20, plus a $12.00 customer charge.

- 500 kWh × $0.16 = $80.00
- 200 kWh × $0.20 = $40.00

- $80.00 + $40.00 + $12.00 = $132.00 for 700 kWh, an average of about $0.189 per kWh
- Cutting 100 kWh removes 100 of the top-tier kWh: 100 × $0.20 = $20.00 saved

Here the average would have understated the saving. Find your marginal price on your rate schedule, or add up every per-unit rate on your bill.

Rule 3: Count each reduction once

One change can save several things. A shorter shower saves water, sewer charges if your sewer is billed on water use, and the energy to heat that water. Those are three real, separate savings. Counting the water-heating energy again under a different name, such as "a lower water-heater bill", counts the same saving twice. Keep one line for each thing you buy: electricity, gas, water and sewer.

Rule 4: Adjust for weather and days before taking credit

A lower bill after a change does not prove the change worked. The weather may simply have been milder.

Illustrative example: Case B, the Okafor family. The Okafors are a fictional family of four in a 1962 house with a gas furnace. Gas costs $1.30 per therm.

	Last January	This January
Gas used	150 therms	135 therms
Heating degree days	1,000	850
Gas per degree day	0.150 therms	0.159 therms
Usage charges at $1.30 per therm	$195.00	$175.50

The bill fell $19.50, but the house used about 6% more gas for each degree day of cold. This January was milder, and the mild weather hid a problem, perhaps a clogged filter or a thermostat schedule that no longer fits. Section 4.7 picks up from here.

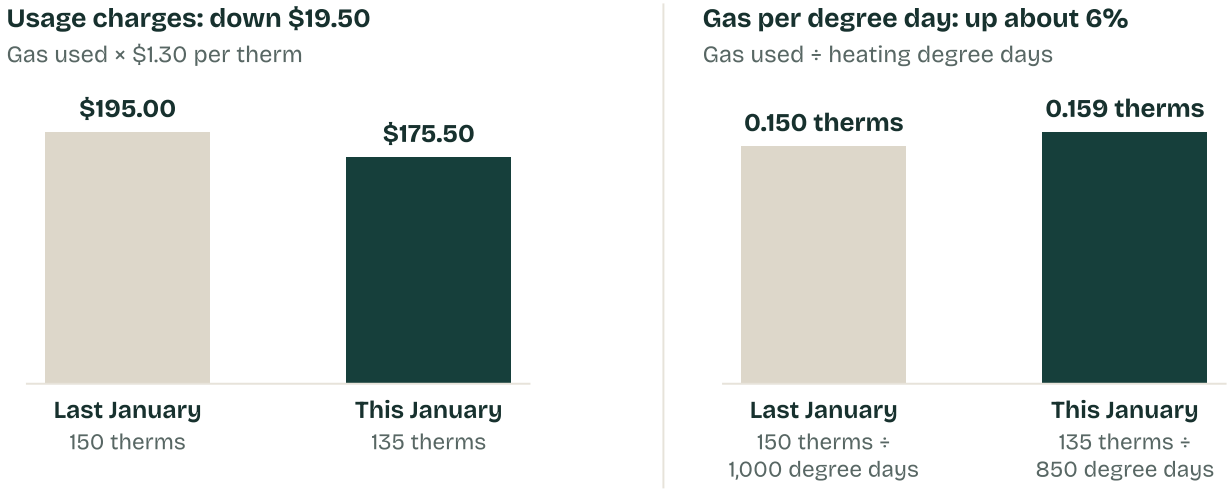


Figure 1.3. Use per degree day: the Okafors' usage charges fell, but their gas use per degree day rose. Illustrative: Case B, gas at $1.30 per therm.

Getting your degree days: Look on your bill first. Some gas bills print the heating degree days for the billing period. If yours does not, NOAA publishes them free. The sources below count from a base of 65 degrees F. Use heating degree days for a heating bill and cooling degree days for a cooling bill.

1. **For your exact billing dates:** go to weather.gov/wrh/climate, or your local National Weather Service office's climate page, open NOWData and pick the station nearest your home. Choose "Daily data for a month" and add up the degree days for the days your bill covers. If the bill spans two months, do both months.
2. **For a quicker, rougher figure:** NOAA's Climate Prediction Center posts weekly and monthly totals for about 200 cities and for each state. A monthly state total does not match your dates or your town.
3. **Use one source for both bills.** Take last year's figure from the same station or the same table. The figures are preliminary, and different NOAA products can give somewhat different totals.

If you cannot get them, compare use per day for the same month last year, and treat the result as rough.

Rule 5: Say how sure you are

Every saving in this book carries one of three labels. Use them in your own records too.

Label	Meaning	Example
Illustrative	Made-up numbers that show how the math works	The Anytown sample bill
Estimated	Your own bills, plus reasonable assumptions, not yet confirmed	Sam's $5.58 in section 1.6, based on one bill and no weather adjustment
Verified	Confirmed by your own bills over several billing periods, after adjusting for billing days, weather and price, with no other big change in the home. For a flat-price service, one full bill at the new price is enough	Use per degree day that stays lower through a whole winter

An honest record of small, verified savings is worth more than a list of large estimated ones. The 90-Day Household Savings Scorecard (B4) keeps the two in separate columns.

The Billix 15-Minute Bill Audit

This audit turns the whole chapter into one routine you can run on any metered bill. You need this bill, the same month's bill from last year (or the usage chart), last month's bill, a calculator and a pen. Compare use with last year, as 1.6 explains. Last month's bill shows a new charge, a price change or a catch-up reading as soon as it starts.

Purpose: to find out whether a bill changed because of your use, the price, the calendar or a mistake.

Instructions

Do the steps in order. A wrong reading, extra days, a new price or a new fixed charge can explain a change before you look for a cause in your home.

- ☐ **Minutes 1 to 2: confirm the account (1.7).** Check the name, service address and account number. Check that the meter number matches the number on your meter.
- ☐ **Minutes 3 to 4: check the reading (1.5).** Is it actual or estimated? Write down the current and previous readings. If last month's reading was estimated and this one is actual, part of this bill may be a catch-up.
- ☐ **Minutes 5 to 6: put it on a per-day basis (1.4).** Count the days. Divide the use by the days, and the total due by the days. Do the same for last year's bill.
- ☐ **Minutes 7 to 8: find the price (1.8, Rule 2).** Add up the usage charges and divide by the units used. Compare with last year. On a tiered bill, compare each tier's rate too. If none changed, a higher average price comes from more of your use falling in a higher tier, not from a price rise.
- ☐ **Minutes 9 to 10: total the fixed charges (1.3).** Add up every charge that is not per unit. Did any rise since last year, or appear for the first time since last month?

- ☐ **Minutes 11 to 12: split the change.** Put a dollar figure on the days, the price, the fixed charges and your use, with the box below.
- ☐ **Minutes 13 to 14: look for warning signs (1.7).** Late fees, a past-due balance, duplicate charges, services you do not recognize, missing credits.
- ☐ **Minute 15: read the messages.** Look for notices of rate changes, promotions ending or plan renewals.

Split the change: Put a dollar figure on each cause, using this bill and last year's. Leave taxes out. A part can be a minus number.

1. Days: the extra days × last year's use per day × last year's price.

2. Price: last year's use per day × this bill's days × the change in price per unit.

3. Use: the change in use per day × this bill's days × this bill's price.

4. Fixed charges: this bill's fixed charges minus last year's.

The four parts add up to the change in charges before taxes. If the change in the amount due differs from them by more than the taxes and a few cents of rounding, look for a credit, a late fee, a past balance or an error (1.7).

Illustrative check, Sam's bills in 1.6: days, 1 × 15.0 kWh × $0.17 = +$2.55; price, 15.0 kWh × 31 × $0.01 = +$4.65; use, −1.0 kWh × 31 × $0.18 = −$5.58; fixed, $12.00 − $11.00 = +$1.00. Together: +$2.62, the whole rise from $87.50 to $90.12. The use part matches the $5.58 that Sam's lower use saved.

Record sheet

The blank record sheet is printed once, as B2 in Appendix B, because Chapters 10 and 14 use the audit too. It is also in the companion worksheet pack. It has a row for each line you check, with columns for this bill, last month's bill and the same month last year.

Reading your results

What you found	What it points to	Where to go next
Use per day rose 5% or more	Something in the home: weather, people, equipment or a leak	2.4 (electricity), 3.3 (gas), 5.3 (water) or 6.3 (sewer). For weather, 1.8, Rule 4
Price per unit rose	A rate change, a rider or a plan or contract change	2.3 and 2.11 (electricity), 3.2 (gas) or 5.2 (water). For how rates change, 10.4
Only the number of days changed	Nothing is wrong	No action needed
The reading was estimated	A possible catch-up bill later	Section 1.5
A charge you cannot explain	A possible error or unwanted service	10.2, then 10.3
A past-due balance or growing late fees	A payment problem, not a price problem	10.9 and 12.5. With a shutoff notice, 12.8

If more than one row applies, take them in the audit's order: reading, days, price, fixed charges, then use. The 5% line for use per day is a Billix rule of thumb, not an official figure. A smaller change is worth noting and checking again on the next bill.

Run the audit on every bill for three months. After that it goes faster, because you will know what normal looks like for your home.

Chapter 1 review

What you learned

- Who owns your provider decides who sets the price and where you take a problem.
- Every metered bill follows the same seven steps, from service to amount due.
- Fixed charges stay even at zero use. The smaller your bill, the larger their share.
- Billing periods vary, so compare use per day and cost per day, against the same month last year.

- An estimated reading is legal but can lead to a catch-up bill. You can check your own meter.
- A saving is worth the units saved times the marginal price, counted once, after allowing for weather and days.

What you can do today

1. Find the fixed charges on your latest electric, gas or water bill and add them up.
2. Check whether your last reading was actual or estimated.
3. Photograph your meter with the date showing.
4. Run the 15-Minute Bill Audit on one bill.

What to check next month

When the next bill arrives, compare your meter photo with the bill's reading, and record use per day and cost per day next to this month's figures. If either moved by 5% or more, run the audit and work through its results table.

Billix Quick Check

Your electric bill rose from $96.00 to $104.00. The old bill covered 30 days and 540 kWh. The new one covers 33 days and 594 kWh. Did your daily use go up?

Answer: No. 540 kWh ÷ 30 days = 18 kWh a day, and 594 kWh ÷ 33 days = 18 kWh a day. Your cost per day actually fell slightly, from $3.20 to about $3.15. The higher bill came entirely from the three extra days.

Sources and further reading

Last Verified: October 2026. The fastest-changing facts in this chapter are state rules on back-billing, late fees and estimated readings, California's base services charge, the national count of fixed-charge cases, and FPL's rates. Check your own state and utility before you rely on them. The degree-day steps in 1.8 were checked against NOAA's and EIA's pages in October 2026.

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PART II

Energy

2 Electricity

3 Natural Gas, Propane, Heating Oil and Other Fuels

4 Heating, Cooling, Insulation and Home Efficiency

2

PART II · ENERGY

Electricity

Electricity is the one bill nearly every American home pays, and the one with the most moving parts. The average US home used 874 kWh a month in 2025 and paid $150.68, an all-in average of about 17 cents per kWh. EIA expects that price to rise about 5% in 2026.

Two numbers decide your bill: how many **kilowatt-hours** you use, and what each one costs you. This chapter teaches both. The first half explains how electricity is priced and billed. The second half goes appliance by appliance, showing which changes are worth making and which are not.

By the end of this chapter you will be able to:

- Convert any appliance's watts into kWh and dollars
- Read every line of an electric bill, including tiers, seasons, time-of-use and demand charges
- Find the loads that actually drive your bill, and measure them
- Judge whether a new bulb, refrigerator or rate plan will pay for itself
- Compare an electricity supplier's offer with your utility's price, where you have a choice
- Spot energy-saving products that cannot work

Most calculations use the Billix example rate card from the Introduction: $0.18 per extra kWh in the middle case, with $0.13 and $0.26 as low and high cases. Swap in your own marginal price (see 1.8).

START HERE:

- → **Your bill jumped?** Go to 2.4, then Case B.
- → **A time-of-use offer arrived?** Go to 2.10.
- → **A supplier is at the door or on the phone?** Go to 2.11.
- → **Wondering what one appliance costs?** Go to 2.5.

2.1 From power plant to outlet

Electricity takes four steps to reach your outlet, and your bill pays for every one. Knowing them makes sense of the supply and delivery sections on a bill.

Step	What happens	Share of the average US price, 2025
Generation	Power plants, wind farms and solar farms produce electricity	About 56%
Transmission	High-voltage lines carry it long distances to local substations	About 14%
Distribution	Local poles, wires and transformers bring it down your street to your home	About 30%
Metering and service	Your meter records use; the utility bills you and answers your calls	Mostly in fixed charges

Those shares average all customers, including factories and offices. Homes take power at a lower voltage, through more local equipment, so they usually pay a larger share for distribution than the table shows.

Delivery costs are where the money has been growing. Large utilities spent $50.9 billion on local distribution equipment in 2023, 160% more than in 2003, to replace aging poles and lines and to harden them against storms and wildfires. Spending on generation fell over the same period, mainly because fuel got cheaper.

In most of the country, one utility handles all four steps and sends one bill. In states with electricity choice, the utility still delivers the power, but you can buy the electricity itself from another company (see 2.11). Either way, the same wires bring it to your home.

2.2 Watts, kilowatts and kilowatt-hours

Your bill charges for **kilowatt-hours (kWh)**, not watts, and mixing the two up is the most common electricity mistake. Think of watts as a car's speed and kWh as the distance driven.

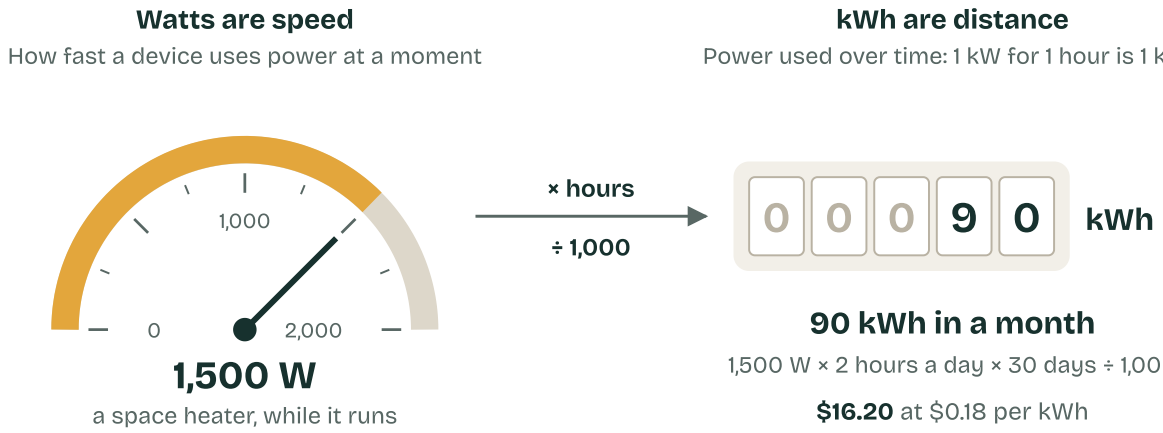


Figure 2.1. Watts versus kWh: watts are how fast a device uses power, kWh how much it has used. Illustrative: a 1,500-watt space heater used 2 hours a day for 30 days.

Term	What it measures	Everyday comparison
Watt (W)	How fast a device uses power at a moment	Speed
Kilowatt (kW)	1,000 watts	Speed in bigger units
Kilowatt-hour (kWh)	Power used over time: 1 kW running for 1 hour	Distance
Volt (V)	The electrical "push". Most US outlets are 120 V; dryers and electric cooktops use 240 V	Water pressure in a pipe

To turn any appliance into kWh and dollars:

> **kWh = watts × hours of use ÷ 1,000**

If a label lists only amps, multiply amps by volts to get watts.

Illustrative example: a 1,500-watt space heater used 2 hours a day for 30 days

- 1,500 W ÷ 1,000 = 1.5 kW
- 1.5 kW × 2 hours = 3 kWh per day
- 3 kWh × 30 days = 90 kWh per month
- At the rate card's low, middle and high prices: 90 kWh × $0.13 = $11.70; × $0.18 = $16.20; × $0.26 = $23.40 a month

This assumes the heater draws its full 1,500 W the whole time. Most heaters cycle off once the room warms, so real use is often lower.

Nameplate watts are a maximum, not an average. A typical refrigerator label might say 725 W, and a common rule of thumb says a fridge runs about a third of the time. That gives 725 W × 8 hours = 5.8 kWh a day. But EIA's survey of real homes found the most-used refrigerator averages about 620 kWh a year, or about 1.7 kWh a day. For appliances that cycle, **measure** rather than calculate (see 2.5).

Small devices add up only when they run for many hours. A 9-watt LED bulb, the usual replacement for an old 60-watt bulb, left on 5 hours a day uses about 16 kWh a year. That is about $3 at $0.18 per kWh.

2.3 Anatomy of an electric bill

An electric bill can carry a dozen kinds of charge. You can influence only three of them: how many kWh you use, when you use them on some plans, and, in some states, who supplies them. The Anytown sample bill in 1.2 showed the lines in place. Here is what each one means.

Charge	What it is	Real example	Can you influence it?
Customer charge	A fixed monthly amount for the meter, billing and connection	$10.52 at Florida Power & Light (October 2026); about $12 is typical (Billix estimate)	Rarely: only by choosing a different plan
Supply or generation charge	The electricity itself, per kWh	Illinois' largest utility's supply price is 8.413 cents per kWh from October 2026 to May 2027	By use; in choice states, also by supplier
Transmission and distribution (delivery)	The wires, per kWh and sometimes partly fixed	Oncor, in Texas: $1.48 customer charge, $2.58 metering charge and 6.03 cents per kWh	By use only; you cannot switch delivery company
Riders and adjustments	Separate charges for fuel, storm repairs, efficiency programs and more	FPL's fuel charge is 2.893 cents per kWh; Central Maine Power's storm charge was 2.5 cents in 2025	Only by using less; they can rise or fall
Taxes and fees	State and local taxes, franchise fees, regulatory fees	Usually a few percent of the bill	No
Budget billing	Your yearly cost spread into equal payments, settled once a year	Offered by many utilities	It changes when you pay, not how much
Aggregation	A city or county buys supply for its residents as a group	Most of California's 5.8 million choice customers are in community programs	Check whether your area has one, and whether you can leave it
Green or renewable plans	Optional plans that cost more to fund renewable energy	Varies by utility and supplier	Your choice; read what the premium actually pays for

When the price depends on how much, or when

Some plans change the price per kWh as you use more, by season or by hour. These change your marginal price, the number that decides what any saving is worth (see 1.8).

- **Tiered rates** charge more after a set amount. At Florida Power & Light, every kWh past 1,000 a month costs about 16% more than the first 1,000.
- **Seasonal rates** charge more in summer. Georgia Power charges 15.1 cents for summer use above 1,000 kWh, against 8.2 cents for any use in winter. That means a hot July raises both your use and your price.
- **Time-of-use rates** charge more at peak hours, usually late afternoon and evening. About 18.4 million US homes, roughly 13%, were in time-based programs in 2025, counting peak-time rebates. Section 2.10 covers these.

Demand charges

A few utilities offer residential plans that also charge for your **demand**: your highest burst of use in the month, measured in kW, not kWh. If you see a charge "per kW", this is it.

Arizona Public Service's R-3 plan, for example, charges $19.585 per kW in summer, based on your single highest hour between 4 and 7 p.m. on weekdays. Georgia Power has a similar optional plan, but it counts your highest hour at any time of day, not just at peak. SRP has one for customers with solar.

Illustrative example using APS's published summer rate. Energy charges on this plan are billed separately.

- One weekday at 6 p.m., the air conditioner, dryer and oven run together, using 8 kW for an hour.

- 8 kW × $19.585 = $156.68 demand charge for the month
- If the dryer and oven wait until after 7 p.m., the peak drops to 5 kW.
- 5 kW × $19.585 = $97.93
- The difference is $58.75 for the month, with no change in total use.

On a demand plan, *when* big appliances run together matters as much as how much electricity you use. One careless hour can set the charge for the whole month. APS's rates are under review in a pending rate case, so check the current tariff.

Find it on your bill: the name of your rate plan or schedule. Then look it up on your utility's website. The rate schedule tells you whether you are on tiered, seasonal, time-of-use or demand pricing.

2.4 Why electric bills rise

A higher electric bill comes from more kWh or pricier kWh, once you allow for extra days and fixed charges. "Split the change" in the 15-Minute Bill Audit (after 1.8) puts a dollar figure on each; Case B below does the same on a tiered bill. Weather, people at home, new equipment and plan changes are in 1.6's table. This section lists the causes particular to electricity.

When you used more

Cause	Typical sign	How to confirm
A second refrigerator or freezer	A small, steady year-round rise	A plug-in meter (2.5)
A dehumidifier, space heater or pool pump	Big seasonal jumps	Run time and watts (2.2)
Failing or neglected equipment	Use creeps up with no change in habits	A clogged filter or an aging AC; see 4.7
A stuck water heater element or a hot-water leak	A sudden rise that runs day and night	Hourly data showing use with nobody home

In 2020, air conditioning was about 19% of US home electricity use, space heating 12% and water heating 12%. EIA notes those shares shift from year to year with the weather. Section 2.5 gives the typical yearly use of each appliance.

When each kWh cost more

Cause	Typical sign	How to confirm
A base-rate increase	The per-kWh rate or customer charge rises, usually after a notice	Bill messages and inserts; your state commission's website
A rider or fuel adjustment	A line changes, often several times a year	Compare the rider lines with last year's bill
A tier or season	The same use costs more in summer, or above a threshold	Your rate schedule (2.3)
A plan change	A promotion ends, or a supplier contract rolls over to a variable rate	Your contract and renewal notices (2.11)

Rate requests are at record levels. PowerLines, a consumer-education nonprofit, says electric and gas utilities requested nearly $31 billion in rate increases in 2025, double the 2024 total. Those are requests, and commissions usually approve less.

Sidebar: download your own data. If you have a smart meter, your utility's website often shows your use hour by hour. DOE notes that smart-meter customers can usually see "hourly consumption on a nearly real-time basis" in their online account. Look at a night when everyone is asleep. That low point is your home's always-on load, and if it is high, something is running that you may not know about. Many utilities also let you download the data as a spreadsheet.

2.5 Appliance by appliance

The appliances that heat or cool things, whether air, water or food, dominate most electric bills. Gadgets rarely do. The table shows EIA's averages from its 2020 survey of real US homes, counting only homes that have each appliance, from largest to smallest.

Appliance	Typical use a year (kWh)	About a month	A year at $0.18 per kWh	Best first step
Electric water heating	2,705	225	$487	Use less hot water; check the temperature setting (5.11)
Electric space heating	2,485	207	$447	Thermostat and air sealing (4.4, 4.5)
Pool pump	2,380	198	$428	Ask a pool professional for the shortest run time that keeps the water safe
Electric vehicle charging	2,360	197	$425	Charge off-peak if your plan rewards it (2.10)
Air conditioning	2,320	193	$418	Thermostat, filter and maintenance (4.4, 4.7)
Hot tub heater	1,480	123	$266	Keep the cover on; lower the temperature when it is not in use
Dehumidifier	1,000	83	$180	Run it to a humidity target, not all the time
All refrigerators in the home	840	70	$151	Check whether a second fridge is needed (2.8)
Electric clothes dryer	680	57	$122	Full loads, air drying (2.9)
Lighting	654	55	$118	LEDs where bulbs are used most (2.6)
TVs, game consoles, streaming boxes and audio	633	53	$114	Standard picture mode; stream on a laptop or tablet
Separate freezer	570	48	$103	Check the door seal and temperature (2.8)
Cooking (range and oven)	290	24	$52	Rarely worth targeting
Ceiling fans	226	19	$41	Switch off in empty rooms (4.8)
Dishwasher (machine only)	111	9	$20	The hot water costs more (2.9)
Microwave	106	9	$19	Rarely worth targeting
Clothes washer (machine only)	67	6	$12	The hot water costs more (2.9)

These are averages, and your home may differ a lot. Air conditioning alone averages about 4,280 kWh a year in hot, dry parts of the Southwest, and about 1,310 in cold climates. Real use depends on the model, its age, its settings, how full it is, and the room it sits in. Dollar figures are illustrative at $0.18 per kWh.

A few appliance facts are worth knowing:

- **TVs and consoles.** DOE says switching a TV to "home" or "standard" picture mode can cut its energy use 18% to 30%. It also says game consoles use about 10 times more power to stream video than a tablet or laptop does.
- **Computers.** An older federal nameplate list puts a laptop at about 50 W and a desktop at about 120 W when awake. Modern devices vary, so measure your own. Both use far less asleep, so sleep settings matter more than shutting down.
- **Modems and routers.** These run all day, but we found no reliable federal figure for their use, so measure yours if you are curious. Do not switch them off if anyone in the home depends on the connection for work, health or safety.
- **Space heaters.** At 750 to 1,500 W, one space heater can rival a refrigerator in a single month of evenings. It makes sense only for heating one room while the rest of the house stays cool.

How to measure an appliance

1. **Use a plug-in electricity monitor.** Plug the monitor into the outlet, then plug the appliance into the monitor. Many libraries and utility programs lend them free; most Maine public libraries do, for ex-

ample. Most plug-in monitors work only with standard 120-volt outlets, so dryers, ranges and central AC need another method.

2. **Run it long enough.** Measure appliances that cycle, like refrigerators, dehumidifiers and freezers, for at least a full day, then multiply the daily kWh by 365.
3. **Use your hourly data for big loads.** Switch the appliance off, or on, for a few hours and watch the change in your utility's hourly chart.
4. **Use the label as a ceiling.** Nameplate watts times hours gives the most it could use, not what it does use (2.2).

Reading the EnergyGuide label

The yellow EnergyGuide label, required by the Federal Trade Commission, shows an estimated yearly energy cost. It is required on refrigerators, freezers, dishwashers, water heaters, clothes washers, room and portable air conditioners, TVs, ceiling fans and pool heaters, but not on dryers, ranges, microwaves or dehumidifiers.

Most labels made since 2022 assume **14 cents per kWh**, and labels for room and portable air conditioners 13 cents. Both are below today's national average, and older labels may use a different price. Rescale the label's cost to your own price:

- A label says $60 a year.
- $60 × ($0.18 ÷ $0.14) = about $77 a year at $0.18 per kWh.

Use the label to compare two models on the same terms, not to predict your bill.

2.6 Lighting

Swapping an old incandescent bulb for an LED is one of the few upgrades that pays for itself in months. Swapping a bulb that is rarely on, or one that is already efficient, often is not worth it until the old one fails. Where the bulb is matters more than the bulb.

Buy by lumens, not watts

Lumens measure brightness; watts measure power. LEDs give the same light for far fewer watts, so match the lumens of the bulb you are replacing.

Old incandescent	Brightness to look for
40 W	450 lumens
60 W	800 lumens
75 W	1,100 lumens
100 W	1,600 lumens
150 W	2,600 lumens

An 800-lumen LED uses only about 9 watts, according to the FTC. ENERGY STAR says certified LEDs use up to 90% less energy than incandescents for the same brightness, and DOE says they last 15 to 25 times longer.

The FTC's **Lighting Facts** label on every bulb package shows the lumens, estimated yearly energy cost, expected life, watts and **color temperature**. Color temperature tells you how warm or cool the light looks: 2,700 to 3,000 K is warm, like an old incandescent; 3,500 to 4,100 K is cool white; 5,000 to 6,500 K is daylight. If the bulb will go on a dimmer switch or in a closed fixture, check that the package says it is suitable.

Most incandescent bulbs can no longer be sold. A federal efficiency standard took effect in 2022, with full enforcement in 2023, and a stricter one follows in 2028. Many homes still have older bulbs in lamps and closets, though, and those are where the savings are.

Does a new bulb pay for itself?

Illustrative examples at $0.18 per kWh, assuming a $3.00 LED.

A: a 60 W incandescent in a living-room lamp, on 4 hours a day

- Old bulb: 60 W × 4 hours × 365 days ÷ 1,000 = 87.6 kWh a year
- LED: 9 W × 4 hours × 365 days ÷ 1,000 = 13.1 kWh a year
- Saved: 74.5 kWh × $0.18 = $13.40 a year ($9.68 at $0.13, $19.36 at $0.26)
- Payback: $3.00 ÷ $13.40 = about 3 months

B: the same bulb in a closet, on 10 minutes a day

- Saved: about 3.1 kWh a year, worth $0.56
- Payback: more than 5 years. Replace it when it burns out.

C: a working 14 W compact fluorescent (CFL) in the living-room lamp

- Saved by switching to a 9 W LED: 5 W × 4 hours × 365 days ÷ 1,000 = 7.3 kWh, worth $1.31 a year
- Payback: about 2.3 years. Reasonable, but there is no hurry. Replacing an LED with a newer LED saves nothing worth counting.

The lesson is to start with the bulbs that are on the most hours. DOE estimates that replacing a home's five most-used bulbs or fixtures with ENERGY STAR products saves about $75 a year.

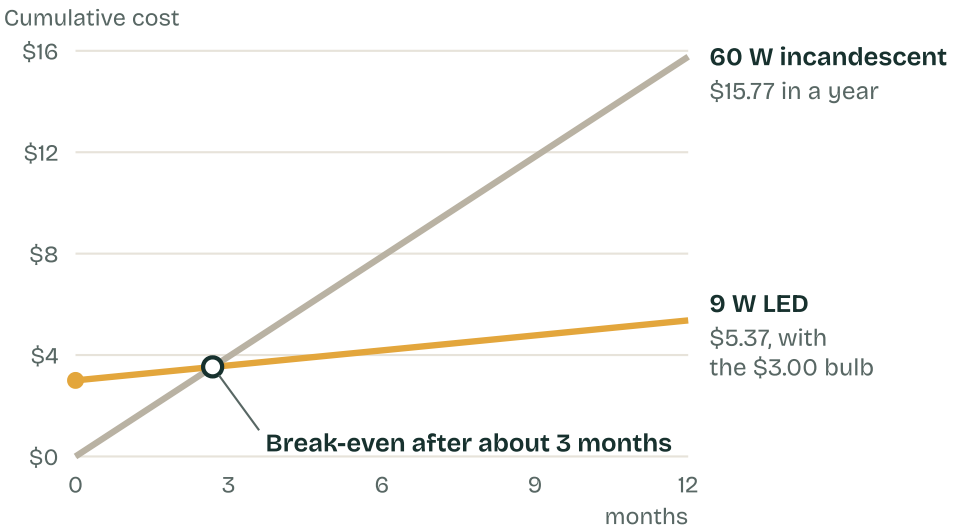


Figure 2.2. LED payback: the cumulative cost of a 60 W incandescent and a $3.00, 9 W LED, on 4 hours a day at $0.18 per kWh. Illustrative: scenario A in 2.6.

What to record: each bulb you change, its old and new watts, and its hours a day. Lighting savings are too small to see on a bill, so count them as Estimated, not Verified (1.8).

2.7 Standby power

Standby power is electricity used by devices that are switched off or asleep but still plugged in. It is real but usually modest. The goal is to find the few devices that draw a lot, not to unplug everything.

Estimates vary. DOE says always-on electronics could add 5% to 10% to a monthly bill. Lawrence Berkeley National Laboratory has put standby at about 5% of US home electricity. NRDC, an environmental group, says always-on "idle" load averaged about 23% of household electricity in a 2015 study of northern California homes. That figure counts everything running all the time, not just standby.

The math shows why device size matters more than device count.

Illustrative examples: devices drawing power around the clock, at $0.18 per kWh.

Always-on draw	Yearly use	Yearly cost
0.5 W (the most an ENERGY STAR TV may use asleep)	4.4 kWh	$0.79
5 W	43.8 kWh	$7.88
30 W	262.8 kWh	$47.30

The math for the last row: 30 W × 24 hours × 365 days ÷ 1,000 = 262.8 kWh, and 262.8 kWh × $0.18 = $47.30.

Unplugging a phone charger saves pennies. Finding one forgotten 30-watt device saves real money.

Finding the big ones

1. **Look at your overnight use.** In your utility's hourly data (2.4), find a quiet hour around 3 a.m. That figure, in kW, is your home's always-on load. Subtract what you know must run, such as the refrigerator.
2. **Measure suspects with a plug-in monitor.** Common culprits are older set-top boxes, game consoles left in an "instant-on" mode, desktop computers that never sleep, and old audio equipment.
3. **Fix the settings first.** Turn on sleep and power-saving modes, and switch consoles to an energy-saving mode where offered.
4. **Use a smart power strip** for a TV or computer setup. It cuts power to accessories when the main device is switched off.

SAFETY WARNING: Never switch off or unplug refrigerators and freezers, medical equipment, smoke or carbon monoxide alarms, security systems, sump pumps or thermostats to save power. Keep the modem and router on if anyone in the home relies on them for work, health monitoring or emergency calls. The savings are not worth the risk.

What to record: your overnight always-on load from hourly data, before and after any change. If it drops and stays down, you have a Verified saving.

2.8 Refrigerators and freezers

Refrigerators run around the clock, so small problems add up, but replacing a working fridge for energy savings alone rarely pays. The biggest win is usually an old second fridge you do not really need.

A second refrigerator averages about 495 kWh a year in US homes that have one (EIA). That is about $89 a year at $0.18 per kWh, or $64 to $129 across the rate card's range. A second fridge in a hot garage works harder than one indoors. DOE suggests recycling an old second fridge, especially one in an unheated or uncooled space. Some utilities offer pickup or recycling programs, so ask yours.

Settings and upkeep

Check	What to do	Why
Temperature	Fridge at 35 to 38 degrees F, freezer at 0 degrees F (DOE)	Colder than needed wastes energy; warmer than 40 degrees F is unsafe for food
Door seal	Close the door on a sheet of paper. If it slides out easily, clean the gasket or have it replaced	A leaky seal makes the fridge run longer
Coils	Clean them as the owner's manual directs	Dusty coils shed heat poorly
Placement	Keep it away from the oven and direct sun, with the airflow space the manual specifies	Heat around the fridge makes it work harder
Door opening	Keep it brief	Cold air escapes, though the effect is small on a bill

Replace it or keep it?

ENERGY STAR says a refrigerator over 15 years old could cost about $95 a year more to run than a new certified model. Separately, it says a certified new model saves about $150 over a 12-year life compared with a new model that only meets the federal minimum. Here is how to run the numbers for your own fridge.

Illustrative example: an old fridge, measured with a plug-in monitor for a week, against a new model's EnergyGuide label.

- Old fridge: 1.9 kWh a day × 365 = 694 kWh a year
- New model's label: 400 kWh a year
- Saved: 294 kWh × $0.18 = $52.92 a year
- Payback on a $1,200 fridge: $1,200 ÷ $52.92 = about 23 years

That is longer than a fridge lasts. Replacing a working fridge for energy alone does not pay here. When the old one fails, though, the running cost of the new one becomes a fair part of the choice. Section 13.3 shows how to compare models on total cost.

What to record: a week of plug-in monitor readings for each fridge and freezer. If you remove a second fridge, its yearly kWh is your Estimated saving. Confirm it against next year's bill for the same months.

2.9 Laundry and dishwashing

In laundry and dishwashing, the machines themselves use little electricity. The cost is in heat: hot water in the washer and dishwasher, and hot air in the dryer.

Washing

DOE says most of the energy a clothes washer uses goes to heating water, and that washing in warm instead of hot water can cut a load's energy use in half. Cold washes save more. ENERGY STAR washers use about 25% less energy and 33% less water than standard models.

That saving shows up on whichever bill heats your water: electric, gas or propane. Count it there, once (1.8, Rule 3).

Drying

An electric dryer averages about 680 kWh a year in homes that have one (EIA), about $122 at $0.18 per kWh. Dryers are one of the few large loads you can cut to near zero.

Illustrative example: air-drying half of all loads

- 680 kWh × 50% = 340 kWh saved a year
- 340 kWh × $0.18 = $61.20 a year ($44.20 at $0.13, $88.40 at $0.26)

This assumes every load uses about the same energy. Air drying needs space and time, and may not suit people with allergies, small spaces, or leases or HOA rules against outdoor lines.

Other dryer savings: run full loads, use moisture-sensing settings instead of timed drying, and clean the lint filter as the manual directs, which is also a fire-safety step. When a dryer needs replacing, heat-pump dryers use 20% to 60% less energy than conventional ones, according to DOE. ENERGY STAR puts the figure higher, at about 70%.

Dishwashing

ENERGY STAR says a certified dishwasher uses about half the energy of washing by hand, and saves about 8,400 gallons of water and $220 a year. Your own result depends on how you wash by hand now. Section 5.8 covers the water side.

Two easy changes: run full loads, and skip the heated-dry setting. DOE notes that if your dishwasher has its own booster heater, you can keep the water heater at 120 degrees F (see 5.11).

What to record: how many dryer loads you run a week, before and after a change. Multiply the loads you no longer machine-dry by your dryer's energy per load, if you have measured it, or use the EIA average as an Estimated figure.

2.10 Time-of-use and other rate plans

On a **time-of-use (TOU)** plan, the same kWh costs more at peak hours and less at others. That can save money without using any less electricity, but only if you can move use out of the peak. Switching to TOU without changing anything can cost more.

TOU plans divide the day into periods:

- **Peak:** the highest price, usually late afternoon and evening, when demand is highest
- **Off-peak:** the lowest price, usually overnight and often weekends
- **Shoulder:** an in-between price on some plans

About 18.4 million US homes, roughly 13%, were enrolled in time-based programs in 2025, including TOU plans and **peak-time rebates**, which pay a credit for cutting use during announced peak events.

The basic math

Illustrative example: a peak price of $0.30 per kWh and an off-peak price of $0.12

- 100 kWh at peak: 100 × $0.30 = $30.00
- The same 100 kWh off-peak: 100 × $0.12 = $12.00
- Difference: $18.00, if no other charges change

The whole-bill test

The real question is whether the whole bill falls. There are two ways to test it.

On paper: ask the utility. Before switching, ask your utility whether it can compare your last 12 months of use on each plan. Some can, and it is the most reliable way to decide.

With hourly data: compare your peak share with the break-even share.

- **Your peak share** = kWh used in peak hours ÷ all kWh, from a month of your hourly data (2.4).
- **The break-even share** = (current price − off-peak price) ÷ (peak price − off-peak price). For the current price, use your usage charges ÷ kWh, as in the audit after 1.8, not your marginal price: the new plan reprices every kWh, not only the last few. If the plan's peak and off-peak prices cover only some charges, compare them with the same charges on your bill.
- If your peak share is below the break-even share, the plan costs less with no change in habits. If it is above, the plan pays only if you move enough use out of the peak.

Illustrative example: 800 kWh a month, 30% of it at peak. The flat plan costs $0.18 per kWh; the TOU plan costs $0.30 at peak and $0.13 off-peak. Fixed charges are the same on both.

	Flat plan	TOU, no changes	TOU, 100 kWh moved off-peak
Peak kWh		240	140
Off-peak kWh		560	660
Energy cost	800 × $0.18 = $144.00	$72.00 + $72.80 = $144.80	$42.00 + $85.80 = $127.80

Here the break-even share is ($0.18 − $0.13) ÷ ($0.30 − $0.13) = 29%. At 30% at peak, switching alone saves nothing; this household would pay 80 cents more. Moving 100 kWh out of the peak, for example by running the dryer, dishwasher and EV charger later, brings the peak share to 17.5% and saves $16.20 a month against the flat plan.

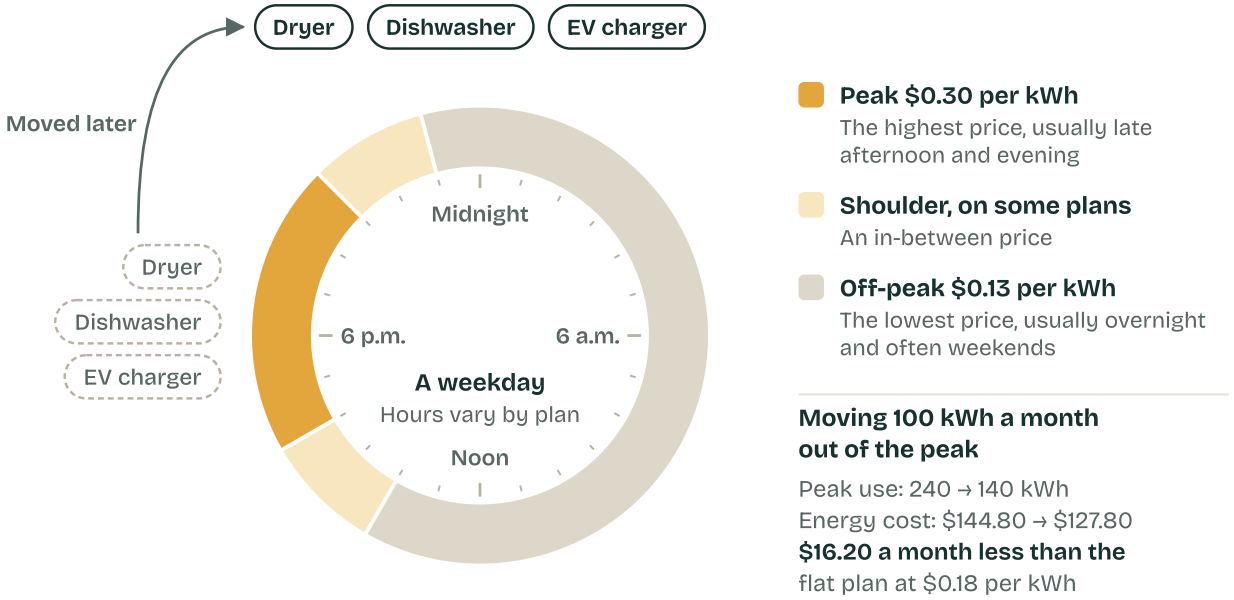


Figure 2.3. A day on a time-of-use plan. Running the dryer, dishwasher and EV charger later moves 100 kWh a month out of the peak. Illustrative: the 2.10 example; peak hours vary by plan.

Does TOU suit you?

TOU tends to pay if you	TOU tends not to pay if you
Can run big appliances late at night or early in the morning	Are home and cooling or heating hard every evening
Charge an electric vehicle at home	Cannot shift laundry, cooking or EV charging
Use little at peak already	Need power at peak for medical or care reasons

Buying from a supplier? Switching can end your utility's TOU rate, or leave only its delivery part. Read "Time-of-use and a supplier" in 2.11 before you combine them.

Other plans to know

- **Electric vehicle plans.** Some utilities offer TOU plans designed for overnight charging. Note that the federal tax credit for new EVs ended for vehicles acquired after September 30, 2025, and the home charger credit ended for equipment placed in service after June 30, 2026.
- **Prepaid plans.** Some providers let you pay in advance and send alerts as your balance runs down. Before signing up, read what happens when the balance reaches zero, and what fees apply.

- **Demand plans.** See 2.3.

> **SAFETY WARNING:** Do not give up cooling during a heat wave, or heating in a cold snap, to avoid peak prices. This matters most for older adults, babies and anyone who is ill. Follow the manufacturer's guidance before running dryers or other appliances overnight or while you are away.

What to record: each month on the new plan, what your actual kWh would have cost at the old plan's rates, minus what the new plan charged for them. That is the month's saving, or a cost if it is negative. Enter it in the 90-Day Household Savings Scorecard (B4).

2.11 Choosing an electricity supplier, where allowed

In some states you can buy the electricity itself from a company other than your utility. Whether you save depends on one comparison: the offer against your utility's own supply price, over the whole contract. Switching can cost you more if you compare only the first few months.

Where and how it works

In 2025, about 14.6 million homes in 15 states and Washington, D.C., bought supply from a separate company. The states are California, Connecticut, Delaware, Illinois, Maine, Maryland, Massachusetts, Michigan, Montana, New Hampshire, New Jersey, New York, Ohio, Pennsylvania and Rhode Island. In California, most of those homes are enrolled through community programs rather than by shopping themselves (2.3). Most of Texas has its own competitive market, where retail providers bill for both supply and delivery. If your state is not on this list, this section likely does not apply to you.

When you switch, only the **supply** part of your bill changes. The utility still delivers the power, fixes outages and charges the same delivery rates whoever your supplier is. On a utility time-of-use rate, switching can change more; see "Time-of-use and a supplier" below.

Your benchmark is the **price to compare**: the utility's own supply price per kWh, often printed on your bill. In Pennsylvania, the regulator defines it as the price your utility charges if you do not choose a supplier. Some states include other charges in it. Illinois' figure for its largest utility, for example, combines supply and transmission, about 10.1 cents per kWh from October 2026 to May 2027. Check what yours includes before you compare.

Find it on your bill: the supply or generation charge, which with a supplier carries the supplier's price per kWh. It sits in the supply section, apart from delivery, as on the Anytown sample in 1.2. The price to compare, if your bill prints it, is what the utility would charge you instead; check what it includes, as above. Each month, (price to compare − supplier price) × your kWh is what the supplier saved you, and a minus result is a cost. At 874 kWh, each cent per kWh is $8.74 a month. **Record it in** B4.

Read the contract, not the headline

Term	What to check
Fixed or variable rate	A fixed rate stays the same for the contract term. A variable rate can change, often monthly.
Introductory rate	How long it lasts, and what the rate becomes afterward
Contract length	The end date. Put a reminder in your calendar 60 days before it.
Early termination fee	Whether there is one, and how much. Pennsylvania's regulator says some plans charge one, but it is waived in the last 30 days of a contract.
Renewal	What happens at the end. In Pennsylvania, customers get renewal notices before the term ends, and those who do not respond stay with the supplier month to month, most likely at a different rate.
Renewable claims	What the premium buys, and how the supplier backs the claim

Compare the whole contract

Illustrative example: 874 kWh a month, the 2025 US average, against a utility price to compare of $0.115 per kWh for 12 months.

	12 months of supply	Against the utility
Utility price to compare: 10,488 kWh × $0.115	$1,206.12	
Offer A, fixed $0.109 for 12 months: 10,488 kWh × $0.109	$1,143.19	Saves $62.93
Offer B, $0.089 for 3 months, then a variable $0.149: (2,622 × $0.089) + (7,866 × $0.149)	$1,405.39	Costs $199.27 more

Offer B's headline rate is the lowest of the three, but it costs the most over a year. The utility's price to compare can also change during the year; in Pennsylvania, for example, it changes on set dates. So treat Offer A's saving as Estimated.

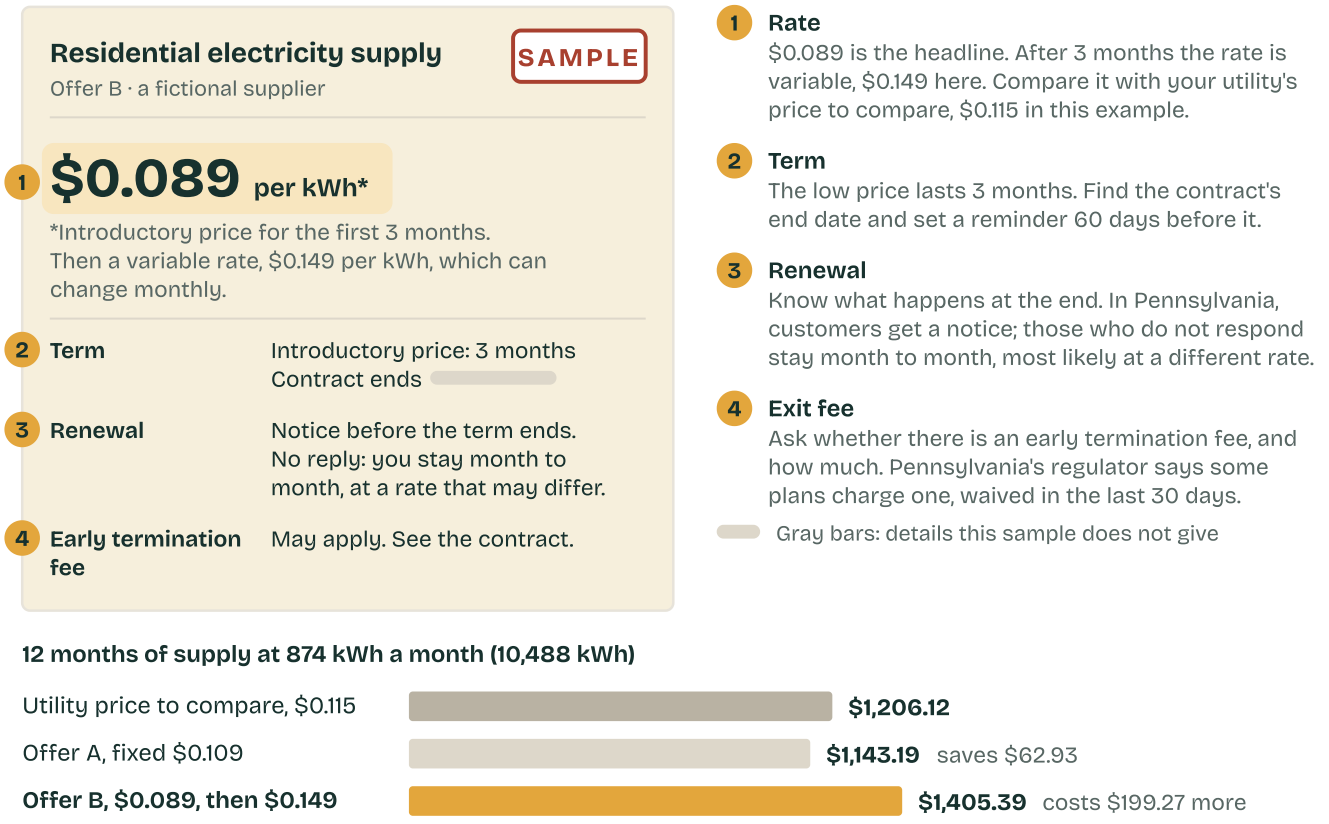


Figure 2.4. Supplier offer decoder: Offer B from 2.11, with the rate, term, renewal and exit fee called out, and its 12-month cost against the utility's price to compare and Offer A at 874 kWh a month. SAMPLE: a fictional offer (Illustrative).

Time-of-use and a supplier

Before you combine a utility time-of-use rate with a supplier, ask the utility which part of the bill its rate changes.

- **If TOU is part of the supply rate,** switching to a supplier ends it, and you pay the supplier's price instead. At FirstEnergy's Pennsylvania utilities, for example, TOU is a generation-only rate, and a supplier's customer must switch back to FirstEnergy to join. In April 2025, Pennsylvania's regulator noted that no supplier had posted a TOU offer yet.
- **If TOU is on the delivery charge,** it can stay when you buy from a supplier, as in Maryland and at Con Edison in New York. The supplier sets the supply price, often a flat one, so moving use off-peak saves only on delivery.

Questions to ask:

1. If I buy from a supplier, which part of the TOU rate stays?
2. Does the supplier offer its own TOU price, and do its peak hours match the utility's?
3. Is there a minimum stay, a wait to come back, or an exclusion? FirstEnergy Pennsylvania excludes budget-billing customers and makes customers who leave wait 12 months to return; Con Edison makes them wait 18 months.
4. What does each combination cost? Run the whole-bill test in 2.10 with both the supply and delivery prices, then check the contract terms above.

Before you switch

1. Find your price to compare and what it includes.
2. Get the offer's rate, rate type, term, fees and renewal terms in writing.
3. Work out 12 months of supply cost under both, as above.
4. If you switch, note the contract's end date. Switching can be quick; Pennsylvania says it can take as little as three business days.
5. Never give your account number to a door-to-door or phone seller you did not contact yourself. Section 10.7 covers sales pressure and scams.

What to record: that monthly figure from the box above, for every month of the contract, in the 90-Day Household Savings Scorecard (B4). Use each bill's own price to compare, since it can change during the year.

2.12 Electricity-saving myths

Any product that claims to cut your bill without changing what you heat, cool, light or run deserves suspicion. Your meter counts kWh, and kWh fall only when a device uses less power or runs for less time.

Claim	What the evidence says
A plug-in "power saver" box cuts bills by 40% to 90%	It does not lower a home bill. These boxes claim to correct "power factor" or "clean" the current. The National Institute of Standards and Technology tested this type of device: current in the wiring can drop, but the real power the home uses, which is what the meter bills, stays the same. Homes are billed for kWh; power-factor charges apply to some businesses, not households.
A magnet on the meter slows it down	Meter tampering is dangerous and illegal. Never touch or attach anything to a meter.
Smart meters run fast	Regulator-ordered tests have found smart meters accurate (1.5). A first bill after a smart meter is installed can be high if earlier bills were low estimates.
Unplugging chargers will save hundreds	Standby is usually a small share of a bill, and a charger draws very little. Find and fix the few big always-on devices instead (2.7).
Turning lights on and off uses more energy than leaving them on	Energy is power multiplied by time. A light that is off uses none.
A space heater always saves money	Only if it heats one room while you turn the central heat down. Used alongside the furnace, it adds cost.
Setting the thermostat way down at night always saves	Not with every system. Large setbacks on a heat pump can switch on expensive backup heat (4.4).

How to judge any energy-saving product

The FTC warns consumers to be wary of products that promise drastic or extreme energy savings, and to look for independent information on how they perform. It also requires energy-savings claims to be backed by competent and reliable scientific evidence. Before you buy, ask:

1. **What does it change?** Does it reduce the watts a device draws, or the hours it runs? If neither, it cannot reduce kWh.

2. **Is there independent testing?** Testimonials, "engineer approved" badges and money-back guarantees are not evidence.
3. **Is the claim even possible?** A device that plugs into one outlet cannot cut the use of your air conditioner, water heater and refrigerator, which are among the largest loads in most homes.
4. **Can you measure it?** Note your overnight always-on load and use per day before and after. A real saving shows up; a fake one does not.

If a seller's claims look false, you can report them at ReportFraud.ftc.gov.

Safety sidebar: power outages and generators

No saving is worth a safety risk, and an outage is when people take the worst ones. Carbon monoxide (CO) from generators and indoor burning kills people every year.

SAFETY WARNING: CARBON MONOXIDE. CDC says more than 400 Americans die each year from unintentional carbon monoxide poisoning not linked to fires. CPSC says nearly 100 a year die from portable-generator CO alone (2017 to 2019 average).

- Run a portable generator **outdoors only**, at least 20 feet from the house, with the exhaust pointing away. Never run one in a garage, basement, crawlspace, shed, porch or carport, even with doors open.
- Never heat a home with a gas oven or range, or use a charcoal grill or camp stove indoors.
- Put battery or battery-backup CO alarms on every level and outside sleeping areas, and test them monthly.
- If an alarm sounds, or anyone has a headache, dizziness, nausea or confusion, get everyone outside and call 911 or Poison Control at 1-800-222-1222.

Food. A refrigerator kept closed keeps food safe for about 4 hours without power. A full freezer holds about 48 hours, a half-full one about 24 (CDC). Throw out perishable food that has been at 40 degrees F or warmer for 2 hours or more (Ready.gov).

Medical equipment. If anyone in your home relies on powered medical equipment, the FDA advises telling your electric company and fire department, and asking the device maker whether it can run on batteries or a generator. Many states also protect these households from shutoffs (12.8).

Space heaters. Plug them straight into a wall outlet, never a power strip (CPSC). Keep anything that can burn 3 feet away, and switch them off when you leave the room or go to sleep (US Fire Administration). Portable heaters cause a small share of home heating fires but a large share of the deadly ones: 3% and 41% respectively in 2017 to 2019.

Light. Use flashlights or battery lanterns instead of candles.

CASE B

The Okafors' $275 August bill

Illustrative example. The Okafors are a fictional family of four in a 1962 house of 1,900 square feet, with a gas furnace and gas range, a 20-year-old central air conditioner and a 50-gallon electric water heater. All numbers are made up to show the method.

The Okafors' August electric bill came to $275.06, up from $230.72 the August before. They were sure the utility was overcharging them. Here is how they found out what actually happened.

Step 1: Is the bill right?

Check	Result
Reading type	Actual (smart meter)
Meter number	Matches the meter
Days	31 both years
Use	1,405 kWh, up from 1,240 (13.3% more)
Rates	Match the utility's published summer schedule: $0.17 per kWh for the first 1,000 kWh, $0.21 above that, a $12.00 customer charge and 3% taxes and fees. Last summer was $0.165, $0.20 and $11.00.

The bill was calculated correctly. The utility was not overcharging; prices had gone up a little, and use had gone up more.

Step 2: How much is price, and how much is use?

First, price last August's use at this year's rates:

- 1,000 kWh × $0.17 = $170.00, plus 240 kWh × $0.21 = $50.40
- $220.40 + $12.00 = $232.40, plus 3% = $239.37

So the $44.34 increase splits into **$8.65 from higher prices** ($239.37 - $230.72) and **$35.69 from higher use** ($275.06 - $239.37). Most of the problem is in the house.

Step 3: Where did the extra 165 kWh go?

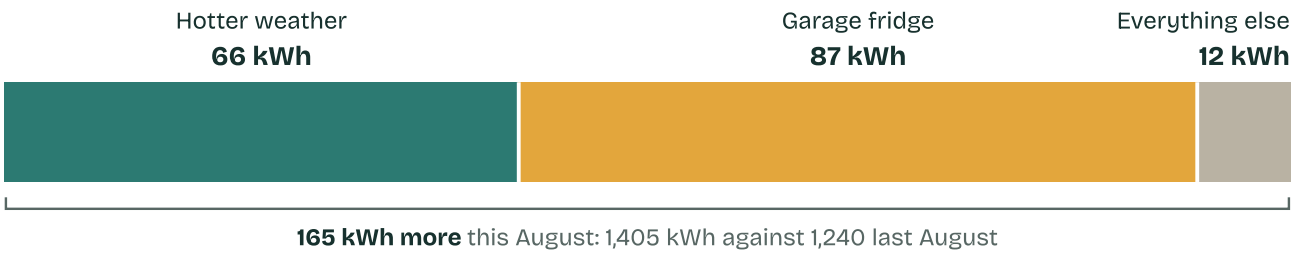


Figure 2.5. Case B: where the 165 kWh went. Illustrative: Case B, the Okafors, this August against last August.

Possible cause	How they checked	Finding
Hotter weather	Cooling degree days on the utility's website: 430 this August, against 380 last year (13% more)	If the air conditioner used about 500 kWh last August, 13% more cooling explains about 66 kWh
A new load	The old kitchen fridge went to the garage in June, when they bought a new one. A plug-in monitor showed 2.8 kWh a day.	2.8 kWh × 31 days = about 87 kWh
Everything else	What is left	About 12 kWh, within normal variation
An estimated or wrong reading	Already ruled out in Step 1	None

All of the extra use fell above 1,000 kWh, so every extra kWh cost the top-tier price: $0.21 plus 3%, or about $0.216. Outside summer, the Okafors pay about the rate card's $0.18 for extra electricity, the price later chapters use for them.

Step 4: What is worth doing?

Action	Cost tier	Estimated saving	Verdict
Empty and unplug the garage fridge, or recycle it	No-cost	About $18.77 in August (86.8 kWh × $0.2163). About 1,022 kWh a year, or $133 to $266 across the rate card's range	Worth doing now
Raise the thermostat a few degrees when nobody is home, and use ceiling fans in occupied rooms	No-cost	DOE says fans let you raise the thermostat about 4 degrees F with similar comfort	Worth doing now
Replace the AC filter, and book a tune-up for the 20-year-old system	Low to moderate	Unknown until measured	Worth investigating (4.7)
Try a time-of-use plan	No-cost to switch	Depends on whether they can move laundry and dishwashing off-peak	Ask the utility for a 12-month comparison first (2.10)
Replace the air conditioner	Major investment	Unknown	Not yet. Measure first, then use 13.2's payback method.
Replace the electric water heater	Major investment	It is the home's largest steady load	A decision for when it nears the end of its life (Chapter 13's Case B)
Unplug phone chargers	No-cost	Pennies	Not worth the effort
Buy a plug-in power-saver box	Low-cost	None (2.12)	Do not buy

Step 5: Talking to the utility

The Okafors called with three questions, not a complaint:

1. "When did the summer rates change, and is there anything else scheduled?"
2. "Can you compare my last 12 months on your time-of-use plan?"
3. "Do you offer fridge recycling, or rebates on air-conditioner tune-ups?"

Step 6: Checking the result

The garage fridge drew a steady 0.12 kW (2.8 kWh ÷ 24 hours). After unplugging it, the Okafors' overnight always-on load in the hourly data should drop by about that much. If it does, that part of the saving is Verified. The rest waits for next August, compared per day and against the cooling degree days.

TOOL 2.1

The Home Electricity Consumption Audit

Purpose: to rank the appliances in your home by what they cost you, so you spend your effort on the top few. Tool 2.1 also serves as the Appliance Energy Calculator.

Instructions

1. List every appliance that heats, cools or runs many hours a day. Section 2.5 lists the usual ones.
2. Find its watts: from a plug-in monitor if you have one, otherwise from the label or nameplate (a ceiling, not an average). For appliances that cycle, enter measured kWh a day instead of watts and hours.
3. Estimate the hours it runs each day, and the days each month.
4. Calculate monthly kWh: watts × hours a day × days ÷ 1,000.
5. Enter your **relevant rate**: the marginal price of a kWh you could save (1.8). On a tiered or seasonal plan, it is usually the top tier you reach.
6. Multiply kWh by the rate for the monthly cost, then sort the list from largest to smallest.
7. For the top three, write one proposed action, and mark it no-cost, low-cost, moderate or major.

Blank form

Appliance	Watts (or measured kWh a day)	Hours a day	Days a month	Estimated kWh a month	Relevant rate	Monthly cost	Proposed action

Sample: the Okafors in August

Illustrative. The relevant rate is their top-tier summer price, $0.21 plus 3%, about $0.2163 per kWh.

Appliance	Watts (or measured kWh a day)	Hours a day	Days a month	Estimated kWh a month	Relevant rate	Monthly cost	Proposed action
Central AC	Estimated from degree days (Case B)		31	566	$0.2163	$122.43	Filter, thermostat, tune-up (moderate)
Garage fridge	Measured: 2.8 kWh a day	24	31	86.8	$0.2163	$18.77	Empty and unplug (no-cost)
Electric dryer	3,000 W	1	20	60.0	$0.2163	$12.98	Air-dry half of loads (no-cost)
TVs and game console	200 W	5	31	31.0	$0.2163	$6.71	Standard picture mode (no-cost)
LED lighting, 20 bulbs	180 W combined	4	31	22.3	$0.2163	$4.83	None needed

Reading your results

- If one or two rows are much larger than the rest, start there. That is where any saving will show on a bill.
- If your listed appliances add up to far less than your bill's kWh, something is missing. Check the water heater, a well pump, a second freezer or the always-on load (2.7).
- Treat every figure here as Estimated until a later bill, compared per day and adjusted for weather, confirms it.

Chapter 2 review

What you learned

- You pay for kWh: watts times hours, divided by 1,000. Nameplate watts are a ceiling; measured use is the truth.
- Heating, cooling, hot water and refrigeration are the biggest parts of most electric bills. Small gadgets rarely matter.
- Tiered, seasonal, time-of-use and demand plans change your marginal price, and with it the value of every saving.
- A higher bill comes from more kWh or pricier kWh. Split the two before you act.
- Replace the bulbs that burn the most hours first. Replace a working fridge only when the numbers say so.

- A supplier offer is worth only what it saves over the whole contract.
- If a product claims to cut your bill without reducing watts or hours, it cannot work.

What you can do today

1. Find your rate plan's name and look up its schedule (2.3).
2. Check your overnight always-on load in your utility's hourly data, if you have it (2.4).
3. Swap the most-used incandescent bulb still in your home for an LED (2.6).
4. Unplug any second fridge or freezer you do not need (2.8).
5. Start Tool 2.1, the Home Electricity Consumption Audit, with your three biggest appliances.

What to check next month

Compare next month's use per day with the same month last year, and look at your overnight always-on load again. If you removed a steady load, such as a second fridge, the always-on load should be lower by roughly its average draw.

Billix Quick Check

A 500-watt dehumidifier runs 12 hours a day for 30 days, and you pay $0.18 per kWh. What does it cost for the month?

Answer: 500 W ÷ 1,000 = 0.5 kW. 0.5 kW × 12 hours = 6 kWh a day. 6 kWh × 30 days = 180 kWh. 180 kWh × $0.18 = $32.40. That assumes it runs the full 12 hours; if it cycles off at its humidity target, it costs less. Measure it to find out.

Sources and further reading

Last Verified: October 2026. The fastest-changing facts in this chapter are the federal EV and home-charger credits (both ended), APS's demand rate (under review in a rate case), Illinois's supply price to compare, California's base services charge, supplier-choice rules, whether a utility's time-of-use rate carries over when you buy from a supplier, and the 2028 bulb standard. Check current rules before you rely on them.

All pages were accessed in October 2026. [CO] marks a utility or company source, and [ADV] an advocacy or trade group.

Prices, bills and the grid

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PART II · ENERGY

3 Natural Gas, Propane, Heating Oil and Other Fuels

For most homes in cold or mixed climates, heating is the largest seasonal bill, and the fuel you burn matters as much as how much heat you need. EIA's forecast for this winter, October 2026 to March 2027, shows how much the average home heated by each fuel is expected to spend. These averages mix different climates and home sizes, so they are not a price comparison; section 3.4 shows how to compare fuels fairly:

Main heating fuel	Forecast winter spending per home	Change from last winter
Heating oil	$2,115	Up 21%
Propane	$1,246	Down 3%
Electricity	$1,196	Up 4%
Natural gas	$640	Down 9%

Natural gas is the main heat in about 46% of US homes, and electricity in more than 40%. Heating oil heats about 3%, mostly in the Northeast, and propane is common in rural areas. Many homes also use the same fuel for hot water and cooking.

This chapter covers the fuel itself: how it is measured, priced, billed and delivered, and how to compare fuels fairly. Chapter 4 covers the house and the heating equipment, which decide how much fuel you need.

By the end of this chapter you will be able to:

- Read a gas bill, and convert between therms, CCF and Mcf
- Tell a change in gas price from a change in weather
- Compare the real cost of heat from gas, propane, oil and electricity
- Judge a propane or heating-oil contract, including tank terms and fees
- Know what your state requires of fuel dealers, in the states that have rules
- Recognize a gas leak or carbon monoxide danger, and know what to do

Calculations use the Billix example rate card. In the middle case, natural gas is $1.30 per therm, propane $2.60 per gallon and heating oil $5.25 per gallon, close to EIA's forecasts for this winter.

START HERE:

- → **Your gas bill jumped?** Go to 3.3, starting with "Is my use normal?"
- → **A propane or oil delivery looks expensive?** Go to "Check your delivery ticket" in 3.6.
- → **A prebuy or capped-price offer arrived?** Go to "How the price is set" in 3.6, then Part 2 of the case study.
- → **Thinking of switching fuels or getting a heat pump?** Go to 3.4, then Tool 3.1.
- → **You smell gas or propane?** Leave the building first, then call from a safe distance (3.7).

3.1 The common heating fuels

The four common heating fuels differ in how they reach you, how they are measured and how much of their energy becomes heat in your home. Those three differences decide what a unit of heat really costs.

Fuel	How it reaches you	Measured in	Heat per unit (EIA)	Common equipment
Natural gas	A pipe from the street, metered	Therms, CCF or Mcf	1 therm = 100,000 Btu; 1 cubic foot is about 1,036 Btu	Furnaces, boilers, water heaters, ranges
Propane	Truck deliveries to a tank on your property	Gallons	91,452 Btu per gallon	Furnaces, boilers, water heaters, ranges
Heating oil	Truck deliveries to a tank, often in the basement	Gallons	138,500 Btu per gallon	Boilers and furnaces
Electricity	Wires, metered	kWh	3,412 Btu per kWh	Heat pumps, baseboard and other resistance heaters

A **Btu** (British thermal unit) is a standard measure of heat. It lets you compare fuels sold in different units on equal terms.

Efficiency: how much fuel becomes heat

Furnaces and boilers burn fuel, and some heat escapes up the flue. Their efficiency is rated by **AFUE** (annual fuel utilization efficiency), the share of the fuel's energy that becomes heat in the home over a year.

Equipment	Typical efficiency	Source
Old, low-efficiency furnaces and boilers	56% to 70% AFUE	DOE
Mid-efficiency	80% to 83% AFUE	DOE
High-efficiency (condensing)	90% to 98.5% AFUE	DOE
New gas furnace, federal minimum	80% AFUE	DOE rule
New gas hot-water boiler, federal minimum	84% AFUE	DOE rule
New oil furnace and oil hot-water boiler, federal minimums	83% and 86% AFUE	DOE rule
ENERGY STAR gas furnace	At least 95% in the South, 97% in the North	ENERGY STAR
Electric resistance heat (baseboard, space heaters)	100%: all the electricity becomes heat	Physics
Heat pump	2 to 4 units of heat per unit of electricity (a COP of 2 to 4)	DOE

A **heat pump** moves heat from outdoors to indoors instead of making it, so it can deliver more heat than the electricity it uses. That ratio is its **coefficient of performance (COP)**. COP falls as the outdoor temperature drops. NEEP, a regional efficiency nonprofit, sets a cold-climate standard requiring a COP of at least 1.75 at 5 degrees F. 4.1 and 4.10 cover heating equipment in depth.

Other fuels

Some homes heat with wood, pellets or kerosene. This edition does not cover them in detail. The cost comparison method in 3.4 still works for any fuel once you know its price, its heat per unit and the efficiency of the appliance that burns it.

3.2 Reading a gas bill

A gas meter measures **volume**, but you pay for **heat**. Your bill converts one into the other with a small number called the therm factor, then charges for delivery and for the gas itself.

The units

Unit	What it is	How it converts
CCF	100 cubic feet of gas	About 1.036 therms at EIA's average heat content
Mcf	1,000 cubic feet of gas	About 10.36 therms
Therm	100,000 Btu of heat	The unit most bills charge by
Therm factor (or heat content, or Btu factor)	The heat in each CCF this month	Printed on many bills, often a little above 1

The therm factor changes slightly with the gas's makeup, so the same CCF can be a few percent more or fewer therms from month to month. It also varies by place: at high altitude it can be below 1 (EIA's 2020 survey figures work out to about 0.89 therms per CCF in the northern Mountain states). So use the factor printed on your bill rather than the 1.036 average.

Therms = CCF used × therm factor

A sample gas bill

Illustrative example: Case B, the Okafors' January gas bill. Anytown Gas Co. is fictional, and the bill is marked SAMPLE.

Line	Detail	Amount
Meter readings	4,630 now, 4,500 last month: 130 CCF over 31 days	
Therm factor	1.0385, so 130 CCF × 1.0385 = 135.0 therms	
Customer charge	Fixed, per month	$15.00
Distribution (delivery) charge	135.0 therms × $0.45	$60.75
Gas cost adjustment (supply)	135.0 therms × $0.85	$114.75
Charges before taxes		**$190.50**
Taxes and fees	3%	$5.72
Total due		**$196.22**

Together, delivery and supply come to $1.30 per therm, the rate card's middle price. That is also the Okafors' marginal price before taxes: each therm they save takes $1.30 off, plus 3% tax. The $15.00 customer charge stays even in July, when the furnace is off and only the range uses gas.

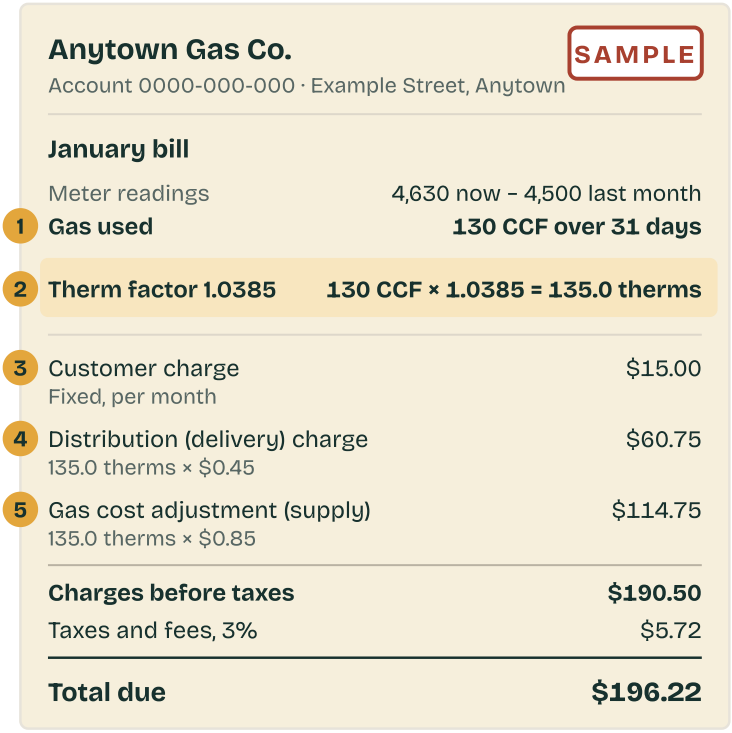


Figure 3.1. Annotated gas bill: the Okafors' January bill from Case B, from CCF and the therm factor to delivery, the gas cost adjustment and the customer charge. SAMPLE: Anytown Gas Co. is fictional.

What the charges mean

- **Customer charge:** a fixed monthly amount, like the customer charge on an electric bill (1.3).
- **Distribution or delivery:** the cost of the local pipes, priced per therm. It usually changes only through a rate case.
- **Gas cost adjustment:** the price of the gas itself. Utilities call it different things: purchased gas adjustment, gas cost recovery or gas supply charge. It usually follows wholesale prices and can change monthly or seasonally, without a rate case. Like an electric fuel charge, it is generally passed through to customers at cost and corrected later if the estimate was off.
- **Taxes and fees:** state and local charges, a percentage or per therm.

Some states also let homes buy gas from a supplier other than the utility. If yours does, the same rules as electricity apply: compare the offer with the utility's gas cost over the whole contract, and read the renewal terms (2.11).

The summer price trap

Do not judge the price of gas from a summer bill. With little gas used, the fixed charges are spread over very few units. EIA's average US residential gas price was $14.95 per Mcf in February 2026, but $26.45 in July 2026, mostly for that reason. Use a winter bill, or a whole year, to find your real price per therm.

Find it on your bill: the CCF used, the therm factor, the per-therm delivery and supply rates, and the customer charge. Add the two per-therm rates to get your marginal price.

3.3 Why gas bills rise

Winter gas bills rise mostly because of cold weather, so the first job is always to separate the weather from everything else. After that, look at the price, the equipment and the rest of the gas you use year-round.

Is my use normal? Compare it with your own home's history. Divide the therms by the days in the billing period, and compare with the same month last year. If the weather differed, compare heating therms per degree day instead (below; for degree days, see 1.8). If that figure has risen, check the filter and book a service (4.7).

A rough yardstick: in EIA's latest national survey (2020), homes heated mainly by natural gas used about 650 therms of gas a year for everything. By region, it was about 750 in the Northeast, 800 in the Midwest, and about 510 to 520 in the South and West. Heating alone averaged about 450 therms in homes that heat with gas, from about 190 to 210 in hot climates to about 610 in very cold and cold ones. To compare, add up a year of your bills' therms. Size, age, insulation, thermostat settings and the winter all move your number, so your own use per degree day, year over year, is the better test.

Cause	Typical sign	How to check
Colder weather	Use rises in step with the cold	Compare use per heating degree day with last year (1.8, Rule 4)
A higher gas cost adjustment	The supply rate per therm rises; use is normal	Compare the supply line with last month's and last year's
A base-rate increase	The delivery rate or customer charge rises, usually after a notice	Bill inserts and messages
Equipment problems	Use per degree day creeps up	A dirty filter, an overdue tune-up or an aging furnace (4.7, 4.10)
More hot water	Use rises in every season, including summer	More people, longer showers, a failing water heater (5.6, 5.11)
A longer billing period or a catch-up read	One bill jumps, then the next falls back	Days in the period and the reading type (1.4, 1.5)

Prices can move in either direction. EIA expects homes heating with natural gas to spend about 9% less this winter than last, mostly because of lower prices. Rate requests, though, are high: PowerLines, a consumer-education nonprofit, says gas and electric utilities together requested nearly $31 billion in increases in 2025.

Separating heating from everything else

If gas also heats your water or runs your stove, a summer bill shows that year-round base. Subtract it from a winter bill to see the heating alone, then divide by degree days.

Illustrative example: a home with a gas furnace, gas water heater and gas range.

- July bill: 18 therms, with the furnace off. That is the base for hot water and cooking.
- January bill: 140 therms.
- Heating alone: 140 - 18 = 122 therms
- January had 900 heating degree days: 122 ÷ 900 = 0.136 therms per degree day

Next January, do the same sum. If the figure per degree day has risen, the house or the furnace is using more gas for the same cold. If it has fallen, a change such as air sealing or a new thermostat schedule is working. This is a sharper version of the comparison in 1.8, because it keeps hot water and cooking out of the heating figure.

Budget billing

Many gas utilities offer **budget billing**, which spreads a year's expected cost into equal monthly payments, then settles the difference once a year. It makes winter easier to plan for, but it does not lower the total. If the winter is colder or prices rise, the settlement bill can be large. Compare your actual use with

the plan every few months, and ask the utility to adjust the monthly amount if they drift apart.

3.4 The real cost of heat

The price on a fuel bill is not the price of heat. To compare fuels fairly, convert each one to the cost of a million Btu of heat actually delivered into your home, after the equipment's losses. That single number settles most "which fuel is cheaper" arguments.

Cost per MMBtu delivered = price per unit ÷ (MMBtu per unit × efficiency)

An **MMBtu** is one million Btu. The heat in each unit: a therm of gas is 0.1 MMBtu, a gallon of propane 0.091452, a gallon of heating oil 0.1385 and a kWh 0.003412. Efficiency is the AFUE as a decimal (0.95 for 95%), or the COP for a heat pump.

Illustrative example: one gas furnace at 95% AFUE, with gas at $1.30 per therm

- 0.1 MMBtu × 0.95 = 0.095 MMBtu of heat delivered per therm
- $1.30 ÷ 0.095 = $13.68 per MMBtu delivered

The fuels side by side

Illustrative, at the rate card's middle prices: gas $1.30 per therm, propane $2.60 per gallon, heating oil $5.25 per gallon, electricity $0.18 per kWh. Lowest cost first.

Fuel and equipment	Cost per MMBtu delivered	Multiply your price by
Gas furnace or boiler, 95%	$13.68	10.526
Gas furnace or boiler, 80%	$16.25	12.500
Heat pump, COP 3.0	$17.58	97.694
Heat pump, COP 2.5	$21.10	117.233
Propane furnace or boiler, 95%	$29.93	11.510
Heat pump at COP 1.75 (a cold-climate unit at 5 degrees F)	$30.15	167.476
Propane furnace or boiler, 80%	$35.54	13.668
Oil boiler or furnace, 86%	$44.08	8.396
Oil boiler or furnace, 80%	$47.38	9.025
Electric resistance (baseboard, space heaters)	$52.75	293.083

To skip the division, multiply your own price per therm, gallon or kWh by the number in the last column. Gas at $1.30 a therm in a 95% furnace: $1.30 × 10.526 = $13.68.

At these prices, efficient gas is the cheapest heat, oil and electric resistance the most expensive, and heat pumps fall in between depending on their COP. Your local prices can reorder the table. In the Northeast, for example, EIA's forecast prices put gas at a 95% furnace around $16 per MMBtu and a heat pump at COP 3.0 around $26. But heating oil at the Oct. 5, 2026 New England price of about $6.04 a gallon, burned at 80%, comes to about $54.

Break-even: when a heat pump beats your fuel

The **break-even COP** is the COP a heat pump needs to match your current fuel's cost per MMBtu. At the rate card prices:

Current heat	Break-even COP
Oil at 80%	1.11
Propane at 95%	1.76
Gas at 80%	3.25
Gas at 95%	3.86

Against oil, almost any working heat pump clears the bar. Against propane, most do, except in the coldest weather or when backup heat runs. Against efficient gas, many do not. That matches a New York study of homes that switched to heat pumps: it found "no cost savings" where they replaced gas, but savings at some homes that replaced oil, at 2020 prices.

Cautions before you compare

- **Use your own prices.** Take your marginal price from a winter bill (1.8, 3.2), not EIA averages and not a summer bill.
- **Heat pump COP changes with the weather.** A federally sponsored field study found compressor COPs from 1.8 at minus 15 degrees F to 3.5 at 45 to 55 degrees F. Counting electric backup heat, whole-system COPs ranged from 1.0 to 4.5.
- **Count every fee.** Tank rentals, delivery fees and a gas customer charge belong in the comparison. A gas customer charge disappears only if you leave gas entirely.
- **This compares running costs only.** Switching fuels or equipment costs thousands of dollars. 13.2 and 13.3 show how to weigh purchase cost against savings.
- **EIA itself warns** that its average spending by fuel should not be used to compare fuels for one home, because the homes using each fuel differ in climate and size. Use the method above instead.

Tool 3.1, the Fuel Cost Comparison at the end of this chapter, runs these sums for your own home.

3.5 Lowering gas use

The biggest gas savings come from heating the house less when no one needs it, and from losing less heat. Both apply to propane and oil too. The house itself is covered in 4.3 to 4.6; this section covers what you can do with the fuel and the controls.

Action	Cost tier	Expected effect	Where to read more
Set the thermostat back at night and when nobody is home	No-cost	About 1% of heating for each degree F set back over 8 hours, and as much as 10% a year (DOE)	4.4
Check the furnace filter monthly and change it at least every 3 months	Low-cost	Keeps airflow and efficiency up (ENERGY STAR)	4.7
Seal air leaks	Low to moderate	10% to 20% off heating and cooling bills (DOE)	4.5
Set a gas water heater to 120 degrees F (right for most homes; see 5.11), and use less hot water	No-cost	Lowers the year-round base	5.6, 5.11
Have the furnace or boiler serviced as the manufacturer recommends	Moderate	Catches problems that raise use per degree day	4.7
Ask your utility about efficiency programs, audits and rebates	No-cost	Varies by utility	12.5, 13.5

What a setback is worth

Illustrative example: Case B, the Okafors, whose gas furnace burns about 600 therms over a heating season.

- They set the thermostat back 7 degrees F for 8 hours every night.
- At about 1% per degree, that is about 7% of heating.
- 600 therms × 7% = 42 therms saved
- 42 therms × $1.30 = $54.60 a season ($42.00 at $1.00 per therm, $67.20 at $1.60)

Treat this as a ceiling, not a promise. DOE's figure is "as much as", and the real saving depends on the house, the climate and how long the setback lasts. Large setbacks are not recommended for heat pumps, which can switch on expensive backup heat to recover (4.4). And never set a home so cold that it is unsafe for the people in it, especially babies, older adults and anyone who is ill.

Cooking and pilot lights

Gas ranges use relatively little gas, so cooking is rarely worth targeting. Some older furnaces, boilers, water heaters and fireplaces have a **standing pilot light** that burns all the time. New gas boilers may no longer be sold with one. Ask a qualified technician whether an old pilot can safely be turned off in summer, and have the technician do it.

What to record: heating therms per degree day for each winter month (3.3), before and after any change. A lower figure through a whole winter is a Verified saving.

3.6 Propane and heating oil

With propane and heating oil, you are not a utility customer but a buyer, and your contract decides most of what you pay. No federal rule covers tank leases, delivery fees or prepaid plans; a few states have strong rules, and most leave it to the contract. Read it before winter.

Check your delivery ticket

Find it on your delivery ticket: the gallons delivered, the price per gallon, and each fee on its own line, such as a delivery, small-delivery or after-hours fee. Connecticut, for example, requires the price, the gallons and any fee on every ticket. **Record it in** Tool 3.1, using the all-in price below as your price per unit.

A low price with a fee can cost more than a higher price without one, so work out the all-in price:

All-in price per gallon = (gallons × price per gallon + fees) ÷ gallons

Illustrative example: a will-call ticket for 210 gallons at $3.19 a gallon, plus a $45 delivery fee.

- 210 gallons × $3.19 = $669.90
- $669.90 + $45.00 = $714.90
- $714.90 ÷ 210 gallons = $3.40 a gallon, all in

Then get two other dealers' prices and fees in writing, for the same number of gallons, and work out each all-in price the same way. Ask what size of delivery avoids the fee. This book's rate card is for examples only, not a benchmark. Also write the gauge reading before and after each fill on the ticket (see the case study).

For a rough check, use EIA's Heating Oil and Propane Update (source 5). Each week from October to March, and monthly from April to September, it gives average home prices for heating oil and propane, with regional and US averages. Heating oil prices cover 21 states and D.C. on the East Coast and in the Midwest, and propane prices 38 states on the East Coast and in the Midwest, Gulf Coast and Rocky

Mountains. EIA asks for the regular credit price, preferably for automatic (keep-full) delivery, before taxes and without cash or volume discounts. It does not say whether separate delivery fees are included, and it notes that prices vary with the amount, method and frequency of delivery and the dealer's location. So compare your all-in price before tax with care: if it is above EIA's figure, see how much of the gap is the fee. A small gap is common. A big one is a reason to call other dealers, not proof of wrongdoing.

Who owns the tank

	Leased tank	Tank you own
Who can fill it	Usually only the company that owns it. Maine, Minnesota and Pennsylvania, for example, allow only the owner or someone the owner authorizes	Any dealer, after a safety check
Switching dealers	You may owe removal or pump-out fees, unless your state limits them	Easier; a new dealer must check the system first (Wisconsin's consumer agency describes this)
Upkeep	Usually the dealer's responsibility	Yours
Typical cost	A yearly rental fee, or a minimum-use requirement	The purchase price

Owning your tank lets you shop on price, which can matter a lot in a region with several dealers. In Connecticut, tank leases must be in writing and let you buy the tank.

Automatic or will-call delivery

- **Automatic (or keep-full) delivery:** the dealer schedules refills based on the weather and your past use. You are less likely to run out, but unless you have a prebuy or capped plan, you pay whatever the dealer charges on the day of delivery.
- **Will-call delivery:** you order when the tank is low. You control timing and can compare prices, but you risk running low in a cold snap, and small or after-hours deliveries can carry fees.

How the price is set

Plan	How it works	The trade-off
Market price	You pay the dealer's price on the day of delivery	No commitment, full exposure to winter price spikes
Prebuy (prepaid)	You pay ahead, usually in summer or fall, for a set number of gallons at a locked price	Protects against spikes; costs more if prices fall, and you are trusting the dealer with your money
Capped price	You pay the market price, but never more than the cap, often for a fee	Some protection both ways, at a price
Budget billing	Your expected yearly cost spread into equal payments	Smooths payments; does not lower the total

State protections

Last Verified: October 2026. State propane and heating oil rules change. Check your state attorney general or consumer protection office for current rules.

Vermont, New Hampshire, Maine, Connecticut, New York, Minnesota and Pennsylvania have rules on tank fees and removal, refunds, prepaid contracts or deliveries. D4 in Appendix D summarizes them.

Fees to ask about

Minnesota's attorney general suggests asking about each of these before you sign:

- ☐ Low-use or no-use fees
- ☐ Special-trip or after-hours delivery fees
- ☐ Partial-fill fees
- ☐ Tank drop-off and pickup fees
- ☐ Leak test and reconnection fees

- ☐ Early termination fees
- ☐ Pump-out or restocking fees
- ☐ Equipment removal fees

Also ask for the price per gallon in writing, the contract length, any minimum purchase, and what happens at renewal. New York's consumer guide recommends getting all of these in writing.

Heating oil specifics

Heating oil prices swing more than gas or electricity. EIA's forecast for this winter is $5.26 a gallon, against a forecast of about $3.48 a year earlier. That makes the pricing plan unusually important for oil homes this year.

Basement heating-oil tanks are not covered by federal underground storage tank rules. State and local fire codes govern them, including how a tank must be removed if you switch fuels. If you are considering a switch, ask your local fire or building department what removal requires.

3.7 Safety

Saving money on fuel never involves working on gas lines, tanks, regulators or burners yourself. A suspected leak is an emergency, not a repair job.

SAFETY WARNING: IF YOU SMELL NATURAL GAS. Do not touch light switches, use phones or create any flame or spark. Leave the building right away, and call 911 or the gas utility's emergency line from well away (PHMSA, the federal pipeline safety agency). Do not go back inside until the utility or fire department says it is safe.

SAFETY WARNING: IF YOU SMELL PROPANE. Make no sparks, and get everyone out. If you can do so safely, turn off the main valve on the tank by turning it clockwise. Call your propane dealer or 911 from a neighbor's home or a safe distance. Do not return until a professional has checked for leaks (Propane Education & Research Council, a trade group; Connecticut's utility regulator gives similar advice).

Carbon monoxide

Every furnace, boiler, water heater and range that burns gas, propane or oil can produce carbon monoxide if it is faulty or poorly vented. CDC says more than 400 Americans die each year from unintentional carbon monoxide poisoning not linked to fires.

- Put carbon monoxide alarms on every level of the home and outside sleeping areas, and test them monthly (CPSC).
- Never block or seal vents, flues or the openings that bring combustion air to a furnace or water heater, even to stop a draft. 4.5 explains which openings must stay open.
- Never heat a home with a gas oven or range (CDC).
- Have fuel-burning equipment serviced as its manufacturer recommends, by a qualified technician.

Running out of propane or oil

Letting a tank run empty can cost more than the fuel. Dealers may charge for a special trip, a leak test and relighting appliances, the kinds of fees Minnesota's attorney general lists, and a cold snap can mean waiting for a delivery. If you buy will-call, order well before the tank runs low, and ask your dealer what gauge level they recommend.

SAFETY WARNING: If a propane tank runs empty, do not relight pilot lights or turn the gas back on yourself. Connecticut's utility regulator says never to turn the gas back on yourself after a shutoff, and the propane industry's safety council says to have the whole system leak-checked before you use any propane appliance. Call your dealer, and wait until the system has been checked.

Call a professional: for any work on gas piping, regulators, tanks, burners, venting or controls. Homeowners can safely change furnace filters, keep vents and the area around equipment clear, and call for service if they see soot, smell something unusual or hear an alarm.

Case study: gas against propane, and a propane contract

Illustrative example. Case B, the Okafors, have a 1962 house with a gas range and an 80% gas furnace that burns about 600 therms a heating season. Case E, the Lindqvists, are a fictional family of three in a 2,200-square-foot rural farmhouse with a 95% propane furnace, a propane water heater and a leased 500-gallon tank. All prices come from the rate card or are marked as illustrative.

Part 1: Whose heat costs more?

The Lindqvists assumed their newer, 95% furnace must heat more cheaply than the Okafors' old 80% one. To compare fairly, put both on the same amount of delivered heat.

The Okafors' season:

- 600 therms × 0.1 MMBtu per therm × 0.80 = 48 MMBtu of heat delivered
- 600 therms × $1.30 = $780 in usage charges

The same 48 MMBtu from the Lindqvists' propane furnace:

- 48 MMBtu ÷ 0.95 = 50.5 MMBtu of propane burned
- 50.5 MMBtu ÷ 0.091452 MMBtu per gallon = about 552 gallons
- 552 gallons × $2.60 = about $1,435

For the same heat, propane costs about 1.8 times as much, even burned in a more efficient furnace. The fuel's price outweighs the furnace's efficiency. The cost-per-MMBtu table in 3.4 shows the same thing: $29.93 for propane at 95% against $16.25 for gas at 80%.

What this comparison does not tell you is which family actually pays more for heat. A fair comparison of two homes would also need:

Factor	Why it matters
Climate	Heating degree days can differ greatly from one region to another
House size and insulation	2,200 square feet of farmhouse may need far more heat than 1,900 square feet
Thermostat habits	A setback schedule changes the heat each home needs (3.5)
Other fuel uses	The Lindqvists' propane also heats water; the Okafors' water heater is electric, and their gas range uses only a little gas
Fixed costs	A $15 monthly gas customer charge ($180 a year) against a tank rental
Delivery fees	Will-call and small-delivery fees add to propane's cost

The cost per MMBtu delivered is the right way to compare **fuels**. Comparing **homes** also needs each home's heat need, which is what 4.2 works out.

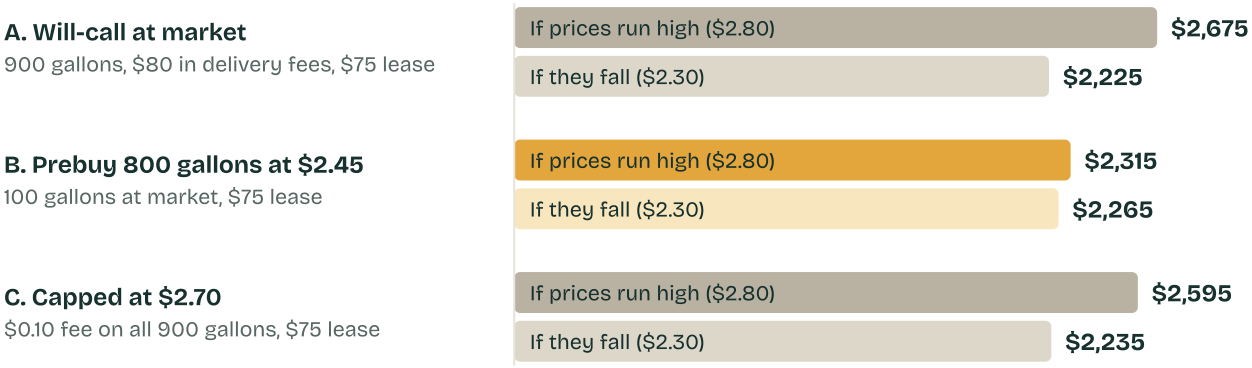
Part 2: The Lindqvists' contract decision

The Lindqvists use about 900 gallons a year for heat and hot water. They have always bought will-call at the market price. This year their dealer offers a prebuy and a capped plan. They lease their tank for $75 a year (illustrative).

Illustrative prices: a winter market price of $2.80 a gallon if prices run high, or $2.30 if they fall; a prebuy price of $2.45; a cap of $2.70 with a $0.10-a-gallon fee. Will-call means two small deliveries a year at $40 each. The prebuy and capped plans are assumed to have no delivery fees.

Plan	If winter prices run high ($2.80)	If they fall ($2.30)
A. Will-call at market: 900 gallons, plus $80 in delivery fees and the $75 lease	$2,675	$2,225
B. Prebuy 800 gallons at $2.45, plus 100 gallons at market and the lease	$2,315	$2,265
C. Capped at $2.70, plus the $0.10 fee on all 900 gallons and the lease	$2,595	$2,235

The math for plan B if prices run high: (800 × $2.45) + (100 × $2.80) + $75 = $1,960 + $280 + $75 = $2,315.



Plan B, the one the Lindqvists chose: $360 less than will-call if prices run high, $40 more if they fall.

Figure 3.2. The Lindqvists' yearly propane cost under three plans, if winter prices run high ($2.80) or fall ($2.30). Illustrative: Case E.

Plan B saves $360 against will-call if prices run high, and costs only $40 more if they fall. The Lindqvists chose it, with three precautions:

1. **They prebought 800 gallons, not 900,** so a mild winter would not leave them paying for propane they do not use.
2. **They asked how the prepaid money is protected.** In Maine, New Hampshire and Connecticut the law requires it to be secured (D4); elsewhere, they asked the dealer directly and got the answer in writing.
3. **They got every term in writing:** price, gallons, fees, delivery method and what happens to unused gallons.

Measuring the result

At each delivery, the Lindqvists write down the gallons, the tank gauge reading before and after the fill, and the heating degree days since the last fill (1.8). A fill does not always top up the tank, so they count the propane used, not delivered: the gauge's drop since the last fill, as a share of their 500-gallon tank. Gallons used per degree day tells them whether the house is using more or less fuel than last winter. They also ask the dealer for its market price on each delivery date, in writing. At the end of the season, they price their actual gallons at those prices, to see whether the prebuy really saved money.

TOOL 3.1

The Fuel Cost Comparison

Purpose: to compare the running cost of heating your home with different fuels or equipment, using your own prices and your own home's heat need.

Instructions

1. **Find your heat need.** Take the fuel you burned for heating last season (heating only; see 3.3) and convert it to heat delivered: units × MMBtu per unit × your equipment's efficiency.
2. **Enter each option's price per unit.** Use your marginal price from a winter bill, the all-in price from a recent delivery ticket (3.6), a dealer quote, or the rate card if you have nothing better.
3. **Enter the heat per unit:** 0.1 MMBtu per therm, 0.091452 per gallon of propane, 0.1385 per gallon of heating oil, 0.003412 per kWh.
4. **Enter the efficiency:** AFUE as a decimal, or a realistic seasonal COP for a heat pump, not its best-case rating.
5. **Calculate the cost per MMBtu delivered:** price ÷ (heat per unit × efficiency).
6. **Multiply by your heat need** for the season's fuel cost.
7. **Add yearly fixed costs and fees** that depend on the choice: a gas customer charge you would drop, a tank rental, delivery fees not already in an all-in price.
8. **Compare totals.** Then take the best options to 13.3 and Tool 13.1, which add the purchase cost.

Blank form

Your heat need: _____ MMBtu delivered per season

Fuel or system	Price per unit	MMBtu per unit	Efficiency	Cost per MMBtu delivered	Season's fuel cost	Yearly fixed costs and fees	Total a year

Sample: the Okafors

Illustrative. Heat need: 600 therms × 0.1 × 0.80 = 48 MMBtu delivered per season. The Okafors also cook on a gas range, so they keep paying the $15 monthly gas customer charge ($180 a year) with any heating system, unless they replace the range too.

Fuel or system	Price per unit	MMBtu per unit	Efficiency	Cost per MMBtu delivered	Season's fuel cost	Yearly fixed costs	Total a year
Current gas furnace	$1.30 per therm	0.1	0.80	$16.25	$780.00	$180	$960.00
New gas furnace	$1.30 per therm	0.1	0.95	$13.68	$656.84	$180	$836.84
Heat pump, seasonal COP 2.5	$0.18 per kWh	0.003412	2.5	$21.10	$1,012.90	$180	$1,192.90
Heat pump, seasonal COP 3.0	$0.18 per kWh	0.003412	3.0	$17.58	$844.08	$180	$1,024.08

Reading your results

- For the Okafors, a new 95% gas furnace is the cheapest to run. It costs about $187 a year less than a heat pump averaging COP 3.0, because the gas range keeps the customer charge on the bill either way. A heat pump averaging 2.5 costs more to run than the furnace they already have. If they also replaced the range with an electric one, the gap with the COP 3.0 heat pump would shrink to about $7 a year, before counting the electric range's own running cost. At that point the deciding factors would be purchase cost, the age of the current furnace, whether the home also needs cooling, and how cold their winters get. 13.3 and Tool 13.1 add those.

- A small change in price or COP can reorder close results. Run the sums with the rate card's low and high prices as well.
- Treat the totals as Estimated. Real heat need varies with the weather from year to year.

Chapter 3 review

What you learned

- A gas meter measures volume; you pay for heat. Therms = CCF × therm factor.
- A gas bill has a fixed customer charge, a delivery charge and a gas cost adjustment that follows wholesale prices.
- Judge gas prices from winter bills, not summer ones, when fixed charges are spread over little gas.
- Separate heating from hot water and cooking, then compare heating per degree day.
- Compare fuels by cost per MMBtu delivered. Fuel price often matters more than furnace efficiency.
- With propane and oil, the contract decides most of the price. Read the tank, delivery and fee terms, and know your state's rules.
- A suspected gas or propane leak means leave first, then call from a safe distance.

What you can do today

1. Find your gas bill's therm factor and per-therm rates, and add the rates to get your marginal price (3.2).
2. Check that a carbon monoxide alarm works on every level of your home (3.7).
3. If you buy propane or oil, find your contract and check the tank, fee and renewal terms against the list in 3.6.
4. Run the Fuel Cost Comparison (Tool 3.1) for your current system.

What to check next month

Record your heating fuel use and the month's heating degree days (see "Getting your degree days" in 1.8). Work out use per degree day, and keep the figure. By next winter you will have last winter's figures to compare with, which show whether any change you make is working.

Billix Quick Check

One home burns heating oil at $5.25 a gallon in an 80% boiler. Another burns propane at $2.60 a gallon in a 95% furnace. Which pays less per MMBtu of heat delivered?

Answer: Oil: $5.25 ÷ (0.1385 × 0.80) = $47.38. Propane: $2.60 ÷ (0.091452 × 0.95) = $29.93. Propane costs less, about 37% less per unit of heat, even though a gallon of propane holds less heat than a gallon of oil.

Sources and further reading

Last Verified: October 2026. The winter price and spending forecasts are EIA's October 2026 figures and change monthly. The federal minimum for new gas furnaces is 80% AFUE today; a DOE rule adopted in 2023 would raise it to 95% for most gas furnaces made from December 2028. Check that rule's current status before you buy. What EIA's weekly heating oil and propane prices cover, and its 2020 gas-use figures, were checked in October 2026; EIA does not say whether separate delivery fees are in its weekly prices.

All pages were accessed in October 2026. [ADV] marks an advocacy or trade group.

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PART II · ENERGY

4 Heating, Cooling, Insulation and Home Efficiency

Chapters 2 and 3 covered the price of energy. This chapter covers how much of it your home needs, and that depends on the house, the heating and cooling equipment, and how you run them. It is where most homes have the most room to save.

In 2020, air conditioning was about 19% of US home electricity use, and space heating another 12%, before counting homes that heat with gas, propane or oil. DOE says that combining a new heating or cooling system with insulation, air sealing and better thermostat settings can cut heating and cooling bills by about 30%.

The chapter works from the cheapest steps to the most expensive. Settings and maintenance come first, then sealing and insulation, then equipment. That order matters, because a new furnace or air conditioner in a leaky house with clogged filters will not deliver what its label promises.

By the end of this chapter you will be able to:

- Explain how your heating and cooling system works, and what makes it use more energy
- Set a thermostat schedule that saves money without risking comfort or health
- Find air leaks, and know which openings must never be sealed
- Decide what you can maintain yourself and what needs a technician
- Judge whether a smart thermostat, new windows or new equipment will pay for itself
- Troubleshoot a high heating or cooling bill, starting with the steps that cost nothing

Check your lease: If you rent, many of the steps in this chapter, such as sealing, insulation and equipment changes, need your landlord's permission. Tool 11.1, the Housing Action Matrix, sorts the steps by who can take them.

START HERE:

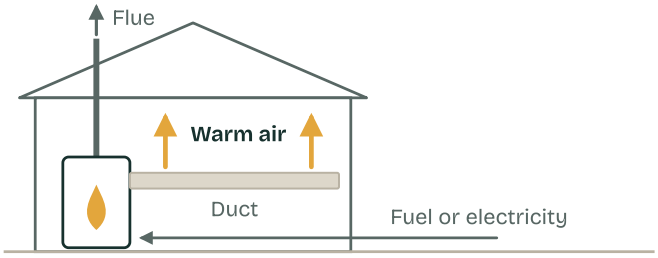
- → **Your heating or cooling bill jumped?** Run the 15-Minute Bill Audit after 1.8. If the price rose, go to 2.4 or 3.3. If your use rose, go to 4.2, then Tool 4.1 and Case C.
- → **Told to replace a system that still works?** Read Case C first, then 4.10.
- → **Want the biggest saving that costs nothing?** Go to 4.4.
- → **Drafty rooms?** Go to 4.5, and read "What must stay open" before you seal anything.

4.1 How heating and cooling systems work

Every heating or cooling system either makes heat by burning fuel or running electricity through a resistance coil, or moves heat from one place to another. Systems that move heat, such as heat pumps and air conditioners, can deliver more heat than the energy they use, which is why they matter so much for bills.

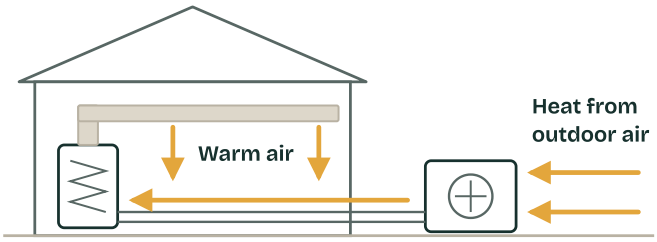
Furnace

Makes heat by burning gas, propane or oil, or with electric coils, then blows warm air through ducts.



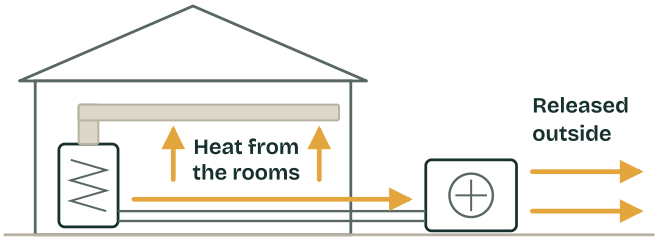
Heat pump in winter

Moves heat in: pulls it from outdoor air into the home, 2 to 4 units of heat per unit of electricity.



Heat pump in summer

Runs as an air conditioner: carries heat from the rooms to the outdoor unit, which releases it outside.



Ductless mini-split

A heat pump with no ducts. Each indoor unit serves its own room or zone.

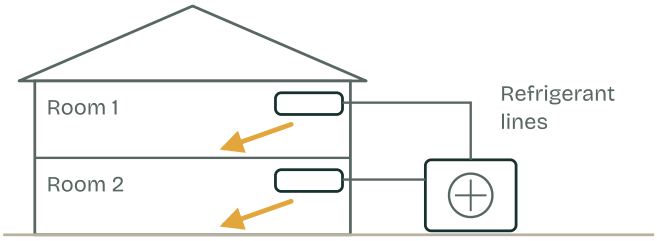


Figure 4.1. How each system moves heat, simplified: a furnace, a heat pump in winter and in summer, and a ductless mini-split.

System	How it works	Fuel	Worth knowing
Furnace	Burns fuel, or uses electric coils, to heat air, then blows it through ducts	Gas, propane, oil or electricity	Efficiency is rated by AFUE (3.1). Filters and ducts affect how much heat arrives.
Boiler	Heats water or steam, which flows to radiators, baseboards or floor tubing	Gas, propane or oil	No ducts. Often also heats the home's hot water.
Central air conditioner	Refrigerant carries heat from an indoor coil to an outdoor unit, which releases it outside; a fan moves cooled air through ducts	Electricity	Shares ducts and the indoor fan with a furnace in many homes.
Window or room air conditioner	The same process, in one box, for one room	Electricity	Cheaper to buy; useful for cooling only the rooms in use.
Heat pump	An air conditioner that also runs in reverse, pulling heat from outdoor air (or the ground) into the home in winter	Electricity	Delivers 2 to 4 units of heat per unit of electricity (3.1). Many have electric backup coils, called auxiliary or emergency heat, that cost far more to run.
Ductless mini-split	A heat pump with an outdoor unit and one or more indoor wall or ceiling units, no ducts	Electricity	Suits homes without ducts, such as homes with boilers. Each indoor unit serves its own room or zone, so homes with many closed-off rooms may need several.
Electric resistance heat	Baseboard heaters or an electric furnace turn electricity directly into heat	Electricity	100% efficient, but expensive per unit of heat (3.4). DOE says heat pumps can cut the electricity used for heating by about 50% compared with electric furnaces or baseboard heaters.
Portable heaters	Resistance heaters for one room	Electricity	Only economical when the rest of the house is kept cooler; see the space heater rules in the safety sidebar after 2.12.

A **thermostat** controls all of these. Manual thermostats hold one setting, programmable ones follow a schedule, and smart thermostats learn or are controlled by phone (4.4, 4.9).

Size matters

Equipment that is too big or too small for the house works badly. ENERGY STAR notes that oversized and undersized systems both struggle to keep a home comfortable. ENERGY STAR advises asking any contractor who proposes new equipment to size it with **Manual J**, the industry's standard calculation of a home's heating and cooling needs, rather than matching the old unit's size. ENERGY STAR also says improper installation can cut a system's efficiency by up to 30%.

Find it in your home: the label on your furnace or boiler, and on the outdoor unit of your air conditioner or heat pump. Note the make, model number and year of manufacture. You will need them for maintenance, rebates and any repair-or-replace decision (4.10).

4.2 What drives heating and cooling costs

Heating and cooling costs come from five things: the weather, the temperature you ask for, how fast the house loses or gains heat, how efficiently the equipment works, and how many hours someone is home. You control the last four, at very different costs.

Factor	How it affects cost	What you can do
Weather	Colder winters and hotter summers mean more degree days and more energy	Nothing, but you can measure it (1.8)
Thermostat setting	The gap between indoors and outdoors drives heat loss and gain	Set it for comfort when you are home, and back when you are not (4.4)
The house	Air leaks, thin insulation, sunny windows and leaky ducts let heat in or out	Seal, insulate, shade (4.5, 4.6)
Equipment efficiency and condition	Dirty filters and airflow problems can cut efficiency by up to 15%, and duct leaks can lose 20% to 30% of the air (ENERGY STAR)	Maintenance (4.7), duct sealing (4.5), and eventually replacement (4.10)
Hours at home	A home that is occupied all day needs conditioning all day	Zone the rooms in use; see Case C
Humidity	In summer, the air conditioner must also remove moisture, which takes energy	Fix leaks that let humid air in; use bathroom and kitchen fans

Finding your heating and cooling share

You can estimate how much of your bill goes to heating or cooling from your bills alone. Find a mild month, usually in spring or fall, when you barely heat or cool. That month's use per day is your **mild-month use**. Anything above it in winter or summer is mostly heating or cooling.

Illustrative example: Case B, the Okafors' electricity.

- April, a mild month: 728 kWh over 30 days, or about 24.3 kWh a day. That is the home's mild-month use before the garage fridge was added in June (Case B in Chapter 2).
- August: 1,405 kWh (Case B, Chapter 2)
- Mild-month use for August's 31 days: 728 ÷ 30 × 31 = about 752 kWh, plus about 87 kWh for the garage fridge = 839 kWh. Extra in August: 1,405 - 839 = 566 kWh, mostly air conditioning
- That matches the roughly 566 kWh estimate from degree days in Case B (Chapter 2). A load added since the mild month, like the garage fridge, belongs in the mild-month figure; leave it out and the cooling share looks bigger than it is.

Always work per day, because months have different lengths (1.4). The same method works for gas: a summer gas bill shows the use for hot water and cooking alone (3.3).

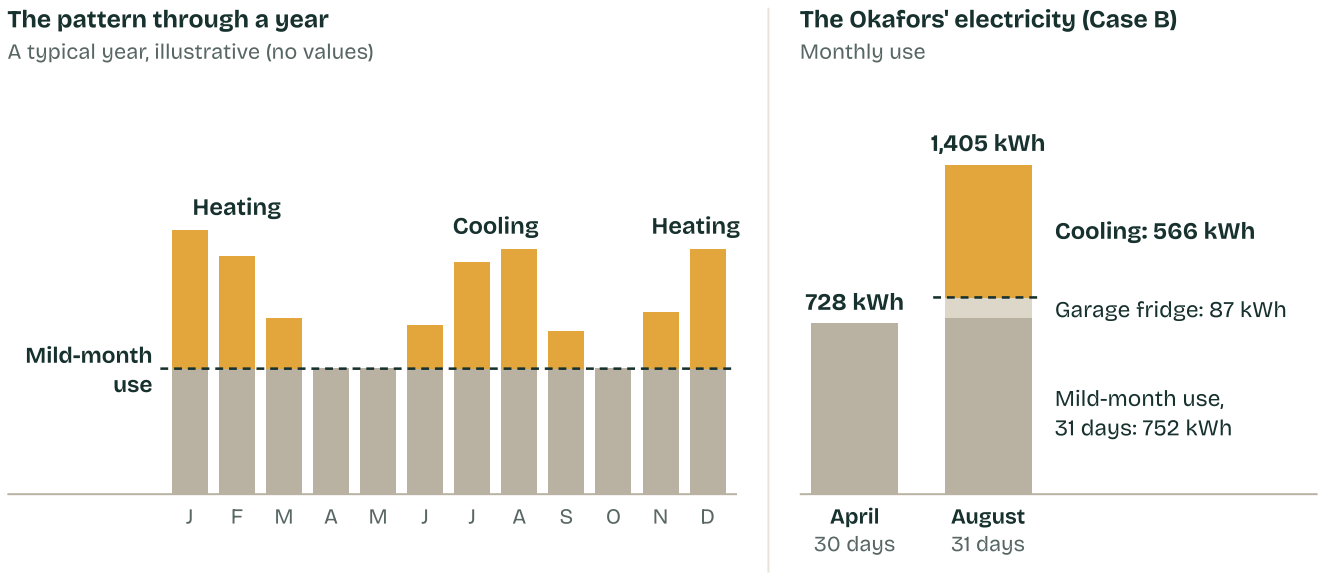


Figure 4.2. Mild-month use and the seasons: use above the mild-month line is mostly heating or cooling. Illustrative: left, the shape of a typical year, with no values; right, Case B, the Okafors' April and August electricity.

Once you know the size of the heating and cooling share, you know the most any heating or cooling improvement could ever save. A 10% improvement on a 566 kWh cooling load saves about 57 kWh, about $10 a month at $0.18 per kWh, not 10% of the whole bill.

Find it on your bill: the usage chart (many bills show the last 12 or 13 months), to pick your mild month, and each month's kWh or therms and the days in the billing period. You also need your price per unit, which on a tiered or seasonal plan is the top tier you reach (2.3). For degree days, see "Getting your degree days" in 1.8. **Record it in** Tool 4.1, in the measurement log.

4.3 How heat moves through a house

Heat always moves from warm to cold, in four ways: through solid materials such as a cold window pane (**conduction**), and carried by moving air (**convection**). It also moves through gaps as outdoor air seeps in and indoor air leaks out (**air leakage**, or infiltration), and as sunlight (**radiation**). Insulation slows the first, air sealing the next two, and shades the last. Every efficiency step in this chapter slows one of them.

In winter, warm air rises and leaks out through the top of the house, pulling cold air in through gaps near the bottom. This is called the **stack effect**, and it is why ENERGY STAR's sealing and insulation advice starts with the attic, floors over crawl spaces and the basement rim, rather than the middle of the walls.

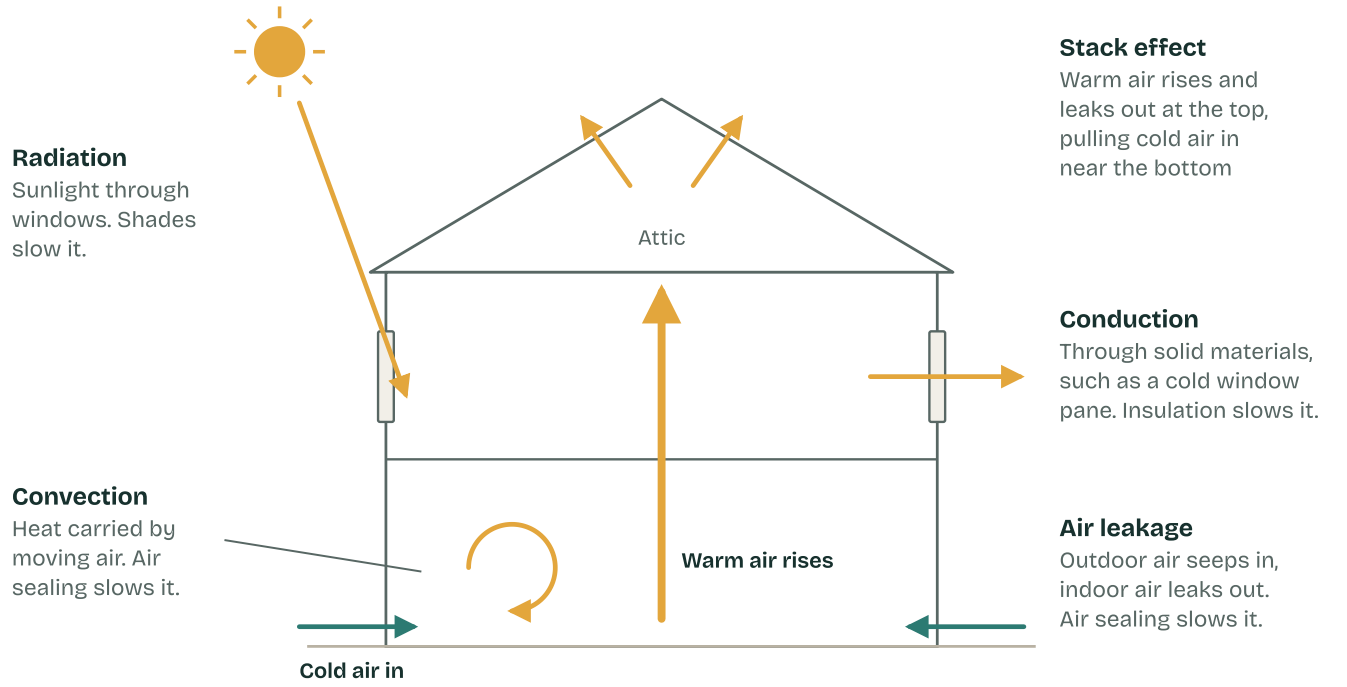


Figure 4.3. How heat moves through a house in winter: conduction, convection, air leakage and radiation, with the stack effect.

In summer the flow reverses. Hot attics, sunny windows and leaky ducts running through hot spaces add heat that the air conditioner must remove.

No house is airtight, and none should be. Homes need fresh air, and fuel-burning appliances need air to burn safely and to vent their exhaust. Section 4.5 explains how to seal the leaks that waste energy without blocking the openings that keep you safe.

4.4 Thermostat management

The thermostat is the cheapest heating and cooling tool you have. DOE says you can save as much as 10% a year on heating and cooling. The method is to turn the thermostat down 7 to 10 degrees F for 8 hours a day in fall and winter, and up in spring and summer. For heating, it puts the saving at about 1% for each degree you set back over an 8-hour period.

The idea is simple: heat or cool the house to your comfortable setting when you are home and awake, and let it drift when you are asleep or away. The house loses heat more slowly when it is closer to the outdoor temperature.

A sample schedule

Illustrative. Use your own comfortable temperatures and times.

Period	Winter	Summer
Waking up to leaving home	Your comfort setting	Your comfort setting
Away, about 8:30 a.m. to 5 p.m.	7 to 10 degrees F cooler	Several degrees warmer, with the system still on
Evening at home	Your comfort setting	Your comfort setting
Asleep, about 10:30 p.m. to 6 a.m.	7 to 10 degrees F cooler, if everyone is comfortable	A little warmer, with ceiling fans (4.8)

In summer, DOE advises leaving the system running at a higher setting when you are away rather than switching it off, to keep moisture from building up indoors.

Heat pumps need a gentler approach

A heat pump recovering from a deep setback may switch on its electric backup coils, the auxiliary or emergency heat, which cost far more to run (3.4). DOE says not to set a heat pump back manually if that brings on backup heat, and to use a programmable or smart thermostat designed for heat pumps. Those are built to bring the temperature back up without leaning on backup heat. If your thermostat shows "aux" or "emergency heat" often in mild weather, have the system checked (4.7).

Common thermostat mistakes

Mistake	What actually happens
Setting it much higher to warm up faster	Most furnaces deliver heat at the same rate whatever the setting, so the house just overshoots. On a heat pump, it can bring on backup heat.
Switching cooling off completely in a humid summer	Moisture can build up indoors (DOE)
Placing the thermostat near a sunny window, a lamp or a drafty door	It reads the wrong temperature and runs the system more or less than needed
Setting it back but overriding it every evening	The schedule saves nothing if it never runs

SAFETY WARNING: Never set the temperature so low in winter, or so high in summer, that it is unsafe for the people in the home. Babies, older adults, and people who are ill or take certain medicines are most at risk from cold and heat. Their needs come before any saving.

Check your lease: Most renters control their own thermostat, but some buildings set limits. Read your lease, or ask your landlord before changing a building-wide system.

What to record: your schedule, and heating or cooling use per degree day before and after you start it (1.8, 3.3).

4.5 Air leaks and insulation

Sealing leaks and adding insulation are usually the best-value improvements to the house itself. DOE says sealing uncontrolled air leaks can save 10% to 20% on heating and cooling bills. ENERGY STAR says air sealing and insulating attics, floors over crawl spaces and basement rim joists saves an average of 15% on heating and cooling costs.

What it could be worth

Illustrative example: Case B, the Okafors. Their heating costs about $780 a year (600 therms at $1.30), and their cooling about $306 (an estimated 1,700 kWh a year at the rate card's $0.18), for about $1,086 in total. At their summer top-tier price of about $0.216, the cooling would be about $367.

Saving	Yearly value
10% (low end of DOE's range)	$108.60
15% (ENERGY STAR's average)	$162.90
20% (high end of DOE's range)	$217.20

The saving applies only to the heating and cooling share of the bill, not the whole bill (4.2).

Where houses leak

The biggest leaks are often hidden. Check these first:

- The attic hatch or pull-down stairs

- Holes where pipes, wires and vents pass into the attic
- Recessed ceiling lights below the attic
- The rim where the house frame sits on the foundation, in the basement or crawl space
- Around doors and windows, and their frames
- Outlets and switches on outside walls
- Ducts running through the attic, garage or crawl space

On a cold, windy day, hold a strip of tissue paper near each spot and watch for movement. For a full picture, an **energy audit** (also called an energy assessment) uses a blower-door test to depressurize the house and find leaks. Many utilities offer audits free or at reduced cost, and income-eligible households can get them through the Weatherization Assistance Program (12.3).

SAFETY WARNING: Attic work has hidden hazards. Do not cover or foam over recessed lights unless they are rated for contact with insulation and airtight. Keep insulation off old knob-and-tube wiring and clear of chimneys, flues and other hot surfaces. Work in the cool part of the day, step only on the joists, and wear a mask and eye protection. If you find old wiring, loose fill that may contain vermiculite, or anything you are unsure about, stop and call a professional.

What you can do yourself

Job	Cost tier	Notes
Weatherstrip doors and operable windows	Low-cost	Replace worn strips; check that doors close tightly
Caulk fixed gaps around window and door trim	Low-cost	Use caulk rated for the location, indoor or outdoor
Foam gaskets behind outlet and switch covers on outside walls	Low-cost	Turn off the circuit first
Weatherstrip and insulate the attic hatch	Low-cost	Often one of the largest single leaks
Seal accessible duct joints with mastic or foil tape	Low to moderate	ENERGY STAR says 20% to 30% of duct air is lost to leaks; never use cloth duct tape
Add attic insulation	Moderate to major	Often a professional job; check ENERGY STAR's recommended levels for your climate

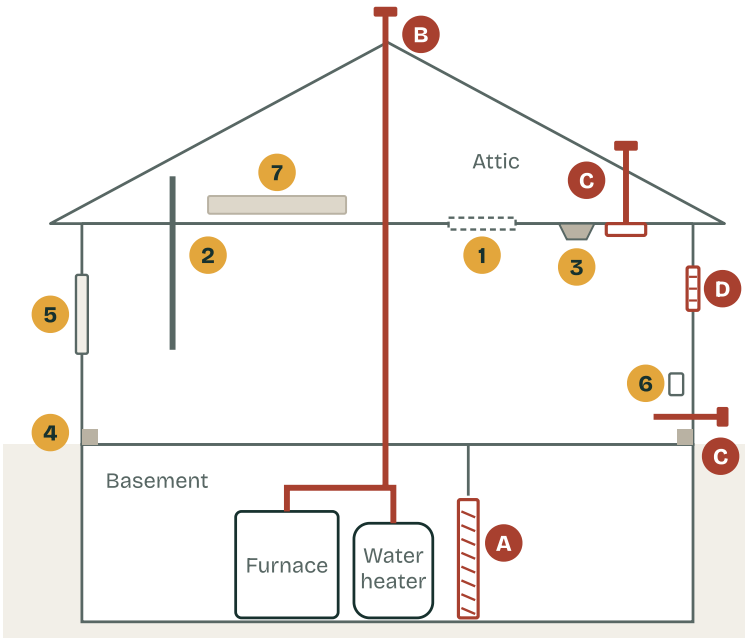
DOE says duct sealing should be tested by a qualified professional to make sure fuel-burning appliances still vent safely.

What must stay open

SAFETY WARNING: Sealing the wrong opening can trap carbon monoxide or moisture in your home. Never seal or block:

- Combustion-air openings or louvered doors for a furnace, boiler or water heater in a closet or basement
- Flues, chimneys and vent pipes
- Dryer vents, and bathroom and kitchen exhaust fans
- Fresh-air intakes on ventilation systems

After any significant air sealing in a home with fuel-burning appliances, have a qualified professional test that they still vent safely.



Seal these leaks

1. Attic hatch or pull-down stairs
2. Pipe, wire and vent holes into the attic
3. Recessed lights below the attic
4. Rim where the frame meets the foundation
5. Around doors and windows
6. Outlets and switches on outside walls
7. Ducts in the attic, garage or crawl space

Never seal or block

- **A** Combustion-air openings or louvered doors for a furnace, boiler or water heater
- **B** Flues, chimneys and vent pipes
- **C** Dryer vents, and bathroom and kitchen exhaust fans
- **D** Fresh-air intakes on ventilation systems

After any significant air sealing in a home with fuel-burning appliances, have a qualified professional test that they still vent safely.

Figure 4.4. Where houses leak (gold, numbered) and the openings that must never be sealed or blocked (red, lettered).

Call a professional: If your attic has loose, pebble-like vermiculite insulation, do not disturb it, because older vermiculite can contain asbestos. Also call a professional for sealing around chimneys and flues, which needs fire-rated materials, and for any work near wiring.

Check your lease: Renters can usually add removable weatherstripping and draft stoppers. Caulking, insulation and duct work need the landlord's permission.

4.6 Windows and sunlight

Windows let heat out in winter and sunlight in during summer, but replacing them is one of the most expensive ways to save energy. Start with what you hang on them and how you use them.

The cheap steps

- **In winter,** close curtains at night to hold heat in (DOE), and open them on sunny windows during the day to let the sun's warmth in.
- **In summer,** close curtains, blinds or shades on sunny windows (DOE), especially those facing west and south in the afternoon.
- **On drafty windows,** insulated cellular shades help, and weatherstripping on the sashes cuts air leaks (4.5).
- **Outside,** awnings, exterior shades and shade trees block summer sun before it reaches the glass.

The Attachments Energy Rating Council (AERC) rates window coverings and storm windows for energy performance, so you can compare products.

Storm windows or replacements?

Option	What the sources say	Typical use
Low-e storm windows (an extra pane added inside or outside)	DOE: they can save 10% to 30% on heating and cooling costs, depending on the window already installed	Older single-pane windows in sound condition
Replacement windows	ENERGY STAR: replacing single-pane windows lowers household energy bills by an average of up to 13%	Windows that are failing, rotted or impossible to repair

DOE says replacing single-pane windows "may be cost effective", but suggests considering storm windows too. The numbers usually decide it.

Illustrative example: the Okafors' 1962 house has 15 original single-pane windows. They spend about $3,210 a year on energy: about $2,200 on electricity and $1,010 on gas (heating, range and the customer charge). Heating and cooling come to about $1,086 of that. Prices are illustrative.

	Replacement windows	Low-e storm windows
Installed cost	15 × $800 = $12,000	15 × $250 = $3,750
Yearly saving	Up to 13% of $2,886, the bills without fixed customer charges = $375.18	10% to 30% of $1,086 = $108.60 to $325.80
Simple payback	$12,000 ÷ $375.18 = about 32 years	11.5 to 34.5 years

Even at the top of each range, neither choice pays for itself quickly on energy alone, and replacement windows take decades to repay. Replacement can still be the right choice when windows are failing, for comfort, noise or safety, or to sell the house. Treat any energy saving as a bonus, not the reason. 13.2 explains simple payback and its limits.

SAFETY WARNING: Never block a window that is a bedroom's emergency exit with permanent coverings, storm panels or furniture.

Check your lease: Curtains, blinds and removable film are usually fine for renters. Storm windows and replacements are the owner's decision.

4.7 HVAC maintenance

A neglected system quietly uses more energy every month. DOE says regular maintenance can cut heating and cooling costs by 10% or more, and ENERGY STAR says airflow problems alone can reduce efficiency by up to 15%. The cheapest part, the filter, is also the one most often forgotten.

What you can do, and what needs a technician

Task	Who	How often
Check the air filter; replace or clean it	You	Check monthly, especially in winter and summer; change at least every 3 months (ENERGY STAR)
Keep supply registers and return grilles clear of furniture and rugs	You	Whenever you rearrange a room
Keep the outdoor unit clear of leaves, snow and plants	You	Seasonally
Replace thermostat batteries, and test carbon monoxide alarms	You	As the manufacturer directs; alarms monthly (CPSC)
Check refrigerant charge, clean coils, test electrical parts	Technician	A yearly tune-up before the cooling season

Task	Who	How often
Inspect the burner, heat exchanger and venting, and test combustion	Technician	A yearly tune-up before the heating season
Test and seal ducts	Technician	When an audit or high bills suggest leaks (4.5)

Turn off power to the system before opening any access panel to change a filter, and follow the manufacturer's instructions. Note the filter size printed on its frame so you buy the right one.

Call a professional: for anything involving refrigerant, gas or oil burners, electrical components, or venting. Never bypass a safety switch or ignore a carbon monoxide alarm.

Case B follow-up: the Okafors' January

Illustrative. In Chapter 1 (1.8, Rule 4), the Okafors' gas use per heating degree day rose from 0.150 to 0.159 therms, about 6%, even though their bill fell.

Working down the table above, the Okafors found the furnace filter had not been changed in a year and was clogged. They replaced it, a low-cost fix, and booked a heating tune-up.

	Last January	This January	This February, new filter
Gas per heating degree day	0.150 therms	0.159 therms	0.149 therms

If the 6% increase had lasted all season, it would have cost them:

- 600 therms × 6% = 36 therms
- 36 therms × $1.30 = $46.80 a year

One month back at the old figure is a good sign, but not yet proof. If use per degree day stays near 0.150 through the rest of the winter, the fix is Verified (1.8).

4.8 Ceiling fans

A ceiling fan does not cool a room. It cools the people in it, with a wind-chill effect. That is why a fan saves money only when someone is under it and the thermostat goes up.

DOE says a ceiling fan lets you raise the thermostat about 4 degrees F without losing comfort. It also says to switch fans off when you leave a room, because a fan running in an empty room only uses electricity.

The fan's own cost is small:

Illustrative example: a 75-watt fan running 8 hours a day for 30 days, at $0.18 per kWh

- 75 W × 8 hours ÷ 1,000 = 0.6 kWh a day
- 0.6 kWh × 30 days = 18 kWh a month
- 18 kWh × $0.18 = $3.24 a month

That is worth spending only if the thermostat setting goes up. How much the higher setting saves depends on the home and the climate; Case C shows how to measure it.

Tip	Why
In summer, run the fan counterclockwise, seen from below	It pushes air down for a cooling breeze (DOE)
In winter, run it clockwise on low	It gently moves warm air down from the ceiling (DOE)
Switch it off when you leave the room	It cools people, not rooms
When replacing a fan, look for ENERGY STAR	ENERGY STAR says certified ceiling fans are up to 44% more efficient than conventional ones

Typical ceiling fans draw about 65 to 175 watts at full speed (Virginia Cooperative Extension). Use the label on yours, or a plug-in monitor for a fan that plugs in.

4.9 Smart thermostats and utility thermostat programs

A smart thermostat saves money only by doing what section 4.4 describes: setting the temperature back when no one needs it. If you already do that, it adds convenience more than savings. If you never set back, it can pay for itself quickly. ENERGY STAR says a certified smart thermostat can save about $100 a year in homes with high heating and cooling bills, or homes that sit empty much of the day.

What they add over a basic programmable thermostat

- Schedules you can change from a phone, and adjustments when plans change
- Occupancy sensing or learning, so setbacks happen even when you forget
- Usage reports that show heating and cooling run time
- Recovery designed for heat pumps, avoiding backup heat (4.4)

Before buying, check that the model works with your system: a heat pump, a boiler, a multi-stage furnace or older low-voltage wiring can each limit your choices. Some utilities offer rebates on certified models.

A payback model

Illustrative example: the Okafors spend about $1,086 a year on heating and cooling (4.5). Assume a $200 thermostat, installed themselves.

Scenario	Yearly saving	Simple payback
They already set back by hand most days, so it adds little (about 2%)	$21.72	About 9 years
ENERGY STAR's figure for high-bill or often-empty homes	About $100	About 2 years
They never set back, and now reach DOE's "as much as 10%"	$108.60	About 1.8 years

The thermostat does not create the saving; the setback does. A $30 programmable thermostat, used faithfully, can capture most of the same savings. A smart model earns its extra cost when the household's schedule is irregular, or when nobody would program a basic one.

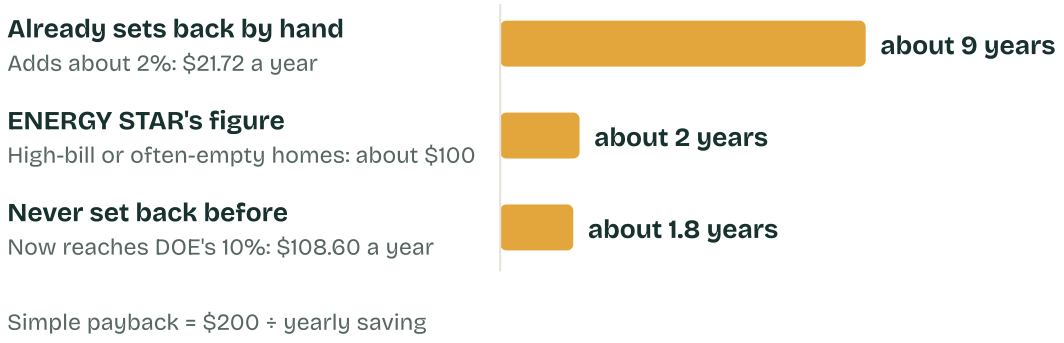


Figure 4.5. Smart thermostat payback in three cases, for a $200 thermostat in the Okafors' home. Illustrative.

Utility thermostat programs

Some utilities offer bill credits or rebates if you let them adjust your thermostat by a few degrees during periods of peak demand, often on hot summer afternoons. These are a form of demand response, related to the peak-time rebates in 2.10. Before enrolling, ask:

- How many events a year, how long they last, and how far the setting changes

- Whether you can override an event, and whether overriding costs you the credit
- What the credit or rebate is worth, and when it is paid
- Whether the program allows exemptions for medical needs

SAFETY WARNING: Do not enroll if anyone in the home could be harmed by a few degrees of heat during an event, unless you can override it at any time.

4.10 Repair or replace

First, has a technician found a fault? If not, and the system still works, do 4.2, Tool 4.1 and a tune-up before you act on a replacement quote, as Dana does in Case C. If so, get the fault and the repair price in writing, ideally from a second technician too, then use the table below and 13.3.

When heating or cooling equipment breaks, the energy saving from a new unit is rarely the deciding factor. Safety, reliability and the repair quote usually are. Work through these questions with your technician before you agree to either.

Question	Points toward repair	Points toward replacement
Is it safe?	The problem is a part, not the burner, heat exchanger or venting	A technician finds a safety defect, such as a cracked heat exchanger or a venting fault
How old is it, and how much life is left?	The technician expects years of service after the repair	It is near the end of its expected life (see the typical lives below), and repairs are becoming regular
How does the repair compare with replacement?	A modest repair on a sound system	A large repair on an old, inefficient system
How efficient is it now?	Already mid- or high-efficiency	Old, low-efficiency equipment: AFUE in the 56% to 70% range (DOE); for air conditioners and heat pumps, see "Efficiency ratings on a quote" below
Does it keep the home comfortable?	Yes, once fixed	Rooms stay too hot or cold even when it works
Are incentives available?		State or utility rebates may cut the cost; federal tax credits for new equipment ended after 2025 (13.5)
Can you afford it now?	A repair buys time to save and compare quotes	An emergency replacement leaves no time to shop

How long equipment lasts: in DOE's analyses, the average central air conditioner is retired after about 21 years (about 18 in hot, humid areas). The average air-source heat pump is retired after about 15 years, and the average gas furnace after about 21. These are averages: many units fail years sooner, and some last longer. For working equipment in the last third of its life, see the rule in 13.3.

The energy saving alone rarely decides it

Your numbers: yearly cooling kWh (4.2) × your summer price per kWh × the expected saving = yearly saving. The quote ÷ the yearly saving = simple payback.

Illustrative: the Okafors' 20-year-old central air conditioner needs a $900 repair, against a $7,500 replacement. An assumed 30% saving on their 1,700 kWh of cooling, at $0.18 to $0.216 per kWh, is about $92 to $110 a year: a payback of about 68 to 82 years.

On energy savings alone, replacing a working air conditioner rarely pays for itself; here it would take decades. The real question is whether the $900 repair keeps a reliable system running, or patches one that will fail again soon. If the technician says the system has years left, repair, then check the cost per year in 13.3: the answer turns on how long a new unit would last. If repairs are becoming regular, get quotes now, so the replacement is planned rather than an emergency.

When you do replace

- Get at least two or three quotes, and ask each contractor to size the system with a Manual J calculation (4.1).
- Fix duct leaks and airflow problems at the same time; ENERGY STAR says poor installation alone can cut efficiency by up to 30%.
- Compare running costs with Tool 3.1, the Fuel Cost Comparison, and total costs with 13.3.
- Ask about state rebates, utility incentives and any assistance programs you may qualify for (12.3, 12.5, 13.5).

Efficiency ratings on a quote

New central air conditioners and heat pumps are rated in **SEER2** (cooling over a season), **EER2** (cooling on the hottest days) and **HSPF2** (heat pump heating). Higher is better.

- **Don't compare old SEER with SEER2.** The new test is tougher, so the same system scores about 5% lower in SEER2 than in SEER, and about 15% lower in HSPF2 than in HSPF.
- **Minimums since January 2023:** a new split-system air conditioner must be at least 13.4 SEER2 in the North and 14.3 SEER2 in the Southeast and Southwest. In those two regions, units of 45,000 Btu/h (about 3.75 tons) or more need at least 13.8 SEER2, and the Southwest has an EER2 minimum too. A new split-system heat pump must be at least 14.3 SEER2 and 7.5 HSPF2 everywhere. The Southwest is Arizona, California, Nevada and New Mexico. The Southeast is Alabama, Arkansas, Delaware, Florida, Georgia, Hawaii, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, D.C., Puerto Rico and the other territories. The North is every other state.
- **Ask for the certified rating of the whole matched system** (the outdoor unit with its indoor coil or air handler), not the outdoor unit's "up to" figure. Also ask for the estimated yearly kWh saving in writing. Check that saving with "Your numbers" above.

CASE C

Dana's high summer cooling cost

Illustrative example. Dana is a fictional remote worker who owns a 1,400-square-foot townhouse heated and cooled by a heat pump, with ducts in the attic. Dana works from home five days a week in an upstairs office facing west. Electricity costs $0.18 per kWh plus a $12.00 customer charge; taxes are left out. All results are made up to show the method.

Dana's July electric bill was $235.20, and a contractor had already suggested replacing the heat pump. Dana started with the free steps instead.

Step 1: Measure the problem

- July: 1,240 kWh over 31 days. 1,240 × $0.18 = $223.20, plus $12.00 = $235.20
- April, a mild month: 450 kWh over 30 days, or 15.0 kWh a day
- Mild-month use for 31 days: 15.0 × 31 = 465 kWh
- Cooling: 1,240 - 465 = 775 kWh, or about $139.50
- July had 420 cooling degree days: 775 ÷ 420 = 1.85 kWh of cooling per degree day

Dana's hourly data showed the heat pump running almost nonstop from about 1 p.m. to 7 p.m., when the sun hits the office, with the thermostat at 72 degrees F all day and night.

Step 2: The no-cost and low-cost fixes

Change	Cost tier
Raise the setting to 76 degrees F during the day, with a ceiling fan in the office (4.8)	No-cost
Close the west blinds from about 1 p.m. to 6 p.m. (4.6)	No-cost
Raise the setting a little overnight, with a fan in the bedroom	No-cost
Replace the air filter, which was dirty (4.7)	Low-cost, about $15
Clear leaves from around the outdoor unit	No-cost

Over the next two weeks, cooling use fell to 1.48 kWh per degree day, about 20% less.

Step 3: Seal the leaks

Dana weatherstripped the attic hatch and sealed the accessible duct joints in the attic with mastic, for about $40 in materials (4.5). Dana worked early in the morning, before the attic heated up, and stepped only on the joists. Cooling use fell again, to 1.40 kWh per degree day.

Step 4: A professional tune-up

A technician checked the refrigerant charge, cleaned the coils and tested the electrical parts for about $150. Everything was within specification. The heat pump was healthy; it had been working against a hot, leaky, sun-baked room.

Step 5: Decide what not to do

Option	Decision	Why
Replace the heat pump	No	The system tested healthy, and the cheap steps cut cooling use by about 24%
Replace the windows	No	The payback would run to decades (4.6)
Exterior shades for the west windows	Maybe, next year	Worth pricing if afternoons stay hot
A time-of-use plan	Ask the utility	Dana's daytime use makes this a close call (2.10)

The result

August had about the same weather as July, 420 cooling degree days.

- Cooling: 1.40 kWh × 420 = 588 kWh, against 775 in July
- Saved: 187 kWh × $0.18 = $33.66 for the month

Over a cooling season of about 1,500 degree days, the saving would be about (1.845 - 1.40) × 1,500 = about 668 kWh, or about $120. Dana spent about $205 on the filter, sealing materials and tune-up, so the steps pay for themselves in under two summers. Because the August figure is one month and the weather happened to match, the saving is Estimated. A full season at the lower figure per degree day would make it Verified.

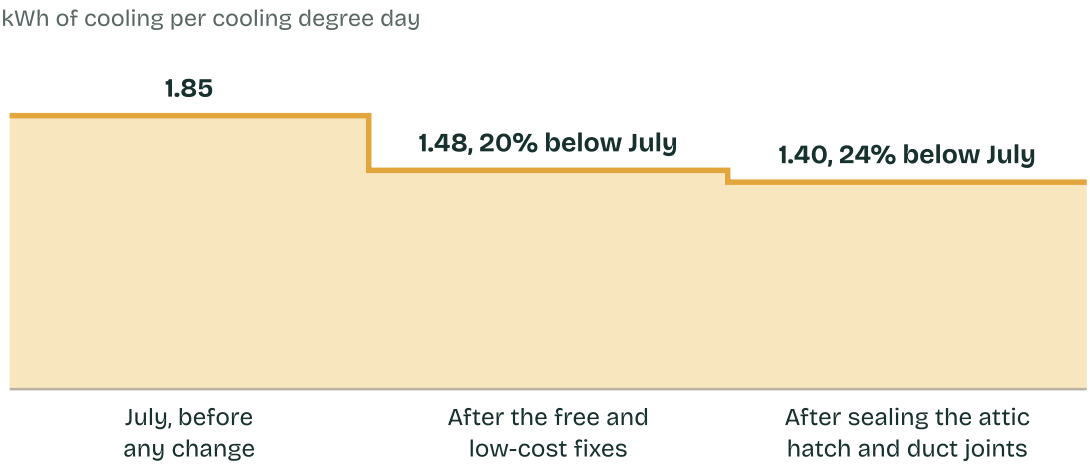


Figure 4.6. Dana's cooling use per cooling degree day after each set of fixes. Illustrative: Case C.

SAFETY WARNING: Dana is a healthy adult. In a home with babies, older adults or anyone ill, keep indoor temperatures safe first, and save in other ways.

TOOL 4.1

The Heating and Cooling Checklist

Purpose: to work through a home's heating and cooling from the free steps to the expensive ones, and to record whether each change worked. Run it at the start of each heating and cooling season, and whenever a bill looks high.

Settings (no-cost)

- ☐ The thermostat has a schedule with setbacks while you sleep or are away (4.4)
- ☐ A heat pump's schedule avoids triggering backup heat (4.4)
- ☐ The thermostat is away from sun, lamps and drafts
- ☐ Ceiling fans run only in occupied rooms, in the right direction for the season (4.8)
- ☐ Curtains and blinds are used by season: closed on sunny summer windows, closed at night in winter (4.6)

Airflow and maintenance (no-cost to low-cost)

- ☐ Filter checked this month, and changed within the last 3 months (4.7)
- ☐ Supply registers and return grilles are open and unblocked
- ☐ The outdoor unit is clear of leaves, snow and plants
- ☐ Carbon monoxide alarms tested this month (homes with fuel-burning appliances)
- ☐ A seasonal tune-up is booked or done (technician)

The house (low-cost to moderate)

- ☐ Doors and operable windows have sound weatherstripping (4.5)
- ☐ The attic hatch is weatherstripped and insulated
- ☐ Accessible duct joints are sealed with mastic or foil tape, not cloth duct tape
- ☐ No combustion-air openings, flues or exhaust vents have been blocked (4.5)
- ☐ An energy audit has been requested from the utility, if available

Bigger decisions (major)

- ☐ Equipment age, model and efficiency are recorded (4.1)
- ☐ Repair-or-replace questions are answered with a technician (4.10)
- ☐ Running costs are compared with the Fuel Cost Comparison (Tool 3.1)

Measurement log

Month	Heating or cooling degree days	Heating or cooling use (therms, gallons or kWh above mild-month use)	Use per degree day	Changes made this month

Reading your results

- A falling figure per degree day after a change means it is working. One month is Estimated; a whole season is Verified (1.8).
- A rising figure with no change in habits points to equipment or the house. Start with the filter, then book a technician.
- If the figure is already low and bills are still high, the price may be the problem, not the house. Go back to 2.4 for electricity, 3.3 for gas, or 3.6 for propane and oil.

Chapter 4 review

What you learned

- Heating and cooling costs depend on the weather, your settings, the house, the equipment and the hours you are home. You control all but the weather.
- Work out your heating and cooling share from your mild-month use. No improvement can save more than that share.
- Thermostat setbacks are the cheapest saving: about 1% of heating per degree over 8 hours, and as much as 10% a year (DOE). Heat pumps need a gentle schedule.
- Air sealing can save 10% to 20% of heating and cooling (DOE), but combustion-air openings, flues and exhaust vents must stay open.
- Filters and maintenance protect efficiency; airflow problems alone can cost up to 15% (ENERGY STAR).
- Fans cool people, not rooms. Smart thermostats save only through the setbacks they make happen.
- Replacement windows and new equipment rarely pay back on energy alone. Fix the house and the habits first.

What you can do today

1. Check your furnace or air-handler filter, and change it if it is dirty (4.7).
2. Set a thermostat schedule with setbacks for sleep and time away (4.4).
3. Find your mild-month use and your heating or cooling share from last year's bills (4.2).
4. Weatherstrip one leaky door or the attic hatch (4.5).

5. Run the Heating and Cooling Checklist (Tool 4.1).

What to check next month

Record the month's degree days (see "Getting your degree days" in 1.8) and your heating or cooling use above mild-month use, and divide one by the other. Compare it with the same figure before any change. Keep the measurement log going through the season.

Billix Quick Check

A home's heating costs $900 a year. A neighbor says new windows will cut that by 30%, and quotes $15,000. Even at 30%, how long is the simple payback?

Answer: 30% of $900 = $270 a year. $15,000 ÷ $270 = about 56 years. And DOE's 10% to 30% range is for low-e storm windows on the right existing windows, not a promise for replacements. Seal leaks and use coverings first.

Sources and further reading

All pages were accessed in October 2026.

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2. DOE, Consumer guide to home heating and cooling, DOE/EE-2464 https://www.energy.gov/sites/default/files/2021-08/ES-HomeHeatingandCooling_081221.pdf. The 1% per degree figure, maintenance savings, and summer away settings.
3. DOE, home heating systems fact sheet https://www.energy.gov/sites/default/files/2014/01/f6/homeHeating.pdf, 2014. AFUE ranges.

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4. Heating and cooling https://www.energystar.gov/saveathome/heating-cooling. Filter guidance and smart thermostat savings.
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9. Ceiling fans https://www.energystar.gov/products/ceiling_fans.
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14. Attachments Energy Rating Council (AERC). Ratings for window coverings and storm windows, as cited by DOE.

Ratings and equipment lives

15. DOE, 2023 central air conditioner and heat pump standards FAQ https://www1.eere.energy.gov/buildings/appliance_standards/pdfs/2023_CAC_Standards_FAQ_10-5-2022_Final.pdf, October 5, 2022. SEER2, EER2 and HSPF2, the 2023 minimums and the three regions.
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18. DOE, consumer furnace standards, final rule, 88 FR 87502 https://www.govinfo.gov/content/pkg/FR-2023-12-18/html/2023-25514.htm, December 18, 2023. Average furnace life.

Cross-references to Chapter 3 rely on its sources for fuel prices, AFUE minimums and heat pump field data.

PART III

Water

5 Water

6 Sewer, Wastewater and Stormwater

5

PART III · WATER

Water

Water is usually the smallest of the big utility bills, but it has risen much faster than prices overall. Federal price data show water and sewer charges up about 155% since 2006, against about 64% for prices overall, and up 4.9% in the year to August 2026. The money mostly goes to pipes and treatment plants: EPA estimates the country needs $625 billion over 20 years for drinking water systems alone.

Water also hides the most expensive surprise in household bills: the silent leak. EPA says household leaks waste nearly 1 trillion gallons a year nationwide, and that the average home's leaks waste more than 9,300 gallons a year. A running toilet can double a bill before anyone notices.

Unlike energy, every gallon you save can cut up to three bills at once: water, sewer if it is charged on water use, and the energy to heat it if it was hot. This chapter shows how to count all three, and how to count each one only once.

By the end of this chapter you will be able to:

- Read a water bill, and convert between gallons, cubic feet and CCF
- Work out your household's gallons per day per person, and tell when it is too high
- Find leaks with two free tests, and know when to call a plumber
- Judge which fixture upgrades pay for themselves, and which do not
- Count the full saving from hot-water changes, without double counting
- Ask your utility for a leak adjustment after a spike

Calculations use the Billix example rate card: water at $6.00 and sewer at $8.00 per 1,000 gallons, electricity at $0.18 per kWh and gas at $1.30 per therm. Sewer charges get their own chapter, Chapter 6.

START HERE:

- → **Your bill jumped?** Run the 15-Minute Bill Audit after 1.8, then work out use per person per day (5.3) and do the meter test (5.10). Case F, at the end of this chapter, goes through every step, up to the leak adjustment.
- → **Your bill is in CCF?** Go to 5.2 to convert gallons and prices.
- → **Wondering whether a new toilet or showerhead pays?** Go to 5.4 or 5.6, then Tool 5.3.
- → **On a private well?** Go to the sidebar in 5.1.

5.1 Where household water comes from

Most American homes get water from a public system, and pay for it through a metered bill. Water is drawn from rivers, lakes, reservoirs or underground aquifers, treated to drinking standards, stored, and pumped through miles of pipe to your meter.

That network is where the money goes. EPA's latest survey of drinking water needs estimates $625 billion over 20 years, and about two thirds of that, $422.9 billion, is for the pipes that carry water to homes.

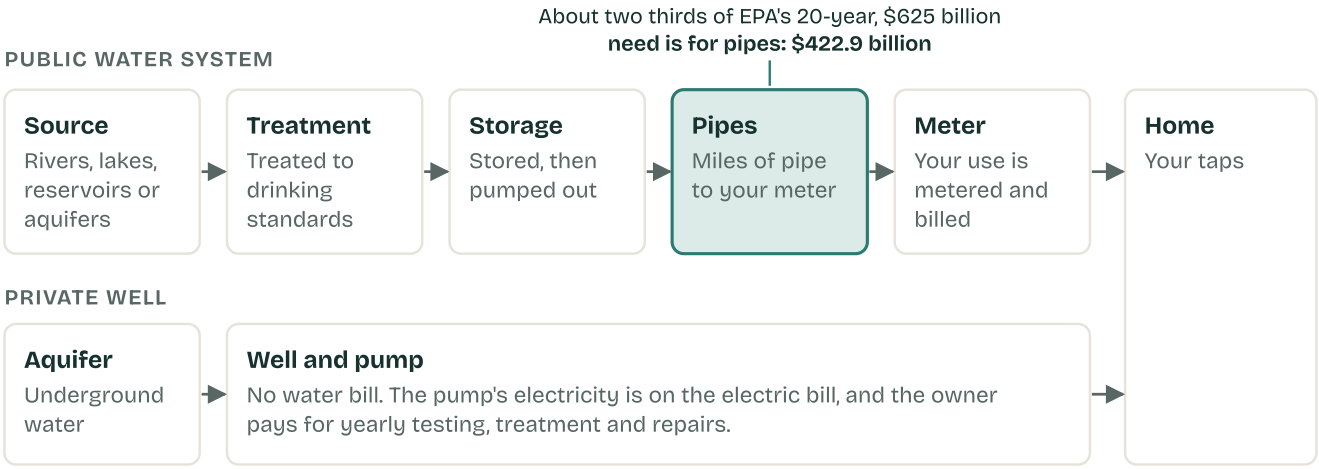


Figure 5.1. From source to tap, with the private-well path below. Source: EPA, 7th Drinking Water Infrastructure Needs Survey and Assessment, for the $625 billion 20-year need.

Most water systems are owned by a city, county or water authority, whose council or board sets the rates. A private water company's rates are usually set by the state utility commission (10.8 shows who handles a dispute).

Sidebar: private wells

More than 23 million households, over 43 million people, rely on private wells, according to EPA. No federal rule protects the water from a private well, and most states do not regulate it either. The owner is responsible for its safety.

Well households have no water bill, but they still pay for water:

- **Electricity** for the well pump, which shows up on the electric bill. A sudden jump in electric use can be a sign the pump is running too often, perhaps because of a leak.
- **Testing.** EPA and CDC advise testing every year for total coliform bacteria, nitrates, total dissolved solids and pH. Test more often if the home has children, older adults, or someone pregnant or nursing. Test right away after flooding, nearby land disturbance or a well repair, or if the water's taste, odor or color changes. Use a state-certified laboratory; some local health departments test for free.
- **Treatment and repairs,** such as filters, softeners, pump replacement and maintenance.
- **Septic systems,** common in homes with wells (6.5).

The leak-finding methods in this chapter work for wells too. Instead of a meter, listen for the pump switching on when no one is using water.

5.2 Reading a water bill

A water bill works like the bills in 1.2 and 1.3: a fixed base charge, a usage charge per unit, and often sewer and stormwater charges on the same bill. What makes water confusing is the units, which differ from one utility to the next.

The units

Unit	Equals	Notes
Gallon		The everyday unit
Cubic foot	7.48 gallons	Some meters read in cubic feet
CCF (100 cubic feet), also written Ccf or HCF	748 gallons	A common billing unit, alongside gallons
1,000 gallons	1.337 CCF	Many bills charge per 1,000 gallons

Gallons = CCF × 748

A real bill

DC Water publishes a typical home's bill. For a home using 5.42 CCF (4,054 gallons) a month, its rates from October 2, 2026 come to $155.88:

Line	Type	Amount
Water and sewer usage	Per CCF: water $6.49 for the first 4 CCF and $8.40 above, sewer $12.91	$107.86
Clean Rivers Impervious Area Charge	Fixed, by roof and paved area (6.6)	$25.50
Customer metering fee	Fixed, by meter size	$7.75
Water system replacement fee	Fixed	$6.30
City fees	Per CCF	$4.44
Stormwater fee	Fixed	$4.03
Total		**$155.88**

Three things stand out. The water rate is **tiered**: each CCF above 4 costs more. Sewer costs more per unit than water and is charged on water use. And about 28% of the bill, $43.58, is fixed. That is one city, not a national average. For comparison, an academic survey of 414 utilities found an average combined water and sewer bill of $86.41 a month at 4,800 gallons in 2025, with sewer averaging about 15% more than water.

A sample bill

Illustrative example: Case F, the Brooks household's usual bill, every two months. Anytown Water & Sewer is fictional, and the bill is marked SAMPLE. Rates follow the Billix rate card.

Line	Detail	Amount
Service period	60 days	
Meter readings	10,800 gallons used (about 14.4 CCF)	
Water base charge	Fixed, two months	$15.00
Sewer base charge	Fixed, two months	$15.00
Water usage	10.8 thousand gallons × $6.00	$64.80
Sewer usage	10.8 thousand gallons × $8.00, billed on water use	$86.40
Total due		**$181.20**

For the Brooks household, every 1,000 gallons saved takes $14.00 off the bill: $6.00 of water plus $8.00 of sewer. That is their marginal price (1.8). The $30.00 of base charges stays whatever they use.

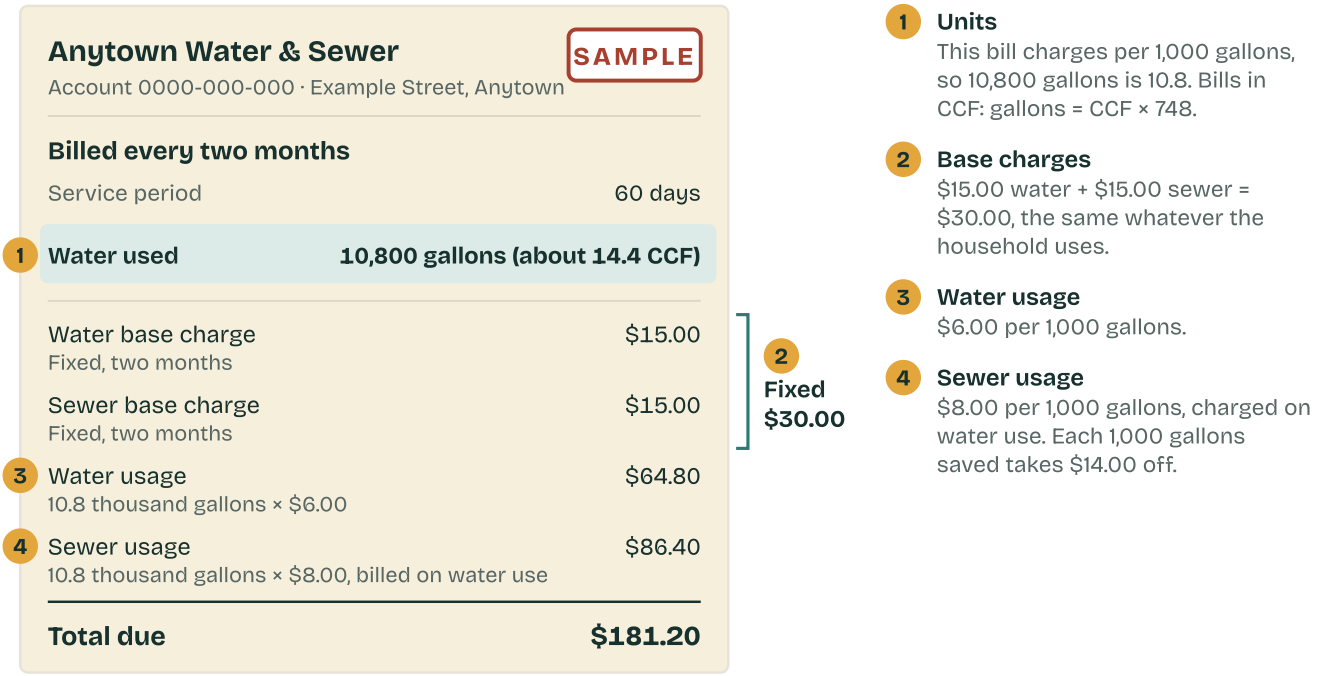


Figure 5.2. Annotated water bill: the Brooks household's usual two-month bill from Case F. SAMPLE: Anytown Water & Sewer is fictional; rates follow the Billix rate card.

Other charges you may see

- **Minimum charges** that include a set amount of water, so the first few thousand gallons cost nothing extra
- **Seasonal or drought rates,** which charge more at times of high demand or shortage
- **Infrastructure or replacement fees,** like DC Water's, that fund pipe replacement
- **Stormwater and sewer district charges,** covered in 6.2 and 6.6

Find it on your bill: the billing unit (gallons, cubic feet or CCF), the rate for each tier, the base charges, and whether sewer is charged on your water use. Add the water and sewer rates for your top tier to get your marginal price. If your rates are per CCF, convert the prices too: price per CCF × 1.337 = price per 1,000 gallons, so $4.49 per CCF is about $6.00. **Record it in** Tool 5.3, lines B and C.

5.3 Finding unexpected use

The quickest way to tell whether your home uses too much water is **gallons per person per day**. It strips out the length of the billing period and the size of the household, so you can compare your home with national figures and with itself over time.

Gallons per person per day = gallons on the bill ÷ (days in the period × people in the home)

Illustrative example: the Brooks household's usual bill

- 10,800 gallons ÷ 60 days = 180 gallons a day
- 180 gallons ÷ 3 people = 60 gallons per person per day

What is typical

Benchmark	Gallons per person per day	What it covers
Indoor use (Water Research Foundation, 2016 study and 2026 update)	About 40 to 60	Indoors only, single-family homes
All use at home (EPA, citing USGS 2015)	About 82	Indoors and outdoors
EPA's winter warning sign	About 100	Indoors in January or February: more than 12,000 gallons a month for a family of four, which EPA says probably means a serious leak

These measure different things, so match like with like. A winter bill, when there is little outdoor watering, compares with the indoor figures. A summer bill includes lawns and gardens.

Indoor use has been falling for decades as toilets, washers and showerheads become more efficient. If your home uses much more than the indoor benchmarks in winter, look for old fixtures and leaks first.

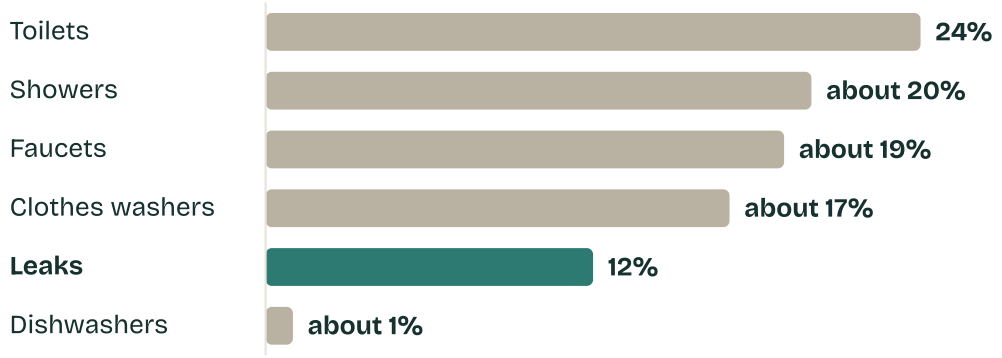


Figure 5.3. Where indoor water goes: each use's share of indoor water use. Source: Water Research Foundation, Residential End Uses of Water, Version 2, 2016.

Find your winter indoor use

1. Take a winter bill, when outdoor use is near zero. Its gallons per person per day is your **winter indoor use**.
2. Compare summer bills with it. The extra is roughly your outdoor use (5.9).
3. Compare every new bill with the same season last year, per person per day.

A jump in a season with no outdoor watering points to a leak or a change in the household. A jump in summer can be a leak or extra watering. To tell which:

1. Compare gallons per person per day with the same bill last year.
2. Turn the irrigation off and run the meter test (5.10). If the meter moves, you probably have a leak; the table in 5.10 helps you find it.
3. If the meter stays still, run each irrigation zone and look for leaks (5.9).
4. What remains above your winter indoor use is watering. Compare your watering schedule, and the rainfall, with last summer's.

Sidebar: watch the meter. Some utilities now have smart water meters and show daily or hourly use online. If yours does, check overnight use: in a home where everyone is asleep, it should be close to zero. Steady overnight use, when nothing should be running, usually points to a leak. Rule out irrigation timers, water softeners and ice makers first.

5.4 Toilets

Toilets are the largest single indoor use of water, and a common source of silent leaks. In the Water Research Foundation's 2016 study, toilets were 24% of indoor use, at an average of 2.6 gallons per flush and about 5 flushes per person per day. The 2026 update reports the average flush has fallen to about 1.84 gallons as old toilets are replaced.

Toilet	Gallons per flush
Older toilets	As much as 6 (EPA)
Federal maximum for new toilets	1.6
WaterSense-labeled toilets	1.28 or less, 20% below the federal standard

To find your toilet's flush volume, look for a number marked "gpf" or "Lpf" stamped on the bowl behind the seat or inside the tank.

The dye test

A leaking flapper or fill valve can run water into the bowl all day without a sound. EPA's test takes 10 minutes:

1. Remove the tank lid.
2. Put a few drops of food coloring in the tank. Do not flush.
3. Wait 10 minutes.
4. If color appears in the bowl, the toilet is leaking.
5. Flush afterward so the dye does not stain the tank.

A common cause is a worn **flapper**, the rubber seal at the bottom of the tank, which is a low-cost part. Turn off the supply valve behind the toilet before replacing it. If water runs into the overflow tube instead, the **fill valve** may need adjusting or replacing.

Check your lease: Renters should report a running toilet to the landlord in writing. Your lease decides who pays for the repair and who pays for the wasted water.

Is a new toilet worth it?

EPA says replacing old, inefficient toilets with WaterSense models can save a family nearly 13,000 gallons a year. One of its pages puts the money saved at $130 a year, and another at more than $170. Whether it pays depends on what you are replacing.

Illustrative example: the Okafors, four people at 5 flushes each a day, or 7,300 flushes a year. They pay $14.00 per 1,000 gallons for water and sewer. Two new WaterSense toilets cost $600 installed (illustrative).

Replacing	Saved per flush	Gallons saved a year	Yearly saving	Simple payback
3.5 gallon-per-flush toilets	2.22 gallons	16,206	$226.88	About 2.6 years
1.6 gallon-per-flush toilets	0.32 gallons	2,336	$32.70	About 18 years

The math for the first row: (3.5 - 1.28) × 7,300 = 16,206 gallons, and 16,206 ÷ 1,000 × $14.00 = $226.88.

Replacing a toilet that uses well over 1.6 gallons a flush often pays quickly. Replacing one that already meets the 1.6-gallon standard rarely does; wait until it needs replacing anyway.

What to record: each toilet's flush volume, and the date of each dye test.

5.5 Faucets

Faucets were about 19% of indoor water use in the 2016 Water Research Foundation study. The savings come from two places: lower-flow aerators on bathroom faucets, and fixing drips.

Faucet	Maximum flow
Federal maximum for new bathroom and kitchen faucets	2.2 gallons per minute (gpm) at 60 psi
WaterSense-labeled bathroom faucets and aerators	1.5 gpm

EPA says WaterSense bathroom faucets or aerators save a family about 700 gallons a year. An **aerator**, the screen that screws into the faucet tip, is often all you need to change, and it is a low-cost part.

Kitchen faucets are different. Filling a pot takes the same water at any flow rate, so a lower-flow kitchen aerator mainly slows you down. Focus on bathroom faucets, where water runs while you wash.

Measure your flow

1. Hold a 1-gallon container under the faucet, turned fully on.
2. Time how many seconds it takes to fill.
3. Divide 60 by the seconds to get gallons per minute.

Illustrative example: a container fills in 20 seconds. 60 ÷ 20 = 3 gpm, well above today's federal maximum. A new aerator is worth trying.

Measure a drip

1. Put a measuring cup under the dripping faucet.
2. Note how long it takes to collect 1 cup, which is one-sixteenth of a gallon.

Illustrative example: 1 cup every 30 minutes

- 1/16 gallon × 2 per hour × 24 hours = 3 gallons a day
- 3 gallons × 365 days = 1,095 gallons a year
- 1,095 ÷ 1,000 × $14.00 for water and sewer = $15.33 a year

A hot-water drip also wastes the energy to heat it (5.11). Drips are often fixed with a new washer or cartridge, a low-cost part, though older faucets may need a plumber.

Check your lease: Renters can usually swap an aerator and put the original back when they move out. Report drips to the landlord.

5.6 Showers: one shower, two bills

Showers are where water and energy savings meet. Every gallon of hot water you do not use saves the water, the sewer charge if sewer is billed on water use, and the energy to heat it. The energy is often worth more than the water.

In the 2016 Water Research Foundation study, showers were about 20% of indoor use, and the average shower lasted 7.8 minutes. Hot water was about a third of all indoor use, mostly for showers and faucets.

Showerhead	Maximum flow
Federal maximum since 1994	2.5 gpm at 80 psi
WaterSense-labeled showerheads	2.0 gpm

EPA says a WaterSense showerhead saves an average family about 2,700 gallons and more than 330 kWh a year. Note that in 2025, DOE dropped its definition of "showerhead", so fixtures with several nozzles no longer have to stay under 2.5 gpm combined. A multi-head shower can use far more.

What a gallon of hot water costs

Heating water takes about 8.34 Btu per gallon for each degree F, because a Btu is the heat that warms one pound of water by one degree, and a gallon weighs about 8.34 pounds.

Illustrative assumptions: water arrives at 50 degrees F and is heated to 120, a 70-degree rise. An electric tank is about 92% efficient and a gas tank about 60%.

- Heat per gallon: 8.34 × 70 = 584 Btu
- **Electric:** 584 ÷ 0.92 ÷ 3,412 Btu per kWh = 0.186 kWh per gallon, or about 3.3 cents at $0.18 per kWh
- **Gas:** 584 ÷ 0.60 ÷ 100,000 Btu per therm = 0.0097 therms per gallon, or about 1.3 cents at $1.30 per therm

At $14.00 per 1,000 gallons, the water and sewer come to 1.4 cents a gallon. For a home with an electric water heater, the energy in a gallon of hot water costs more than twice the water itself.

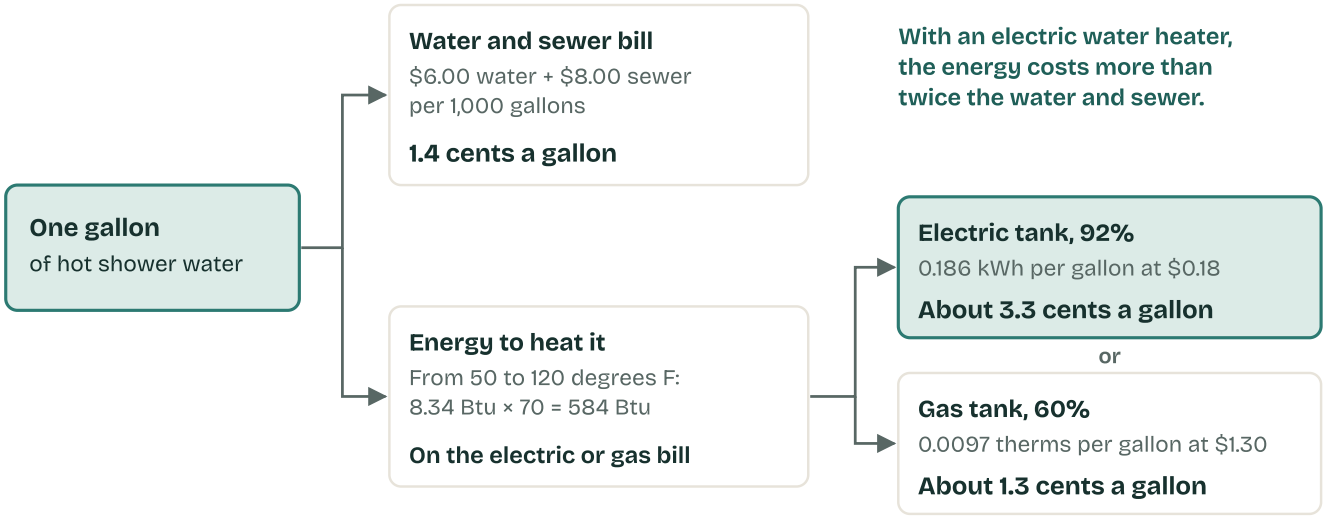


Figure 5.4. One shower, two bills: what a gallon of hot water costs on the water and sewer bill and on the energy bill. Illustrative: rate card prices; water heated from 50 to 120 degrees F.

Case B: a new showerhead for the Okafors

Illustrative. Four people each shower 7.8 minutes a day, the study's average, under 2.5 gpm heads. The Okafors have an electric water heater. Assume 70% of shower water is hot.

- Now: 4 × 7.8 minutes × 2.5 gpm = 78.0 gallons a day
- With 2.0 gpm heads: 4 × 7.8 × 2.0 = 62.4 gallons a day
- Saved: 15.6 gallons a day × 365 = 5,694 gallons a year

Bill	Saving	Value
Water and sewer	5,694 gallons × $14.00 per 1,000	$79.72
Electricity for hot water	70% of 5,694 = 3,986 hot gallons × 0.186 kWh = 741 kWh × $0.18	$133.38
Total a year		**$213.10**

Two WaterSense showerheads at $30 each (illustrative) pay for themselves in about 3.4 months. As a cross-check, this works out to about 0.13 kWh saved per gallon, close to the 0.12 implied by EPA's figures.

A one-minute-shorter shower for each person, at 2.0 gpm, saves 2,920 gallons a year: $40.88 of water and sewer, plus 380 kWh of water heating, or $68.40, for $109.28 in all.

When this advice does not apply

- If your showerheads already flow at 2.0 gpm or less, a new one saves little.
- If water pressure is low, a lower-flow head may give a weak shower. Try one, and keep the receipt.
- With a gas water heater, the energy saving is smaller: about 38.7 therms, or about $50, in the Okafors' example.

Count these savings once. The energy saved here is a water-heating saving; do not count it again in 5.11 or under the electric bill (1.8, Rule 3).

5.7 Laundry

Clothes washers were about 17% of indoor water use in the 2016 Water Research Foundation study, at an average of 31 gallons a load, down from 41 in 1999. The water matters, but the bigger lever is temperature. DOE says most of the energy a washer uses goes to heating water, and that washing in warm instead of hot can cut a load's energy use in half.

What to change

Change	Cost tier	Effect
Wash full loads, or set the water level to match the load	No-cost	Fewer loads, less water and energy per item
Wash in cold or warm instead of hot	No-cost	Cuts the hot water, which is most of the energy (DOE)
Use the high-speed spin setting	No-cost	Drier clothes need less time in the dryer (2.9)
When the washer needs replacing, choose ENERGY STAR	Moderate to major	ENERGY STAR washers use about 25% less energy and 33% less water than standard models (DOE)

Illustrative example: switching hot washes to cold. Assume a hot wash uses 15 gallons of hot water and the household runs 100 hot loads a year.

- Electric water heater: 15 gallons × 0.186 kWh = 2.79 kWh a load, about $0.50 at $0.18 per kWh
- 100 loads × $0.50 = about $50 a year
- With a gas water heater: 15 × 0.0097 = 0.146 therms a load, or about $18.92 a year at $1.30 per therm

The water use does not change, only the energy to heat it. Count it once, as a water-heating saving.

When this advice does not apply

- Some items, such as heavily soiled work clothes, bedding during illness, or cloth diapers, may need hot water. Follow the care label and any health guidance.
- A new washer for water savings alone rarely pays back if the old one works; replace it when it fails (13.3).

What to record: loads a week and the temperature setting, before and after a change.

5.8 Dishwashing

Modern dishwashers use little water: an average of 6.1 gallons a load in the 2016 Water Research Foundation study, down from 10 in 1999, and only about 1% of indoor use. Whether a dishwasher beats hand washing depends on how you wash by hand.

ENERGY STAR says a certified dishwasher uses less than half the energy of washing by hand, and can save about 8,400 gallons of water and $220 a year. Your own result depends on how you wash by hand now. The same 2016 study found that homes without a dishwasher did not use more faucet water than

homes with one, so the saving in a real home may be much smaller. This book does not claim that a dishwasher always uses less water.

Getting the most from either method

If you use a dishwasher	If you wash by hand
Run it only when full	Wash in a filled basin or sink rather than under a running tap
Skip the heated-dry setting (DOE)	Rinse in a second basin, or with a brief spray
Scrape dishes rather than pre-rinsing them under the tap	Use the lowest water temperature that gets dishes clean
If it has a booster heater, the home's water heater can stay at 120 degrees F (DOE; see 5.11)	Fix a dripping kitchen faucet (5.5)

5.9 Outdoor water

Outdoor watering can be the largest water use in summer, and the easiest to waste. EPA says the average American family uses more than 300 gallons a day at home, with roughly 30% of it outdoors. Common sources of outdoor waste are watering too often, watering at the wrong time, and leaks you cannot see.

What to check

Problem	What to do	Cost tier
Watering on a fixed schedule, rain or shine	Adjust the timer to the season, and skip watering after rain	No-cost
Watering in the heat of the day	Water early in the morning, when less evaporates	No-cost
Sprinklers spraying pavement	Aim or adjust the heads	No-cost
Broken heads or leaking lines	Run each zone and look for pooling, geysers or dry patches	Low-cost repairs
No rain sensor	Add one, so the system skips watering when it rains	Low-cost
Thirsty lawn in a dry climate	Replace some lawn with plants suited to the local climate; use drip lines for beds	Moderate to major

Check local rules first. Many cities set watering days or limits during droughts, and fines can apply.

Does outdoor water carry a sewer charge?

Water used outdoors never reaches the sewer, but whether you pay sewer on it depends on your utility. Some bill sewer on winter use only, some cap it, and some offer separate irrigation or deduct meters (6.4). This matters a great deal for any outdoor saving, because it decides your marginal price.

Is a smart irrigation controller worth it?

A smart controller adjusts watering to the weather. Whether it pays depends on how much you water outdoors and what each gallon costs you.

Illustrative example: a home uses 8,000 gallons a month more in four summer months than in winter (5.3). Assume a smart controller cuts outdoor use by 20%, and costs $150.

- Outdoor use: 4 × 8,000 = 32,000 gallons a summer
- Saved: 20% × 32,000 = 6,400 gallons

If sewer is	Price per 1,000 gallons	Yearly saving	Simple payback
Not charged on outdoor water	$6.00	$38.40	About 3.9 years
Charged on all water	$14.00	$89.60	About 1.7 years

The 20% is an assumption for this example, not a tested result. A controller earns its keep only if you were overwatering to begin with. Check its effect by comparing summer use per day with last summer's, allowing for rainfall.

5.10 Leaks

Leaks are a common cause of a sudden water bill spike, and the cheapest to fix once found. EPA says the average household's leaks waste more than 9,300 gallons a year, and that 9% of homes have leaks wasting 50 gallons or more a day. In the 2016 Water Research Foundation study, leaks averaged 12% of indoor use, and a few homes leaked as much as 600 gallons a day.

Illustrative, at $14.00 per 1,000 gallons for water and sewer:

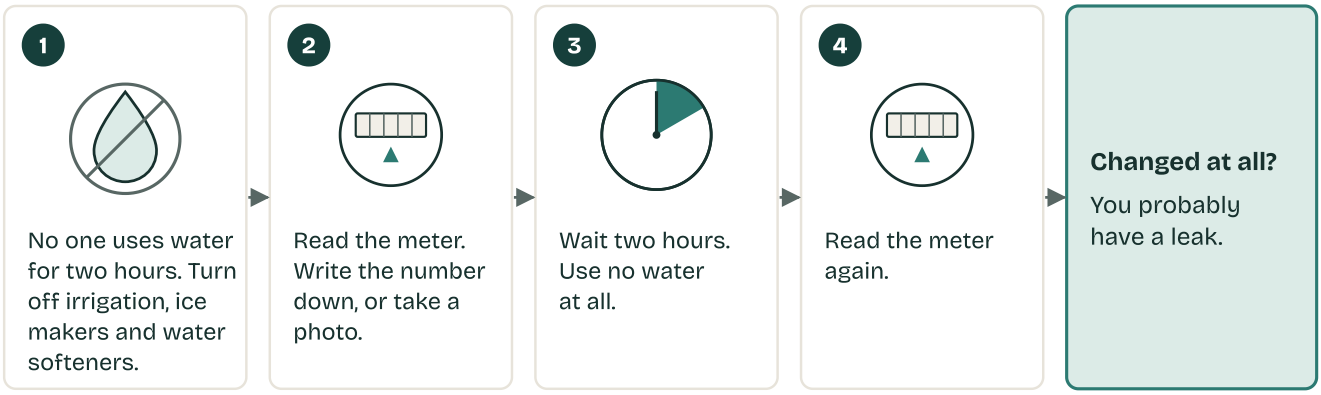
Leak	Gallons a year	Cost a year
EPA's average household leakage	9,300	$130.20
A 50-gallon-a-day leak	18,250	$255.50

The meter test

EPA's meter test finds leaks anywhere in the plumbing, including ones you cannot see.

1. Make sure no one will use water for two hours, and turn off automatic uses such as irrigation, ice makers and water softeners.
2. Read the water meter and write down the number, or take a photo.
3. Wait two hours without using any water.
4. Read the meter again. If the number has changed at all, you probably have a leak.

Many meters also have a small dial or symbol that spins whenever water flows, which can show a leak in minutes.



Tip: many meters also have a small dial or symbol that spins whenever water flows, which can show a leak in minutes.

Figure 5.5. The meter test (EPA): read, wait two hours without using water, read again.

Narrowing it down

Where	How to check
Toilets	The dye test (5.4)
Faucets, showerheads and tub spouts	Look and listen for drips; measure with a cup (5.5)
Outdoor faucets and hoses	Check connections and the faucet itself while it is off
Irrigation system	Run each zone; look for pooling and soggy spots (5.9)

Where	How to check
Water heater	Look for water around the base or from the relief valve pipe; call a professional if you find any
Water softener	Check that it is not regenerating constantly
The service line from the meter to the house	Close the house's main shutoff valve and watch the meter. If it still moves, the leak is in the pipe between the meter and the house

If the leak is in the service line, it is often the homeowner's responsibility. In New York City and Philadelphia, the owner is responsible for the line out to the city main. In Washington, D.C., the owner owns the whole pipe, but DC Water maintains it out to the property line. Rules vary by city, so ask your utility.

SAFETY WARNING: Find your home's main water shutoff valve before you need it. If water is pooling, coming through a ceiling, or running inside a wall, close the main valve and call a plumber. Keep away from water near electrical outlets, panels or appliances, and turn off power to the area if you can do so safely.

Ask about a leak adjustment

Many utilities, though not all, reduce a bill after a leak has been found and fixed. Policies differ, so ask what yours covers, what proof it needs and how often it can be used. Before you call, have:

- The bill with the spike, and a normal bill for comparison
- Proof of the repair: a receipt, a plumber's invoice or a photo of the replaced part
- Meter readings showing use back to normal after the repair

Case F shows a full example.

5.11 Water-heating costs and safe temperatures

Heating water is one of a home's largest energy uses. In homes that heat water with electricity, EIA's survey found it averaged about 2,705 kWh a year, about $487 at $0.18 per kWh. Every saving in 5.4 to 5.8 that cuts hot water lowers this bill too; this section covers the water heater itself.

Types of water heater

Type	How it works	Worth knowing
Storage tank (gas, propane or electric)	Keeps a tank of water hot all the time	Loses heat through the tank walls even when no one uses hot water, called standby loss. DOE puts the average life of a gas or electric tank at about 15 years; many fail sooner.
Tankless (on-demand)	Heats water only as it flows	No standby losses (DOE). Typically delivers 2 to 5 gallons a minute, so a shower and a dishwasher at once can hit its limit. DOE says most last more than 20 years. Gas models still need electricity to run.
Heat pump water heater	Moves heat from the surrounding air into the tank	ENERGY STAR says it can save a family of four about $550 a year compared with a standard electric tank. Usually the strongest option when replacing an electric tank (13.4, and Case B in Chapter 13).

ENERGY STAR says a certified gas tankless heater saves a family of four about $95 a year on gas compared with a standard gas tank. Tankless units usually cost more to install, so the payback can be long. 13.3 shows how to compare options on total cost.

The right temperature

DOE recommends setting a water heater to 120 degrees F. Hotter water wastes energy and can scald: CPSC says water at 140 degrees F can cause a serious burn in about 6 seconds, against about 5 minutes at 120 degrees F.

There is a trade-off. Legionella bacteria, which can cause a serious pneumonia, grow in water between 77 and 113 degrees F. CDC's guidance for large buildings calls for storing hot water above 140 degrees F, with mixing valves to prevent scalding at the tap. For homes, New Jersey's Department of Health suggests at least 120 degrees F, and notes that higher settings may help households at high risk, if a mixing valve guards against scalding.

Household	A reasonable approach
Most homes	120 degrees F, as DOE recommends
Homes with young children or older adults	Keep tap water at 120 degrees F or below to prevent scalds
Anyone in the home at higher risk of infection, such as a person with a weakened immune system	Ask a doctor and a plumber about a higher tank setting with a mixing valve
Dishwashers without a booster heater	Check the dishwasher's manual for the water temperature it needs

Many tank thermostats are not marked in degrees. Run hot water at the tap closest to the heater for a minute, measure it with a kitchen thermometer, and adjust in small steps.

SAFETY WARNING: On an electric water heater, the thermostats sit behind covers next to live 240-volt wiring. Switch off the heater's circuit breaker before you remove a cover, or ask a plumber or electrician to make the change. On a gas water heater, turn only the temperature dial on the gas control.

Call a professional: for any work on a water heater's gas supply, venting, wiring or temperature and pressure relief valve. Water leaking from the relief valve pipe or the tank needs prompt attention.

Other steps

Step	Cost tier	Notes
Lower the setting to 120 degrees F	No-cost	See the table above first
Fix hot-water drips	Low-cost	They cost energy as well as water (5.5)
Insulate accessible hot-water pipes near the heater	Low-cost	Keep insulation clear of a gas heater's flue
Ask about timer or on-demand controls on a hot-water recirculation pump	Low-cost to moderate	Recirculation delivers hot water faster but can use extra energy if it runs all the time
Replace an old tank with a heat pump water heater, when it fails	Major	13.4; Case B in Chapter 13

5.12 Putting the savings math together

A water saving can show up on up to three bills, and each must be counted once, at its own marginal price.

Saving = (gallons saved × water and sewer price per gallon) + (hot gallons saved × energy per gallon × energy price)

Use these inputs:

- **Water price:** your top-tier rate, since savings come off the last gallons you use (5.2)
- **Sewer price:** add it only if your sewer charge rises with your water use in the months the saving happens (6.2)
- **Energy per hot gallon:** about 0.186 kWh for an electric tank, or 0.0097 therms for a gas tank, under the assumptions in 5.6

- **Energy price:** your marginal electricity or gas price (1.8, Rule 2)

A savings ledger

Keep one line for each change and one column for each bill. A change appears once, even if it saves on several bills.

Illustrative example: the Okafors make three changes. Their two toilets are older 3.5 gallon-per-flush models. Water and sewer cost $14.00 per 1,000 gallons, and electricity $0.18 per kWh.

Change	Gallons saved a year	Water and sewer	Electricity for hot water	Total
Two WaterSense showerheads (5.6)	5,694	$79.72	$133.38	$213.10
Two WaterSense toilets (5.4)	16,206	$226.88		$226.88
Bathroom faucet aerators, at EPA's 700 gallons a year, assuming 70% hot	700	$9.80	$16.38	$26.18
Total	**22,600**	**$316.40**	**$149.76**	**$466.16**

The showerheads and aerators cost about $70 and pay back within months. The toilets cost about $600 and pay back in under three years. These are Estimated savings; the bills over the next year will show whether they are real.

Rules that keep the ledger honest

- **Do not count water heating twice.** If the Okafors later replace their water heater (Case B in Chapter 13), its saving is calculated on the smaller amount of hot water they now use.
- **Fixed charges do not change.** Base charges, meter fees and stormwater charges stay the same whatever you save (5.2).
- **Check the sewer rule.** If sewer is set on winter use, an indoor saving made in winter can lower sewer charges for the whole year, while an outdoor saving in summer may not lower them at all. The line C note in Tool 5.3 shows how to count this (6.2).
- **Use the right tier.** On a tiered rate, value saved gallons at your top tier's rate until your use falls below that tier's threshold, then at the next tier's rate. Check whether the thresholds are set per month or per billing period.
- **Measure.** Compare gallons per person per day with the same season last year (5.3). A saving becomes Verified when your bills show it over several billing periods, adjusted for days and season (1.8).

CASE F

The Brooks household's $160 water spike

Illustrative example. The Brooks household is a fictional family of three in a suburban house with city water and sewer, billed every two months. Sewer is charged on water use. Rates follow the Billix rate card: $6.00 for water and $8.00 for sewer per 1,000 gallons, plus $30.00 of base charges per bill.

The winter bill arrived at $340.80. The Brooks household usually paid about $181.20. Their first thought was a billing error.

	Usual bill	This bill
Days	60	62
Gallons used	10,800	22,200
Water usage	$64.80	$133.20
Sewer usage	$86.40	$177.60
Base charges	$30.00	$30.00
Total	**$181.20**	**$340.80**

Step 1: Verify the billing period

The bill covered 62 days instead of 60. Two extra days at their usual 180 gallons a day explain only about 360 gallons, or $5.04. The period was not the problem.

Step 2: Check the usage

- Usual: 10,800 gallons ÷ 60 days ÷ 3 people = 60 gallons per person per day
- This bill: 22,200 gallons ÷ 62 days ÷ 3 people = 119 gallons per person per day

Use had doubled, in winter, with no watering and no new people in the house. That is above EPA's winter warning level of about 100 gallons per person per day (5.3).

Step 3: Actual or estimated reading?

The bill showed an **actual** reading, and so did the one before it. Their photo of the meter matched the current reading. That ruled out a catch-up after estimates (1.5). The water had really gone through the meter.

Step 4: Look for leaks

One evening, with everyone home and no water running, they did the meter test (5.10). Over two hours the meter moved by 15 gallons, roughly 180 gallons a day. They did the dye test on both toilets (5.4). In the upstairs toilet, color reached the bowl within minutes: a worn flapper. A new flapper cost about $8 (illustrative), and they fitted it in 15 minutes.

Step 5: Compare meter readings after the repair

A second two-hour meter test showed no movement. For a week they read the meter each morning, and use averaged 182 gallons a day.

- Leak: 358 gallons a day during the bill - 182 after the repair = about 176 gallons a day
- Over 62 days: about 10,900 gallons, or about $152.82 at $14.00 per 1,000 gallons

The leak explains almost all of the $159.60 increase. The two extra days and normal variation explain the rest.

Step 6: Contact the provider

They called the utility with the bills, the flapper receipt and the week of readings in hand:

> "Our last bill was $340.80, against our usual $181.20 for a similar two-month period. We found a leaking toilet flapper and replaced it on [date]. Our meter readings since then are back to about 182 gallons a day, and I have the receipt. Does the utility offer a leak adjustment, and how do we apply?"

They wrote down the date and the representative's first name, but did not ask for a reference number. Case F in Chapter 10 shows what that cost them.

Step 7: Request the leak adjustment

Their utility's policy, which is illustrative here because policies vary, credits half of the water and sewer charges above normal use, once in any 12 months.

- Normal use for 62 days: 180 gallons × 62 = 11,160 gallons
- Excess: 22,200 - 11,160 = 11,040 gallons
- Excess charges: 11,040 ÷ 1,000 × $14.00 = $154.56
- Credit at 50%: $77.28

The adjustment cut their net cost for the leak roughly in half. Had the utility refused, their next step would be the utility's complaint process, then its board or the city council (10.8).

What they changed afterward

- A dye test on every toilet twice a year, and a meter test whenever a bill looks high
- A quick look at the meter each month, so a leak shows up in weeks rather than at the next bill
- A note in their records of the main shutoff valve's location

The lesson: with sewer billed on water use, a leak cost them twice. A two-hour meter test, a 10-minute dye test and an $8 part found and fixed a problem that would have kept adding about $150 to every bill.



Figure 5.6. Case F: how the Brooks household traced a $160 water spike to a worn flapper and a leak adjustment. Illustrative.

Tools for this chapter

Tool 5.1: the Household Water Audit

Purpose: to see where your water goes, room by room, and which fixtures are worth changing.

Instructions: For each fixture, enter its flow rate (gallons per minute, measured as in 5.5) or its volume per use (gallons per flush or per load). Estimate uses a day and minutes per use. Multiply to get gallons a day, and mark whether the water is hot. Compare your total with your bill's gallons per day (5.3); a large gap points to a leak or outdoor use.

Area	Fixture	Flow or volume	Uses a day	Minutes per use	Gallons a day	Hot water?	Action
Bathroom 1	Toilet	gpf				No	
Bathroom 1	Shower	gpm				Mostly	
Bathroom 1	Faucet	gpm				Partly	
Bathroom 2	Toilet, shower, faucet						
Kitchen	Faucet	gpm				Partly	
Kitchen	Dishwasher	gallons a load				Yes	
Laundry	Clothes washer	gallons a load				Depends on setting	
Outdoors	Irrigation or hose	gpm				No	
Total							
Bill, gallons a day (5.3)							

Tool 5.2: the Water Leak Inspection Checklist

Purpose: to find leaks safely and record what you found. Run it twice a year, and whenever a bill jumps.

- ☐ Main shutoff valve located, and everyone in the home knows where it is (5.10)
- ☐ Meter test: reading at start _____, after 2 hours _____ (5.10)
- ☐ Dye test on each toilet: Toilet 1 _____ Toilet 2 _____ (5.4)
- ☐ Faucets and showerheads checked for drips; drip rate measured if found (5.5)
- ☐ Tub spouts and shower valves checked for drips when off
- ☐ Outdoor faucets and hose connections checked
- ☐ Irrigation zones run and inspected for pooling or broken heads (5.9)
- ☐ Floor around the water heater and the relief-valve pipe checked for water (5.11)
- ☐ Water softener checked for constant cycling
- ☐ Service line test, if the meter still moves with the house valve closed (5.10)
- ☐ Ceilings, walls and floors checked for stains, damp spots or the sound of running water

Record the date, what you found and what you did. Keep receipts for any repair; you may need them for a leak adjustment.

Tool 5.3: the Water Savings Calculator

Purpose: to value a water saving on every bill it affects, once (5.12).

Input	Your figure	Okafor showerhead example (5.6)
A. Gallons saved a year		5,694
B. Water price per 1,000 gallons, top tier (on a CCF bill: price per CCF × 1.337)		$6.00
C. Sewer price per 1,000 gallons, if charged on this water (on winter averaging, see the note below)		$8.00
D. Water and sewer saving: A ÷ 1,000 × (B + C)		$79.72
E. Share of the saved water that was hot		70%
F. Hot gallons saved: A × E		3,986
G. Energy per hot gallon: 0.186 kWh (electric) or 0.0097 therms (gas)		0.186 kWh
H. Energy price		$0.18 per kWh
I. Energy saving: F × G × H		$133.38
J. Total yearly saving: D + I		**$213.10**
K. Cost of the change		$60
L. Simple payback in months: K ÷ J × 12		About 3.4

Notes for lines B and C:

- **Tiered rates:** use the top tier's rate until the saving takes your use below that tier's threshold, then the next tier's rate (5.12).
- **Winter averaging:** enter 0 in line C for a saving made only outside the averaging months. For a saving made in them, the sewer part is gallons saved a month in those months × 12 ÷ 1,000 × the sewer price, because the average sets sewer for the whole year. Add that to the water part, A ÷ 1,000 × B, to get line D (see the worked example in 6.2).

Treat the result as Estimated until a year of bills, compared per person per day, confirms it.

Chapter 5 review

What you learned

- Water and sewer prices have risen far faster than prices overall, mostly to pay for pipes and treatment.
- One CCF is 748 gallons. Your marginal price is usually the top-tier water rate plus the sewer rate.
- Gallons per person per day, compared season to season, shows when something is wrong.
- The dye test and the two-hour meter test find most leaks for free.
- New fixtures pay quickly when they replace old, high-flow ones, and slowly when the old ones already meet today's standards.
- Hot water is two bills in one; with an electric water heater, the energy can be worth more than the water.
- 120 degrees F is DOE's recommended setting, with extra care for households at risk of scalds or infection.
- Count each saving once, on each bill it affects, at that bill's marginal price.

What you can do today

1. Find your main water shutoff valve (5.10).
2. Do the dye test on every toilet (5.4).
3. Read your meter before bed and again in the morning, with no water used overnight (5.10).

4. Work out your gallons per person per day from your last bill (5.3).
5. Measure your showerhead's flow with a bucket and a timer (5.5, 5.6).

What to check next month

Record gallons per person per day from your next bill and compare it with the same season last year. If you made a change, enter it in Tool 5.3, the Water Savings Calculator, and keep the result as Estimated until two more bills confirm it.

Billix Quick Check

A two-person household's bill shows 9 CCF used over 30 days in January. How many gallons per person per day is that, and is it worth investigating?

Answer: 9 CCF × 748 = 6,732 gallons. 6,732 ÷ 30 days ÷ 2 people = about 112 gallons per person per day. In January, with no outdoor watering, that is well above the indoor benchmarks of about 40 to 60 gallons, and above EPA's winter warning level of about 100. Run the meter test and the dye test.

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6

PART III · WATER

Sewer, Wastewater and Stormwater

On many bills, the sewer charge is larger than the water charge. An academic survey of 414 utilities found sewer bills averaged about 15% more than water bills in 2025. In Washington, D.C., the sewer rate per unit is about one and a half to two times the water rate. Yet most people never look at how the sewer charge is worked out.

Sewer is usually not metered. Your utility cannot measure what goes down the drain, so it estimates it, most often from your water use. How it does that decides whether a water saving also cuts your sewer bill, and whether water on the lawn is charged as if it went down the drain.

The costs behind the charge are large. EPA's most recent survey of wastewater and stormwater needs, sent to Congress in 2024, estimated about $630 billion over 20 years.

By the end of this chapter you will be able to:

- Find out how your sewer charge is calculated, and which kind of rate you are on
- Tell whether a water saving will also lower your sewer charge
- Judge whether an irrigation or deduct meter is worth the cost
- Plan for septic system costs if your home is not on a public sewer
- Read a stormwater fee, and find out whether your city offers credits

Calculations use the Billix example rate card: water at $6.00 and sewer at $8.00 per 1,000 gallons. The sewer figure is a placeholder; real sewer rates vary widely.

START HERE:

- → **Your sewer charge is high, or stays high all year?** Go to 6.3, then the worked example in 6.2.
- → **Not sure whether a water saving cuts sewer?** Find your method in 6.2.
- → **Thinking about an irrigation or deduct meter?** Go to 6.4.
- → **On a septic system?** Go to 6.5 and Case E.
- → **A stormwater fee on your bill?** Go to 6.6.

6.1 How wastewater systems work

Everything that goes down a drain or toilet in a sewered home flows through a pipe called the **lateral** to a public sewer main. From there it goes to a treatment plant, which cleans it before releasing it to a river, lake or the ocean. Homes not connected to a public sewer use a septic system instead (6.5).

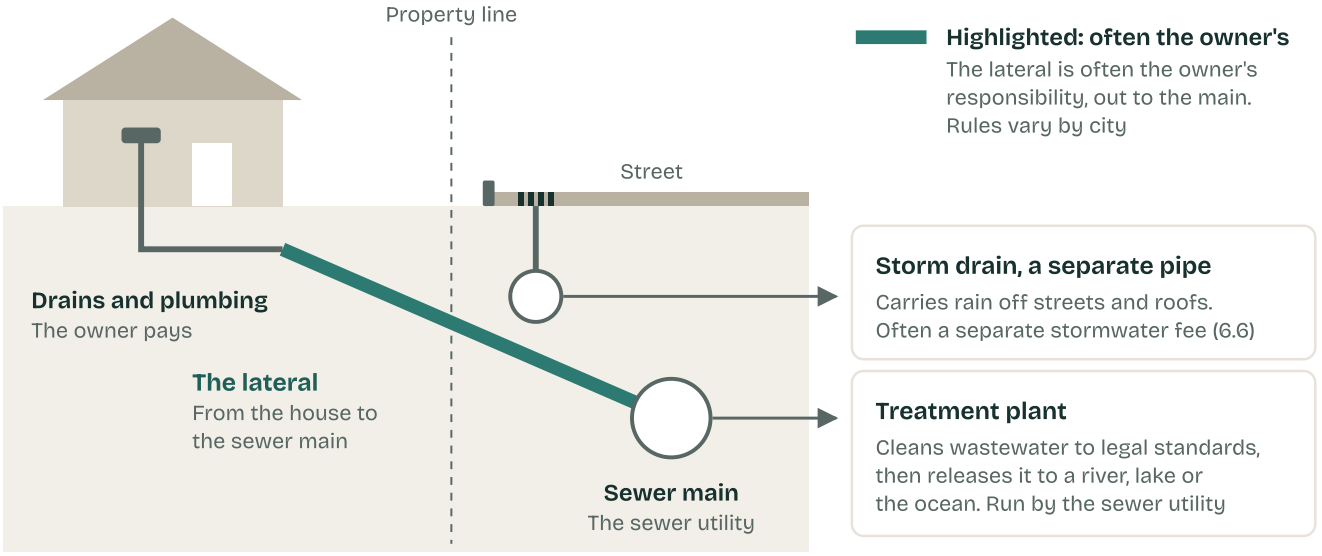


Figure 6.1. Where wastewater goes. House drains feed the lateral, which runs to the sewer main and on to the treatment plant; rain goes down storm drains, in separate pipes. The lateral is often the owner's responsibility, out to the main.

Part	What it does	Who usually pays for it
Drains and household plumbing	Carries wastewater out of the house	The owner
The lateral	The pipe from the house to the public sewer main	Often the owner, out to the main. New York City and Philadelphia, for example, make the owner responsible for it; rules vary by city
Sewer mains and pumping stations	Carry wastewater across the city	The sewer utility, through sewer charges
Treatment plant	Cleans wastewater to legal standards	The sewer utility
Storm drains	Carry rainwater off streets and roofs	Often a separate stormwater fee (6.6)

Combined sewers

In many older cities, the same pipes carry both wastewater and rainwater. In heavy rain these **combined sewers** can overflow, sending untreated sewage into rivers. Federal enforcement has required many cities to fix this, and the cost shows up on customers' bills.

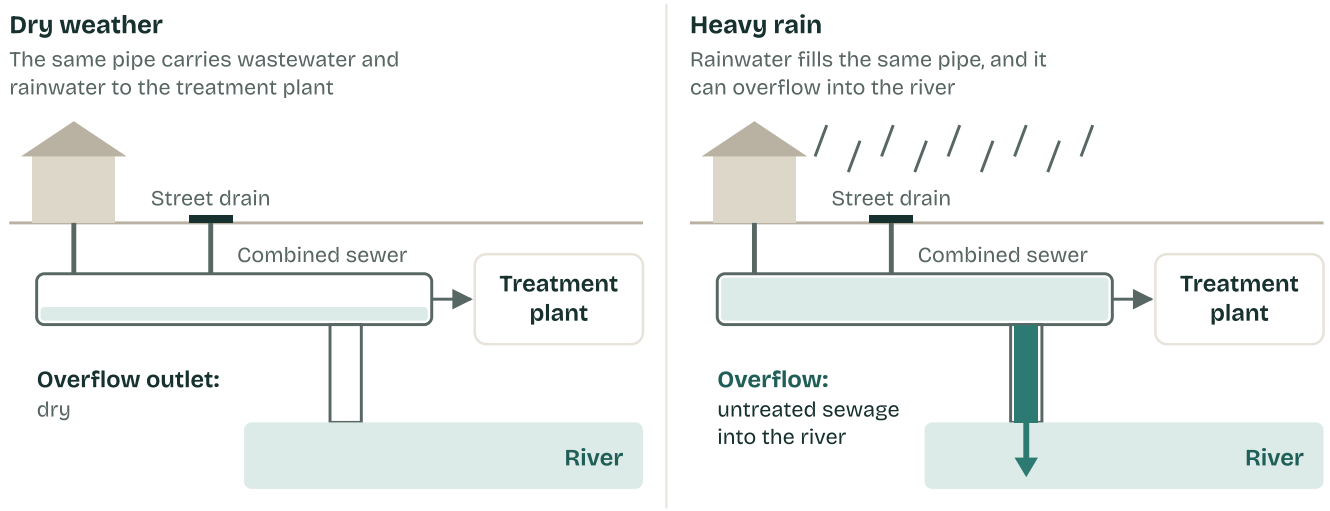


Figure 6.2. A combined sewer in dry weather and in heavy rain. The same pipe carries wastewater and rainwater; in heavy rain it can overflow, sending untreated sewage into the river.

Washington, D.C.'s Clean Rivers Project, required under a 2005 federal consent decree, is now described by DC Water as a $3.27 billion project. Its cost is recovered partly through an impervious-area charge on every bill (6.6).

If your bill has grown faster than your use, a project like this may be one reason. Your utility's website or annual report usually explains its major projects.

6.2 How sewer is charged

Because sewer flow is not metered, every utility picks a stand-in for it. There is no national rule, and the method makes a big difference to what any water saving is worth.

Method	How it works	Real example	Does a water saving lower sewer?
Billed on all water use	Sewer charged per unit of metered water	DC Water: $12.91 per CCF of water used	Yes, every gallon, indoors or out
A percentage of the water charge	Sewer is a set multiple of the water bill	New York City's Water Board sets wastewater charges at 159% of water charges	Yes, in proportion
Winter averaging	Your water use in winter months sets the sewer volume for the year	Austin bills sewer all year on the average of three winter months	Only savings made in the averaging months
Lower of winter average or actual	Outside winter, you pay on whichever is lower	Portland, at $14.24 per CCF in 2026-27; Minneapolis, which uses its December to March average	Yes in winter; in summer only if use falls below the winter average
A monthly cap	Sewer charged on water use up to a limit	Charlotte caps sewer at 16 CCF a month, about 12,000 gallons, assuming water above that goes outdoors	Yes, below the cap
Fixed sewer fee	A flat monthly amount, often by meter size	Charlotte, $6.75 a month; Minneapolis, $9.30 for a standard 5/8-inch meter, on top of usage charges	No
Flat rate	A set charge with no link to use	Used for some unmetered accounts; New York City has about 3% of accounts on flat rates	No

Many utilities combine a fixed fee with one of the usage methods. Some areas also have a separate sewer district, or a sewer charge on the property tax bill instead of the water bill.

Homes with a private well but a city sewer connection have no water meter to bill on. Charlotte, Minneapolis and Portland, for example, use an assumed or average volume for these homes. Ask yours how it bills.

Find out which method you are on

1. Look at the sewer lines on your bill. Is there a rate per unit, a fixed charge, or both?
2. Check whether the sewer volume matches your water volume each bill. If it stays the same all year, you are probably on winter averaging or a flat rate.
3. Read your utility's rate page, or call and ask: "Is my sewer charge based on all my water use, a winter average, a cap or a flat rate? And which months set the average?"

Find it on your bill: the sewer volume and the sewer rate. Write down the method; you will need it for every water saving you calculate (5.12). **Record it in** Tool 5.3, line C.

Worked example: one saving, four sewer rules

The same water saving can be worth more than twice as much in one city as in another, depending only on how sewer is charged. Check your method, above, before you value any saving.

Illustrative example: a household cuts its water use by 2,000 gallons a month, for example by fixing a leaking toilet. Water costs $6.00 per 1,000 gallons. Where sewer is usage-based, it costs $8.00 per 1,000 gallons.

Sewer rule	Water saving a month	Sewer saving a month	Total a month	Total a year
Usage-based on all water	2 × $6.00 = $12.00	2 × $8.00 = $16.00	$28.00	$336.00
Flat or fixed sewer charge	$12.00	$0.00	$12.00	$144.00
A percentage of the water charge, at 159%	$12.00	159% × $12.00 = $19.08	$31.08	$372.96
Winter averaging, with the saving made all year	$12.00	$16.00, because the lower winter average sets every month's sewer	$28.00	$336.00

Two cases need care:

- **Under winter averaging, timing matters.** A saving made only in summer, such as watering the lawn less, cuts the water charge but not the sewer charge, because summer use does not set the average. Four summer months at $12.00 a month is $48.00 a year. A leak fixed before the averaging months begin, though, lowers sewer for the next 12 months.
- **Under a flat or fixed charge, only the water line falls.** The household still saves $144 a year, but less than half of what the same change saves on a usage-based rate.

This is why the Water Savings Calculator (Tool 5.3) asks whether sewer is charged on the water you save, and has a note for winter averaging. Get that answer wrong, and the calculated saving can be off by more than half.

6.3 Why sewer costs rise

A higher sewer charge comes from the same two sources as any utility bill: more units billed, or a higher price per unit. With sewer, the units are usually your water, so a water problem becomes a sewer problem too.

Why is my sewer charge high in summer? On winter averaging, extra water use in summer cannot raise your sewer charge. If it rose, compare three things with last year's bills: the sewer rate, the number of billing days, and your water use in the averaging months. A leak or an extra person in those months raises sewer for the whole year (see the worked example in 6.2).

Cause	Typical sign	What to do
More water use, on a usage-based sewer rate	Water and sewer volumes rise together	Find the reason: a leak, more people, outdoor use (5.3, 5.10)
A leak during the winter averaging months	Sewer stays high all year, even after the leak is fixed	Fix the leak, then ask the utility whether it will recalculate your average
A default average for a new customer	Sewer volume is the same every bill from the day you moved in	Austin, for example, starts new customers at 5,000 gallons a month until a winter average is set. Check when yours will be reset
A rate increase	The rate per unit or the fixed fee rises, usually after a public notice	Bill inserts and the utility's rate page
New charges for big projects	A new line appears, or an existing one grows quickly	Read the project explanation on the utility's website
A wrong meter size or customer class	Fixed fees higher than neighbors'	Check the meter size and class on the bill against your home
A longer billing period	Volume up, use per day unchanged	Compare per day (1.4)

The bigger picture

Lower water sales can push rates up, though the evidence is mixed, so treat "saving water raises my rates" as a possible effect over years, not a reason to stop saving. Your own bill still falls when you use less on a usage-based rate.

Sidebar: when sewer costs you twice. In Case F (Chapter 5, Illustrative), sewer was charged on every gallon, so the leak raised both the water and the sewer lines. More than half of the extra cost, $8 of every $14 per 1,000 gallons, was sewer.

6.4 Outdoor water and sewer

Water used on lawns and gardens never reaches the sewer, but on a straight usage-based rate you pay sewer on it anyway. Utilities handle this in four ways, and which one yours uses decides whether a separate meter is worth buying.

Approach	How it keeps outdoor water out of sewer charges	Example
Winter averaging	Sewer is set from winter use, when there is little outdoor watering	Austin, Portland, Minneapolis (6.2)
A monthly cap	Use above the cap is assumed to go outdoors	Charlotte, 16 CCF a month
A separate irrigation meter	Water through it is billed for water only	Charlotte does not bill sewer on irrigation-meter water
A deduct (sewer credit) meter	A second meter measures outdoor water, which is subtracted from the sewer volume	Bettendorf, Iowa

If your utility already uses winter averaging or a cap, a separate meter may save you little or nothing. Check first.

Is a deduct meter worth it?

In Bettendorf, a homeowner installs the second meter at their own cost, with a plumber and a permit, and pays a $100 annual fee. Sewer charges on the outdoor water are then credited. Readings are due once a year.

Illustrative example: a home on a usage-based sewer rate of $8.00 per 1,000 gallons, with a Bettendorf-style $100 yearly fee. Assume the meter costs $400 installed.

Outdoor water a year	Sewer charged on it	Less the yearly fee	Net saving	Simple payback
32,000 gallons (the 5.9 example)	$256.00	$100.00	$156.00	About 2.6 years
10,000 gallons	$80.00	$100.00	A $20 loss	Never

The math for the first row: 32,000 ÷ 1,000 × $8.00 = $256.00, less the $100 fee, leaves $156.00. $400 ÷ $156 = 2.6 years.

A deduct or irrigation meter pays only for heavy outdoor users on a usage-based sewer rate. Before applying, ask your utility:

- Does it offer irrigation or deduct meters at all?
- What are the installation, permit, connection and yearly fees, and is there a second base charge?
- Who reads the meter, and how often?
- Does it already exclude outdoor water through winter averaging or a cap?

For light outdoor users, the cheaper fix is the outdoor checklist in 5.9.

6.5 Septic systems

About 1 in 5 US homes uses a septic system instead of a public sewer, according to EPA. There is no monthly sewer bill, but there is a real cost: regular pumping and inspection, and a very large bill if the system fails. The money question is planning, not saving.

Item	EPA guidance
Professional inspection	At least every 3 years for a typical household system; every year for systems with pumps, float switches or other mechanical parts
Pumping	Typically every 3 to 5 years
Routine maintenance cost	About $250 to $500 every 3 to 5 years
Repairing or replacing a failed conventional system	About $5,000 to $15,000, and more for alternative systems
Additives	Not needed

Case E: a septic budget for the Lindqvists

Illustrative. The Lindqvists' farmhouse has a private well and a septic system. No one knows when the tank was last pumped.

They started with an inspection, then set a monthly amount aside so pumping would never be a surprise.

- A $500 pumping every 4 years: $500 ÷ 4 = $125 a year
- Monthly: $125 ÷ 12 = about $10.42

This covers pumping only. EPA also advises an inspection at least every 3 years, so ask what one costs and add it to the monthly amount.

Against a $5,000 to $15,000 failure, routine maintenance is the cheapest insurance a septic home has. The Lindqvists added the $10.42 to their monthly budget, next to their propane prebuy (3.6).

Habits that protect the system

EPA's advice:

- Flush only human waste and toilet paper. Keep out wipes, grease, diapers, cat litter, coffee grounds, medicines and household chemicals.

- Limit use of a garbage disposal.
- Fix leaks. EPA notes that a leaky or running toilet can add up to 200 gallons a day to a septic system, which can overload it (5.4).
- Keep cars, buildings, trees, and roof or sump drainage off the drainfield.

Call a professional: if drains back up or run slowly through the whole house, if sewage odors appear indoors or outside, or if the ground over the drainfield is soggy or unusually green. These can be signs of a failing system, and waiting usually makes the repair larger. Never enter a septic tank: the gases inside can kill.

Check your lease: In a rented home with a septic system, the lease should say who pays for pumping and repairs. Ask before you sign.

6.6 Stormwater fees and credits

A **stormwater fee** pays for the drains, pipes and ponds that carry rain off roofs, driveways and streets. Unlike sewer, it is not based on your water use. Most are based on how much of your lot is covered by hard surfaces that rain cannot soak into, called **impervious area**. Some cities offer credits if you manage rainwater on your own property; others do not.

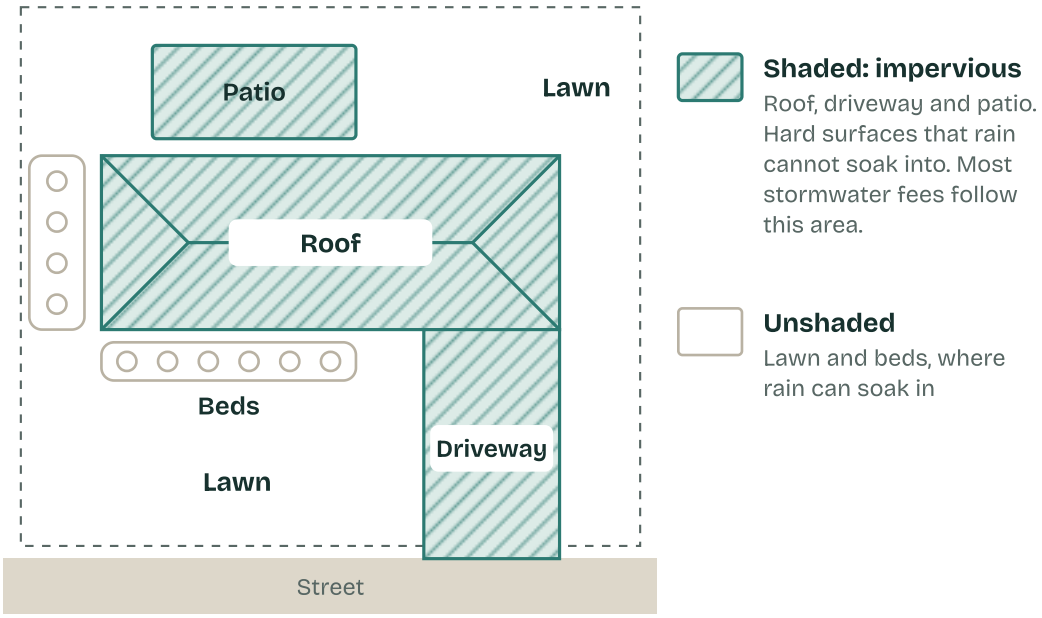


Figure 6.3. Impervious area seen from above: the roof, driveway and patio are hard surfaces rain cannot soak into; the lawn and beds are not. Illustrative.

EPA describes a stormwater utility as a dedicated fee, separate from property taxes, that tax-exempt properties pay too. It also notes that fees based on water meters are "often not equitable", because water use has nothing to do with runoff. In 2008, the latest national count we found, EPA reported that more than 80% of stormwater utilities billed in **equivalent residential units (ERUs)**, where one ERU is the impervious area of a typical single-family lot.

How two cities charge, and what they credit

City	How the fee is set	Credits
Washington, D.C.	Two charges. DC Water's Clean Rivers charge is $25.50 per ERU a month in 2026-27, by tiers of impervious area. A home with 700 to 2,099 square feet is 1 ERU. The District's own stormwater fee is $4.03 on DC Water's typical bill.	RiverSmart Rewards: up to 55% off the District fee and up to 20% off the Clean Rivers charge, for rain barrels and cisterns, rain gardens, trees, permeable paving, green roofs and disconnected downspouts. Discounts last 3 years and can be renewed.
Philadelphia	A flat residential charge based on the citywide average lot	No residential credits at present. Rain barrels do not lower the bill. A side yard or garage lot may qualify for an exemption.

What a credit could be worth

Illustrative example: a 1-ERU home in Washington, D.C., at 2026-27 rates, earning the maximum credits.

- Clean Rivers charge: $25.50 a month × 20% × 12 = $61.20 a year
- District stormwater fee: $4.03 a month × 55% × 12 = $26.60 a year
- Most it could save: $87.80 a year

The actual credit depends on how much area you manage, so many homes would get less. In a city like Philadelphia with no residential credit, these features may still be worth doing for the garden or the neighborhood, but not for the bill.

Find it on your bill: a stormwater, drainage or impervious-area line, the units it is billed in (ERUs, a tier or an area), and the impervious area your utility has on record for your property. **Record** the monthly fee and that area before you ask for any change.

Lowering a stormwater fee

1. **Check the area or tier on record.** If it looks wrong, for example if it counts a patio you removed, ask the utility to review it.
2. **Look for a credit program.** Search your utility's website for stormwater credits, the practices it accepts and how long a credit lasts. Washington, D.C., Portland and Minneapolis run credit programs; Philadelphia has none for homes at present.
3. **Check for exemptions.** In Philadelphia, for example, a side yard or garage lot may qualify.

Before you pay for a feature to earn a credit, work out the payback:

- Yearly credit = monthly fee × credit % × 12
- Payback in years = the feature's cost, less any help from a city program, ÷ the yearly credit

Illustrative example: the 1-ERU home above spends $300 on a rain garden (an illustrative cost). If it earned the full 20% off the Clean Rivers charge, $25.50 × 20% × 12 = $61.20 a year, the payback would be $300 ÷ $61.20 = about 4.9 years. At half that credit, it would take about 9.8 years. Because D.C.'s discounts last 3 years, a payback this long depends on renewing them.

Chapter 6 review

What you learned

- Sewer is rarely metered, so utilities charge it from your water use, a winter average, a cap, a percentage of the water bill or a flat rate.
- The method decides whether a water saving also cuts your sewer charge, and by how much.
- Under winter averaging, a leak or extra use in the averaging months can raise sewer charges for a whole year.
- Irrigation and deduct meters pay off only for heavy outdoor users on a usage-based sewer rate.

- Septic homes have no sewer bill, but need inspection every 3 years and pumping every 3 to 5 (EPA), and should budget for it.
- Stormwater fees usually follow your property's impervious area. Some cities give credits for managing rainwater on site, and some give none.

What you can do today

1. Find out which sewer method your utility uses, and which months set any winter average (6.2).
2. If you are on winter averaging, run the dye and meter tests before the averaging months begin (5.4, 5.10).
3. Check the impervious area your utility has on record for your property (6.6).
4. If you have a septic system, find out when it was last pumped and inspected, and set aside a monthly amount (6.5).

What to check next month

Compare the sewer volume on your next bill with your water volume. If they match, every water saving cuts sewer too. If the sewer volume stays fixed, note when it will next be reset. After the averaging months, check that the new sewer volume matches your average water use in those months. After an area review or a credit, check the stormwater line on the next bill.

Billix Quick Check

Your utility sets sewer charges from your average use in December, January and February. In March you find and fix a leaking toilet that had been wasting 1,500 gallons a month since November. What happens to your bills?

Answer: Your water charge falls from the next bill: 1.5 × $6.00 = $9.00 a month at the rate card's price. But your sewer charge stays based on the inflated winter average until it is reset next winter: 1.5 × $8.00 = $12.00 a month too much, or $144.00 over 12 months. In general, the cost is the extra gallons a month in the averaging months ÷ 1,000 × the sewer rate × 12. Ask the utility, with your repair receipt and meter readings, whether it will recalculate the average now.

Sources and further reading

Last Verified: October 2026. The city sewer and stormwater rates in 6.2 and 6.6 are 2026 or 2026-27 rates, and the stormwater credit programs (RiverSmart Rewards, Clean River Rewards and Minneapolis's credits) change their terms from time to time. Bettendorf's deduct-meter terms come from a July 2022 page. Check your own utility's current rates and programs.

All pages were accessed in October 2026. [CO] marks a utility's own page; most utilities here are public agencies.

National picture

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Sewer billing methods

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Stormwater credits

4. DC Department of Energy and Environment, RiverSmart Rewards https://doee.dc.gov/node/171142.
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PART IV

Connections and Waste

7 Home Internet and TV

8 Mobile Phone Service

9 Trash and Recycling

7

PART IV · CONNECTIONS AND WASTE

Home Internet and TV

Home internet and TV are not utilities in the way electricity and water are. No state commission sets their prices. The companies do, and they change them often: promotions expire, fees appear, and the price you pay can drift well above what a new customer is offered for the same service. That makes these the household bills where a phone call, or a switch, can save the most for the least effort.

There is more choice than there used to be. The FCC reported that about 94% of Americans could get internet at its benchmark speed of 100 Mbps down and 20 up over a wired connection by mid-2025, and about 97% counting fixed wireless. And since 2024, providers have had to publish a standard **broadband label** for each internet plan they sell on its own, showing the full monthly price, fees and speeds.

The federal Affordable Connectivity Program, which once took up to $30 a month off internet bills, ended on June 1, 2024. The help that remains, including Lifeline, is in 12.4.

By the end of this chapter you will be able to:

- Read an internet or TV bill and find its true monthly cost, including every fee
- Use broadband labels to compare plans on equal terms
- Work out how much speed your household actually needs
- Ask for a better rate, with a script for areas with competition and one for areas without
- Decide whether to rent or buy your modem and router
- Judge a bundle, and know what the FCC's all-in pricing rule requires of cable and satellite TV

Internet prices in this chapter are illustrative examples, not averages.

START HERE:

→ **Your bill went up when a promotion ended?** Go to 7.3, then 7.7 and Case C.
→ **A fee you don't recognize?** Go to "What the lines mean" in 7.3.
→ **Comparing plans or providers?** Go to 7.4, then Tool 7.1.
→ **Offered a bundle?** Go to 7.9.
→ **Paying a monthly equipment fee?** Go to 7.8.

7.1 How home internet works

An **internet service provider (ISP)** connects your home to the internet over one of five kinds of connection. Which ones reach your address decides your choices, and having more than one is your biggest bargaining chip.

Connection	How it reaches your home	Worth knowing
Fiber	Glass fiber lines carrying light	Often offered with similar download and upload speeds; availability is still spreading
Cable	The same coaxial lines as cable TV	Widely available; upload speeds are often much lower than download
DSL	Telephone lines	Older technology; speeds fall with distance from the provider's equipment
Fixed wireless	A radio signal from a tower to a receiver at your home, including newer 5G home internet	Can compete with wired service where the signal is strong
Satellite	A dish on your home	Reaches almost everywhere; can have higher latency and data limits

The FCC reported that, as of mid-2025, about 94% of Americans could get service at its benchmark of 100 Mbps download and 20 Mbps upload over a wired connection. Counting fixed wireless, the figure was about 97%, and counting satellite, about 99.7%. Those are availability figures. They do not mean every home has a choice of providers at that speed.

How internet differs from your utilities

	Electricity, gas and water	Home internet and TV
Who sets the price	Usually a state commission or city board	The company
Can you switch?	Usually not, except electricity or gas supply in some states	Yes, wherever more than one provider serves your address
Do prices change with promotions?	Rarely	Often; promotions end and prices rise
Where to complain	The utility, then the state commission	The provider, then the FCC (10.8)
Required price disclosure	The approved rate schedule	The FCC's broadband label (7.4)

That difference cuts both ways. Nobody protects you from a price rise the way a rate case does, but nothing stops you from asking for a better deal or leaving for one.

Find it for your address: which providers serve your home. Provider websites let you check by address, and the FCC's broadband label for each plan (7.4) shows what each one charges.

7.2 Speed explained

Internet plans are sold mainly on download speed, but four other numbers decide whether a connection feels fast. Knowing them stops you paying for speed you cannot use.

Term	What it means	Why it matters
Mbps (megabits per second) and **Gbps** (1,000 Mbps)	How much data the connection carries each second	Plans are priced by it
Download speed	Data coming to you	Streaming, browsing, downloading files
Upload speed	Data you send	Video calls, sending large files, cloud backups, livestreaming
Latency (milliseconds)	The delay before data starts to arrive	Video calls and online games feel laggy when it is high
Reliability	How often the connection drops or slows	Matters most if you work from home
Data allowance ("data cap")	How much data the monthly price includes	Going over can mean charges or slower speeds

The broadband label (7.4) shows each plan's **typical** download speed, upload speed and latency. The FCC notes these can differ from advertised speeds and from what a speed test shows, and that the more devices are connected, the less speed each one gets.

Data caps

No federal rule limits data caps. The FCC opened an inquiry into them in 2024, but it asked questions only and set no rules. What the FCC does require is disclosure: the label must show the "Data Included with Monthly Price", and any charge or slowdown for using more. If you use a phone hotspot as backup, the FCC also advises checking your mobile plan's data limit first.

Wi-Fi or wired

Slowness blamed on the plan often comes from Wi-Fi inside the home. The FCC's home network advice:

- A wired Ethernet connection between the router and a device "will provide the highest speeds", and frees up Wi-Fi for everything else.
- Routers broadcast on 2.4 GHz, which reaches farther but is slower, and 5 GHz, which is faster over shorter distances.
- Put the router in a central, open spot, not in a cabinet or a corner.
- In larger homes, a mesh system or extenders can fill dead zones.

Test before you upgrade

Before paying for a faster plan, run a speed test with a computer plugged into the router by cable, at the busiest time of day. If the wired result is close to your plan's speed but Wi-Fi is slow, the fix is inside the home, not a pricier plan. If the wired result is well below the label's typical speed, call the provider (7.7).

7.3 Reading an internet bill

An internet bill is short, but the number that matters, the total you pay each month, is often higher than the plan price you remember. Fees, equipment and the end of a promotion account for most of the gap.

A sample internet bill

Illustrative example: Case C, Dana's internet bill. Anytown Broadband is fictional, and the bill is marked SAMPLE.

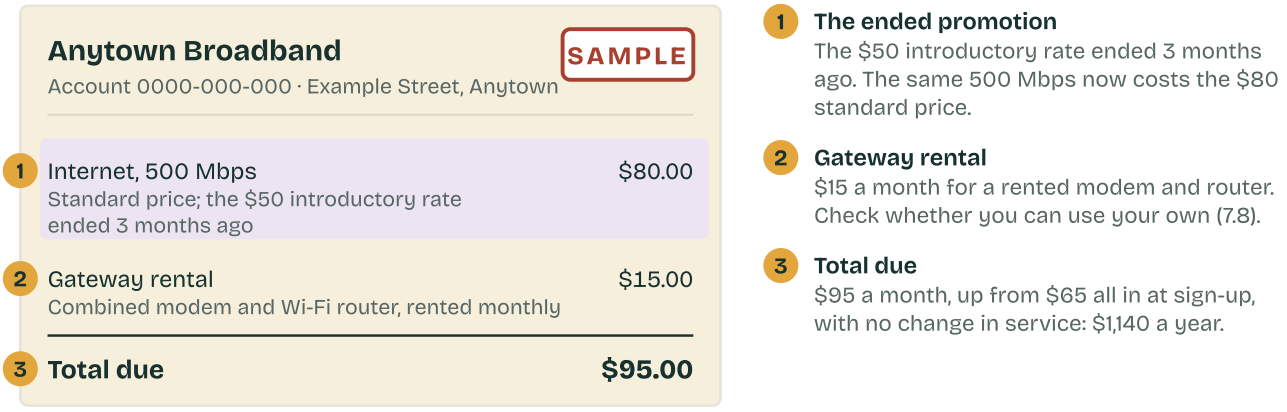


Figure 7.1. Annotated internet bill: Dana's bill three months after the promotion ended (Case C). SAMPLE: Anytown Broadband is fictional.

Dana signed up at $65 a month all in. Three months ago the promotion ended, and the bill rose to $95 without any change in service. That is $1,140 a year.

Find it on your bill: the plan line, with its speed; the date any promotion ends; every monthly fee and equipment rental; and the total. The broadband label (7.4) also shows when a promotion ends and the price afterward. Divide the total by the plan price. At about 1.2 or more, fees and equipment add a fifth or more to what you pay: see 7.8 for equipment, and ask about fees (7.7). Promotion ended? Compare labels (7.4), then call (7.7). **Record it in** Tool 7.1, in the "Current plan" column.

What the lines mean

Line	What it is	What to check
Promotional price	A lower rate for new customers, usually for 12 or 24 months	When it ends, and the price afterward. Both are on the broadband label (7.4)
Standard (regular) price	What the plan costs after the promotion	Whether new customers are being offered less for the same plan
Equipment rental	Monthly charge for a modem, router or gateway	Whether you can use your own (7.8)
Installation and activation fees	One-time charges when service starts	Whether they can be waived
Provider fees	Monthly charges the provider adds, sometimes named "network" or "broadcast" fees	Every one is listed on the label (7.4). A monthly provider fee there is part of the price: count it in your all-in cost, and ask to drop it. A pass-through fee is for a government program. Not on the label? Ask in writing (A4)
Taxes and pass-through fees	Government charges, which vary by location	The label says taxes vary; ask for your local total
Data overage charges	Charges for going over the plan's data allowance	The allowance and overage rate on the label
Autopay or paperless discount	A monthly credit for automatic payment or e-billing	Whether you get it, and the overdraft risk of autopay (10.6)
Contract and early termination fee	A commitment period, and a charge for leaving early	The end date before you switch
Price guarantee	A promise to hold the price for a period	Exactly what it covers; some exclude fees and taxes

7.4 Broadband labels

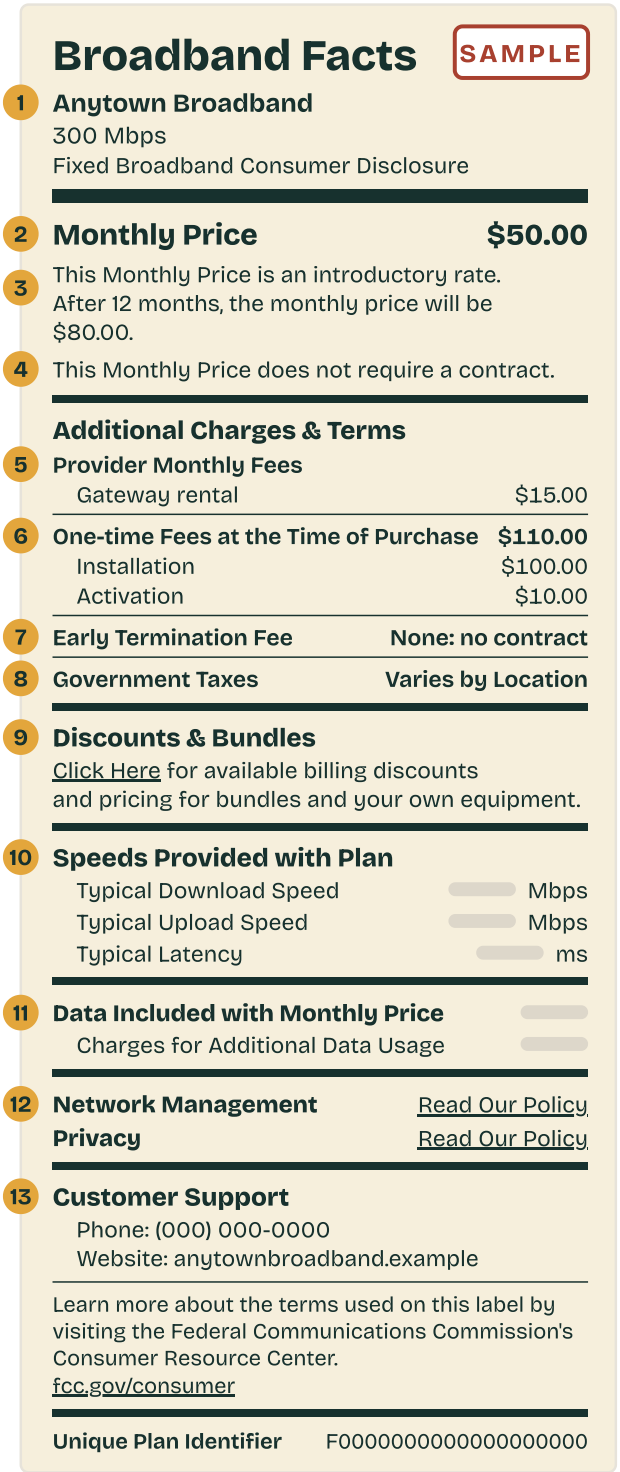
Since 2024, every home and mobile internet provider has had to publish a standard label for each plan it sells on its own, called **Broadband Facts**. It works like a nutrition label: the same boxes in the same order, so plans can be compared on equal terms. It is the single most useful tool for shopping for internet.

What the label shows

Under the FCC's rules, each label lists:

- The monthly price before taxes, and any **introductory rate**, how long it lasts and the price afterward
- The contract length, with a link to its terms, and any **early termination fee**
- Every **monthly provider fee**, such as equipment rental, listed separately
- **Pass-through fees** for government programs that the provider chooses to pass on
- **One-time fees**, such as installation, activation and equipment
- A statement that taxes vary by location
- **Typical** download speed, upload speed and latency
- The data included in the monthly price, and any charges or slowdowns beyond it
- Links to the provider's network management and privacy policies, and its customer support contacts

Labels became required on April 10, 2024 for most providers, and on October 10, 2024 for providers with 100,000 or fewer subscriber lines. Labels must appear in English and in any other language the provider markets in.



1. **Provider and plan**
   One label for each plan sold on its own, with the same boxes in the same order.
2. **Monthly price**
   The price before taxes: $50.00 here.
3. **Introductory rate**
   How long it lasts and the price after: $80.00 from month 13. Set a calendar reminder.
4. **Contract**
   None here. Otherwise the length, with a link to its terms.
5. **Provider monthly fees**
   Every monthly fee, listed separately. Count the $15.00 gateway in your all-in cost.
6. **One-time fees**
   Installation $100 and activation $10: $110. Spread them over the time you keep the plan.
7. **Early termination fee**
   What leaving early would cost. With no contract, there is none.
8. **Government taxes**
   They vary by location. Ask for your local total.
9. **Discounts and bundles**
   A link to discounts and bundle prices. Bundles need no label of their own.
10. **Typical speeds**
    Download, upload and latency. They can differ from advertised speeds and from a speed test.
11. **Data**
    The data in the monthly price, and any charges or slowdowns beyond it.
12. **Policies**
    Links to the provider's network management and privacy policies.
13. **Support and plan ID**
    Support contacts, the FCC's page explaining the label's terms, and the plan's unique identifier.

All in over two years: $890 + $1,140, about $84.58 a month, not the $50 headline.

Gray bars: details this sample does not give

Figure 7.2. Annotated broadband label: the illustrative Anytown Broadband 300 Mbps plan from 7.4, in the FCC's Broadband Facts layout, with each box explained. SAMPLE: a fictional provider; the layout is the FCC's. Gray bars mark values the example does not give.

Where to find a label

The FCC says to look for labels "at any point of sale, including online and in stores." Labels must also be available in your online account, if the provider has one. Bundles, such as internet and TV together, do not need their own label; providers may link to bundle details from the label's discounts section instead. If a provider shows no label, or one with wrong prices or fees, you can complain to the FCC.

Last Verified: October 2026. In July 2026 the FCC adopted changes to how the label is shown. Phone sales staff could summarize it instead of reading it in full, location-based fees could be shown as an "up to" total, and a link or icon could replace the full label. Those display changes have been delayed indefinitely and were not in effect as of October 2026, so the 2022 rules still apply. Check the FCC's broadband labels page for updates.

Turning a label into a true cost

The monthly price on a label is not what a plan costs. Add the fees, and spread the one-time charges over the time you expect to keep it.

Illustrative label, SAMPLE: Anytown Broadband 300 Mbps. Introductory price $50 a month for 12 months, then $80. Gateway rental $15 a month. Installation $100 and activation $10, one time. No contract.

	Year 1	Year 2
Service	12 × $50 = $600	12 × $80 = $960
Equipment	12 × $15 = $180	12 × $15 = $180
One-time fees	$110	$0
Total	**$890**	**$1,140**

Over two years, the plan averages ($890 + $1,140) ÷ 24 = about $84.58 a month, not the $50 in the headline. Compare plans on this two-year figure, and set a calendar reminder for the month the promotion ends.

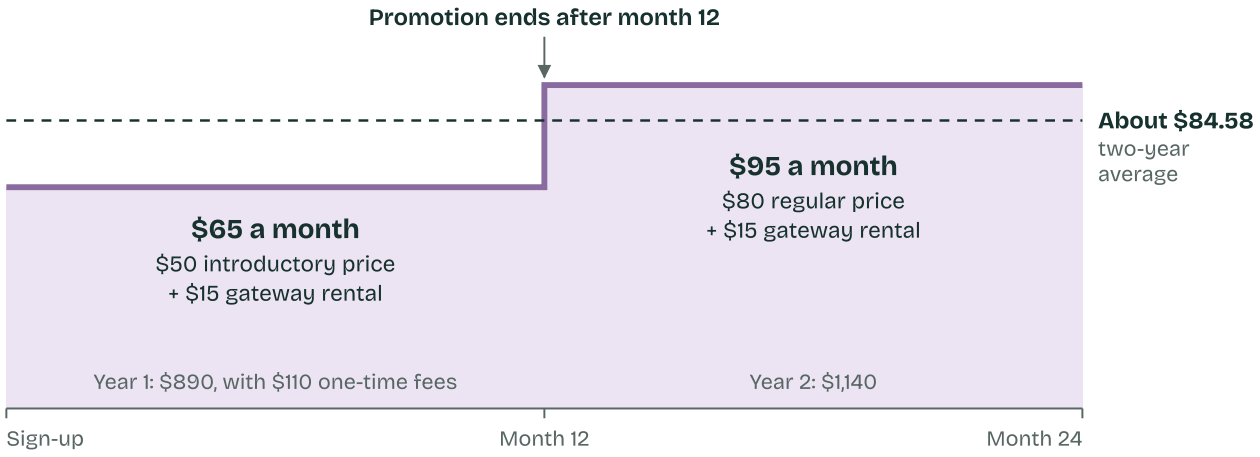


Figure 7.3. The promotion cliff: the SAMPLE Anytown Broadband 300 Mbps plan from 7.4, with the gateway rental, costs $65 a month for 12 months, then $95. With $110 of one-time fees, it averages about $84.58 a month over two years. Illustrative.

7.5 How much speed you need

It is easy to pay for far more speed than you use. The right plan is set by what your household does at the same time, at its busiest hour, plus some headroom, not by the number of people or the plan a salesperson suggests.

What common activities need

The FCC's consumer guide gives these minimum download speeds for one activity at a time. It calls them "rough guidelines", not the result of FCC testing, and they date from 2022.

Activity	Minimum download speed (FCC guide)
Email and web browsing	1 Mbps
HD video call, personal	1.5 Mbps

Activity	Minimum download speed (FCC guide)
HD video teleconferencing for work	6 Mbps
Working from home or schoolwork	5 to 25 Mbps
HD video streaming	5 to 8 Mbps
4K video streaming	25 Mbps
Online multiplayer gaming	4 Mbps
Downloading files	10 Mbps

The FCC's benchmark of 100 Mbps down and 20 up is different. It is the yardstick the FCC uses to judge whether broadband is being deployed fast enough, not a recommendation for what a household should buy.

A four-step framework

1. **List your busiest hour.** Write down everything that runs at once on a typical busy evening or workday.
2. **Add up the speeds.** Use the table above, or your own measured use.
3. **Allow headroom** for updates, new devices and the drop-off in Wi-Fi speed around the home. Our rule of thumb: a plan whose typical download speed is a few times your total.
4. **Check upload too,** if anyone makes video calls, sends large files or backs up to the cloud. The label shows typical upload speed; it is often much lower than download.

Illustrative example: Dana, who works from home five days a week, and a partner who works from home some days.

Busiest hour	Activities	Total download
Workday	Two work video calls (6 + 6), plus browsing and phones (2), plus a file download (10)	24 Mbps
Evening	A 4K stream (25), an HD stream (8), plus browsing and phones (2)	35 Mbps

Their busiest hour needs about 35 Mbps. A plan with a typical download speed of 100 to 300 Mbps leaves ample headroom. Dana is paying for 500 Mbps.

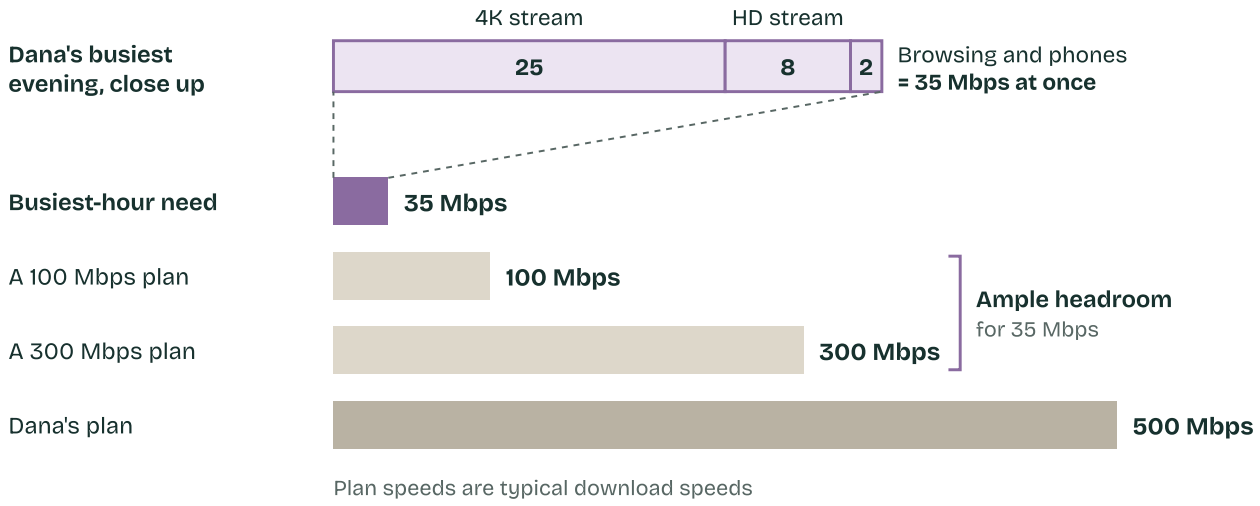


Figure 7.4. Dana's busiest evening needs about 35 Mbps (25 + 8 + 2), a sliver of a 100, 300 or 500 Mbps plan. Illustrative: Case C.

When this advice does not apply

- Households that regularly upload large video files, livestream or host game servers may need much higher upload speeds.
- Homes where several people game online at once may care more about latency than speed.
- If your connection is unreliable, a faster plan will not fix it. Ask the provider to check the line, and compare reliability before you switch.

The real test is experience. If your provider lets you change plans without a fee, try a lower tier for a month and run a wired speed test at your busiest hour (7.2).

7.6 Lowering your internet bill in ten steps

Most internet savings come from three moves: dropping to a plan that fits, removing fees, and making the provider compete for you. These ten steps take about an hour, spread over a week.

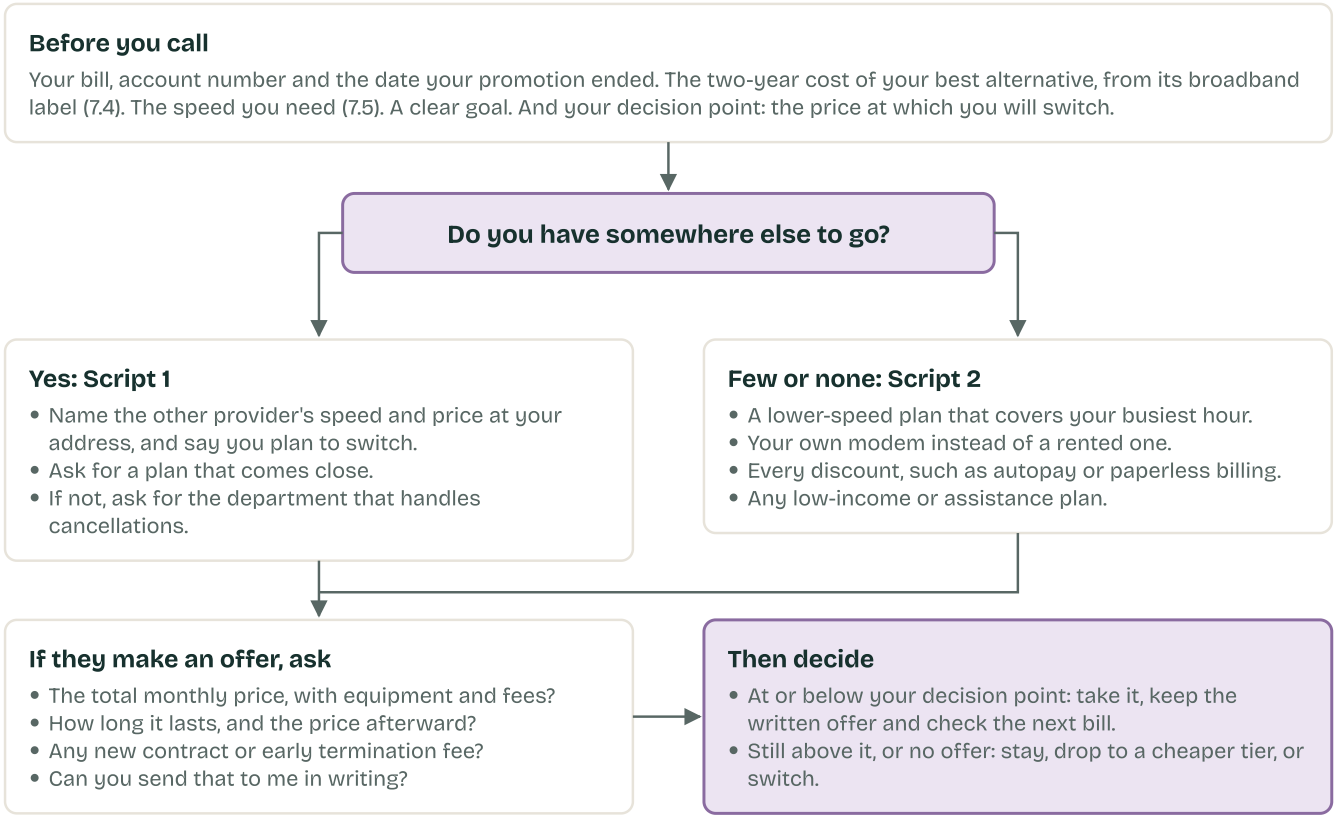
1. **Review your current bill.** Find the plan name, speed, every fee and the total (7.3).
2. **Work out your all-in monthly cost,** including equipment and fees, and multiply by 12. This is the number to beat.
3. **Work out the speed you need** at your busiest hour (7.5).
4. **Compare every provider at your address,** using their broadband labels. Calculate each plan's two-year cost (7.4).
5. **Check when promotions end,** yours and theirs. Note the price each plan rises to.
6. **Look at lower-cost plans,** including lower speed tiers, Lifeline and providers' own low-cost plans for eligible households. The list in 12.4 names the largest, with prices, speeds and who qualifies. Ask each provider at your address whether it has one, as Script 2 in 7.7 does.
7. **Ask about discounts** for automatic payment, paperless billing or bundling, and check whether you already receive them.
8. **Consider owning your equipment** if you rent a modem or router (7.8).
9. **Add up the cost of switching:** installation and activation fees, any early termination fee, a few days of overlap, and the hassle of returning equipment.
10. **Check the first bill after any change** against what you were promised, and call at once if it differs.

Steps 4 and 9 tell you whether a better deal is real. Step 7 costs nothing to ask. Step 10 makes sure you get what you agreed.

What to record: each plan's two-year cost, the date your current promotion ends, and the name, date and reference number for every call (the Provider Call Log, A2).

7.7 Asking for a better rate

Because internet prices are set by the company, they can often be changed by the company, but only if you ask, and nothing guarantees a yes. Your leverage depends on one thing: whether you have somewhere else to go.



A billing error or a missing broadband label is different: take it to the FCC (10.8).

Figure 7.5. Asking for a better rate: what to ask with a real alternative and without one, what to check in any offer, and when to switch.

Before you call

- Your current bill, account number and the date your promotion ended
- The two-year cost of your best alternative, from its broadband label (7.4)
- The plan speed you actually need (7.5)
- A clear goal: a lower price, a lower tier, removing the equipment fee, or all three
- A note of your decision point: the price at which you will switch

Write these down on the Provider Negotiation Script (A1).

Script 1: when you have a real alternative

"Hi, I've been a customer for [number] years. My promotion ended and my bill went from [old amount] to [new amount]. [Other provider] offers [speed] at my address for [amount] a month, with [fees], and I'm planning to switch. Before I do, can you offer a plan that comes close to that? If not, could you transfer me to the department that handles cancellations?"

If they make an offer, ask:

- "What is the total monthly price, including equipment and fees?"
- "How long does it last, and what will it be afterward?"
- "Is there a new contract or an early termination fee?"
- "Can you send that to me in writing, by email or in my account?"

Script 2: when you have few or no alternatives

"Hi, my bill has gone up to [amount] a month and that's more than I can keep paying. I'd like to lower it without losing service. Do you have a lower-speed plan that would cover [your busiest-hour needs]? Can I use my own modem instead of renting one? Am I getting every discount available, such as for autopay or paperless billing? And do you offer any low-income or assistance plans?"

Without competition, the realistic goals are a lower tier, removing fees and any assistance plan you qualify for, rather than a new-customer price.

Tips that help

- Be polite, specific and brief. The person on the phone may have limited authority; ask who has more.
- If the first answer is no, it is fair to call back another day.
- Do not accept a new contract without reading its early termination fee.
- Write down the date, the representative's name, the offer and any reference number.
- Check the next bill. If it does not match, call with your notes.

A retention offer is a business decision, not a right. If a provider will not move, the decision is yours: stay, drop to a cheaper tier, or switch. A billing error or a missing broadband label is different; those you can take to the FCC (10.8).

7.8 Modems and routers: rent or buy

An equipment rental fee is one of the easiest internet charges to remove, but only if your provider lets you use your own equipment and you are comfortable maintaining it.

Device	What it does
Modem	Connects your home to the provider's network
Router	Shares that connection with your devices, by Wi-Fi and cable
Gateway	A modem and router in one box; often what providers rent

The math

Illustrative example: a $15 monthly gateway rental against buying a compatible modem and router for $180.

- Rental: $15 × 12 = $180 a year
- Payback on buying: $180 ÷ $15 = 12 months
- Over three years: $540 renting against $180 buying, a $360 difference, if the equipment lasts

Before you buy

Question	Why it matters
Does the provider allow customer-owned equipment for your type of connection?	Some connections require the provider's own device
Is the model on the provider's approved list?	An incompatible modem may not work, or may not reach your plan's speed
Will the provider still support you?	Some providers limit help for equipment they did not supply
Who keeps it secure?	You become responsible for updating its software and changing default passwords
What happens to the rented device?	Return it promptly, get a receipt, and check that the rental charge disappears from the next bill

When renting can make sense: if you expect to move or switch providers soon, if you want the provider to replace failed equipment for free, or if your connection type requires the provider's device anyway.

Check your lease: In some apartment buildings, internet service and equipment come with the unit. Ask the landlord before changing anything.

7.9 TV, streaming and bundles

TV costs come in three forms: cable or satellite packages, streaming subscriptions, and bundles that combine TV with internet or phone. Each hides its true cost differently.

Cable and satellite: the all-in pricing rule

Since December 19, 2024 (March 19, 2025 for small cable companies), an FCC rule has required cable and satellite TV providers to show an **all-in price** for video programming. Bills must show it as a single line, and any ad that quotes a price must use it. The rule also requires disclosure of the price that applies after a promotion ends.

The all-in price must include charges such as broadcast retransmission and regional sports fees, which used to be listed separately. It does not have to include taxes, equipment rental, administrative fees or franchise fees, and it does not cover streaming services. The rule was still in effect in October 2026.

Bundles: compare the parts

A bundle can save money, but only if you would buy every part anyway. Compare it with buying the parts separately, over two years.

The two-year bundle rule: two-year cost = (promotion months × promotion price) + ((24 − promotion months) × the price after the promotion), plus 24 months of box rental, plus taxes. The all-in price leaves both of those out. Compare the result with your internet plan's two-year total from Tool 7.1, plus only the TV you would buy anyway. Bundles need no broadband label (7.4), so ask for the label of the internet plan on its own. A bundle can take its own column in Tool 7.1.

Illustrative example

Option	Monthly cost	Two-year cost
Internet and TV bundle: $150 a month for 12 months, then $190	$150, then $190	(12 × $150) + (12 × $190) = $4,080
Internet alone at $60, plus two streaming services at $40 together	$100	24 × $100 = $2,400

In this example, separate services save $1,680 over two years, but only if the household is happy with streaming in place of the TV package. Before dropping part of a bundle, ask what the remaining service will cost alone. The internet price sometimes rises when the TV part is removed.

Streaming subscriptions

Streaming services are cheaper one at a time, but they add up. List every one you pay for, including any charged through an app store or bundled into a phone plan, and cancel those nobody watched last month.

Other subscriptions, beyond TV, are outside the scope of this book.

Home phone lines

Some bundles include a home phone, often delivered over the internet connection (VoIP). If you keep a landline for emergencies, a medical alert device or a home security system, check how it works during a power or internet outage before you drop or change it.

Cancelling and complaints

Last Verified: October 2026. The FTC's "click-to-cancel" rule was struck down by a federal appeals court on July 8, 2025, on procedural grounds, and the FTC has restarted the rulemaking. A federal law, ROSCA, still requires clear terms and a simple way to stop recurring charges for services sold online. Whether it reaches phone, cable and internet providers is unsettled. Some state automatic-renewal laws, including California's and Minnesota's, exempt services regulated by the FCC or the state utility commission, so do not count on them to cancel phone, cable or internet service.

When you cancel, use the service cancellation checklist (A6): ask for a confirmation number, return every piece of equipment with a receipt, and check the final bill. For billing problems with internet, phone or TV, contact the provider first, then file a complaint with the FCC. If the FCC forwards your complaint to the provider, the provider must reply in writing within 30 days (10.8). For billing disputes about stand-alone cable or satellite TV, the FCC points to your state commission (10.8).

CASE C

Dana's $95 internet plan

Illustrative example. Dana works from home and shares a townhouse with a partner who also works from home some days. All plans, prices and providers are fictional and SAMPLE; Anytown Broadband is cable and Anytown Fiber is fiber. Taxes are left out.

Dana was paying $95 a month: $80 for 500 Mbps and $15 for gateway rental, three months after a promotion ended (7.3). That is $1,140 a year.

Step 1: What Dana actually needs

Dana's busiest hour needs about 35 Mbps of download (7.5). The household also makes two work video calls at once and backs up files to the cloud, so upload speed matters.

Step 2: The options

Option	Monthly cost	One-time costs	Typical upload (from the label)	Year 1	Year 2
Stay as is	$95	None	20 Mbps	$1,140	$1,140
Retention offer from Anytown Broadband: 300 Mbps at $55 plus the $15 gateway for 12 months, then back to $95	$70, then $95	None	20 Mbps	$840	$1,140
Lower tier and own equipment: Anytown Broadband 100 Mbps at $50, with a $180 modem and router	$50	$180	10 Mbps	$780	$600
Switch to Anytown Fiber: 300 Mbps at $60 all in, equipment included, price locked for 24 months, no contract	$60	$50 installation, plus about a week of overlap on the old bill, $23.75	300 Mbps	$793.75	$720

The math for the fiber option's first year: 12 × $60 = $720, plus $50 installation, plus $23.75 of overlap ($95 ÷ 4), comes to $793.75.

Step 3: The savings

Option	Saved in year 1	Saved in year 2	Two-year total
Retention offer	$300	$0	$300
Lower tier and own equipment	$360	$540	$900
Fiber	$346.25	$420	$766.25

The simple version, $95 - $60 = $35 a month or $420 a year, is the fiber plan's year-2 saving. In year 1, installation and overlap cut it to $346.25. If the fiber price had not been locked and rose to $80 after a year, year 2 would save only $180.

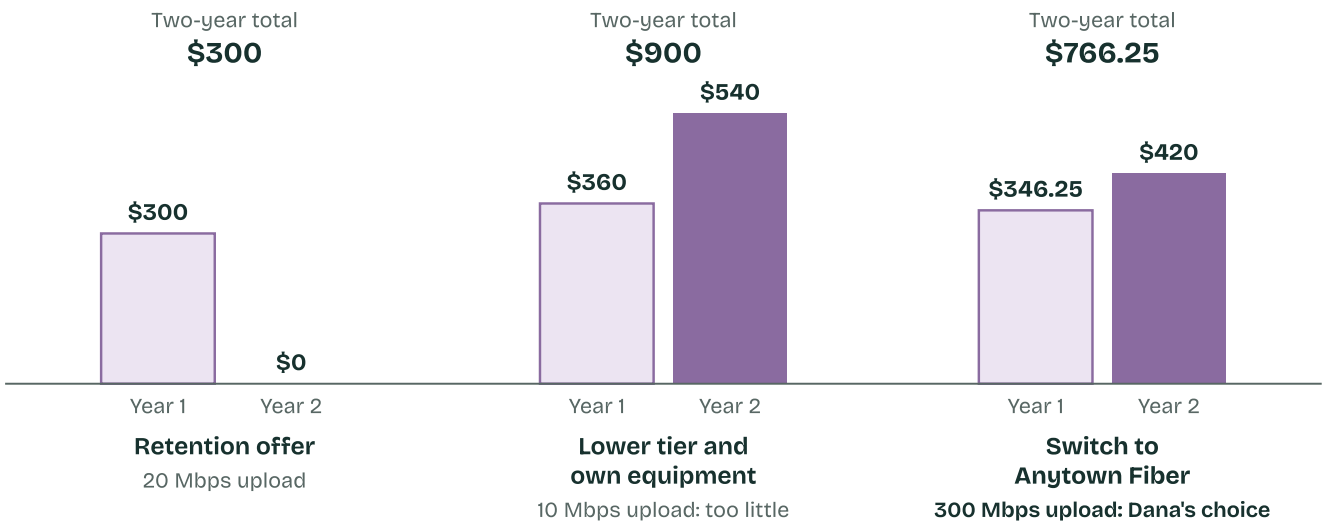


Figure 7.6. Case C: what each option saves against Dana's $95 plan, in year 1 and year 2. The lower tier saves most on paper, but its 10 Mbps upload is too little; Dana chose fiber. Illustrative, SAMPLE providers.

Step 4: The decision

The lower-tier plan saves the most on paper, about $134 more over two years. But its label shows a typical upload of only 10 Mbps, too little for two video calls and cloud backups at once, and Dana's work depends on the connection. Dana chose the fiber plan: it costs $60 all in, uploads at 300 Mbps and has a written two-year price lock.

Dana did three more things:

1. Kept the old service until the fiber was installed and tested, to avoid a gap.
2. Returned the rented gateway the same week, with a receipt, so no unreturned-equipment charge appeared.
3. Set a calendar reminder for month 22, two months before the price lock ends, to shop again.

Step 5: Checking the result

The first fiber bill was $60.00 plus the $50 installation, as quoted. A wired speed test at the busiest hour showed close to the label's typical speeds. The final Anytown Broadband bill showed the gateway returned. The $420 yearly saving from year 2 is Estimated until a full year of bills confirms it.

Tools for this chapter

Tool 7.1: the Internet Plan Comparison Worksheet

Purpose: to compare internet plans on their true two-year cost, not their headline price. Fill in one column per plan, using each plan's broadband label (7.4).

Item	Current plan	Plan A	Plan B
Provider and plan name			
Typical download / upload speed			
Typical latency			

Item	Current plan	Plan A	Plan B
Data included each month			
A. Monthly price now			
B. Months at that price			
C. Monthly price afterward			
D. Monthly equipment fees			
E. Other monthly provider fees			
F. One-time fees (installation, activation, equipment purchase)			
G. Early termination fee to leave your current plan			
H. Contract length and price guarantee			
Year 1 cost			
Year 2 cost			
Two-year total			

How to calculate: for a promotion of 12 months or less, Year 1 = (B × A) + ((12 − B) × C) + 12 × (D + E) + F + G, and Year 2 = 12 × (C + D + E). Count G only if you are leaving a contract. No promotion? Put the same price in A and C. For a promotion or price lock longer than 12 months, Year 1 = 12 × (A + D + E) + F + G, and Year 2 = ((B − 12) × A) + ((24 − B) × C) + 12 × (D + E). A bundle takes its own column: compare its two-year total with an internet plan's total plus 24 months of the TV you would buy anyway (7.9). Our rule of thumb: count about a week of overlap on your old bill if you switch.

Reading your results: pick the lowest two-year total among plans that meet your busiest-hour needs (7.5), including upload. If two plans are close, prefer the one with a written price guarantee and no contract.

A1: the Provider Negotiation Script

The blank script is printed once, as A1 in Appendix A, because other chapters use it too. It is also in the companion worksheet pack. It holds the two openings, the questions to ask and a call record. Scripts 1 and 2 in 7.7 show it in use.

Chapter 7 review

What you learned

- Internet and TV prices are set by the companies, not regulators, so they can be negotiated and, where there is competition, shopped.

- Promotions end. The true cost of a plan is its two-year total, including equipment, fees and one-time charges.
- Broadband labels show every plan's price, fees, speeds and data allowance in the same format. They are the best shopping tool you have.
- It is easy to buy more speed than you need. Size the plan to your busiest hour, and check upload as well as download.
- Renting a modem or router can cost more than buying one within a year, if your provider allows your own.
- Cable and satellite TV must show an all-in price under an FCC rule. Bundles save money only if you would buy every part anyway.
- Federal click-to-cancel rules are not in force, and some state laws exempt phone, cable and internet. Get a confirmation number when you cancel.

What you can do today

1. Find the date your internet promotion ends, and the price it rises to (7.3).
2. Find your plan's broadband label, and one competitor's (7.4).
3. Run a wired speed test at your busiest hour (7.2).
4. List every streaming subscription you pay for, and cancel one nobody used last month (7.9).
5. Fill in Tool 7.1, the Internet Plan Comparison Worksheet, for your current plan.

What to check next month

Check the next bill against the plan and price you agreed, line by line. If you switched, confirm the old account is closed, the equipment return is recorded, and no final charges are left.

Billix Quick Check

Plan 1 costs $40 a month for 12 months, then $75, plus $12 a month for equipment and a $99 installation fee. Plan 2 costs $60 a month with everything included. Which costs less over two years?

Answer: Plan 1: (12 × $52) + (12 × $87) + $99 = $624 + $1,044 + $99 = $1,767. Plan 2: 24 × $60 = $1,440. Plan 2 costs $327 less, even though Plan 1's headline price is $20 lower.

Sources and further reading

All pages were accessed in October 2026.

Broadband labels

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2. FCC, Report and Order FCC 26-48 https://docs.fcc.gov/public/attachments/FCC-26-48A1.pdf, released July 23, 2026. Label rule changes and compliance dates.
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Speed, availability and home networks

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8

PART IV · CONNECTIONS AND WASTE

Mobile Phone Service

A mobile phone bill usually combines three different things: the service plan, the payments on the phone itself, and a layer of add-ons, insurance, fees and taxes. Many people know their plan price but not their total, and fewer know how much of it is the phone. That mix is what makes mobile bills hard to compare, and easy to overpay.

The biggest costs often hide in offers that look like savings. A "free" phone is usually paid for through bill credits spread over two or three years, and under many offers the credits stop if you change plans or leave. Price guarantees may not cover fees. And unlimited plans differ in what "unlimited" means.

The tools for comparing are better than they were. Since 2024, mobile broadband plans have had to carry the same FCC broadband label as home internet (7.4), showing price, fees and typical speeds. And federal rules let you keep your phone number when you switch.

By the end of this chapter you will be able to:

- Read a cell phone bill and separate service, device payments, add-ons and fees
- Explain unlimited, high-speed data, throttling and deprioritization
- Decide between prepaid and postpaid, and between a carrier and a smaller provider that uses its network
- Work out the real cost of a "free" phone over 24 and 36 months
- Switch carriers without losing your number or service, and protect your account from fraud

All plan and phone prices in this chapter are illustrative. Help with phone bills for low-income households, including Lifeline, is in 12.4.

START HERE:

- → **Your bill seems too high?** Go to 8.3, then 8.4.
- → **Offered a "free" phone?** Go to 8.5.
- → **Partway through phone credits and thinking of leaving?** Go to "Already on credits?" in 8.5, then Case G.
- → **Ready to switch carriers?** Go to 8.6 before you cancel anything.

8.1 How cell plans work

Most mobile plans run on one of a small number of national networks, but you can buy service in several different ways. The way you buy it, more than the network, decides most of the price.

Type	How it works	Typical trade-offs
Major carrier, postpaid	You use the service, then get a monthly bill. Usually needs a credit check.	The most phone financing deals, perks and in-store support. Often the highest monthly prices.
Major carrier, prepaid	You pay before each month of service.	Usually cheaper, with no credit check. Phones are often bought outright, and may stay locked longer (8.6).
Smaller provider using a carrier's network (an MVNO, or mobile virtual network operator)	A company that buys network capacity from a major carrier and resells it under its own name, usually prepaid	Often much cheaper for the same network. Its customers' data may be slowed first when the network is busy, and support and perks are usually thinner.
Family or multi-line plan	Several lines on one account, with a lower price per line	Cheaper per line, but every line is tied to one account holder and one bill

The network is what decides coverage. A smaller provider on the same network as a major carrier usually reaches the same places, though its customers may get lower priority when the network is congested (8.2). Ask any provider which network it uses, and check its coverage map at your home, your work and the places you travel.

Home phones

Some households still keep a landline, often delivered over the internet (VoIP) as part of a bundle (7.9). If you depend on a home line for a medical alert device or a security system, check how it works in a power or internet outage before you change it. If you move a landline number to a mobile phone, the FCC warns that 911 may not be able to call you back while the move is under way. Low-income households may qualify for a Lifeline discount on either a mobile or a home phone (12.4).

Find it on your bill: your provider's name, whether you are prepaid or postpaid, and how many lines are on the account.

8.2 Plan terms decoded

Mobile plans use a small vocabulary that hides big differences. Two "unlimited" plans can differ widely, both in price and in what you actually get.

Term	What it means	What to check
Unlimited data	No hard cap on data, but usually not unlimited full-speed data	How much high-speed data is included
High-speed (premium) data	The amount of data at full speed each month	The amount, and what happens after it is used
Throttling	Your speed is cut, often sharply, after a set amount of data	The threshold and the reduced speed
Deprioritization	When the network is busy, your data may be slowed before other customers'	Whether it applies always, or only after a threshold
Hotspot allowance	Data you can share from your phone to a laptop or tablet	The amount at full speed
Domestic roaming	Using another network where yours has no coverage	Whether it is included, and any limit
International roaming	Using your phone abroad	Daily or monthly passes, and per-minute or per-megabyte charges
Coverage	Where the network works well	The provider's map, and real experience at home and work
Device financing	Paying for a phone in monthly installments, often over 24 or 36 months	The total price, the term, and what happens if you leave early
Trade-in or promotional credits	A discount paid as monthly bill credits, not upfront	How many months, and what ends them (8.5)
Account and line fees	Charges per line, activation fees, or account-level fees	Whether the advertised price includes them
Autopay discount	A monthly credit for automatic payment	Whether it requires a debit card or bank account

Mobile broadband plans carry the FCC's broadband label (7.4). It shows the monthly price, fees and typical speeds, plus the data included and any charge or slowdown after it. That is the quickest way to compare two unlimited plans.

How much data do you use?

Your phone's settings show your monthly data use, and most carriers show it in your account. Look at the last three months. If you use far less than your plan includes, a cheaper plan may cover you. If you are on Wi-Fi at home and work most of the day, your mobile use may be lower than you think. Measure yours rather than guessing.

8.3 Reading a cell phone bill

A cell phone bill mixes the service with the phone and the extras. Separate them, and you can see what each one really costs.

A sample bill

Illustrative example: Case G, Jordan and Casey's shared account with a major carrier. Anytown Wireless is fictional, and the bill is marked SAMPLE.

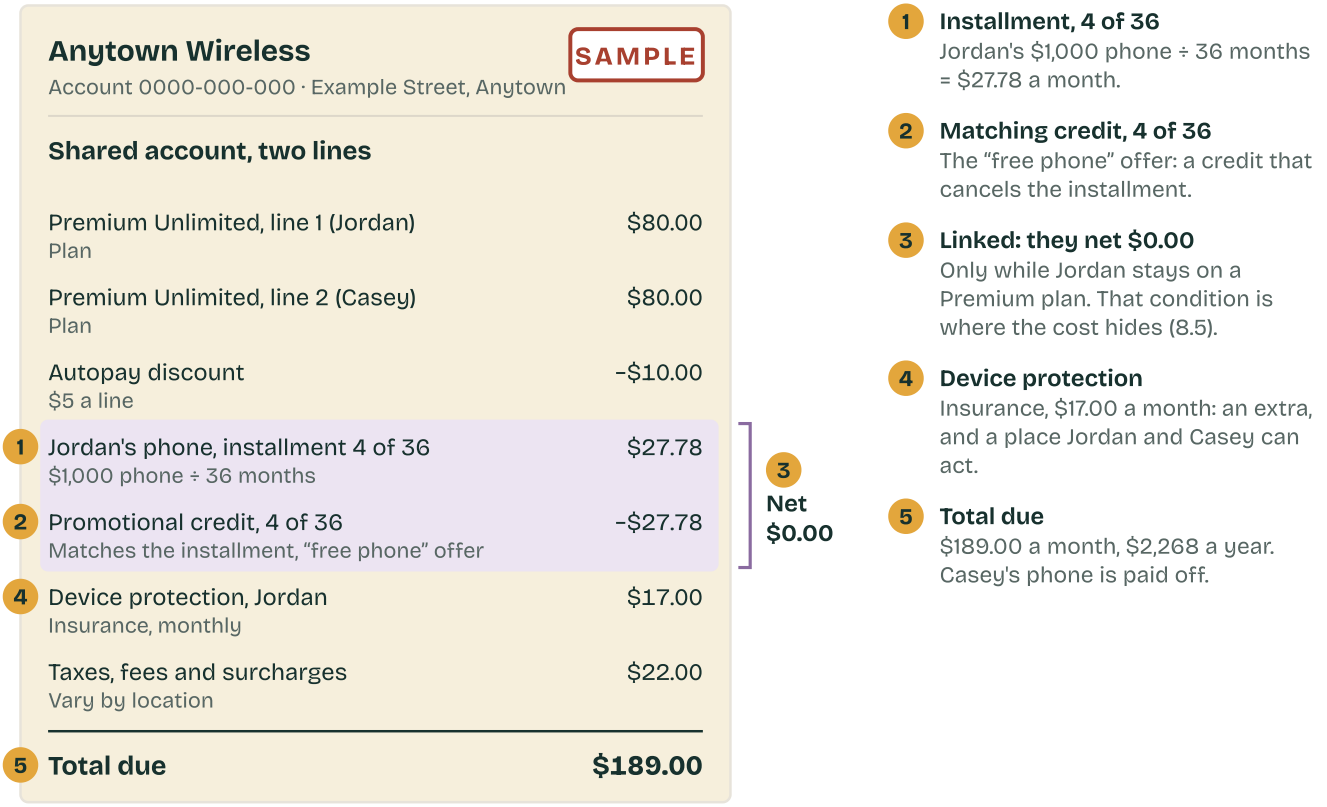


Figure 8.1. Annotated phone bill: Jordan and Casey's shared account from Case G, with the phone installment and its matching credit linked. SAMPLE: Anytown Wireless is fictional.

Casey's phone is paid off. The "free" phone shows as two lines that cancel out, but the credit lasts only while Jordan stays on a Premium plan. That condition is where the cost hides (8.5).

Sort the bill into four parts

Part	Lines	Monthly total
Service	Both plans, less autopay	$150.00
Phones	Installment less credit	$0.00, conditional on staying on Premium
Extras	Device protection	$17.00
Taxes and fees	Government and carrier charges	$22.00

That is $2,268 a year. The service and extras are where Jordan and Casey can act.

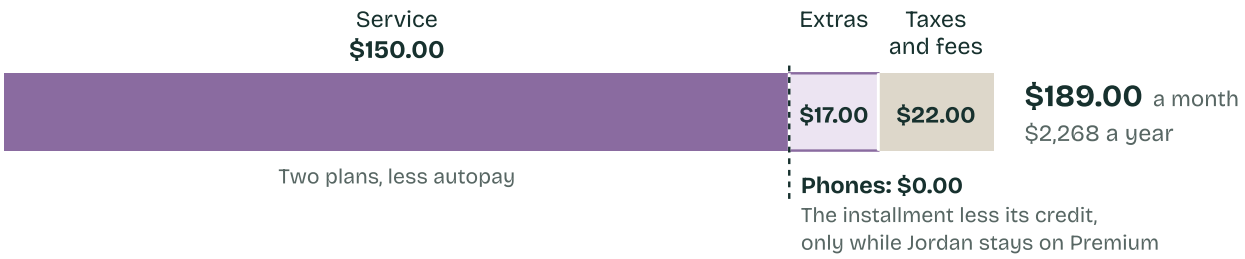


Figure 8.2. The four parts of Jordan and Casey's $189.00 monthly bill. The phone shows as $0.00 only while Jordan stays on Premium. Illustrative, SAMPLE: Case G.

What else to look for

- **Add-ons** you did not choose or no longer use: streaming bundles, international passes, extra cloud storage, premium voicemail
- **Third-party charges** billed to your phone account, such as games or subscriptions; ask your carrier how to block them
- **Lines you no longer need,** such as an old tablet or smartwatch line
- **The installment count:** how many payments remain, and the balance you would owe if you left today
- **A price guarantee:** what it covers. In October 2026, New York's attorney general reached a settlement with a carrier whose "price guarantee" customers saw increases of about $5 a line; read what any guarantee excludes

Find it on your bill: the plan price per line, every installment and credit with its count (such as "4 of 36"), any insurance or add-ons, and the total.

8.4 Lowering your mobile costs

Two moves often matter most: paying for the plan you use, not the one you were sold, and keeping a phone longer. Start with the free steps.

Step	Cost tier	What to do
Check your real data use	No-cost	Look at the last three months in your phone or account (8.2), and compare with what your plan includes
Remove add-ons and unused lines	No-cost	Cancel extras you do not use, and lines for devices you no longer need (8.3)
Reconsider device protection	No-cost	Compare the monthly charge, and any deductible, with the cost of repairing or replacing your phone
Ask about discounts	No-cost	Autopay, employer or group discounts, and any discounts for which you qualify
Compare prepaid and smaller providers on the same network	No-cost to compare; leaving may mean paying off a financed phone (8.5)	Use broadband labels and coverage maps; a trial month is the best test
Ask your carrier to match	No-cost	Tell your carrier about the better offer and ask it to match, using the Provider Negotiation Script (7.7, A1). Ask whether a match restarts a contract or changes a credit
Compare multi-line plans	No-cost	Lines on one account can cost less each, if everyone trusts the account holder
Move only the lines without credits	No-cost to switch, usually	On a multi-line account, move the lines that have no credits now, and keep the financed line until its last credit. First ask whether the credit needs a minimum number of lines or a plan tier, and what the remaining line will cost on its own
Keep your phone longer	No-cost	Each extra year you keep a paid-off phone avoids a new phone's cost (8.5)

Step	Cost tier	What to do
Read financing terms before upgrading	No-cost	Know the total price, the term, and what ends any credits
Check for Lifeline if your income is low	No-cost	See 12.4

Is device protection worth it?

Illustrative example: device protection at $17 a month.

- Over 24 months: 24 × $17 = $408
- Over 36 months: 36 × $17 = $612

Compare that with what you would pay to repair a cracked screen or replace the phone, plus any deductible the insurance charges per claim. Some people self-insure instead: they keep a sturdy case on the phone and put the monthly amount into savings. If you have broken or lost phones often, protection may still make sense. Check whether a credit card or home insurance policy already covers your phone before paying twice.

Check coverage before you switch

A cheaper plan is no bargain if it does not work at home. Before switching:

- Check the new provider's coverage map at your home, work and regular routes
- Ask neighbors or coworkers which network they use and how it performs
- If you can, try the new service first, for example with a prepaid month on a second SIM or eSIM, if your phone supports it

What to record: your plan's monthly cost per line, data used per month, and any add-ons removed, with the date.

8.5 The real cost of a "free" phone

A "free" phone is usually a phone you pay for in installments, offset by monthly bill credits. The credits typically last as long as the installments, often 36 months, and only while you keep a qualifying plan. Under many offers, leaving early or moving to a cheaper plan stops the credits while the balance stays. Read your promotion's terms.

Whether the deal is good depends on what the qualifying plan costs compared with the plan you would otherwise choose.

A three-year comparison

Illustrative example, one line. A carrier offers a $1,000 phone "free" through 36 monthly credits of $27.78, if you stay on its Premium plan at $90 a month. Its Basic plan, which meets your needs, costs $60 a month. Taxes and fees are left out.

Option	Monthly	36-month total
A. Premium plan with the "free" phone	$90	36 × $90 = $3,240
B. Basic plan, and buy the same phone outright	$60, plus $1,000 once	(36 × $60) + $1,000 = $3,160
C. Basic plan, and keep your current paid-off phone	$60	36 × $60 = $2,160

The "free" phone costs $80 more than buying it, because the Premium plan costs $30 a month more for 36 months, $1,080, and the phone was worth $1,000. Keeping the old phone saves $1,080 against the "free" deal.

If you leave after 24 months

After 24 of 36 credits, a third of the phone is still unpaid: $1,000 ÷ 3 = $333.33.

Option	24-month total
A. Premium plan, then pay the remaining $333.33 to leave	(24 × $90) + $333.33 = $2,493.33
B. Basic plan, phone bought outright	(24 × $60) + $1,000 = $2,440.00
C. Basic plan, keep the old phone	24 × $60 = $1,440.00

The phone also stays locked to the carrier until it is paid off (8.6), which can make switching harder.

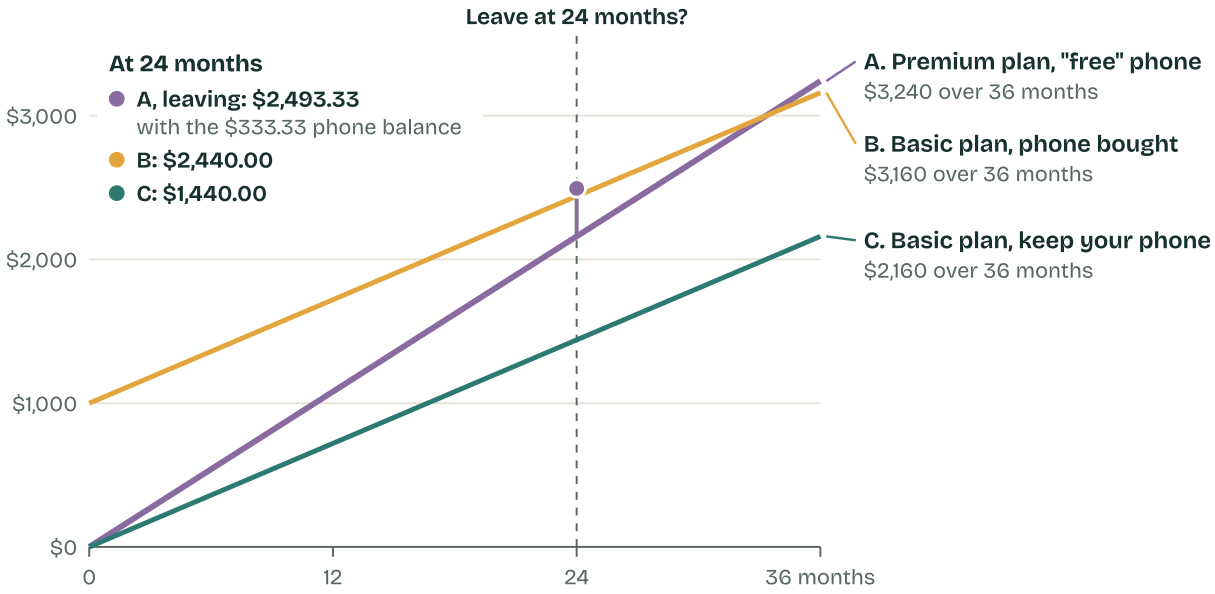


Figure 8.3. The real cost of a "free" phone: cumulative cost of options A, B and C over 36 months, and what each costs if you leave at 24 months. One phone line; taxes and fees left out. Illustrative.

Already on credits?

If you are partway through a "free" phone, judge leaving from today. Under many offers, leaving stops the credits, and you still owe the phone's balance, which your bill or account shows (8.3). Switching now, keeping the same phone, pays if (your monthly total − the new monthly total) × the credit months left is more than that balance. If it is not, set a reminder for the month of the last credit, and switch then.

Illustrative example, Case G: Jordan has 32 credits left and would owe $888.89. Switching saves Jordan and Casey $189 − $70 = $119 a month, and $119 × 32 = $3,808, far more than $888.89. Switching now pays.

On a multi-line account, also consider the middle option in 8.4.

How to judge any phone offer

1. **Ask what plan the offer requires,** and what it costs compared with the plan you actually need.
2. **Multiply the difference by the number of credit months.** If it is more than the phone's price, the "free" phone costs more than buying it.
3. **Ask what happens if you change plans or leave:** do the credits stop, and how much would you owe?
4. **Consider keeping your phone.** It is usually the cheapest option of all.
5. **Check trade-in offers the same way.** A trade-in paid as bill credits ties you to the plan just like a free-phone offer.

We found no specific federal guidance on phone installment plans or bill credits, so read the offer terms carefully and get them in writing.

What to record: the credit amount and count on each bill (such as "4 of 36"), the balance you would owe, and the month of the last credit, in Tool 8.1.

8.6 Switching carriers safely

Switching can cut a mobile bill sharply, and the rules are on your side: you can keep your number, and a paid-off phone can usually be unlocked. The risks are a phone that will not work on the new network, a gap in service, and fraud.

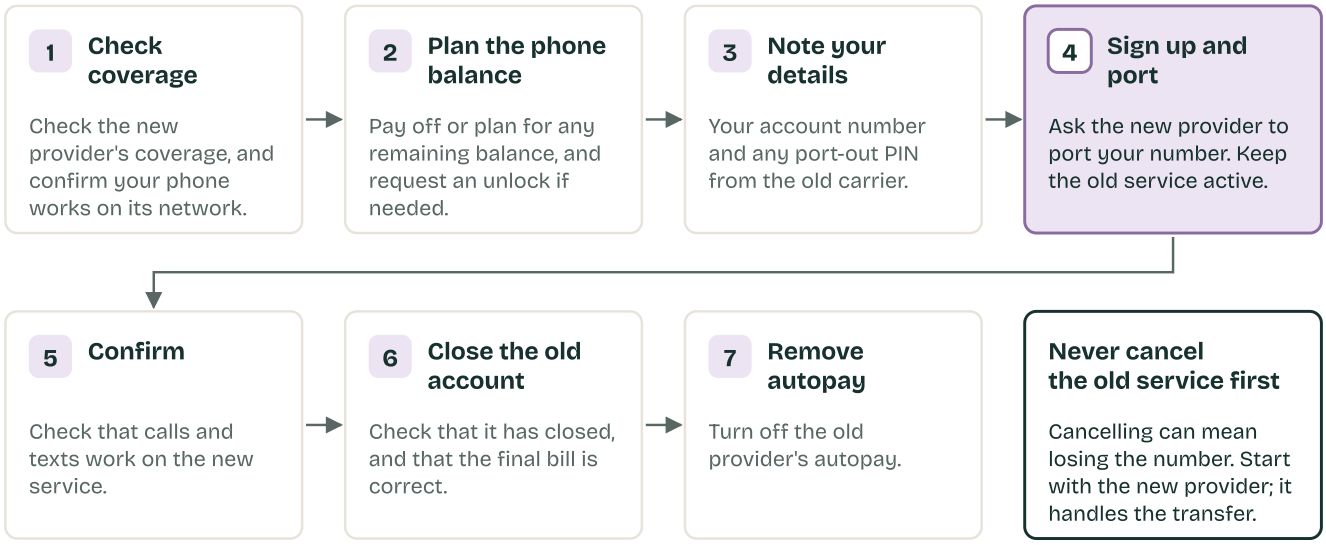


Figure 8.4. Switching without losing your number: the seven steps of the switching checklist, with the old service kept active until the port is done.

Keeping your number (porting)

The FCC's rules let you keep your number when you change providers in the same area. Its guidance:

- **Do not cancel your old service first.** Cancelling can mean losing the number.
- **Start with the new company.** Give it your 10-digit number and the account details it asks for; it handles the transfer.
- **Your old company cannot refuse** to port your number, even if you owe it money. You still owe the balance or any termination fee.
- Simple ports must be completed within one business day; moves between wireless carriers can take only a few hours.
- Some companies charge a porting fee. Ask whether it can be waived.

Unlocking your phone

A locked phone works only on the carrier that sold it. The FCC explains that phones sold at a discount or in installments usually stay locked until they are paid off, and even full-price phones may be locked for a short period, such as 60 days. Prepaid phones may stay locked longer.

Many carriers follow an industry code. **CTIA's code says** carriers will:

- Unlock postpaid phones on request once the contract, financing or termination fee is paid
- Unlock prepaid phones no later than one year after activation
- Tell customers when a phone becomes eligible, or unlock it automatically, at no charge
- Respond to unlock requests within two business days

Last Verified: October 2026. The code is voluntary. In January 2026 the FCC waived a stricter 60-day unlocking rule that had applied to Verizon, which now follows the code; Verizon's policy keeps prepaid phones locked until 365 days of paid service. An FCC proposal to require unlocking within 60 days at all carriers was still pending.

An unlocked phone may still not work on another network for technical reasons. Ask the new provider to check your phone's compatibility before you switch.

Protecting your account from SIM-swap and port-out fraud

In a **SIM swap** or **port-out** scam, a criminal moves your number to a phone they control, then uses texted codes to break into your bank and email accounts. The FCC adopted rules in 2023 requiring carriers to verify identity and notify customers before these changes, but as of October 2026 carriers were not yet required to follow them. Protect yourself now, using FCC and FTC advice:

- ☐ Set a PIN or password on your mobile account, and turn on any port-out or SIM lock your carrier offers
- ☐ Use an authenticator app or security key, not text-message codes, for your bank and email where you can
- ☐ Turn on alerts from your bank and other financial accounts
- ☐ Never give account details to someone who calls you; hang up and call your carrier at a number you trust
- ☐ If your phone suddenly loses service or can only call 911, contact your carrier and then your bank at once

A switching checklist

1. Check the new provider's coverage, and confirm your phone works on its network.
2. Pay off or plan for any remaining phone balance, and request an unlock if needed.
3. Note your account number and any port-out PIN from the old carrier.
4. Sign up with the new provider and ask it to port your number. Keep the old service active.
5. Confirm calls and texts work on the new service.
6. Check that the old account has closed, and that the final bill is correct.
7. Remove the old provider's autopay.

CASE G

Jordan and Casey compare two plans

Illustrative example. Jordan and Casey are fictional roommates sharing a rented house. Their two phone lines are on one account with a major carrier (8.3). Four months ago Jordan took a "free" $1,000 phone on 36 monthly credits; Casey's phone is paid off. All providers and prices are fictional.

Their bill is $189 a month. A smaller provider that uses the same network offers unlimited plans at $35 a line, with taxes and fees included, and no family plan needed.

The two plans

	Plan A: stay with the carrier	Plan B: switch to the smaller provider
Service	2 Premium lines at $80, less $10 autopay = $150	2 lines at $35 = $70, taxes and fees included
Jordan's phone	$27.78 a month, cancelled by matching credits, while on Premium	Pays off the remaining balance to leave
Casey's phone	Paid off	Unlocked and brought along
Device protection	$17 a month for Jordan	None; they set money aside instead
Taxes and fees	$22 a month	Included
Monthly total	**$189**	**$70**

To make the comparison fair, the totals below cover the phone in both plans: Plan A pays for it through the plan, and Plan B pays $1,000 for it at the start. Starting from a fresh $1,000 phone in each case keeps the two on equal terms. For the choice they actually faced, four credits in, start from today's balance, $888.89, as in "Already on credits?" (8.5).

Over 24 and 36 months

Period	Plan A	Plan B	Plan B saves
24 months, then leave Plan A	(24 × $189) + $333.33 remaining phone balance = $4,869.33	(24 × $70) + $1,000 = $2,680.00	$2,189.33
36 months	36 × $189 = $6,804.00	(36 × $70) + $1,000 = $3,520.00	$3,284.00

The $1,000 phone is the same in both plans, so the difference is almost entirely service, insurance and fees: $119 a month. Even if Jordan and Casey stayed on Plan A without leaving at 24 months, Plan B would still save $1,856 by then.

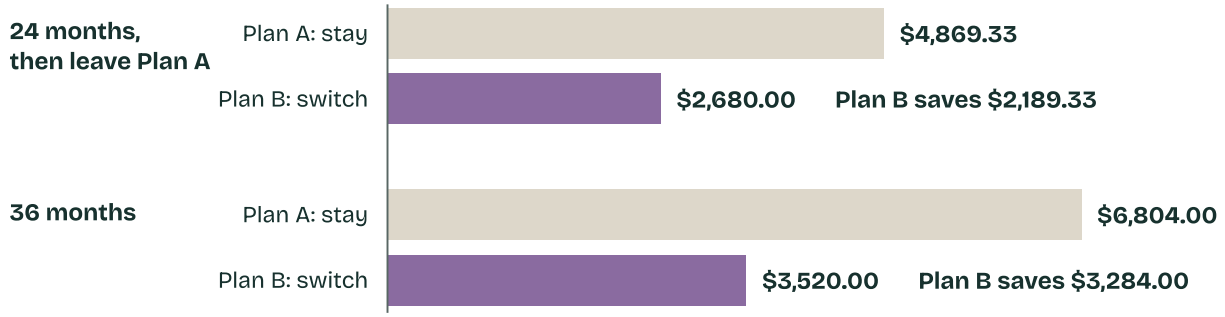


Figure 8.5. Case G: Plans A and B over 24 months, leaving Plan A at the end, and over 36 months. Both include a $1,000 phone. Illustrative.

What Plan B gives up

Trade-off	How they checked
Lower data priority when the network is busy	Read the broadband label, and tried a prepaid month on a second SIM at home and work
No device insurance	They put $17 a month into savings: $612 over 36 months, toward any repair
Fewer perks and no in-store support	Decided they did not use them
International roaming	Casey travels once a year; priced a travel pass for that trip

The decision

They chose Plan B, as two separate accounts. That way neither roommate is responsible for the other's line if one moves out (11.4).

1. Casey confirmed the phone was paid off and asked the carrier to unlock it (8.6).
2. Jordan paid the remaining phone balance, $1,000 × 32 ÷ 36 = $888.89, so the phone could be unlocked. That is about what the phone would cost to buy outright from here.
3. Each signed up with the new provider and ported their number, keeping the old service active until the port finished.
4. They checked the final carrier bill and cancelled autopay.

Checking the result

The first two months' bills were $35 a line, as promised. Calls, texts and data worked at home and work. Their savings are Estimated until a full year of bills confirms them. They set a reminder to recheck plans in a year, since prices change.

TOOL 8.1

The Mobile Plan Comparison Worksheet

Purpose: to compare mobile plans, including phone costs, over 24 and 36 months, so that "free" phones and low headline prices are judged on equal terms.

Instructions

1. Fill in one column for your current plan and one for each option, using bills, offers and broadband labels.
2. Put the same phone, or the same plan to keep your phone, in each column, so you compare like with like.
3. Add every monthly charge, less discounts and credits, for the monthly total.
4. For the 24-month total, add any phone balance you would owe if you left at 24 months. Already part-way through credits? Fill in the credit months left and today's balance too, and use the test in 8.5 ("Already on credits?").
5. Add one-time costs, such as a phone bought outright or activation fees.
6. Note coverage and data priority. A cheaper plan must still work where you live and work.

Form

Item	Current plan	Option A	Option B
Provider and network used			
Number of lines			
Plan price per line			
Account or line fees			
Autopay or other discounts			
Taxes and fees (or "included")			
Phone price and financing term			
Monthly installment			
Monthly credits, and the condition that ends them			
Device protection			
Add-ons			
Monthly total			
One-time costs (phone bought outright, activation, porting)			
Phone balance owed if leaving at 24 months			

Item	Current plan	Option A	Option B
Credit months left, and phone balance owed if leaving now			
24-month total			
36-month total			
High-speed data included			
Data priority when the network is busy			
Coverage at home and work (tested?)			

Reading your results

- Choose the lowest 24- or 36-month total among the plans that work where you need them, depending on how long you expect to stay.
- If a "free" phone option wins, check how much of its advantage disappears if you leave early.
- If two options are close, the one with no long-term credits or contract keeps you free to switch when prices change.
- Treat the totals as Estimated: plan prices and fees can change during the period.

Chapter 8 review

What you learned

- A mobile bill has four parts: service, phones, extras, and taxes and fees. Separate them to see what each costs.
- "Unlimited" plans differ in high-speed data, throttling, data priority and hotspot allowances. The broadband label shows the differences.
- Smaller providers on the same network can cost far less, with trade-offs in data priority, support and perks.
- A "free" phone is paid for through the plan. If the required plan costs more each month, multiplied by the credit months, than the phone's price, it is not free.
- Keeping a paid-off phone is usually the cheapest choice of all.
- You can keep your number when you switch, and your old carrier cannot refuse to port it. Never cancel first.
- As of October 2026, the FCC's 2023 SIM-swap protections were not yet in force. Protect your account with a PIN, a port-out lock and authenticator apps.

What you can do today

1. Sort your latest bill into service, phones, extras and fees (8.3).
2. Check your data use for the last three months (8.2).
3. Cancel one add-on or unused line you do not need.
4. Find the remaining balance on any financed phone, and the condition attached to any credit.
5. Set a PIN or password on your mobile account (8.6).

What to check next month

Confirm that removed add-ons are gone from the bill, and that any new plan was billed at the quoted price. If you switched, check that the old account is closed with no surprise charges. If a line has bill credits and you changed its plan or dropped its insurance, check that the credit still posts. It should be the same amount, with the count one higher ("5 of 36" after "4 of 36").

Billix Quick Check

A carrier offers a "free" $800 phone through 36 monthly credits, if you move to its $75 plan. The plan you need costs $50. Is the free phone a good deal?

Answer: The required plan costs $25 a month more: $25 × 36 = $900. That is $100 more than the phone's price, so buying the phone and taking the $50 plan costs less. Keeping your current phone on the $50 plan saves the full $900.

Sources and further reading

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PART IV · CONNECTIONS AND WASTE

9 Trash and Recycling

The bill many households never see

Most utility bills arrive with your name on them. Trash often does not. Depending on where you live, garbage and recycling may be paid through your property taxes, a line on the city water bill, your rent, your HOA dues, or a private hauler's invoice. When a cost is folded into something else, few people check it.

That matters more than it might seem. In cities that price service by cart size, the cart you chose years ago sets a big part of the bill. In Seattle, for example, the largest garbage cart costs $110.30 a month more than the smallest (9.2). Not every household fits in the smallest can. But some households pay for far more space than they fill.

This chapter is shorter than the energy and water chapters, because trash has fewer moving parts. You will learn how to find out how you pay, how to read the charges, and where savings tend to hide: the cart size, avoidable fees, services your base fee already covers, and duplicate charges. You will also learn what never to do to save money, including how to get rid of hazardous products safely.

The chapter ends with Tool 9.1, the **Waste Service Audit**, a four-week exercise that compares the service you need with the service you pay for.

Last Verified: City rates in this chapter were checked in October 2026. Trash rates change often, in many places every year. Check your own provider's current rate page before you act.

START HERE:

- → **Not sure how you pay for trash?** Go to 9.1.
- → **Two trash bills, or a charge you can't explain?** Go to "Catch duplicate and wrong charges" in 9.3.
- → **Is your garbage cart too big?** Go to Tool 9.1, the Waste Service Audit, then Case D.
- → **Getting rid of a couch, branches or old paint?** Go to "Use the services you already pay for" in 9.3, then 9.4.

9.1 How waste collection is billed

There is no national system for trash. Each city, county or town decides how to collect it and how to charge for it. EPA notes that residents have traditionally paid through property taxes or a fixed fee, no matter how much trash they made. Many places now use other methods. Your household probably pays in one of these five ways.

How you pay	Where it shows up	Real example	What you can control
Property taxes or the general city budget	Inside your property tax bill, or nowhere you can see	EPA describes this as the traditional method	Very little directly. You can still use free services and avoid extra fees.
Flat city fee	A line on your city utility bill, often next to water and sewer	Dallas: $40.09 a month for residential sanitation	Extra carts and add-on services
Volume-based fee (cart size or pay-as-you-throw)	A city utility bill, or bags and tags you buy	Seattle and Austin price by garbage cart size	Cart size, extra bags and how much you throw away
Private hauler subscription	A monthly or quarterly invoice from a company	Unincorporated Carroll County, Maryland, and parts of Kane County, Illinois	Which hauler, which plan and the contract terms
Landlord or HOA	Inside rent, HOA dues or a shared utility charge	HOAs may contract one hauler for a neighborhood	Usually little. Check what you pay and why.

Property taxes. When trash is paid this way, there is no trash line to cut. Renters in these places still pay, indirectly, through rent. Your savings come from avoiding extra fees and using the free services your taxes already fund.

Flat city fee. You pay the same amount whether your cart is full or nearly empty. Dallas lists residential sanitation at $40.09 a month, or $481.08 a year. Its rate page shows conflicting effective dates, so check the city's current page before relying on this figure.

Volume-based fee. You pay more for more space. Seattle and Austin charge by the size of your garbage cart. Other towns use pay-as-you-throw, often shortened to PAYT. EPA describes PAYT as charging per bag or can, and in a few places by weight. EPA says thousands of communities use it. This is the method where your own choices change the bill the most.

Private hauler subscription. In some areas, no government collects household trash. You hire a licensed company yourself. In unincorporated Carroll County, Maryland, residents contract with any licensed hauler or take trash to the county landfill and pay tipping fees. In the Kane County, Illinois, townships without a township contract, owners or tenants choose their own licensed hauler. Kane County advises checking the contract length, price-increase caps, cancellation terms and early-termination penalties before you sign.

Landlord or HOA. No national rule covers how landlords pass on trash costs. Some include it in rent. Some bill it as a separate charge. Carroll County notes that HOAs may contract for a whole neighborhood.

Check your lease: Your lease or HOA documents should say who pays for trash and recycling, and how. If you see a trash charge from both the city and your landlord, ask which one actually covers your home.

How to find out how you pay

1. Look at your city or water bill for a line called solid waste, sanitation, garbage, refuse or environmental services.
2. If you own your home, look at your property tax bill for a solid waste or sanitation line.
3. Search your bank and card statements for a hauler's name.
4. If you rent, read the utilities section of your lease, then ask your landlord.
5. If you are still unsure, call your city or county's public works or solid waste department. Ask who collects at your address and how it is billed.

Find it on your bill: the trash line may be called solid waste, sanitation, garbage, refuse or environmental services. On a city bill it often sits next to water and sewer. It may be one line, or several: a base fee, the garbage cart and its size, and any yard waste or extra-cart lines. Note how often you are billed, monthly or quarterly. Only one line? The city's rate page, or the fee explanation request (A4), shows what it buys. **Record it in** Tool 9.1, Step 2.

9.2 Common charges

A trash bill can be one line or ten. The charges below are the ones you are most likely to see. The real figures come from city rate pages checked in October 2026. They show how charges work, not what you will pay.

The garbage cart

In cart-size cities, the garbage cart is usually the biggest number you control. Here is how Seattle prices it.

Seattle Public Utilities, garbage service (rates effective April 1, 2026)

Garbage cart	Per month	Per year
12-gallon micro can	$29.20	$350.40
20-gallon mini can	$35.80	$429.60
32-gallon cart	$46.55	$558.60
64-gallon cart	$92.90	$1,114.80
96-gallon cart	$139.50	$1,674.00

Austin works very differently. In Seattle, the price climbs with every size, at about $1.45 a month for each gallon of cart from 32 gallons up. In Austin, every size includes the same $24.90 base fee (its rate page shows no effective date). Its 24-gallon cart costs $29.80 a month, and moving up to a 64-gallon cart adds only $8.20. The big jump is the step up to 96 gallons, which adds $31.05.

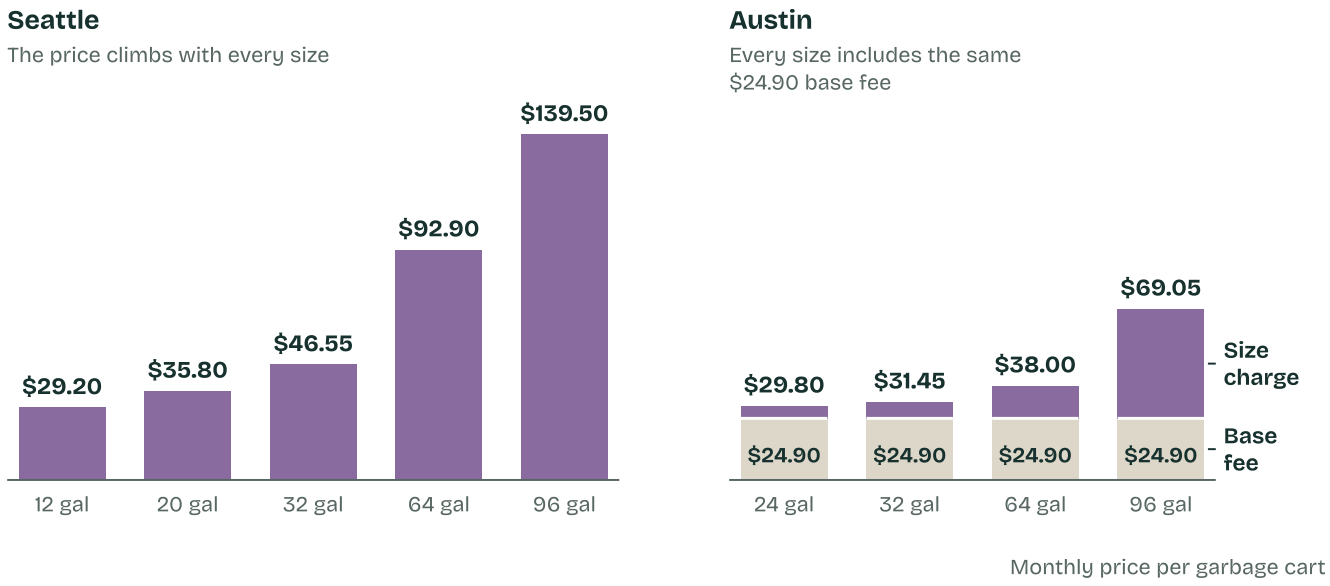


Figure 9.1. Monthly garbage price by cart size in Seattle and Austin; every Austin size includes the same $24.90 base fee. Sources: Seattle Public Utilities, rates effective April 1, 2026; City of Austin, Austin Resource Recovery, rate page with no effective date shown, checked October 22, 2026.

What this means for you: before you downsize, find out how much of your bill actually changes with cart size. A large base fee means a smaller cart saves less.

Base and service fees

Some bills include a fee you pay no matter what cart you choose. Austin's $24.90 base fee pays for recycling, composting, brush, bulk item and household hazardous waste collection. It is charged even if you opt out of those services. Austin also bills a separate $5.65 Clean Community Fee. Think of a base fee as a list of services you have already paid for, and use them.

Extra capacity

- **Extra carts.** Dallas charges $15.51 a month for an extra garbage cart, rising to $16.29.
- **Extra bags.** Seattle charges $14.40 for each extra bag or bundle set out beside the cart.
- **Overfilled carts.** Riverside, California, lists a $46 fee for an overfilled cart.

Changing your service

- **Size changes.** Seattle charges $35.75 if you change cart size more than once a year. Austin lets you downsize for free and charges $15 to upsize.
- **Cart or container lines.** If a private hauler bills a separate cart or container line, ask what it covers and whether a customer-owned option exists.

Bulky items, yard waste and special disposal

These vary more than any other charge. In Austin, brush and bulk item collection are part of the base fee. Elsewhere, a couch pickup, a yard waste cart or a holiday tree collection may cost extra, or may be free on set days. Your provider's fee schedule is the only reliable answer.

Penalty fees

- **Contamination fees.** Riverside bills an $18.02 fee for a contaminated recycling or organics cart. It applies only after warnings. The first and second offenses get a tag on the cart, plus a mailed warning the second time. The fee is billed from the third.
- **Spill cleanup.** Riverside lists a $209 fee for cleaning up a hazardous-material spill.

Taxes

Some cities add sales tax to trash service. Dallas's rate page lists 8.25% sales tax with its proposed rates. Check whether your bill's figures include tax.

9.3 Lowering waste costs

Trash savings are usually smaller than energy savings. They are also some of the easiest to get, because most of them cost nothing and take one phone call.

Right-size your garbage cart

Cost: No-cost in most cities. Some charge a fee to change size.

If you pay by cart size, the question is simple. How full is your cart on pickup day? Some households picked a cart years ago, when more people lived at home, and never looked again.

Before you judge, move everything you can out of the garbage. Flatten boxes into the recycling cart. Put food scraps and yard waste in the organics cart if your city collects them. Seattle Public Utilities gives the same advice: recycle and compost more, so you can use a smaller garbage cart. Then watch the garbage cart for four pickups. The Waste Service Audit (Tool 9.1) gives you a log to fill in.

Worked example: downsizing in Seattle (Illustrative)

- **Assumed:** a household has a 64-gallon garbage cart. After moving recyclables out, the cart is half full or less on most pickups. It would need about four extra bags a year, during holiday weeks.

- **Price check:** the 64-gallon cart costs $92.90 a month. The 32-gallon cart costs $46.55. Extra bags cost $14.40 each.
- **Savings before extra bags:** $46.35 a month, or $556.20 a year.
- **Cost of extra bags:** 4 × $14.40 = $57.60 a year.
- **Estimated net savings:** $556.20 − $57.60 = **$498.60 a year.**
- **The limit:** the switch stops paying if you need more than about three extra bags a month ($46.35 ÷ $14.40 = 3.2).

The same move in Austin saves $6.55 a month, or $78.60 a year, because most of Austin's price is the base fee. Downsizing there is free, so it can still be worth doing. It is just a smaller win.

Stop paying penalty fees

Cost: No-cost.

Recycling rules are local. A plastic container your old city took may be banned from the cart in your new one. The list from your own hauler or city is the only one that counts.

- Print your local accepted-items list and post it where you sort trash.
- When you are not sure about an item, check the list or call. Guessing can contaminate a cart.
- If you find a tag on your cart, read it. In cities like Riverside, tags are the warning before a fee.
- Do not overfill. Riverside's overfilled-cart fee is $46, more than three extra bags in Seattle.

Penalties add up fast. Three Riverside contamination fees in a year would cost $54.06.

Use the services you already pay for

Cost: No-cost.

Read what your base fee or tax-funded service includes. Look for bulky item pickup, brush or yard waste collection, drop-off centers, household hazardous waste days and holiday tree collection. Check this list before you pay a private junk removal company to haul away a couch or a pile of branches.

Catch duplicate and wrong charges

Cost: No-cost.

Trash billing errors are easy to miss because few people read the trash line. Check for these:

- **The wrong cart size.** Compare the cart at your curb with the size on your bill.
- **A cart you returned.** An extra cart you gave back may still be on the bill.
- **Two providers for one home.** You pay the city and a private hauler, or the city and your landlord, for the same service.
- **A hauler you left.** An old provider keeps billing after you move or switch.
- **An add-on you never use.** A yard waste cart or extra recycling cart sits empty for months.

If you find an error, ask for a correction and a credit back to when the error started. Do it in writing and keep a copy. To dispute a charge, see 10.2 and the billing dispute letter (A3). Some cities do not let you opt out of service, so ask before you cancel anything. Austin, for example, charges its base fee even to customers who opt out.

An error, or two services you chose? A wrong cart size, a returned cart still on the bill or a hauler you left is a billing error: ask for the credit. Paying both the city and a private hauler may instead be two services you chose. List what each one covers. If the city fee already includes a service, such as yard waste, and you cannot opt out of it, the private subscription is the one to end. Read its cancellation terms first (see "Compare private haulers" below), and use the service cancellation checklist (A6). If the city does not offer that service, you are not paying twice.

Compare private haulers

Cost: No-cost to compare.

If you hire your own hauler, get quotes from two or three licensed companies. Compare the same service: cart size, pickup frequency and whether recycling is included. Then read the contract. Kane County, Illinois, suggests checking four things:

1. The contract length.
2. Any cap on price increases.
3. The cancellation terms.
4. Early-termination penalties.

Ask your HOA or neighbors whether a neighborhood contract exists. In some counties, such as Carroll County, Maryland, you may also haul trash to the county facility yourself and pay its tipping fee. Count your time and fuel, and secure the load so nothing blows out on the road.

If you pay by the bag

Cost: No-cost.

In a pay-as-you-throw town, every bag you avoid is money kept.

Worked example: pay-as-you-throw (Illustrative)

- **Assumed:** a town charges $3 for each 30-gallon bag. This price is illustrative, not taken from any real town.
- **Before:** 2 bags a week × 52 weeks = 104 bags, or $312 a year.
- **After:** better recycling cuts that to 1.5 bags a week, or 78 bags. That costs $234 a year.
- **Estimated savings:** $78 a year.

An archived EPA bulletin from 2010 reported one study of 228 New England towns, by the firm Green Waste Solutions. Towns with curbside PAYT disposed of 467 pounds of trash per person, against 918 pounds in towns without it. That is about 49% less. Treat this as one older study, not a promise. It does not show how much your own household would change.

9.4 What not to do

Some ways of cutting a trash bill are illegal, unsafe or unfair to your neighbors. They can also cost far more than they save. This handbook never recommends them.

Never dump trash illegally

Do not leave bags, furniture, mattresses, tires or building debris on a roadside, in an alley, on a vacant lot or beside a donation bin. Illegal dumping is against the law and can bring fines and cleanup charges. Even when no one is caught, the cleanup cost lands on your neighbors and your city. If you have a large item, use the bulky pickup, drop-off center or special collection your area offers. Section 9.3 explains how to find them.

Never use other people's carts or dumpsters

A neighbor's cart is tied to their account. Your bags can push them into an overflow fee or a contamination tag they did nothing to earn. A business's dumpster is private property that the business pays for. Use someone else's container only with their clear permission, and only if your local rules allow it.

Never hide garbage in the recycling or organics cart

It may look like free space. It is how contamination fees start. In Riverside, it ends in an $18.02 fee after two warnings.

Dispose of hazardous products safely

Household hazardous waste is leftover product that can catch fire, react, explode, or is corrosive or toxic. EPA's examples include paints, cleaners, oils, batteries and pesticides. These items need their own disposal route.

SAFETY WARNING: Never pour household hazardous waste down a sink, toilet or storm drain, or onto the ground. Never put it in your regular trash. EPA advises taking it to a permanent collection site, a collection event or a designated drop-off day. Keep each product in its original labeled container. Never store it in a food or drink container, and never mix leftovers together.

To find a site, search your local solid waste agency's website for "household hazardous waste," or use the Earth911 search tool that EPA points to. Check the accepted items and opening hours before you load the car.

Unsafe disposal can also cost money. Riverside lists a $209 fee for cleaning up a hazardous-material spill.

Call a professional: If you find containers you cannot identify, containers that leak, or a large stockpile of old chemicals, do not open, mix or move them yourself. Call your local solid waste agency or hazardous waste program and ask how to proceed.

TOOL 9.1

The Waste Service Audit

The Waste Service Audit compares the service your household needs with the service you pay for. It takes four pickup cycles and about 30 minutes of paperwork. A blank version is in the companion worksheet pack as a fillable PDF and a spreadsheet. Case D, after the steps, shows one household working through it.

Your audit, step by step

Step 1: Find out how you pay. Tick one or more:

- ☐ Property taxes or the general city budget
- ☐ A flat fee on a city or utility bill
- ☐ A cart-size or pay-as-you-throw fee
- ☐ A private hauler subscription
- ☐ Rent, HOA dues or a landlord charge

Step 2: List what you pay for. Use your last bill and your provider's rate page.

Service	Size or level	Per bill	Per year (× 12 if monthly, × 4 if quarterly)	Used in the last month?	Notes
Base or service fee					
Garbage cart					
Recycling					
Yard waste or organics					
Extra carts					
Extra bags or tags					

Service	Size or level	Per bill	Per year (× 12 if monthly, × 4 if quarterly)	Used in the last month?	Notes
Bulky item pickups					
Penalty fees, last 12 months					
Taxes and other lines					
Total					

Step 3: Log four pickups. First move everything you can into recycling and organics. Then record each cart on pickup day: empty, a quarter, half, three quarters, full or overflowing.

Pickup	Garbage	Recycling	Yard waste or organics	Extra bags	Notes
1					
2					
3					
4					

Step 4: Compare what you need with what you pay for.

- **Garbage cart half full or less on three of four pickups:** price the next smaller size.
- **Overflow on two or more pickups:** compare a larger cart with what you spend on extra bags.
- **An add-on cart empty for the whole log:** think about the season first. If it is truly unused, ask whether you can pause or drop it, and whether its fee still applies.
- **You paid someone else for a service your base fee covers:** use the city service next time.
- **A line you cannot explain:** ask your provider what it is, in writing.
- **Two providers billing for one home:** if one bills by mistake, ask which is correct and request a credit from the other. If you chose both, list what each covers first (9.3).

Step 5: Price each change.

- Yearly savings = (current monthly price − new monthly price) × 12, minus expected extra bag costs, minus any change fee. Billed quarterly? Divide each quarterly price by 3 first.
- Break-even = monthly savings ÷ price per extra bag. If you would need more extra bags a month than this number, the change does not pay.

Step 6: Make the change and check it.

Change requested	Date	Confirmation number	New monthly price	Shown on bill dated	Last Verified

Check that the right cart arrives at your curb. Recheck your extra bag count after three months.

Case D: the Harris household audits its trash (Illustrative)

The Harris household is a fictional case study. Anytown Public Works and its rates are SAMPLE figures, not real prices.

Starting situation. The Harrises are two retired adults on a fixed income. They live in a small, paid-off house. While they work through the $420 electric balance (Case D in Chapter 12), they are looking for any safe cut. Their city bill has a solid waste line of $57 a month. They have never questioned it.

Household facts. Two people, a small yard, and a 96-gallon garbage cart chosen when their three children lived at home. They fill a yard waste cart from April to November. Last spring they paid a private junk removal company $150 to haul away an old sofa and a mattress.

Sample bill: Anytown Public Works, solid waste (SAMPLE)

Line	Per month
Solid waste base fee (recycling, two bulky item pickups a year, hazardous waste drop-off)	$20.00
Garbage cart, 96 gallons	$28.00
Yard waste cart	$9.00
Total	**$57.00** ($684 a year)

Anytown's SAMPLE prices for the other garbage carts are $16.00 for 64 gallons and $8.00 for 32 gallons. Extra bag tags cost $4.00 each. One size change a year is free. A second costs $20.

Likely source of the extra cost. A cart sized for five people, and a bulky pickup they already paid for but did not use.

Diagnostic steps. For four weeks, they moved all cardboard into the recycling cart. They counted the tied 13-gallon kitchen bags in the garbage cart on each pickup day.

Pickup	Garbage cart	Recycling cart	Yard waste cart	Notes
Week 1	2 bags (about 26 gallons)	Nearly full	Full of leaves	
Week 2	2 bags	Full	Half full	
Week 3	3 bags (about 39 gallons)	Full	Full of leaves	Family visit
Week 4	2 bags	Nearly full	Full of leaves	

In a normal week, their garbage filled about a quarter of the 96-gallon cart.



Figure 9.2. The Harris log: four pickups, with the garbage cart filled to scale and a dashed line at 32 gallons. SAMPLE, Illustrative: Case D, the Harris household.

Solutions.

1. **Move down to the 64-gallon cart.** No extra bags needed.
2. **Move down to the 32-gallon cart.** Normal weeks fit, with about 6 gallons to spare. They estimate 6 heavy weeks a year, for holidays and visits, each needing one $4 tag.
3. **Drop the yard waste cart.** It would save $108 a year. But their log shows the cart is full in the fall, and they have no space to compost at home. They keep it.
4. **Use the free bulky pickups.** Next time a large item goes, they call the city instead of paying a private company.

Cost comparison (Estimated)

	Monthly service	Extra tags per year	Total per year	Savings per year
Today: 96-gallon cart	$57.00	$0	$684	
Option 1: 64-gallon cart	$45.00	$0	$540	$144
Option 2: 32-gallon cart	$37.00	$24 (6 × $4)	$468	**$216**

The $150 they spent on the junk removal company is not in this table. It was a one-time cost, and it is the kind of spending their base fee was meant to replace.

The decision and why. They choose the 32-gallon cart. Their normal week fits, tags cover the rare heavy week, and the change is free. If the small cart proves too tight, moving up to 64 gallons would cost $20. On that cart they would still save $144 a year, less the one-time $20 fee.

How they check the result. On the next bill, the garbage line should read $8.00 and the solid waste total $37.00. They keep a tally of the tags they buy. If they need more than two tags a month, the 64-gallon cart becomes the cheaper choice ($8.00 extra a month ÷ $4.00 per tag = 2 tags). After three bills, they mark the savings **Verified**.

Chapter 9 review

What you learned

- Households pay for trash in one of five ways: property taxes, a flat city fee, a volume-based fee, a private hauler, or a landlord or HOA.
- In cart-size cities, the garbage cart is usually the biggest cost you control. How much a smaller cart saves depends on how much of the price is a fixed base fee.
- A base fee can already pay for bulky pickups, yard waste and hazardous waste drop-off. Using those services is part of getting your money's worth.
- Contamination and overfill fees are avoidable. Your local accepted-items list is the rule that counts.
- Illegal dumping, using other people's containers and unsafe hazardous waste disposal are never ways to save.

What you can do today

- Find your trash charge, using the five steps in section 9.1.
- Compare the cart at your curb with the cart size on your bill.
- Print your local recycling list and post it where you sort trash.
- Look up your area's bulky item pickup and hazardous waste drop-off options.

What to check next month

- Finish the four-pickup log and complete the Waste Service Audit (Tool 9.1).
- If you changed your cart, check the new price on your bill and the new cart at your curb.
- Look for tags on your carts and penalty lines on your bill.

Billix Quick Check

Answer yes or no:

1. Do you know how your household pays for trash?
2. Does the cart size on your bill match the cart at your curb?
3. Was your garbage cart more than half full on most of your last four pickups?
4. Have you avoided penalty fees for the last 12 months?
5. Do you know where to take old paint, batteries and other hazardous products?

A "no" to question 1, 2, 4 or 5 points to a step in this chapter. A "no" to question 3 means your cart may be bigger than you need. Run the Waste Service Audit (Tool 9.1) to find out.

Sources and further reading

Last Verified: All sources below were checked on October 22, 2026. Trash rates change often. The rates are examples of how charges work, not national averages.

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4. City of Dallas Sanitation Services. "FY 2026-27 Rate Changes." https://dallascityhall.com/departments/sanitation/Pages/FY%202026-27%20Rate%20Changes.aspx (The page shows conflicting effective dates.)
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8. Carroll County, Maryland, Office of Recycling. "Trash and Recycling Pickup." No date. https://www.carrollcountymd.gov/government/directory/public-works/office-of-recycling/trash-and-recycling-pickup/
9. Kane County, Illinois, Recycling. "Unincorporated Hauling." No date. https://kanecountyil.gov/Recycling/Pages/UnincorporatedHauling.aspx

Further reading

- Your own city or county solid waste department's rate page and accepted-items list. These are the rules that apply to you.
- Earth911's recycling and disposal search, which EPA points to for finding hazardous waste drop-off sites.

PART V

Protect and Decide

10 Billing Disputes, Negotiation and Consumer Protection

11 Renters, Homeowners and Moving

12 Financial Assistance, Discounts and Hardship Programs

13 Smart Upgrades and Return on Investment

PART V · PROTECT AND DECIDE

10 Billing Disputes, Negotiation and Consumer Protection

Your bill is a claim, not a verdict

A bill tells you what a company says you owe. Most of the time it is right. But bills are built from meter reads, rate tables, plan settings and manual adjustments, and any of those can go wrong. A read can be estimated. A promotion can expire without notice. A credit you were promised can fail to appear.

You have more power here than you might think. Many billing problems are fixed with one well-prepared call. Some need a short letter. A few need a regulator. In each case, the household that keeps good records and asks clear questions usually does better than the one that is angry and unprepared.

This chapter is also about protecting your money from people who are not your provider at all. Scammers pose as utilities, and some door-to-door sellers misstate what they offer. Knowing how to check who you are dealing with is part of managing your bills.

What this chapter covers

- **10.1** A simple monthly review that catches problems early
- **10.2** The billing errors to look for, and the evidence for each
- **10.3** A nine-step method for contacting customer service
- **10.4** What can and cannot be negotiated
- **10.5** Budget billing and the annual settle-up
- **10.6** Autopay: the discounts and the risks
- **10.7** Scams and high-pressure offers
- **10.8** Where to escalate, and which agency handles what
- **10.9** Deposits, late fees, reconnection fees, collections and credit reports

Two households return. The Brooks household, whose $160 water spike you followed in Chapter 5, uses a call log and a letter to get a promised credit onto its bill. The Harris household prepares for the call it has been putting off about its $420 electric balance.

START HERE:

- → **A call or text says your service will be cut off today?** Go to 10.7 before you pay anything.
- → **A charge looks wrong?** Go to 10.2 and 10.3, then send the billing dispute letter (A3).
- → **Behind on a bill, or holding a shutoff notice?** Go to 12.8.
- → **The company said no?** Go to 10.8 to find the right agency.
- → **A budget-billing settle-up arrived?** Go to 10.5.

Last Verified: The rules and agency details in this chapter were checked in October 2026. State rules differ, and they change. Where this chapter gives a state example, it is only an example. Check your own state's rules before you rely on them.

10.1 Reviewing bills systematically

Billing problems are usually easier to fix when you catch them in the first month. They get harder once several bills have gone by. A short review each month is the best protection you have.

The table below is Tool 10.1, the Billix Electric Bill Analyzer. It works for any metered bill: gas, water, propane deliveries and even internet or phone. Track these eleven fields every month.

Field	What it is	What to check
Provider	The company that bills you, and any supplier listed separately	Is it a company you chose? Has a new name appeared?
Account type	Residential, the rate plan or tariff name, any program you are on	Is it still the plan you signed up for?
Billing date	The date the bill was issued	Did it arrive on its usual cycle?
Service period	The start and end dates, and the number of days	Do the dates follow on from last bill, with no gap or overlap?
Usage	Units used: kWh, therms, gallons, gigabytes or minutes	Is the read actual or estimated? How does use per day compare?
Unit rate	The price per unit, including tiers or time periods	Does it match your plan, contract or the utility's posted rate?
Fixed charges	Customer, base, service or equipment charges	Has any fixed charge changed or appeared?
Taxes	Sales taxes, government fees and surcharges	Are they in line with past bills?
Adjustments	Credits, debits, promotions and prior-period corrections	Is every promised credit there, at the right amount?
Total due	The amount you owe now	Does it equal last balance, minus payments, plus new charges?
Payment date	The due date, and when your last payment posted	Was your payment received and applied to the right account?

A blank copy of Tool 10.1 is in the companion pack. The 15-Minute Bill Audit after 1.8 (blank sheet in B2) is the quick check to run alongside it.

The five-minute monthly check

1. Put the new bill next to last month's bill and the same month last year.
2. Count the days in the service period.
3. Divide usage by days to get use per day. This removes the effect of a long or short billing period.
4. Compare the unit rate and fixed charges with your plan or the utility's posted rates.
5. Read every adjustment line, and confirm your last payment posted.

Worked example: use per day (Illustrative)

- **Last month:** 900 kWh over 30 days = 30 kWh a day.
- **This month:** 900 kWh over 33 days = about 27.3 kWh a day.
- **What it means:** the total use is the same, but daily use fell. The three extra days, not extra use, explain any rise in the bill.

10.2 Billing errors

Billing errors come in seven common kinds. Each leaves a trail you can follow, and each calls for its own evidence.

Error	What it can look like	Evidence to gather
Incorrect reading	Usage far above normal, an "estimated" marker, or a reading that does not match your meter	A dated photo of your meter; past bills showing normal use
Wrong service period	Dates that overlap the last bill, a much longer period, or charges after you moved out	Your previous bill; your move-out confirmation and final reading
Duplicate fee	The same fee twice, such as two late fees or two equipment charges	The bill lines; your payment receipts

Error	What it can look like	Evidence to gather
Unexplained price change	A higher rate with no notice you can find, or a promotion that ended early	Your contract or plan confirmation; screenshots of the offer; the broadband label or rate notice
Unauthorized service	An add-on, plan or supplier you did not choose	Your own records; a request for the company's proof that you agreed
Adjustment mistake	A promised credit that is missing, or applied at the wrong amount	Your call log with date, name and reference number; any confirmation email
Wrong deposit or credit	A deposit not returned when it should be, missing deposit interest, or a payment applied to the wrong account	Your deposit receipt; your state's deposit rule; your bank statement

Find it on your bill: the letter E, the abbreviation Est. or the word Estimated beside the meter reading, or a "reading type" marked actual or estimated. It sits next to the current and previous readings. Current minus previous is the use you were billed for. **Record it in** Tool 10.1, the Electric Bill Analyzer, under Usage.

Should you dispute an estimated reading?

Compare a dated photo of your meter with the bill's current reading, allowing for the days between them (1.5).

- **Your meter shows less than the bill.** You were billed for use you have not had. Ask for a rebill now, using your reading (10.3). The gap in units × your marginal price (1.8) is roughly the overcharge. For example, a 300 kWh gap at the rate card's $0.18 per kWh is about $54 (Illustrative).
- **Your meter shows more than the bill.** The estimate ran low, so you used more than you were billed for. There is little to dispute. Expect a catch-up bill at the next actual reading (1.5), and set the gap × your marginal price aside now.

Either way, pay the part you do not dispute by the due date (10.3, step 3).

How to collect evidence

- **Keep one folder per provider,** paper or digital. Keep at least a year of bills, and longer for anything in dispute.
- **Photograph your meter** with the date visible, at the start of a dispute and whenever you move.
- **Screenshot offers and plan pages** when you sign up. Prices and terms on websites change.
- **Log every contact** in the Provider Call Log (A2).
- **Send copies, not originals,** when a company or agency asks for documents.

SAFETY WARNING: Never open, adjust or tamper with a meter to check it. Read it from the outside, or ask the utility to test it.

10.3 Contacting customer service

A good call follows a plan. These nine steps work for a utility, an internet provider, a phone carrier or a hauler.

1	**Define the exact problem**	Write it in one sentence before you call.
2	**Gather the relevant bills**	The disputed bill, the two or three before it and last year's same month.
3	**Identify the disputed amount**	The exact dollar figure. Pay what you do not dispute by the due date.
4	**Request the applicable rate or contract**	A utility's tariff, or your plan terms or the broadband label.
5	**Ask for an explanation**	"Can you walk me through how this charge was calculated?"
6	**Request a correction where warranted**	Say exactly what you want, such as a rebill using an actual reading.
7	**Obtain written confirmation**	An email, a letter or an account note, and always a reference number.
8	**Record the conversation**	Date, time, number called, name or ID, and what was promised (A2).
9	**Follow the escalation process**	A supervisor, then the dispute in writing, then the right agency (10.8).

Figure 10.1. The nine-step call: define, gather, identify, request, ask, correct, confirm, record, escalate.

1. **Define the exact problem.** Write it in one sentence before you call. For example: "My March bill shows 1,850 kWh, about double my usual, and the read is marked estimated."
2. **Gather the relevant bills.** Have the disputed bill, the two or three before it, and the same month last year.
3. **Identify the disputed amount.** Work out the exact dollar figure you question. Pay the part you do not dispute by the due date, and ask how the provider handles the disputed part while it reviews it. Some states protect you here. In Pennsylvania and New York, a utility may not shut off service over a pending dispute as long as you pay the undisputed part. Some states also bar late fees on a disputed amount, or while a complaint is pending (see the late-fee table in 10.9).
4. **Request the applicable rate or contract.** For a regulated utility, ask which rate schedule, or tariff, applies to your account. For a competitive service, ask for your plan terms or the broadband label.
5. **Ask for an explanation.** "Can you walk me through how this charge was calculated?" Listen, and take notes.
6. **Request a correction where warranted.** Say exactly what you want: "Please rebill this period using an actual reading," or "Please remove the second late fee."
7. **Obtain written confirmation.** Ask for a confirmation email, a letter or a note on your online account. Always ask for a reference number.
8. **Record the conversation.** Write down the date, time, number you called, the representative's name or ID, what they said and what they promised. Use the Provider Call Log (A2).
9. **Follow the escalation process.** If the answer is no and you still think the bill is wrong, ask for a supervisor. Then put your dispute in writing. If that fails, go to the right agency (section 10.8).

A note on recording. In this chapter, "record" means write down. If you want to make an audio recording of a call, check your state's law first. Some states require everyone on the call to agree.

Tips that help

- **Call the number on your bill** or the provider's official website, never a number from an unexpected message (section 10.7).
- **Stay calm and specific.** The person on the phone usually did not cause the problem, but may be able to fix it.
- **Ask what they can do,** not only what they cannot.

- **Set a follow-up date.** If a correction is promised, note when it should appear on your bill, and check.

10.4 What can and cannot be negotiated

Some prices are set in public and cannot be changed on a phone call. Others are set by the company and often can.

Regulated utility rates. For investor-owned electric, gas and water utilities, rates are approved by the state utility commission and published in tariffs. A customer service representative cannot lower your price per kWh or per therm. In most states, city-owned utilities and cooperatives set rates through a city council or an elected board. Rates themselves change through public proceedings, often called rate cases, and many commissions accept public comments.

Competitive services. Internet, TV, mobile service, competitive electricity and gas supply, propane, heating oil and private trash hauling are priced by the company. These prices often have room to move, especially when you are willing to switch.

Service	Who sets the price	What you can usually ask for
Investor-owned electric, gas or water utility	The state commission approves the tariff	Corrections, a better-fitting rate plan, budget billing, a payment arrangement, fee waivers, assistance programs
City-owned utility or cooperative	Usually a city council or utility board	The same asks as above
Competitive electricity or gas supplier	The supplier, within state rules	A different plan, or a return to the utility's default service (2.11)
Internet and TV	The provider	A retention offer, a lower plan, dropping rented equipment (7.7)
Mobile phone	The carrier	A cheaper plan, removing add-ons, a loyalty offer (8.4)
Propane and heating oil	The dealer	Price plans and delivery terms (3.6)
Private trash hauler	The hauler	Quotes, cart size and contract terms (9.3)

Legitimate asks of a utility

Even when the rate is fixed, a utility can often do several things for you. Whether it will depends on its rules, your state's rules and your account history.

- **Correct an error** and rebill the period.
- **Move you to a rate plan** that fits your use better, such as a time-of-use plan (2.10).
- **Set up budget billing** (section 10.5).
- **Agree a payment arrangement** for a past-due balance. Section 12.8 covers state rules on payment plans.
- **Waive a late fee** as a one-time courtesy. Some will and some will not.
- **Offer a deposit alternative** or return a deposit you have earned back (section 10.9).
- **Grant a leak adjustment** on a water bill after a repaired leak (5.10).
- **Tell you about assistance programs** and medical protections (12.5 and 12.8).

A useful opening line for a utility call: "I'm not asking you to change the rate. I'd like to know whether any of these apply to my account: a better rate plan, budget billing, a payment arrangement or a waiver of this fee."

For internet, TV and mobile, use the Provider Negotiation Script (A1). It works the same way for any competitive service.

10.5 Budget billing

Budget billing spreads your yearly energy cost into equal or near-equal monthly payments. Utilities call it different things: budget billing, an equal payment plan, levelized billing or averaged payments. The idea is the same. You pay about the same amount in January as in May.

The most important point: budget billing changes when you pay, not how much you pay. Every kWh and therm you use is still charged at the normal rate. Over a year, you pay for exactly what you used.

How the monthly amount is set

Each utility has its own method. Three real examples show the range.

Utility	How the amount is set	How often it changes	The settle-up
PG&E (California)	A rolling average of your costs over the past 12 to 13 months, plus 1/12 of any balance or credit	Recalculated monthly	No separate annual settle-up is described; the balance is updated monthly
FirstEnergy (Ohio, Pennsylvania, New Jersey, New York)	1/12 of your estimated annual usage cost	Reviewed quarterly in Ohio and Pennsylvania	An annual true-up in your anniversary month: a credit is applied, or a balance owed is added to that month
Dominion Energy Virginia	The average of your past 12 months of bills	Every 12 months, or mid-year after a rate change or a big change in use	An overpayment is credited over the next 12 months; a shortfall can be paid at once or spread over 12 months

Rules to join vary, too. PG&E asks for a history of on-time payments, no past-due balance and no payment plan. Dominion requires that you owe no more than your current bill and have 12 months of history at the address.

The settle-up surprise

Because the payment is based on the past, a colder winter, more use or a rate increase can leave a gap.

Worked example: a settle-up (Illustrative)

- **Assumed:** a household's energy cost last year was $1,800. Its budget payment is $1,800 ÷ 12 = **$150 a month.**
- **During the year:** a cold January costs $260, but the household pays $150. A mild May costs $95, and it still pays $150. The plan absorbs both.
- **At year end:** a colder winter and a rate increase raise the year's actual cost to $1,980. The household paid $1,800.
- **The shortfall:** $1,980 − $1,800 = **$180.** Under a Dominion-style rule, it could pay $180 at once or add $15 a month for the next 12 months.

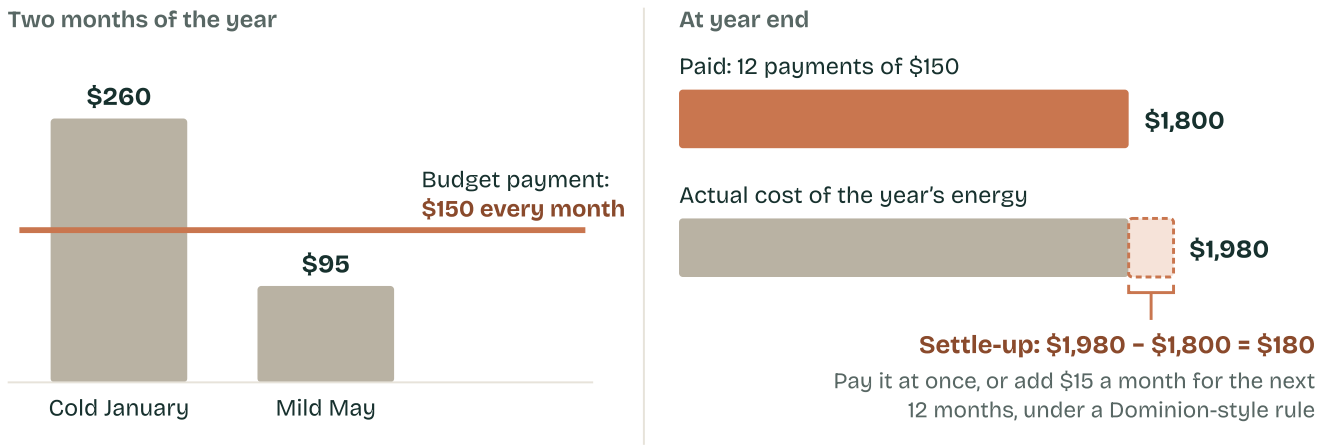


Figure 10.2. Budget billing over a year: the monthly cost moves, the $150 payment does not, and the gap comes due at the settle-up. Illustrative: the worked example in 10.5.

Leaving the plan

If you leave budget billing or move, the difference between what you paid and what you used usually comes due. Dominion says the actual balance is due when you cancel or move. PG&E applies any remaining balance to your final bill. It also says that if you miss or underpay two billing cycles, the full balance is due and you cannot rejoin for six months.

How to use budget billing well

- **Use it for predictability, not savings.** It suits households on a fixed income, like the Harrises, who need the same bill each month.
- **Watch the running balance.** Look for the line that compares your actual charges with your budget payments. If your bill does not show it, ask.
- **Ask how and when your plan settles up,** and whether a shortfall can be spread out.
- **Keep saving energy.** A lower actual cost shows up as a credit at the settle-up, or as a lower payment when the amount is reset.

10.6 Autopay

Automatic payments take the amount due from your bank account or card on a set date. They can help, and they carry risks.

The benefits

- **Discounts.** Some internet and mobile providers offer a monthly credit for autopay or paperless billing (7.6 and 8.4).
- **No missed due dates.** Autopay protects you from late fees caused by forgetting.

The risks

- **Overdraft.** The CFPB explains that an overdraft happens when your account does not have enough money to cover a transaction, and that automatic bill payments are one cause. The bank generally pays the transaction and charges a fee.
- **Not opting in may not protect you.** Federal rules require your consent before a bank charges overdraft fees on ATM and one-time debit card transactions. The CFPB has warned that banks may charge overdraft fees on checks and certain recurring electronic payments even if you did not opt in.
- **Big bills arrive without warning.** A spike, a catch-up bill after estimated reads or a budget-billing settle-up can be far larger than usual.

- **Errors get paid automatically.** Autopay pays a wrong bill as quickly as a right one.

One overdraft fee can cancel out many months of a small autopay discount.

How to use autopay safely

- **Still read every bill** before the payment date, using the five-minute check in section 10.1.
- **Keep a cushion** in the account the payment comes from, or link autopay to an account set aside for bills.
- **Ask whether your provider lets you set a maximum amount** for automatic payments.
- **Consider a credit card** for autopay if you pay the card in full each month. Check first whether the provider charges a card fee.

How to stop an automatic payment

The CFPB lays out these steps:

1. Tell the company you are revoking its permission to take payments.
2. Tell your bank.
3. For a stop-payment order, contact the bank at least three business days before the scheduled payment. You can do this by phone or in writing. The bank may ask for written confirmation within 14 days of a phone request.
4. Watch your account to make sure the payment stops.

Banks commonly charge a fee for a stop-payment order. Stopping a payment does not cancel your contract or the amount you owe, so keep paying the bill another way.

10.7 Scams and high-pressure offers

Some of the most expensive "bills" a household pays are not bills at all. They are payments to people pretending to be a provider, or contracts signed in a hurry at the door.

Fake utility shutoff calls and impersonation

The FTC describes the basic pattern. You get an unexpected call, text, email or visit from someone who says they are from your gas, electric or water company. They threaten to shut off your service unless you pay right away.

Real utilities do contact customers, including about past-due bills. What gives a scammer away is how they ask.

Warning sign	What the FTC and utilities say
A demand to pay now to stop a shutoff today	The FTC says real utilities do not demand immediate payment to avoid same-day disconnection. The industry group Utilities United Against Scams says utilities give multiple notices before interrupting service.
Payment by gift card, wire transfer, cryptocurrency or a payment app	The FTC calls these signs of a scam. With gift cards, the scammer asks you to read the numbers on the back.
A request for your bank details by phone or email	An FTC alert says real utilities do not ask for banking details that way.
Paying by phone as your only option	The same alert says real utilities do not make phone payment your only choice.
A visitor who pressures their way into your home	The FTC lists in-person visits among the tactics scammers use.
A search ad that looks like your utility	The FTC warns that paid ads can lead to fake phone numbers and websites. Scroll past the ads at the top of the results.

How to check who you are dealing with

1. **Stop.** Do not pay, and do not give any account or bank details.

2. **Hang up or close the message.** Do not use any number or link it gives you.
3. **Contact the provider yourself,** using the number on your bill or the company's official website, typed in yourself.
4. **Log in to your online account** to see your real balance and any real notices.
5. **Report it anyway,** even if you did not pay: to your real utility, and to the FTC at ReportFraud.ftc.gov.

SAFETY WARNING: If someone at your door says they are from a utility, you do not have to let them in. Call the utility at the number on your bill to check before you open the door. If you smell gas, leave right away and call 911 or the gas utility's emergency line from well away from the building (3.7).

If you already paid a scammer

Act quickly. The FTC's advice depends on how you paid:

How you paid	What to do
Gift card	Call the card company at the number on the back. Say a scammer got the number and PIN, and ask for a refund. Keep the card and receipt.
Wire transfer	Report it to the wire company and ask for a refund.
Payment app	Report it to the app and ask it to reverse the payment.
Zelle or bank transfer	Report it to your bank or credit union and ask it to reverse the payment.
Cryptocurrency	Contact the exchange or ATM operator, report the fraud and ask for a reversal. The FTC notes crypto has fewer legal protections than cards.
Credit or debit card	Call the card issuer or your bank and ask for a refund.

Then report it to your real utility, to the FTC at ReportFraud.ftc.gov and to your state attorney general. Imposter scams are a large problem. In 2025, the FTC received more than 1 million reports of imposter scams, with more than $3.5 billion in reported losses. That figure covers all kinds of imposters, not only fake utilities.

Door-to-door energy suppliers

In states with energy choice, sales agents may knock on your door with an offer to switch your electricity or gas supplier (2.11). Some offers are legitimate. Some agents misstate who they work for or what you will pay.

- **Know your state's rules.** In Pennsylvania, for example, agents must wear a badge showing the supplier, the agent's photo and name, and a customer service number. They may not wear anything suggesting they work for the utility or a government agency. Right after greeting you, they must say who they are, which supplier they represent and that they are independent of your utility. They must leave if you ask.
- **Know your right to cancel.** Pennsylvania lets you rescind a supplier switch within 3 business days after you receive the disclosure statement. The FTC's Cooling-Off Rule may also apply to many home sales of $25 or more, giving you until midnight of the third business day to cancel.
- **Protect your account number.** Do not hand over a bill or account number at the door. Ask for the offer in writing and compare it later.
- **Compare the full term,** not the first month. Section 2.11 explains fixed and variable supply rates.

Community solar pitches

Community solar is real. The US Department of Energy describes it as a shared solar project whose benefits flow to many customers. Subscribers pay for a share and receive bill credits. It can suit renters and homes with roofs that cannot take panels. But it is available only where state rules allow it, and terms vary.

- **No guaranteed savings.** DOE lists "at least 20% household savings" as a best-practice target, not a promise. It also recommends no sign-up, exit or termination fees, and plain-language terms.

- **"Free" panels are a red flag.** The FTC says "free" or "no cost" solar panel offers are scams, and that the federal government does not install solar systems in homes for free.
- **Pressure is a red flag.** The FTC says anyone who pressures you into a contract or demands immediate payment is a scammer.
- **Ask before you sign:** the price of each credit, the contract length, any exit fee and how the credits appear on your bill. Section 13.6 covers solar in more depth.

Protection plan mailers

Letters offering service-line or appliance protection plans can look official. Some cities do partner with these companies, and many do not. The FTC says home warranties are really service contracts. Before you buy, check the deductibles and per-visit fees, payout caps, exclusions, cancellation fees and whether you already have coverage. Section 11.3 explains how to judge these plans.

Regulators have acted on misleading mailers. In a 2023 consent judgment with the Kansas attorney general, HomeServe USA agreed to pay $500,000 to the state and $350,000 in consumer restitution, without admitting the allegations. It must now say it is independent when it uses another company's logo, and it may not send solicitations that look like bills. If a letter carries your city's or utility's logo, call them at a number you look up yourself.

Fake assistance websites

Some websites pose as government help. The FCC warns that some sites still advertise the Affordable Connectivity Program, which ended on June 1, 2024, or collect personal information for it. Report them at fcc.gov/complaints. If you entered sensitive information on such a site, visit IdentityTheft.gov. Find real programs through your utility or official .gov sites (12.7).

Misleading energy-saving products

Plug-in "power saver" boxes claim to cut electric bills by large amounts. The National Institute of Standards and Technology found that this type of device does not save a typical household money. Homes are billed for the energy they use, in kWh, and the device does not change that. The FTC advises being wary of any product that promises drastic energy savings, and looking for independent test results.

10.8 Escalation: who handles what

Escalate when the provider's own process has failed. That usually means you called, asked for a supervisor, put the dispute in writing and gave the company a fair chance to answer.

No single agency handles every bill. Each one covers certain services and certain problems. Filing with the wrong one wastes weeks, so use this map.

Problem	Where to go	How to reach them	What to know
A bill, deposit, late fee or shutoff dispute with an investor-owned electric, gas or water utility	Your state utility commission (often called the PUC or PSC)	The commission's number is often printed on your bill. NARUC, the national association of commissions, lists every state commission.	Commissions regulate investor-owned utilities. City-owned utilities and cooperatives usually fall outside their billing rules. Commissions may expect you to contact the utility first; Pennsylvania's accepts complaints only from customers who have.
A dispute with a city-owned utility or a cooperative	The utility's own appeal process, then the city council, utility board or co-op board	The utility's website or your bill	Ask in writing for the utility's dispute or appeal procedure.
Phone, internet or TV service	The Federal Communications Commission (FCC)	fcc.gov/complaints, or 1-888-225-5322	Filing is free. Contact the provider first. If the FCC sends your complaint to the provider, the provider must answer in writing within 30 days.

Problem	Where to go	How to reach them	What to know
A fake utility caller or other scam	The Federal Trade Commission (FTC)	ReportFraud.ftc.gov	Also tell your real utility and your state attorney general.
A deceptive sales pitch, such as a warranty mailer or a supplier at the door	Your state attorney general or state consumer protection office	NAAG's "Find my AG" page, or USA.gov's list of state consumer offices	State offices take complaints about businesses and scams.
A credit report error or a debt collector	The credit bureau first, then the Consumer Financial Protection Bureau (CFPB)	consumerfinance.gov/complaint, or (855) 411-2372	The CFPB handles credit reports and debt collection. It does not handle phone or internet billing.
A utility account opened in your name without permission	The FTC's identity theft site	IdentityTheft.gov	Also tell the utility and the credit bureaus.

What agencies can and cannot do

- **A state commission** can review whether a regulated utility followed its tariff and the state's rules. It cannot set your price outside those rules.
- **The FCC** forwards many billing and service complaints to the provider for a written reply. It says not every complaint is actionable, and some are referred to other agencies. For stand-alone cable or satellite TV billing, the FCC's own guidance points to state commissions.
- **The FTC** uses reports to spot patterns and build cases. Its pages do not promise that a report will get your own money back.
- **The CFPB** handles credit reports and debt collection, not your utility or phone bill itself.

How to file a complaint that gets read

1. Name the provider and give your account number.
2. Give a short timeline: the dates of each bill, call and letter.
3. State the amount in dispute and what you think it should be.
4. Say what you have already done, with reference numbers.
5. Say exactly what you want: a corrected bill, a refund, a fee removed.
6. Attach copies of the bills, letters and your call log. Never send originals.
7. Save the complaint number and any reply.

10.9 Deposits, late fees, reconnection fees, collections and credit reports

These charges are where a small billing problem can turn into a large one. State rules matter most here, and they vary widely. The examples below show the range. They are not national rules.

Deposits

A utility may ask for a deposit when it sees a risk that a bill will go unpaid. That can happen when you are new, have no credit history with the utility, or have paid late before.

State	Who can be asked	Maximum	When it comes back
New York	New residential applicants cannot be required to pay one, except seasonal or short-term customers. Existing customers only if delinquent. People 62 or older and known public assistance or SSI recipients are generally exempt.	About 2 months of estimated bills	Promptly after 1 year with no late bills, with interest set by the PSC
Pennsylvania	Applicants who do not meet credit standards	1/6 of the estimated annual bill	After 12 straight months of full, on-time payment
Texas, competitive electricity providers	Applicants without satisfactory credit; existing customers only after more than one late payment in 12 months or a disconnection	The greater of 1/5 of estimated annual billing or 2 months of bills	After 12 straight on-time billings, with interest
Ohio	Applicants who cannot show credit another way	About 130% of an average month's bill	Applied to the final bill, with at least 3% interest a year on deposits held 180 days or more
California, large investor-owned utilities	The commission bars these utilities from requiring deposits to set up or reestablish residential service	None	Not applicable

The pattern: in many states, a deposit is capped at about two months of bills, comes back after about a year of on-time payment, and earns interest at a rate the state sets. Some states protect certain groups, such as older customers.

Ways to avoid or recover a deposit:

- **Ask what the alternatives are.** Ohio, for example, lets applicants show credit by owning the home, passing a credit check, having had similar service in the past 24 months, or using a guarantor. Texas providers must offer alternatives to customers 65 or older with no current delinquency.
- **Mark the date.** Note when your deposit should come back, and ask for it if it does not appear as a refund or bill credit.
- **Check your final bill** when you move. The deposit and any interest should be applied to it.

Late fees

State	Late fee rule
Texas, competitive electricity providers	A one-time penalty of up to 5% of the late bill, never charged twice on the same balance. The due date must be at least 16 days after the bill is issued.
Pennsylvania	Up to 1.5% a month on the overdue balance. Not allowed on disputed estimated bills.
Illinois	Up to 1.5% a month, only on undisputed amounts, and only if the utility's tariff includes the fee. Residential bills must allow at least 21 days to pay.
New York	Up to 1.5% a month on an unpaid balance, but not while a complaint about the bill is pending.
California, water submetered by a landlord	$7 for the first late fee, then $10 for each later bill with an unpaid balance, capped at 10% of the unpaid amount in any 12 months

Worked example: what a late fee costs (Illustrative)

- **Assumed:** a $420 past-due balance, the Harris household's, at a late fee of 1.5% a month. Their state is not named, so the rate is an example.
- **Each month:** $420 × 1.5% = **$6.30.**
- **Six months with no payment:** $6.30 × 6 = **$37.80,** if the fee is charged on the original balance each month.
- **Compare:** a one-time 5% penalty, like the Texas rule, would be $21.00.

A late fee is often not the biggest cost of falling behind. The bigger risks are a shutoff, a reconnection fee, a new deposit and a collection account. That is why it pays to call before the bill is past due, not after.

Reconnection fees

If service is shut off, you may owe a reconnection fee on top of the balance. Florida Power & Light's service-charge page, for example, showed a $4 reconnect charge and a $28 field collection charge in October 2026. Fees change, so check your own utility's tariff. In Pennsylvania, the fee must be based on cost and approved in the utility's tariff. Ohio allows a reconnection charge only if service was actually cut. Section 12.8 covers shutoff protections.

Collections and credit reports

The CFPB says most utility companies do not report your payments to the three nationwide credit bureaus. So paying on time usually does not build your credit. But if an unpaid utility bill goes to a collection agency, it will most likely appear on your credit reports.

- **The numbers:** a 2023 CFPB report, using early-2022 data, counted 7.0 million utility collection accounts on credit reports, with a median balance of $276. Telecom collections were 19.2 million, with a median of $390.
- **How long they last:** most negative information can generally stay on a credit report for 7 years.
- **What the medical debt changes do not cover:** the credit bureaus' removal of medical collections under $500 applies to medical debt only, not to utility or phone debt.
- **A credit bureau you may not know:** NCTUE, the National Consumer Telecom and Utilities Exchange, shares account and payment data among its members. The CFPB puts membership at over 60 telecom, pay TV and utility companies, and members may use the data to decide whether to ask for a deposit. You can get a free NCTUE report every 12 months at 866-349-5185 or nctue.com/consumers.
- **Your free reports:** AnnualCreditReport.com offers free weekly reports from all three nationwide bureaus.

If a utility or phone debt on your report is wrong:

1. Dispute it in writing with each bureau that shows it. Say what is wrong and why, and include copies of your documents.
2. Dispute it with the company that reported it, too: the utility, the carrier or the collector.
3. The bureau generally has 30 days to investigate. If the information is wrong or cannot be verified, it must be corrected or removed.
4. If the problem is not fixed, complain to the CFPB at consumerfinance.gov/complaint.

Before a bill ever reaches collections, call the provider and ask for a payment arrangement. After a move, give a forwarding address so your final bill reaches you. A final bill that goes to an old address is easy to miss.

Case studies

Both households are fictional. The providers, bills and policies are SAMPLE figures, and the outcomes are illustrative. Real companies may decide differently.

Case F: the Brooks household and the missing credit

Starting situation. In Chapter 5, the Brooks household traced a $160 jump on its water bill to a worn toilet flapper. After an $8 repair, its use fell back to 182 gallons a day. On the phone, Anytown Water & Sewer agreed to a leak adjustment of $77.28. Two months later, the next bill arrived without it.

Household facts. Three people in a suburban house, with city water and sewer billed every two months. Anytown Water & Sewer is city-owned.

Sample bills (SAMPLE)

Bill	Water used	Total
Usual bill, 60 days	10,800 gallons	$181.20
The spike, 62 days	22,200 gallons	$340.80 (paid in full)
The next bill, 60 days	10,920 gallons	$182.88, with no adjustment

Likely source of the problem. An adjustment promised on a call but never entered, or entered and then stopped.

Diagnostic steps. They checked their notes from the first call. They had the date and the representative's first name, but no reference number and no confirmation email. On a second call, the representative read the account notes: "Leak adjustment approved, pending repair receipt." No one had told them a receipt was needed.

Solutions.

1. **Send the proof in writing.** Email the receipt for the flapper, the before and after meter readings, and a short letter that refers to the first call.
2. **Ask for written confirmation** of the amount and the bill it will appear on.
3. **If it is refused, appeal.** Because the utility is city-owned, the state commission probably does not handle its billing. The next step would be the utility's own appeal process, then its governing board.

Cost comparison. The credit at stake is $77.28. Getting it costs about 30 minutes and nothing else. Doing nothing costs the full $77.28.

The decision and why. They used the billing dispute letter (A3), attached the receipt and readings, and asked for confirmation by email. Putting it in writing gave the utility everything it needed in one place and created a record for any appeal.

How they checked the result. A confirmation email arrived with a reference number. The following bill showed "Leak adjustment −$77.28." Their new charges of $182.88 less the credit came to $105.60. They logged the result and marked the credit **Verified**.

The lesson: a promise on the phone is not a credit on the bill. Always ask for a reference number and written confirmation, and ask whether anything else is needed from you.

Case D: the Harris household and the double late fee

Starting situation. The Harrises, two retired adults on a fixed income, are $420 behind on electricity. They have been putting off a call to the utility. Their October bill shows two late payment charges posted on the same day.

Household facts. A small, paid-off house with gas heat. One of them uses powered medical equipment. Their electric utility is investor-owned, and its late fee is 1.5% a month on the unpaid balance (an illustrative rate; their state is not named).

Sample bill lines (SAMPLE)

Line	Amount
Past-due balance	$420.00
Late payment charge, October 2	$6.30
Late payment charge, October 2	$6.30

Likely source of the problem. A duplicate fee. At 1.5% of $420, one late charge is $6.30.

Diagnostic steps. They checked the two previous bills. Each showed one $6.30 charge. They found the late payment section of the utility's tariff on its website (SAMPLE). It allows one charge per billing cycle. The second charge is an error.

Solutions.

1. **Ask for the duplicate to be removed.** This is a correction, not a favor.
2. **Ask whether the two earlier fees can be waived** as a courtesy. The utility may say no.
3. **If the correction is refused,** send the billing dispute letter (A3), then file with the state commission, since the utility is investor-owned.

Cost comparison.

Ask	Amount	Likelihood
Remove the duplicate charge	$6.30	Strong: it breaks the tariff's own rule
Waive the two earlier charges	$12.60	Uncertain: at the utility's discretion
Every month the balance stays unpaid	$6.30 more	Ends only when the balance is dealt with (12.8)

The decision and why. They filled in the top of the Provider Call Log (A2) before calling, so they would not freeze up. They asked for the correction first, then the courtesy waiver. The representative removed the duplicate charge. She declined the waiver for now, but said it could be reviewed once they were on a payment arrangement. She offered to talk about one, and about a medical certificate for the equipment. Case D in Chapter 12 follows that conversation.

How they checked the result. They got a reference number and a confirmation email. The next bill showed "Late payment charge reversal −$6.30." They marked it **Verified** in their log.

The lesson: the call they had dreaded took 15 minutes. Preparing a short list of asks made it easier to make.

Tools for this chapter

Tool 10.1, the Billix Electric Bill Analyzer, is the table in 10.1. The five forms this chapter uses are printed once, in Appendix A, because other chapters use them too. Each is also in the companion worksheet pack, where the call log comes as a fillable PDF and a spreadsheet. Replace every item in square brackets with your own details.

- **A2. The Provider Call Log:** for every call or chat with a provider (10.3). Case D used it.
- **A3. The billing dispute letter:** when a call does not fix a wrong charge (10.2). Case F used it.
- **A4. The fee explanation request:** to ask in writing what a charge is for (10.2).
- **A5. The rate or plan review request:** to ask in writing for a cheaper plan or rate (10.4).
- **A6. The service cancellation checklist:** before, during and after you cancel any service. It covers keeping a phone number (8.6) and stopping service when you move (11.5).

A note on cancellation laws. The FTC's 2024 "click-to-cancel" rule was struck down by a federal appeals court on July 8, 2025, before its main requirements took effect. In March 2026 the FTC asked for comments on a possible new rule, but none was in place in October 2026. Some states have their own automatic-renewal laws. California's and Minnesota's exclude services regulated by the FCC or the state utility commission. So for phone, internet, cable and utility service, rely on your contract, the provider's posted process and, if needed, the FCC complaint route. Streaming and other online subscriptions are where these state laws most clearly apply.

Chapter 10 review

What you learned

- A bill is a claim you can check. Eleven fields and a use-per-day figure catch most problems early.
- Billing errors come in seven common kinds, and each has its own evidence.
- A good call follows nine steps and ends with a reference number and written confirmation.
- Regulated utility rates cannot be changed on a call, but a utility can correct errors, change your plan, set up payment arrangements and sometimes waive fees. Prices for competitive services often have room to move.
- Budget billing changes when you pay, not how much. Watch the running balance and the settle-up.
- Autopay brings discounts and fewer late fees, but it can cause overdrafts and pays wrong bills too.
- A demand to pay now by gift card, wire, crypto or a payment app is a scam sign. Check with the number on your bill.
- Each agency handles certain services and problems. Escalate to the right one.
- Deposit, late fee and reconnection rules vary by state. An unpaid bill sent to collections can hurt your credit for years.

What you can do today

- Start a folder for each provider and print or download the Provider Call Log (A2).
- Run the five-minute check on your latest bill.
- Find your state utility commission's contact details and save them.
- Save each provider's official phone number, taken from your bill, in your phone.
- If you use autopay, check the payment dates and that the account has a cushion.

What to check next month

- Compare the new bill with this one, using use per day.
- Confirm that every promised credit appeared, at the right amount.
- If you are on budget billing, check the running balance and your settle-up month.
- Note when any deposit is due back.
- If you have had utility or phone debt problems, request your free credit reports and your NCTUE report.

Billix Quick Check

Answer yes or no:

1. Do you check the service days and use per day on each bill?
2. For your last call to a provider, do you have a reference number?
3. Do you know whether your electric and gas utilities are investor-owned, city-owned or cooperatives?
4. Would you know how to check that a "utility" caller is real?
5. If you use budget billing or autopay, do you know your settle-up month or payment date?
6. Do you know when any deposit you have paid is due back?

Each "no" points to a section of this chapter. Start with the one that could cost you the most.

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11

PART V · PROTECT AND DECIDE

Renters, Homeowners and Moving

Same bills, different levers

Two households can pay the same electric rate and live very different utility lives. A homeowner can add insulation, replace a water heater and pick an internet provider. A renter in the same town may not be allowed to change a showerhead, and may pay for water through a formula set by the landlord.

Neither position is better. Each comes with its own levers. Renters have fewer upgrades to make, but also fewer large bills to pay for. Homeowners can make bigger changes, but they carry the cost of every repair and every replacement.

This chapter sorts out who controls what. It covers what renters can do on their own, what they can ask a landlord for, and how ratio billing works. It covers what homeowners should weigh before spending money. It covers how roommates can split bills fairly, and how anyone can start, move and stop service without paying twice.

Sam, the first-time renter from Chapter 1, finally looks at the water charge on the rent statement. Jordan and Casey, the roommates from Chapter 8, put their shared bills in writing.

Check your lease: Tenant rights differ by state, city and lease. The state rules in this chapter are examples, not a guide to your own rights. Read your lease, and check your state's tenant and utility rules before you act.

START HERE:

- → **Paying a utility share on your rent statement?** Go to 11.2, then Case A.
- → **A shutoff notice on the building over the landlord's debt?** Go to "If the landlord does not pay the utility" in 11.2.
- → **Moving soon?** Go to 11.5 and Tool 11.2.
- → **Sharing bills with housemates?** Go to 11.4.
- → **Not sure a fix is yours to make?** Go to 11.1 and Tool 11.1.

11.1 Why housing status matters

Three questions decide which savings are open to you.

1. **Who owns the equipment?** The furnace, air conditioner, water heater, toilets and windows belong to someone. If they belong to your landlord or your condo association, you cannot replace them yourself.
2. **Who can make changes?** Even small changes, such as swapping a showerhead or adding weatherstripping, may need permission under a lease or HOA rules.
3. **Who chooses and pays the providers?** You may hold the electric account yourself, or the landlord may hold it and bill you a share. Internet, phone and trash may be yours to choose, or may come with the building.

Housing situation	Equipment	Changes	Providers
Renter with your own meters	The landlord's	Small, reversible changes; ask before anything else	You usually hold the electric and gas accounts and choose internet and phone
Renter in a master-metered building	The landlord's	Small, reversible changes; ask before anything else	The landlord holds the utility accounts and may bill you by submeter or formula
Homeowner (single-family)	Yours	Yours, within building codes and permits	Yours to choose where choice exists
Condo or co-op owner	Yours inside the unit; shared systems belong to the association	Inside your unit, within association rules	Some services may come through the association
Shared housing	Depends on whether you rent or own	Agree among yourselves, and with the landlord	Usually one person per account, shared by agreement

The gap between control and payment is where money leaks. A tenant who pays the electric bill cannot fix the drafty windows that drive it up. A landlord who pays the water bill has little reason to fix a tenant's running toilet quickly, and the tenant may not notice it. Knowing who controls each cost tells you who to talk to.

What this means for you: before you spend time or money on a fix, ask whether it is yours to make. If it is not, your best move is a clear, written request to the person who controls it.

11.2 Renters

Renters have fewer levers, but the ones they have are cheap, quick and portable. Most of them leave with you when you move.

The thermostat

If you control your own heating and cooling, the thermostat is your biggest no-cost lever. The US Department of Energy says you can save as much as 10% a year on heating and cooling by setting the thermostat back for 8 hours a day. Turn it down 7 to 10 degrees F in the fall and winter, and up in the spring and summer. A ceiling fan lets you raise the cooling setting about 4 degrees F without feeling warmer, DOE says, but only while someone is in the room.

SAFETY WARNING: Keep your home at a safe temperature, especially for babies, older adults and anyone with a health condition. Never set heat so low that pipes could freeze. If you have a heat pump, do not set it back manually if that turns on its backup electric heat, which costs more. Use a thermostat designed for heat pumps.

If the landlord controls a central system, you cannot change the schedule yourself. You can ask for it to be changed, report rooms that are too hot or too cold, and ask whether radiators or vents can be adjusted.

Portable fixes

These cost little, usually need no permission, and come with you when you leave.

- **LED bulbs.** Replace the bulbs you use most. Keep the originals to put back at move-out.
- **Window coverings.** DOE suggests closing curtains at night in winter and on sunny windows in summer.
- **Removable draft stoppers** for doors and windows. Ask before you attach anything permanent.
- **A clean filter in a window air conditioner,** checked monthly in heavy-use months.
- **Switching things off.** Turn off fans, lights and electronics when no one is using them.

Permitted aerator and showerhead swaps

A new showerhead or faucet aerator is a small change, but it is still a change to the landlord's fixture. Check the lease or ask first. Keep the original and put it back when you move out, unless the landlord wants the new one left.

EPA says WaterSense showerheads use no more than 2.0 gallons a minute, against 2.5 for a standard model. It estimates that one saves an average family about 2,700 gallons a year and more than 330 kWh of water-heating energy.

Who gets the saving depends on who pays. If you pay the electric bill for your own water heater, the energy saving is yours. At the rate card's $0.18 per kWh, 330 kWh is worth about $59 a year for an average family (Illustrative); a one-person home would save less. If the landlord pays for water, or splits it by formula, most of the water saving goes to the building, not to you (see "Master meters, submeters and ratio billing" below).

Reporting leaks

A leak you do not pay for still costs someone, and under ratio billing it can cost you. Report leaks, running toilets and dripping faucets in writing, through the landlord's portal or by email. Note the date. Some states set rules on leaks in submetered buildings. California's include a 21-day leak repair requirement that the landlord must disclose before the lease is signed.

Lease clauses to read

Before you sign, and again when you renew, look for:

- **Which utilities you pay,** and which are included in rent.
- **How shared utilities are billed:** your own meter, a submeter or a formula, and the formula itself.
- **Any billing company or admin fees** added to utility charges.
- **Whether you may make small changes,** such as showerheads, weatherstripping or smart thermostats.
- **Who controls the heating and cooling.**
- **How trash, recycling and internet are handled** (7.8 and 9.1).

Working with management

Written requests work better than hallway conversations. Keep them short and specific: what the problem is, where, when it started and what it is doing. For example: "The toilet in unit 4B has run constantly since March 3. Please repair it." Keep a copy, and follow up if nothing happens within a reasonable time. If you share ratio-billed utilities, say so. Repairs that lower the building's bill help everyone.

Master meters, submeters and ratio billing

In some apartment buildings, one meter serves the whole building. The owner is the utility's customer and passes the cost on to tenants in one of three ways.

Method	How it works	Does your own use matter?
Included in rent	The owner pays and builds the cost into rent	Not directly
Submetering	A meter in each unit measures its use, and the owner bills each tenant from it	Yes. You pay for what your unit uses.
Allocated billing, or RUBS (ratio utility billing system)	The owner splits the building's bill by a formula, such as occupants, square footage or bedrooms	Barely. Your share depends on the formula, not on what you use.

Maryland's law describes RUBS as an allocation method that does not measure each tenant's actual use. The National Consumer Law Center, a consumer advocacy group, points out three risks. The formula covers the whole building and can change at the landlord's discretion. Third-party billing companies may add their own fees. And the state protections that cover utility customers may not apply to RUBS bills.

Find it on your rent statement: a line called RUBS, utility allocation or water and sewer share, listed with any billing company's admin fee (see Case A). It is your share of the building's bill, not your own use. **Record it in** the Annual Household Utility Spending Tracker (B1).

Worked example: how a ratio bill is split (Illustrative)

Sam's building, from the case study later in this chapter. All figures are SAMPLE.

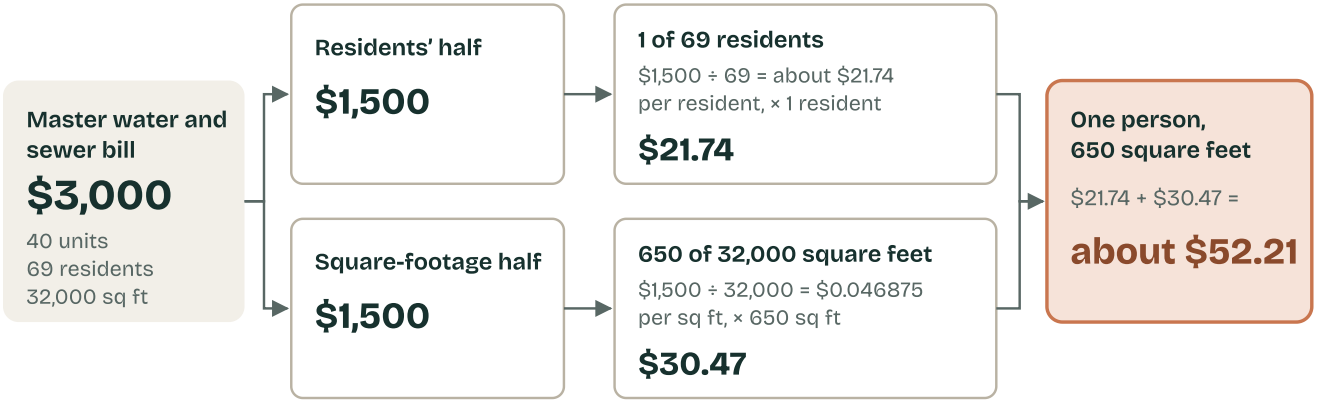


Figure 11.1. How a ratio bill is split: half by residents, half by square footage, then one unit's share. SAMPLE figures from Sam's building (Case A).

- **The building:** 40 units, 69 residents and 32,000 square feet. This month's master water and sewer bill is $3,000.
- **The formula:** half the bill is split by residents and half by square footage.
- **Per resident:** $1,500 ÷ 69 = about $21.74.
- **Per square foot:** $1,500 ÷ 32,000 = $0.046875.
- **A one-person, 650-square-foot unit:** $21.74 + (650 × $0.046875 = $30.47) = about **$52.21.**

Now suppose that tenant saves 500 gallons this month. At the rate card's $14.00 per 1,000 gallons for water and sewer, the building's bill falls by $7.00. But the tenant's share of the building is only about 1.7%, so their own charge falls by about **12 cents.** This is why ratio billing weakens the reason to save water, and why leaks anywhere in the building matter to everyone.

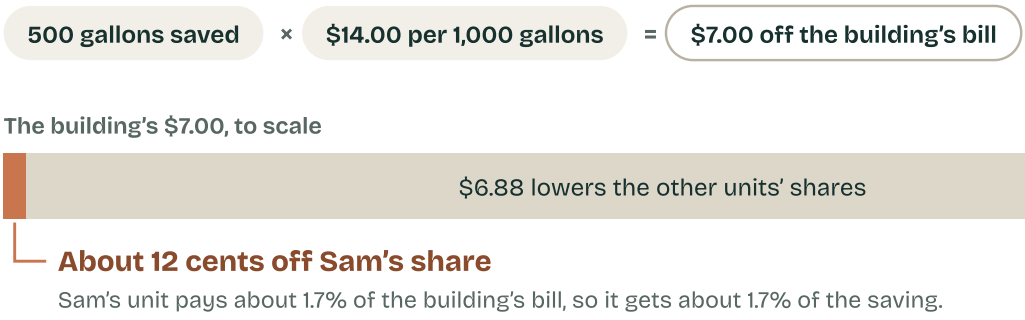


Figure 11.2. The 12-cent problem: saving 500 gallons cuts the building's bill by $7.00, but only about 12 cents of it reaches the tenant's share. Illustrative, using Sam's building (SAMPLE figures).

Is your share right? Five checks

1. **The formula is in your lease,** and the bill follows it. Take out any common-area use, then for each part of the formula: that part of the bill × your count ÷ the building's count. Add any fees.
2. **Your inputs are right:** residents, square feet or bedrooms.
3. **Your share moves with the master bill.**

4. **Common areas come out first,** where required (Texas, below).
5. **Each fee is in the lease** and within any cap (Texas and California, below).

A rough check (Illustrative): people in the unit × about 82 gallons a day (EPA's all-home figure, 5.3) × days × your water-and-sewer price. For the unit above, in a 30-day month, 1 × 82 × 30 = 2,460 gallons, about $34.44 at $14.00 per 1,000 gallons. The master bill also pays for common areas and leaks, so a share above this is not proof of an error. But a share far above it, like Sam's first $73.33 (Case A), is worth questioning.

State rules vary widely

State	Examples of the rules	Applies to
Texas	Only the utility's own water and wastewater charges can be passed through, not its deposits or late fees. On submetered bills, apartment and mobile home park owners may add a service charge of up to 9% of the tenant's water and wastewater charge. It is not allowed for tenants in tax-credit housing or using a Section 8 voucher. The lease must state the percentage. Formula (ratio, RUBS) bills cannot include that charge: only the owner's actual costs pass through, common-area use must be taken out first, and only listed formulas may be used. The lease must show the formula and last year's average, highest and lowest bills. The rules do not limit rent or fees unrelated to water and sewer costs.	Water and wastewater in apartments, condos and mobile home parks
California	Before the lease, the landlord must disclose an estimated monthly bill and the allowed charges. Admin fees are capped. The landlord cannot shut off water for nonpayment.	Submetered water
Maryland	Before leasing, the landlord must give the utilities billed, the last two utility bills, the formula, last year's average bill and any added fees. Without these, a lease clause requiring RUBS payments cannot be enforced.	RUBS for electric, gas, water and sewer

What to ask your landlord for

1. How each utility is billed, and the formula, before you sign.
2. The average monthly bill for similar units over the past year.
3. Copies of the building's master utility bills.
4. For submeters, your readings, where your meter is and how your bill was calculated.
5. Whether a billing company is used, and what it charges.
6. Who handles disputes, and how.

Send the list in writing. To question an admin fee, adapt the fee explanation request (A4).

In Texas, for example, an owner must make billing records kept at the on-site office available within 3 days of a written request.

If the landlord does not pay the utility

Sometimes a landlord falls behind on a master-metered account and the utility threatens to cut off the whole building. Several states protect tenants.

State	Notice	Tenants' options
California	At least 15 days' notice on each unit's door for master-metered buildings	Tenants can become customers without paying the owner's debt, and deduct their payments from rent where rent includes utilities
Pennsylvania	At least 30 days' notice to each unit	Tenants can keep service by paying going forward and deduct those payments from rent. Retaliation by the landlord carries penalties.
New York	15 days' written notice, posted and mailed	Tenants can pay to prevent the shutoff, under rules that keep them from owing more than 2 months of the landlord's arrears
Texas (where the lease says the landlord pays)	When the tenant learns of a notice or a cutoff	The tenant can pay the utility and deduct it from rent, end the lease with 30 days' notice, and recover damages

Call a professional: If your building gets a shutoff notice because of the landlord's debt, contact your local legal aid office or tenant organization quickly. Deadlines are short, and the steps depend on your state.

11.3 Homeowners

Homeowners can make the changes renters cannot. They also pay for every one of them. The skill is spending in the right order: cheap fixes first, big projects only when the numbers and the timing work.

Start with an audit

Begin with your own walk-through, using the Heating and Cooling Checklist (Tool 4.1) and the Household Water Audit (Tool 5.1). Then ask your utility whether it offers a home energy audit. Some offer them free or at low cost (12.5). An auditor can test for air leaks and check insulation and equipment that are hard to judge yourself.

Air sealing, insulation and weatherization

DOE says sealing uncontrolled air leaks can save 10% to 20% on heating and cooling bills. ENERGY STAR says air sealing and insulating attics, floors over crawl spaces and basement rim joists saves an average of 15% on heating and cooling costs. These are usually the best first projects in an older house. Section 4.5 explains how to find leaks. Income-eligible households may qualify for free weatherization through the Weatherization Assistance Program (12.3).

Equipment

The biggest choices are the heating and cooling system and the water heater. The cheapest time to choose a more efficient model is usually when the old one needs replacing anyway, because you pay the extra cost of efficiency, not the whole cost of a new unit. DOE says heat pumps can cut the electricity used for heating by about 50% compared with electric furnaces or baseboard heaters. See 3.4 for fuel choice, 4.10 for equipment and 13.2 for payback.

Call a professional: Gas appliances, furnaces, water heaters, wiring and major plumbing need licensed installers and, in many places, permits. Never work on gas lines yourself.

Plumbing

Old fixtures can be some of the best-paying upgrades in a house. In 5.4, replacing 3.5-gallon toilets paid back in about 2.6 years at the rate card's prices, while replacing 1.6-gallon toilets took about 18. Fix leaks first: they are often the cheapest repair with the biggest saving. Section 5.11 covers safe water-heater temperatures.

Solar

Solar can lower bills, but it is a major purchase with long contracts. Before you sign, know three things.

- **How you would pay.** The FTC explains the options. Buying means you own the system. A lease often runs about 20 years, payments may rise, and ending early can be hard and costly. With a power purchase agreement, you pay for the power at a rate the company sets. With a lease or PPA, the company usually keeps the incentives.
- **The real cost.** Lawrence Berkeley National Laboratory put the median price of a homeowner-owned residential system installed in 2025 at $3.60 per watt before incentives. The median system was 7.7 kilowatts. At that price, it would cost about $27,720 before incentives (Illustrative).
- **How your utility credits extra power.** Some pay the full retail rate for power you send to the grid. Others pay a lower export rate. Ask before you size a system.

Last Verified: The federal Residential Clean Energy Credit (25D) and Energy Efficient Home Improvement Credit (25C) ended after December 31, 2025, under P.L. 119-21. The IRS says 25D does not apply to expenditures made, and 25C to property placed in service, after that date. Checked October 2026. Section 13.5 covers incentives that remain.

Thinking in long-run costs

Compare upgrades by what they cost and save over their life, not by the price tag alone.

- **Simple payback** = the extra cost ÷ the yearly saving.
- **Compare payback with the expected life** of the equipment. A project that pays back in 15 years is a poor deal for equipment expected to last 12.
- **Count only the marginal price** of what you save, as 1.8 explains. Fixed charges stay.

Should I Buy It? (Tool 13.1) does this math for you.

Judging service-line protection plans

Homeowners may get letters offering plans that cover the water or sewer pipe between the street and the house. Whether you need one depends first on who owns that pipe, and that varies by city.

City	Who is responsible	What the city says
New York City	The owner, for the water and sewer service lines out to the city mains	DEP puts typical repairs at $3,000 to $5,000 for a water line and $10,000 to $15,000 for a sewer line, and says most home insurance does not cover them. Its optional plan costs $84.84 a year for water and $143.04 a year for sewer.
Philadelphia	The owner, for the water pipe out to the valve on the main and the sewer lateral to the sewer main	The water department issues a notice when repairs are needed
Washington, D.C.	The owner owns the whole water pipe; DC Water maintains the part in public space	The owner maintains the sewer lateral

Before buying any plan, check:

1. **Who owns the line** at your address, from your utility's own website.
2. **What your homeowner's insurance covers,** and whether an add-on is available.
3. **The plan's terms,** using the FTC's checklist for service contracts: deductibles and per-visit fees, payout caps, exclusions, cancellation fees and the company's complaint record.
4. **Whether your city or utility actually endorses it.** Call them at a number you look up yourself (10.7).

In New York City, both plans together cost $227.88 a year. That is a real cost to weigh against your savings cushion, the age of your pipes and the plan's payout cap.

11.4 Shared housing

Roommates, couples and adult family members share more than rent. Shared bills can cause arguments, and many of them come from things nobody wrote down.

The account holder carries the risk

Most utility accounts are in one person's name. That person owes the whole bill to the provider, whatever the roommates agreed among themselves. If the bill goes unpaid, it is the account holder's credit that suffers (10.9). Where you can, spread the accounts: one person holds electricity, another holds internet. Keep phone lines on separate accounts, as Jordan and Casey did in Chapter 8.

Put it in writing

A one-page household agreement prevents most disputes. Cover:

- **Who holds each account,** and where the bills can be seen.
- **How each bill is split** (see below).
- **When and how to pay the account holder,** at least a few days before the provider's due date.
- **What happens if someone pays late.**
- **Who paid each deposit,** and how refunds are shared.

- **Thermostat ranges and other usage norms** everyone can live with.
- **Guests or partners who stay often,** and whether they add to the share.
- **Moving out:** how much notice, how the final bills are shared, and when accounts are moved or closed.
- **A review date,** such as every six months or each season.

Ways to split a bill

Method	How it works	Works best when	Watch out for
Equal shares	Divide each bill by the number of people	Rooms and habits are similar	Feels unfair if one person uses much more
By room size	Weight shares by bedroom size or rent share	Rooms differ a lot	Does not reflect how much each person uses
Usage adjustment	Split equally, then one person pays extra for a large use only they have	One person runs a space heater, home office or extra freezer	Needs a fair estimate of that use
Fixed and usage parts	Split fixed charges equally and usage by an agreed ratio	People are home at very different times	Takes more effort each month

Worked example: Jordan and Casey's winter split (Illustrative)

Case G, the roommates from Chapter 8. Figures are SAMPLE.

- **The bills:** electricity averages $140 a month in winter, in Jordan's name. Internet is $60 a month, in Casey's name.
- **The difference:** Casey works from home and runs a 1,500-watt space heater about 6 hours each workday.
- **The heater's cost:** 1.5 kW × 6 hours × 22 workdays = 198 kWh a month. At the rate card's $0.18 per kWh, that is **$35.64.**
- **The split:** Casey pays the heater's $35.64. They share the remaining $104.36 equally, $52.18 each.
- **Each month in winter:** Jordan pays $52.18 for electricity and $30 for internet. Casey pays $87.82 for electricity and $30 for internet.
- **In summer,** when the heater is off, they go back to equal shares.

They wrote the rule into their agreement, so neither has to bring it up again each month.

SAFETY WARNING: The US Consumer Product Safety Commission says to plug electric space heaters directly into a wall outlet, never into an extension cord or power strip. Keep them at least three feet from drapes, furniture, bedding and anything else that can burn. Turn a space heater off when you leave the room or go to sleep.

Keep talking

- **Share the bills,** not just the totals. A shared folder or photo of each bill avoids suspicion.
- **Review together once a month** using the five-minute check in 10.1. It takes a few minutes and catches errors early.
- **Agree on changes before you make them,** such as a new internet plan or an energy supplier.

11.5 Starting, moving and stopping service

A move is an easy time to pay twice, pay for someone else's use or lose a deposit. A little planning prevents all three.

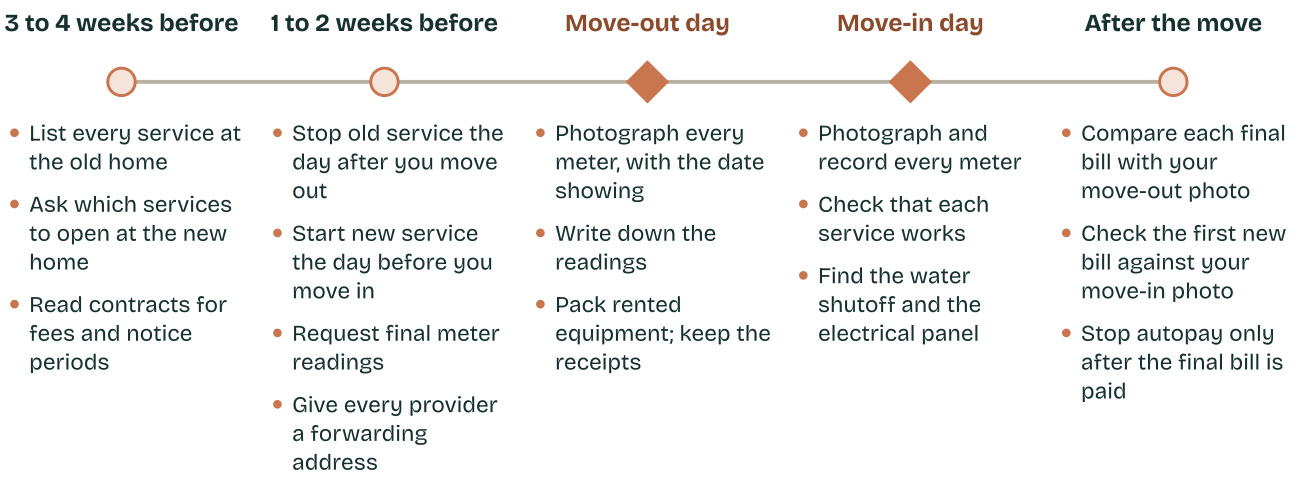


Figure 11.3. Moving timeline: what to do 3 to 4 weeks before, 1 to 2 weeks before, on move-out day, on move-in day and after the move. Tool 11.2 has the full checklist.

Stop and start on the right days

The Missouri Public Service Commission's moving advice is a good general guide. Call every provider and tell them where and when to stop service. Write down the date, the time and the name of the person you spoke with. Calling several weeks ahead is suggested.

Service	At the old home	At the new home	Notes
Electricity and gas	Stop the day after you move out	Start the day before you move in	A short overlap costs little. A gap can leave you without heat or light. If gas service was off, the utility turns it back on and relights pilot lights; do not do it yourself.
Water, sewer and trash	Stop or transfer on move-out day	Start on move-in day, if the account is yours	Renters: ask the landlord which accounts you must open.
A ratio-billed share	Ask in writing how the last month is prorated and billed, and whether it comes out of the deposit	Ask for the formula (11.2)	Settle any corrections before you hand back the keys.
Internet and TV	End or transfer, and return equipment	Book installation early, for the day you arrive	Check what is available at the new address (7.1).
Mobile phone	Update your address	Check coverage at the new address	If you switch carriers, port your number before cancelling (8.6).
Energy supplier, in choice states	Check your contract for what happens when you move	Compare offers at the new address (2.11)	A move may end a supplier contract, or it may not.

The same Missouri guidance carries a warning worth repeating. Do not count on anyone else to take service out of your name. You remain responsible for any account you have not closed.

Final readings and meter photos

- **Ask for a final meter reading** on your move-out day, so your last bill uses an actual reading, not an estimate. Illinois, for example, requires an actual reading when service starts and ends, unless one was taken in the past 60 days. You must give 5 days' notice and access to the meter.
- **Photograph every meter** you can safely see, at move-out and at move-in, with the date showing. Note the readings in your call log.
- **Check that the last occupant's service was closed** with a final reading before you move in, so you do not inherit their use.
- **Give a forwarding address** for every final bill.

Deposits and credit checks at the new address

A new provider may check your credit or ask for a deposit (10.9). Rules vary by state.

- **Ohio:** the utility may start with a quick credit check and may ask for your Social Security number, but it cannot refuse service if you decline. You can also qualify by owning the home, having had similar service in the past 24 months, or using a guarantor.
- **Texas:** retail electric providers accept proof of on-time payment with another provider or utility in the prior 2 years, among other options.
- **New York:** most new residential applicants cannot be required to pay a deposit.

Before you leave, ask your old utility for a letter showing your payment history. Some new providers accept it in place of a deposit.

Getting your old deposit back

Deposits, with any interest owed, are generally applied to the final bill, and any credit left over is refunded. Pennsylvania, for example, requires the balance to be returned within 60 days. Illinois refunds the deposit with the final bill. In Texas, a retail electric provider that stops serving you must transfer or promptly refund the deposit. Note the amount you expect, and follow up if it does not arrive.

Avoid paying twice

- **Overlaps.** Two internet plans running for a month, or a phone line you forgot to cancel, are easy to miss. Use the service cancellation checklist (A6).
- **Gaps.** A late start can mean a reconnection fee, a rush installation charge or a night without heat.
- **Someone else's use.** A missing final reading at your new home can put the last occupant's use on your first bill. Your move-in photo is the evidence.

TOOL 11.1

The Housing Action Matrix

This table sorts the main saving actions in this handbook by who can usually take them. "Usually" matters: your lease, HOA rules, local codes and state law come first.

Key: Yes = usually true. Ask = usually needs landlord or association approval. Pro = needs a licensed professional. Some = depends on the housing.

Action	Renter-friendly	Landlord approval	Homeowner-controlled	Licensed professional	Not available in some housing	Chapter
Thermostat setbacks	Yes, if you control it	Ask, for a central system	Yes	No	Some: central systems	4
LED bulbs	Yes	No	Yes	No	No	2
Window coverings and removable draft stoppers	Yes	Ask before attaching anything permanent	Yes	No	No	4
Permanent caulking and weatherstripping	No	Ask	Yes	No	No	4
Showerhead or aerator swap	Yes, keeping the original	Ask, if the lease limits changes	Yes	No	No	5
Toilet flapper or leak repair	Report it	The landlord's job	Yes	No, for simple repairs	No	5
Toilet replacement	No	Ask	Yes	Sometimes	No	5
Water heater temperature	Only for an in-unit heater you can safely reach; on an electric heater, switch off the breaker first, or ask (5.11)	Ask	Yes	No	Some: shared or central heaters	5
Water heater replacement	No	The landlord's job	Yes	Pro, plus a permit in many places	No	5, 13

Action	Renter-friendly	Landlord approval	Homeowner-controlled	Licensed professional	Not available in some housing	Chapter
Air sealing and insulation	No	Ask	Yes	Sometimes; duct and combustion testing need a pro	Some: condos and shared walls	4
Heating and cooling system replacement	No	The landlord's job	Yes	Pro	Some: central systems	4, 13
Smart thermostat	Sometimes	Ask	Yes	Sometimes, for wiring	Some: central systems	4
Rooftop solar	No	Rarely possible	Yes	Pro	Some: condos, renters, shaded roofs	13
Community solar subscription	Yes, where offered	No	Yes, where offered	No	Some: states without programs	10, 13
Choosing an energy supplier	Yes, if the account is yours	No	Yes	No	Some: master-metered buildings, states without choice	2
Budget billing or a rate plan change	Yes, if the account is yours	No	Yes	No	Some: master-metered buildings	2, 10
Switching internet or phone	Yes	No, unless the building limits providers	Yes	No	Some: buildings with one provider	7, 8
Garbage cart size	Sometimes	The account holder's choice	Yes	No	Some: trash included in rent or taxes	9
Service-line protection plan	No	Not applicable	Yes	No	Some: condos and renters	11
Checking a ratio-billing formula	Yes	Ask for the records	Not applicable	No	Only where RUBS is used	11

Check your lease: If you are not sure whether an action needs permission, ask in writing and keep the reply.

CASE A

Sam checks the water share

Sam is a fictional first-time renter. The building, the billing company and every figure are SAMPLE. The outcome is illustrative, and Sam's state is not named.

Starting situation. In Chapter 1, Sam learned to read the electric bill. But Sam's rent statement has another utility line that Sam has never questioned: a water and sewer share, plus a billing service fee. This month it came to $76.83.

Household facts. One person in a 650-square-foot, one-bedroom apartment. The apartment is all-electric, with a window air conditioner and an in-unit electric water heater. Sam pays the electric bill directly. Water and sewer come through the building's master meter and are split by formula.

Sample rent statement, utility lines (SAMPLE)

Line	Amount
Water and sewer share (ratio billing)	$73.33
Billing service fee	$3.50
Utility charges this month	**$76.83**

Likely source of the extra cost. With ratio billing, Sam's own use barely matters. What matters is the formula and the numbers fed into it.

Diagnostic steps. Sam sent the landlord a written request using the list in section 11.2: the formula, a copy of this month's master bill, and the figures used for Sam's unit. The reply showed:

- **The formula,** from the lease: half the master bill split by residents, half by square footage.

- **The master bill:** $3,000 for a 40-unit building of 32,000 square feet.
- **Sam's inputs:** 650 square feet, and **2 residents.**

Sam lives alone. The second "resident" was Sam's parent, who co-signed the lease but has never lived there. When Sam renewed the lease 8 months ago, the billing company had counted the co-signer as an occupant, which also made the building's total 70 residents instead of 69.

	As billed	Corrected
Residents' half: $1,500 split by residents	2 × ($1,500 ÷ 70) = $42.86	1 × ($1,500 ÷ 69) = $21.74
Square-footage half: $1,500 × (650 ÷ 32,000)	$30.47	$30.47
Sam's water and sewer share	**$73.33**	**$52.21**

Solutions.

1. **Ask for the occupant count to be corrected,** and for a credit for every month it was wrong.
2. **If the landlord refuses,** check the state's rules on ratio billing and lease disclosures, and contact a local tenant organization or legal aid office.
3. **Cut water use anyway, for a different reason.** Under ratio billing, Sam saving 500 gallons would lower Sam's share by about 12 cents. But Sam's water heater is electric and on Sam's own meter. Less hot water means a lower electric bill.
4. **Report building leaks.** A running toilet in the laundry room raises every tenant's share.

Cost comparison (Estimated)

Fix	Effect
Correct the occupant count	$21.12 less each month, or $253.44 a year, while the master bill stays near $3,000
Credit for the 8 months since the lease renewal	8 × $21.12 = $168.96, once, if each of those master bills was near $3,000
A WaterSense showerhead	A small cut in Sam's electric bill from heating less water; the water saving mostly goes to the building
The billing service fee	No change: the lease allows it, and Sam's state rules would decide whether it is capped

The decision and why. Sam emailed the landlord a short letter with the lease page and the statement, asking for the correction and an 8-month credit. The correction was the clear win: it fixed an error, cost nothing and would keep paying every month. With the landlord's written OK, Sam also swapped in a WaterSense showerhead and stored the original to reinstall at move-out.

How Sam checked the result. The billing company changed Sam's unit to 1 resident and applied a $168.96 credit. The next statement listed 1 resident and a share near $52. Sam's share will still move up and down with the building's master bill. So each month Sam checks two things: that the inputs are right, and that the share is in line with the master bill. Sam marked the credit **Verified**.

The lesson: you cannot control a ratio bill by using less, but you can make sure the formula is applied correctly. Ask for the numbers.

TOOL 11.2

The Move-In and Move-Out Utility Checklist

A blank copy is in the companion worksheet pack as a fillable PDF and spreadsheet.

Three to four weeks before the move

- ☐ List every service at the old home: electricity, gas or propane, water, sewer, trash, internet, TV, phone, security and any energy supplier.
- ☐ Find out which services you must open at the new home, and which come with it. Renters: ask the landlord in writing.

- ☐ Check internet providers and plans available at the new address (7.1).
- ☐ Read contracts for early termination fees, notice periods and equipment returns.
- ☐ Ask your old utility for a letter showing your payment history.

One to two weeks before

- ☐ Schedule stop dates at the old home: the day after you move out.
- ☐ Schedule start dates at the new home: the day before you move in.
- ☐ Request final meter readings at the old home, and confirm the new home's readings were closed for the last occupant.
- ☐ Book internet installation for the day you arrive.
- ☐ Give every provider a forwarding address.
- ☐ Log each call: date, time, name and confirmation number.

Move-out day

- ☐ Photograph every meter you can safely see, with the date showing.
- ☐ Write down the readings.
- ☐ Pack rented equipment for return, and keep the receipts or tracking numbers.

Move-in day

- ☐ Photograph and record every meter at the new home.
- ☐ Check that each service you need is working.
- ☐ Find the water shutoff valve and the electrical panel, in case of an emergency.

After the move

Item	Expected	Received	Date	Notes
Final bill, electricity				
Final bill, gas or propane				
Final bill, water and sewer				
Final ratio-billed share, or a deduction from the deposit				
Final bill, internet and TV				
Deposit refunds or credits				
Equipment return confirmed				

- ☐ Compare each final bill with your move-out photo.
- ☐ Check the first bill at the new home against your move-in photo.
- ☐ Stop autopay on closed accounts only after the final bill is paid.
- ☐ Watch the next two bank or card statements for charges from old providers.

Chapter 11 review

What you learned

- Three questions decide your options: who owns the equipment, who can make changes, and who chooses and pays the providers.
- Renters' best levers are cheap and portable: the thermostat, LED bulbs, window coverings, permitted showerhead swaps and prompt leak reports.
- Under ratio billing, your own use barely changes your share. Checking the formula and its inputs can matter far more.
- State rules on submetering, ratio billing and landlord nonpayment vary widely, and some protect tenants strongly.
- Homeowners should audit first, seal and insulate early, replace equipment when it fails, and judge every upgrade by long-run cost.
- Service-line plans depend on who owns the pipe and what your insurance covers.
- Shared bills work best with a written agreement and a clear rule for unequal use.
- A planned move avoids gaps, overlaps, someone else's use and lost deposits.

What you can do today

- Find the utilities section of your lease or HOA documents and read it.
- If you pay a ratio-billed share, ask in writing for the formula and your unit's inputs.
- If you share a home, draft a one-page bill agreement.
- If you own, check who is responsible for your water and sewer lines on your utility's website.

What to check next month

- Compare your ratio-billed share with the building's master bill, if you have it.
- Follow up on any repair request you have made in writing.
- Review your shared-bill split with your housemates.
- If you are moving soon, start the Move-In and Move-Out Utility Checklist (Tool 11.2).

Billix Quick Check

Answer yes or no:

1. Do you know which utilities you pay directly and which come through your rent or HOA?
2. If you pay a ratio-billed share, do you know the formula and the numbers used for your unit?
3. Do you have a written record of your last repair request?
4. If you share a home, is your bill split written down?
5. If you own, do you know who is responsible for the pipe between the street and your house?
6. After your last move, did you get every deposit back?

Each "no" points to a section of this chapter.

Sources and further reading

Last Verified: Sources were checked between October 12 and October 23, 2026. The Texas water rules were rechecked on October 23, 2026. State rules are examples only, and tenant law changes often.

Renters, ratio billing and landlord nonpayment

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PART V · PROTECT AND DECIDE

12 Financial Assistance, Discounts and Hardship Programs

When the bill is more than you can pay

Shutoff notice? Do this first:

1. **Today, call the number on your bill.** Ask the 12.8 questions, starting with how much you must pay, and by when, to avoid disconnection.
2. **For a serious medical need in your home,** ask for the medical certificate form and call the doctor the same day. A certificate may postpone a shutoff (12.8).
3. **Apply for LIHEAP crisis help.** For an eligible household, federal law says some help must come within 48 hours of applying, though it may not cover the whole bill (12.2).
4. **Tell the utility what you applied for,** and ask it to hold the shutoff. Get any hold, its reference number and its end date in writing.
5. **If the utility says no,** contact your state utility commission before the shutoff date, or the appeal process of a city-owned utility or cooperative (10.8).

Protections vary by state and utility. These steps may buy time, but none is sure to stop a shutoff.

Falling behind on a utility bill is common, and it is not a personal failing. A cold month, a job loss, a medical bill or a rate increase can put an essential bill out of reach for anyone.

This chapter is about the help that exists, and how to get it before a past-due bill becomes a shutoff. Help comes from several places: federal programs run by your state, discounts and payment plans from your utility, local charities and state protections that pause shutoffs for vulnerable households.

None of it is automatic. Almost every program has to be asked for, and many have limited money or application windows. The households that get help are usually the ones that ask early.

The Harris household from Chapters 1, 9 and 10 is $420 behind on electricity. In Chapter 10, the utility's representative offered to talk about a payment arrangement and a medical certificate. This chapter follows that conversation.

Last Verified: Program funding, amounts and rules in this chapter were checked in October 2026. Federal funding for 2027 was not settled when this edition was written. Check the official sources listed in each section before you apply.

START HERE:

- → **A shutoff notice arrived?** Do the steps above, then read 12.8 and Case D.
- → **Behind, but no notice yet?** Go to 12.5 for payment plans and discounts, then 12.2 for LIHEAP.
- → **Not sure what you qualify for?** Check the income table in 12.7, then fill in Tool 12.1.
- → **Phone or internet bill too high?** Go to 12.4.
- → **Can't pay the water bill?** Go to 12.6.

12.1 Why assistance exists

Heat in winter, cooling in a heat wave, running water and a working phone are not luxuries. They keep people safe, healthy and connected to work, school and medical care. When a household cannot afford them, the risks go beyond money.

That is why governments, utilities and charities fund help. In preliminary results from its 2024 Residential Energy Consumption Survey, released in March 2026, EIA found that 43.6 million households, about 33%, reported some form of energy insecurity.

Most programs are aimed at households with lower incomes, older adults, people with disabilities, families with young children, and anyone facing a crisis such as a shutoff notice. Each program sets its own rules. A household that does not qualify for one may still qualify for another.

Where help comes from

Type of help	Who runs it	What it does	Section
LIHEAP	States and tribes, with federal funds	Helps pay heating and cooling bills; crisis help to prevent or end a shutoff	12.2
Weatherization Assistance Program	Local agencies, with federal funds	Free efficiency work, such as insulation and air sealing	12.3
Lifeline	The FCC, through phone and internet companies	A monthly discount on phone or internet service	12.4
Utility programs	Your utility, under state rules	Discount rates, payment plans, debt forgiveness, hardship funds, medical and senior protections	12.5
Water assistance	Cities, water utilities and some states	Discounts, payment plans and crisis help on water bills	12.6
Local charities and 211	Nonprofits, often with utility partners	One-time help and referrals	12.7
Shutoff protections	State utility commissions	Pause shutoffs for weather, medical needs and disputes	12.8

What this means for you: if you are behind, ask about all of them. Several can be combined, such as a LIHEAP grant, a utility payment plan and a medical certificate.

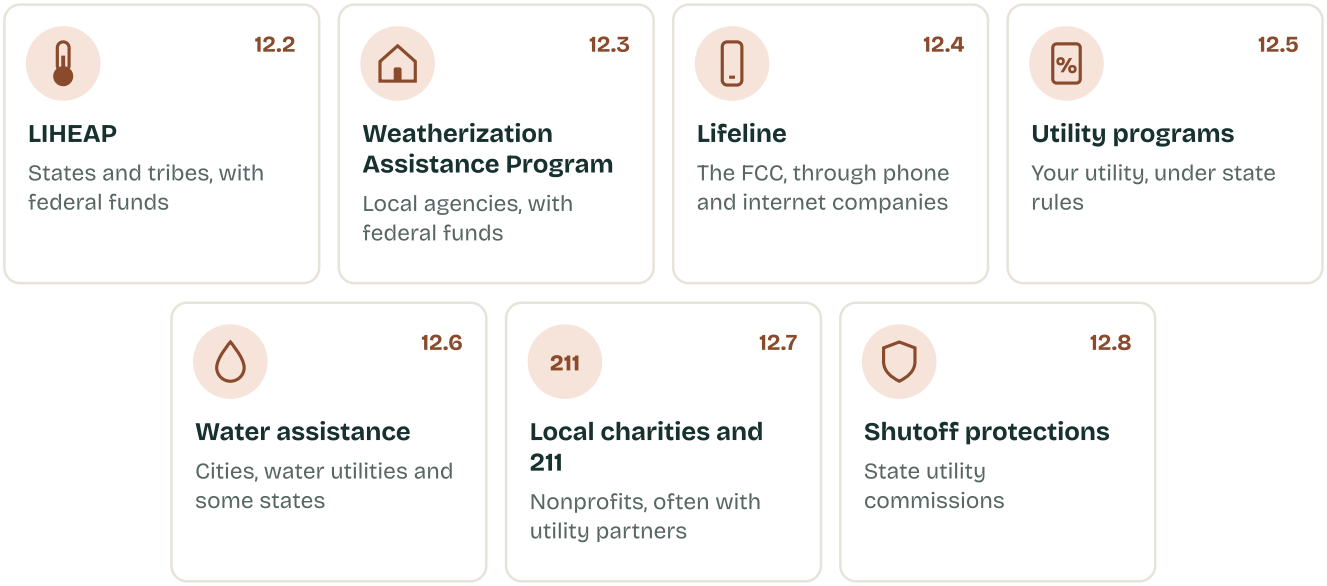


Figure 12.1. Where help comes from: seven sources of help, who runs each and the section that covers it.

12.2 LIHEAP

The Low Income Home Energy Assistance Program, or LIHEAP, is the largest source of help with home energy bills. It is a federal block grant run by the Office of Community Services at the US Department of Health and Human Services. States and tribes receive the money and run their own programs, so the rules, amounts and dates differ from place to place.

What it can pay for

- **Heating and cooling bills,** usually as a one-time or seasonal benefit.
- **Crisis help** when a household faces a shutoff, has run out of fuel or is in danger. Federal law says that for an eligible household, some form of assistance that resolves the crisis must come within 48 hours of applying, or within 18 hours if the situation is life-threatening. That does not always mean the whole bill is paid.

The federal government does not give LIHEAP money directly to individuals. Your state program decides how benefits are paid, often straight to your utility or fuel dealer. If you heat with propane or heating oil, ask whether your state's program covers your fuel.

Who can qualify

Federal law sets the outer limit. States may serve households with income up to the greater of 150% of the federal poverty guidelines or 60% of the state's median income. Households that receive TANF, SSI, SNAP or certain veterans' benefits can qualify without a separate income test. States may set lower limits, but may not turn a household away on income alone if its income is below 110% of the poverty guidelines.

For 150% of the 2026 guidelines by household size, see the income table in 12.7.

How to apply

1. **Find your state's program.** Go to EnergyHelp.us, which leads to the LIHEAP Clearinghouse search tool, or use the state contact list on the HHS Administration for Children and Families website. You can also call the National Energy Assistance Referral line at **1-866-674-6327**, weekdays from 9 a.m. to 7 p.m. Eastern.
2. **Check the dates.** Application dates vary by state. Some programs open in the fall for heating help, and crisis help may follow different dates. Apply as early as you can, because money can run out.
3. **Gather documents.** Programs usually ask for ID, proof of income for everyone in the household and a recent utility bill. Your state office will give you the exact list.
4. **Apply** through your state's online portal or a local community action agency.
5. **Tell your utility you have applied.** In some states, a pending LIHEAP payment commitment, called a pledge, can pause a shutoff. California's large utilities, for example, cannot disconnect while a LIHEAP pledge is pending.

> **SAFETY WARNING:** LIHEAP never charges a fee to apply. The Administration for Children and Families says anyone who asks for a fee should be reported to the HHS fraud hotline at 1-800-447-8477.

Last Verified: October 2026. LIHEAP is open. All federal 2026 money, about $4 billion, has been released to states. On October 9, 2026, HHS released $3.6 billion of 2027 money under a temporary spending law that runs to December 11, 2026. The White House's 2027 budget proposed ending LIHEAP; a House committee bill would fund it. Full-year 2027 funding was not settled when this edition went to press.

12.3 The Weatherization Assistance Program

LIHEAP helps pay this winter's bill. Weatherization helps lower every winter's bill after it. The US Department of Energy's Weatherization Assistance Program, or WAP, pays local agencies, often community action agencies, to make low-income homes more energy efficient, at no cost to the household. It is work on your home, not a cash payment.

What it can involve

The work depends on what the agency finds when it assesses your home. It can include:

- Insulation and air sealing.
- Repair or tuning of the heating system.
- Other efficiency work the agency finds worthwhile.
- Health and safety checks that go with the work.

DOE says the program serves about 32,000 homes a year with DOE funds, and that a national evaluation found households save $372 or more a year on average. More than 7.2 million families have been served since 1976. Your own savings depend on your home and the work done.

Who can qualify

Under federal rules, a home qualifies if the household's income is at or below 200% of the federal poverty guidelines, or if a household member receives SSI or TANF cash assistance. A state can also choose to accept households that qualify for LIHEAP.

For 200% of the 2026 guidelines by household size, see the income table in 12.7.

DOE says priority goes to older adults, people with disabilities, families with children, high energy users and households with a high energy burden. **Renters can apply,** but the agency needs the landlord's written permission before doing the work.

How to apply

1. Check your state's guidelines, using the map of state weatherization offices on DOE's website.
2. Contact your state's weatherization office or the local agency it lists for your county.
3. Provide proof of income, usually for the prior year.
4. If you qualify, expect a waiting list. Ask how long it is, and apply for LIHEAP at the same time for help in the meantime.

Last Verified: October 2026. Weatherization is open and funded. Congress provided $369 million for 2026, and DOE set state allocations on July 9, 2026. The White House proposed ending the program in its 2026 and 2027 budgets; Congress kept it for 2026. Full-year 2027 funding was not settled when this edition went to press.

12.4 Phone and internet help: Lifeline

Lifeline is the federal program that lowers the monthly cost of phone or internet service for eligible households. It is run by the Federal Communications Commission and paid for through the Universal Service Fund, so it does not depend on Congress passing a yearly budget.

What it provides

Service	Monthly discount
Broadband, or broadband bundled with voice	Up to $9.25
Voice-only phone service	Up to $5.25
On qualifying Tribal lands	Up to $34.25, plus up to $100 off a startup fee with participating carriers (Tribal Link Up)

The FCC had planned to phase out voice-only support. A July 2026 order paused that change until December 1, 2027.

Who can qualify

You qualify if your household income is at or below 135% of the federal poverty guidelines. You also qualify if you or someone in your household takes part in SNAP, Medicaid, Federal Public Housing Assistance, SSI, Veterans Pension and Survivors Benefits, or certain Tribal programs. A child's or dependent's participation counts.

For the 2026 limits by household size, see the income table in 12.7.

The rules that trip people up

- **One benefit per household.** A household means the people at an address who share income and expenses. Separate households at the same address can each qualify.
- **Recertify every year,** or the discount stops.
- **Use it.** A free Lifeline plan must be used at least once every 30 days.
- **Fresh documents.** Proof must come from a database check or documents dated within the past 12 months.

How to apply

1. **Go to lifelinesupport.org,** typed in yourself rather than through an ad. It leads to the National Verifier, the FCC's eligibility system, run by the Universal Service Administrative Company (USAC).
2. **Apply online, by mail, or through a participating phone or internet company.** Residents of Texas and Oregon use their state's own process.
3. **Choose a provider** using the "Companies Near Me" tool, and ask it to apply the discount.
4. **For help,** call USAC's Lifeline support line at 1-800-234-9473.

Many internet companies also have their own lower-cost plans for eligible households. These are separate from Lifeline. The largest are listed at the end of this section.

Last Verified: October 2026. Lifeline is open. In February 2026, the FCC proposed changes to Lifeline, including stronger identity checks. No final rule had been adopted when this edition went to press. Check lifelinesupport.org before you apply.

SAFETY WARNING: The Affordable Connectivity Program ended on June 1, 2024, when its funding ran out. It is not an active benefit. The FCC warns that some websites still advertise it or collect personal information for it. Report them at fcc.gov/complaints, and if you gave sensitive information to one, visit IdentityTheft.gov.

Internet companies' own low-cost plans

Several large internet companies run their own lower-priced plans. They are company plans, not Lifeline, and each is offered only where that company provides service. In the table, 200% means 200% of the poverty guidelines (the last column of the 12.7 table).

Plan	Price, October 2026	Speed	Who qualifies, in brief
Xfinity Internet Essentials (Comcast)	$14.95 a month; Internet Essentials Plus $29.95	75 Mbps; Plus 100 Mbps	SNAP, Medicaid, SSI, school lunch, housing assistance and others, or income at or below 200%. Not if you had Xfinity Internet in the past 90 days.
Spectrum Internet Assist (Charter)	$25 a month; an extra discount may bring it to $15. Wi-Fi is $10 a month more.	50 Mbps	Someone in the home gets free or reduced-price school lunch (NSLP or CEP), or SSI
Verizon Forward	Up to $30 a month off a Verizon or Frontier Fiber home internet plan (more in some states), from $20 a month	Depends on the plan	Lifeline, SNAP, Medicaid, WIC, SSI, public housing, a Pell Grant in the past year and others, or income at or below 200%. Can be combined with Lifeline.
Optimum Advantage Internet	$15 a month, plus taxes and fees	100 Mbps	Not listed on Optimum's page; ask
Access from AT&T	Set by address	Set by address	SNAP, Medicaid, SSI, school lunch, LIHEAP, WIC and others, or income below 200%

Verizon's Fios Forward page now leads to Verizon Forward. Charter (Spectrum) and Cox completed their merger in 2026, and Cox's ConnectAssist page now leads to Spectrum Internet Assist. If you are on ConnectAssist, ask Cox what your terms are now.

Compare each plan with the company's regular offers too, including what a promotion costs after it ends (7.6).

Last Verified: October 23, 2026, on each company's own page. These plans change often, and prices, speeds and rules vary by address, so check the company's page before you sign up. The Affordable Connectivity Program ended in June 2024 and is not active.

12.5 Utility company programs

Your utility may be the best single source of help, because it can change what you owe each month, not just pay one bill. Programs, names and rules vary widely by utility and state. Some utilities offer several of the programs below; some offer none. The only way to know is to ask.

Income-based discounts

These lower every bill for as long as you qualify. They come in two main forms: a percentage off the bill, or a payment capped at a share of your income.

Program (state)	How it works	Who qualifies	How to apply
CARE (California)	30% to 35% off electric bills at the larger utilities, 20% at smaller ones, and 20% off gas	Income at or below $43,280 for 1 or 2 people, or $66,000 for 4 (June 2026 to May 2027); or enrolled in Medi-Cal, SNAP, WIC, LIHEAP, SSI or other listed programs	Through your utility
Customer Assistance Programs, or CAP (Pennsylvania)	Payments capped at a share of gross income. For example, 2% to 4% for electric service without electric heat, and 6% to 10% with electric heat, depending on income	Income-eligible customers who are having trouble paying	Call your utility
PIPP Plus (Ohio)	Pay 5% of monthly household income for gas or electric service, or 10% in an all-electric home, with a $10 minimum	Income at or below 175% of the poverty guidelines	energyhelp.ohio.gov or a local agency
Universal Service Fund, or USF (New Jersey)	A monthly credit of up to $200, aiming to keep gas and electric costs near 6% of income	Income at or below 60% of state median income	The LIHEAP application counts: energyassistance.nj.gov or 1-800-510-3102

Worked example: a percentage-of-income plan (Illustrative)

- **Assumed:** an all-electric household with income of $1,800 a month, on a plan like Ohio's that caps payments at 10% of income.
- **Monthly payment:** $1,800 × 10% = **$180.**
- **If its bills average $240 a month,** the plan lowers its payment by $60 a month, or $720 a year.
- **The catch:** these plans have rules. Missing payments or failing to recertify can end the benefit. Read the terms.

Payment arrangements and debt forgiveness

- **Payment arrangements** spread a past-due balance over several months, alongside your new bills. Section 12.8 covers state rules.
- **Arrearage management or debt forgiveness programs** forgive part of an old balance each month you pay your new bill on time. California's large utilities, for example, run an Arrearage Management Plan. Ask whether your utility has one before you set up an ordinary payment plan.

Hardship funds

Some utilities support a fund, often paid for by customer donations and run by a charity such as the Salvation Army. These give one-time help with a bill, usually while money lasts. Ask your utility for the name of its fund, or dial 2-1-1.

Senior and medical programs

- **Senior protections** vary. Michigan's winter protection covers customers 65 and older, and some states limit deposits for older customers (10.9). Ask whether your utility offers a senior discount.
- **Medical rate programs** give extra energy at a lower price to people with certain conditions or life-support equipment. California's Medical Baseline program has no income requirement. At PG&E, it adds about 500 kWh of electricity, 25 therms of gas, or both, each month at the lowest price, and members get extra warning before planned safety shutoffs. Your doctor certifies the need, and you renew every 2 or 4 years depending on the condition.
- **Medical certificates** postpone a shutoff (section 12.8).

Budget billing

Budget billing evens out monthly payments but does not lower the yearly total (10.5). It helps when winter or summer peaks are the problem.

Efficiency programs and free audits

Some utilities fund efficiency programs: free or discounted home energy audits, free LED bulbs or showerheads, rebates on efficient equipment and, for income-eligible customers, free weatherization work. These lower future bills rather than this one. Sections 11.3 and 13.5 explain how to judge them.

Last Verified: The CARE, CAP, PIPP Plus, USF and Medical Baseline terms were checked in October 2026. Programs and limits change, often every year. Check your own utility's current terms.

12.6 Water assistance

Water help works differently from energy help. There is no national water program today. Water assistance is run mostly by local water utilities and cities, and by some states, so you have to find it locally.

Last Verified: October 2026. The federal Low Income Household Water Assistance Program (LIHWAP) has ended. It was a temporary pandemic-era program run by HHS. Its funding expired at the end of fiscal 2023, and the last extensions ran to June 30, 2024. A bill to restore it was introduced in Congress in July 2025, but it had not become law when this edition went to press. Do not apply for LIHWAP, and be wary of any site that offers it.

The need is real. In a December 2024 report to Congress, EPA estimated that 12.1 million to 19.2 million US households lack affordable water service. It also noted that small utilities often cannot afford to run their own assistance programs.

Kinds of local help to ask about

- **Income-based discounts or tiered rates** that lower the bill for qualifying households.
- **Senior or disability discounts.**
- **Payment plans** for past-due balances.
- **Leak adjustments** after a repaired leak (5.10).
- **Crisis or hardship funds,** sometimes run with a local charity.
- **Shutoff protections.** New Jersey's Winter Termination Program, for example, covers water and sewer as well as electricity and gas for eligible customers.

Two local examples

- **Philadelphia's Tiered Assistance Program (TAP)** caps the water bill at 2% to 3% of household income for households at or below 150% of the poverty guidelines, and about 4% for incomes up to 250%. After 24 full payments, debt from before enrollment is cleared, and enrolled customers are protected from shutoff.
- **New York City's Home Water Assistance Program** gives a bill credit to owners of one- to four-family homes who received a HEAP heating benefit or certain senior, disability or veteran property tax exemptions. The credit was $191 on its page in October 2026. It is applied automatically, with no enrollment.

These are examples of what to look for, not programs you can join from elsewhere.

How to research what is open where you live

1. **Find out who bills your water:** a city department, a water district or a private company (5.1).
2. **Search that utility's website** for "customer assistance," "affordability," "discount" or "payment plan."

3. **Call and ask directly.** "Do you have an income-based discount, a payment plan, a senior discount or a crisis fund?"
4. **For a privately owned water company,** check your state utility commission's website. Commissions approve these companies' rates and any low-income programs.
5. **Ask your city or county's human services office, or dial 2-1-1,** for local charities that help with water bills.
6. **If your water is billed through rent** or a ratio formula, ask your landlord, and see 11.2.

Programs open and close, and some have waiting lists or limited funds. Check the program's own page for its current status before you rely on it.

12.7 Finding help safely

Help is spread across agencies, utilities and charities. Finding it takes a method, and so does avoiding the scams that copy real programs.

Where to look

Start here	For	How to reach it
Your utility	Discounts, payment plans, hardship funds, medical and senior programs	The number on your bill, or its official website
Your state LIHEAP office	Bill help and crisis help	EnergyHelp.us, the HHS state contact list, or 1-866-674-6327
Your state weatherization office	Free efficiency work	DOE's map of state weatherization offices
Lifeline	Phone and internet discounts	lifelinesupport.org
211	Local charities, hardship funds and referrals	Dial 2-1-1 or visit 211.org
Your local community action agency	Often LIHEAP and weatherization applications in one place	Through your state office or 211
USA.gov	A federal guide to help with energy bills	usa.gov, or 1-844-872-4681 (1-844-USAGOV1)

Ask your utility one broad question: "What assistance programs, discount rates and payment plans do you offer, and which might I qualify for?" Representatives do not always mention programs unless you ask.

How to tell a real program from a fake

- **Real programs do not charge a fee to apply.** HHS says LIHEAP never does.
- **Real programs do not ask for payment by gift card, wire, cryptocurrency or a payment app** (10.7).
- **Check the web address.** Most government sites end in .gov, but some official ones do not, such as lifelinesupport.org and EnergyHelp.us. Use the addresses in this chapter. Find your state's program from the HHS or DOE contact lists, not from an ad or a message.
- **Scroll past paid ads** in search results. The FTC warns that some ads lead to fake sites and phone numbers.
- **Do not give your Social Security number or bank details** to anyone who contacts you first. Apply through a site or office you found yourself.
- **Be wary of offers of "guaranteed" approval** or of help in exchange for personal information.

Check your eligibility before you apply

Have these facts ready, because almost every program asks for them:

1. **Household size:** everyone who lives with you and shares income and expenses.

2. **Household income:** usually gross income, before taxes, for the last month or the last year. Each program says which.
3. **Benefits you receive,** such as SNAP, SSI, TANF or Medicaid. These can qualify you automatically for some programs.
4. **Your energy type,** and who holds the utility account.
5. **Any crisis:** a shutoff notice, a disconnection, an empty fuel tank, or a medical need for power.

Compare your income with the table below, then check your state's own limits.

Household size	135% (Lifeline, 12.4)	150% (LIHEAP, 12.2)	200% (Weatherization, 12.3)
1 person	$21,546	$23,940	$31,920
2 people	$29,214	$32,460	$43,280
3 people	$36,882	$40,980	$54,640
4 people	$44,550	$49,500	$66,000
Each extra person	Add $7,668	Add $8,520	Add $11,360

Yearly income limits for 2026 in the 48 states and D.C. The 135% column is USAC's Lifeline table, which also covers the territories. The other two columns are Billix calculations from the HHS 2026 guidelines. States and local agencies publish their own tables, and their limits may be higher or lower. Alaska and Hawaii use higher figures.

Check your state's limit States and local agencies publish their own tables, and their limits may be higher or lower. Alaska and Hawaii use higher figures.

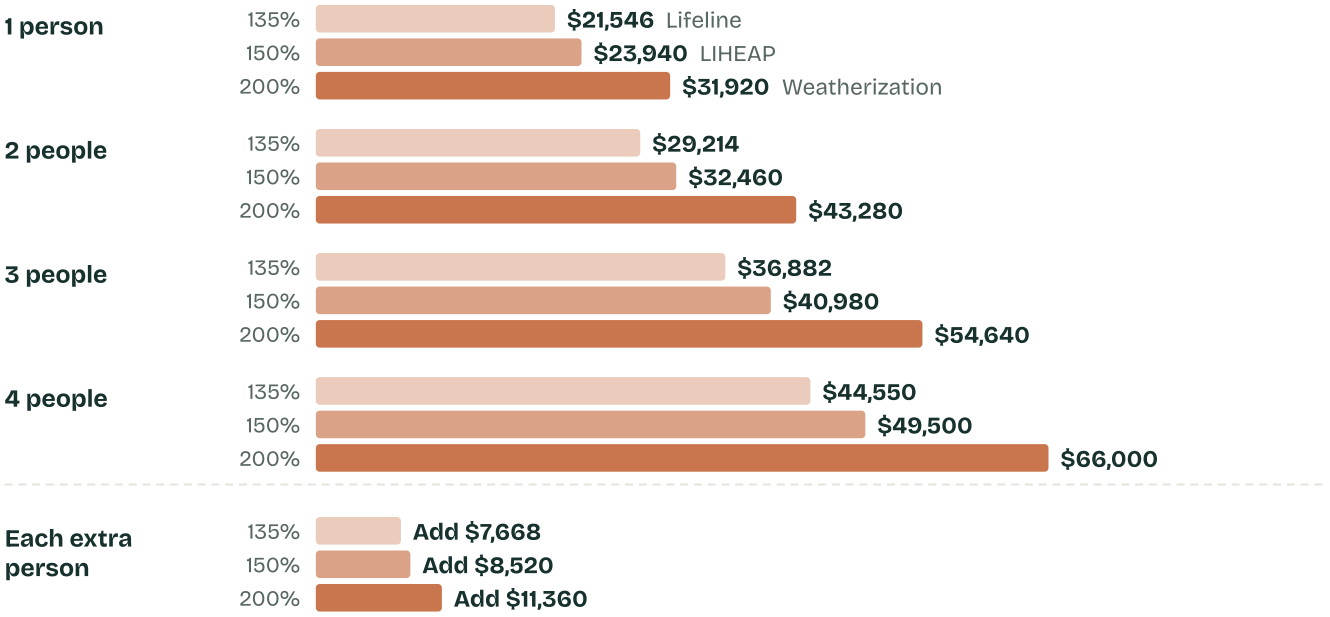


Figure 12.2. Yearly income limits for 2026 by household size, 48 states and D.C.: 135% (Lifeline), 150% (LIHEAP) and 200% (Weatherization) of the poverty guidelines. Check your state's limit. Source: USAC Lifeline table (135%); Billix calculations from the HHS 2026 poverty guidelines (150% and 200%).

Prepare a strong application

Missing paperwork is a common cause of delay. Gather these before you start:

- Photo ID for the applicant.
- Social Security numbers or other ID for household members, if the program asks for them.
- Proof of income for everyone in the household: pay stubs, benefit letters, a Social Security statement or a tax return.

- Your most recent utility bills, and any shutoff notice.
- Proof of address, such as a lease.
- For a medical need, your doctor's name and phone number.

Make copies, and keep the originals. Record each application, its deadline and its status in Tool 12.1, the Utility Assistance Finder Checklist, at the end of this chapter. Then tell your utility what you have applied for, and ask whether it will hold off on collection while you wait.

12.8 If you are facing a shutoff

A shutoff notice is frightening, but it is usually not the last word. In many states there are steps, deadlines and protections between the notice and the shutoff. The most important thing is to act quickly.

Find it on your notice: Look for three lines on the notice itself. The **shutoff date** (or disconnection or termination date) is the first day service may be cut, not the date the notice was printed. The **amount required to avoid disconnection** may be less than your full balance. The **number to call** is the one for payment plans or medical needs. **Record it in** the Provider Call Log (A2), and use the shutoff date as each deadline in Tool 12.1.

Call the provider now

Call the number on your bill today and ask:

1. How much must I pay, and by when, to stop the shutoff?
2. Can I set up a payment arrangement for the rest?
3. Do you have discount rates, hardship funds or debt forgiveness programs?
4. Does anyone in my home qualify for medical, senior or weather protection?
5. If I apply for LIHEAP or another program, will you hold off while it is decided?
6. Will you confirm any hold in writing, with a reference number and the date it ends?

Write down each answer in the Provider Call Log (A2).

How shutoffs usually work

There is no national shutoff rule. Each state's utility commission sets rules for the utilities it regulates. City-owned utilities and cooperatives often set their own policies, and propane and heating oil dealers are usually not covered. The usual sequence is a past-due bill, then a separate written notice, then a waiting period, often a final call or a notice on the door, and a shutoff only on allowed days.

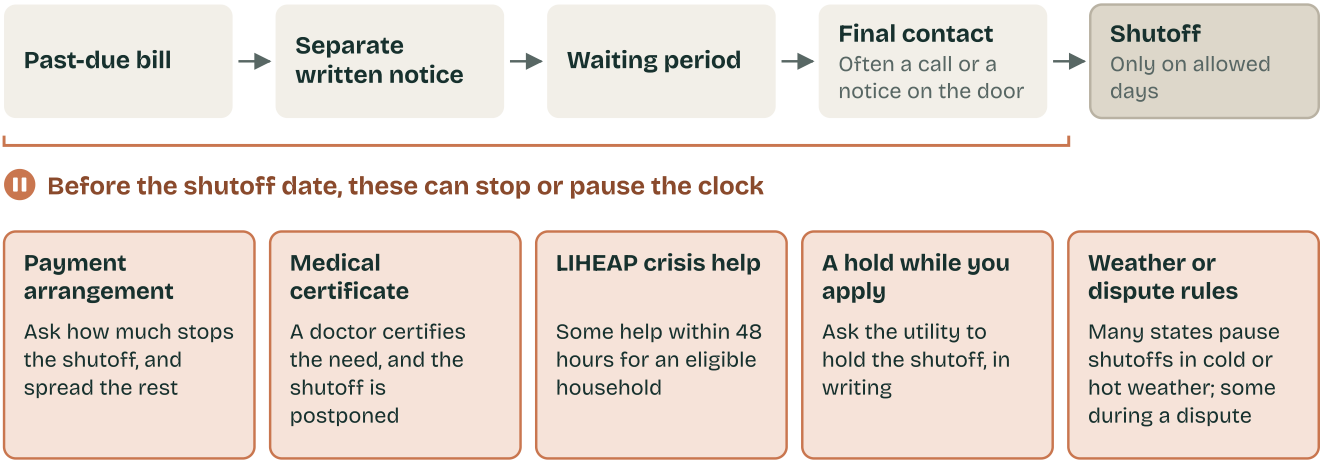


Figure 12.3. From notice to shutoff: the usual sequence, and the help that can stop or pause the clock before the shutoff date. Rules vary by state and utility.

State	Notice before shutoff	When shutoffs are not allowed (examples)
Texas	Shutoff at least 10 days after the notice	Weekends, holidays and the day before either; at or below 32 degrees F; during a heat advisory
Pennsylvania	10 days' written notice, plus a contact attempt at least 3 days before	Friday through Sunday
Illinois	10 days after the notice, plus a call at least 48 hours before	After noon Friday, weekends and holidays; at a forecast of 32 degrees F or below
New York	A final notice at least 20 days after the due date, then 15 days	Outside 8 a.m. to 4 p.m. Monday to Thursday; around holidays
California	10 days, plus a contact attempt at least 24 hours before	Weekends, holidays and when offices are closed
Ohio	At least 14 days, plus personal notice on the day	Extra steps from November 1 to April 15

Weather protections

The LIHEAP Clearinghouse lists 41 states and D.C. with cold-weather shutoff protections, and 20 states and D.C. with hot-weather protections, in tables last updated between September 2025 and July 2026. Some use a temperature, most often no shutoffs at 32 degrees F or below. Others use dates. Who is covered varies widely.

- **Pennsylvania:** December 1 to March 31, for households at or below 250% of the poverty guidelines.
- **New Jersey:** November 15 to March 15, for customers in listed assistance programs or who certify a hardship. You must tell your utility you qualify.
- **Massachusetts:** November 15 to March 15, for heat-related service, for customers who certify financial hardship.
- **Wisconsin:** November 1 to April 15, for service that provides or affects a home's primary heat.
- **Michigan:** November 1 to March 31, for customers 65 or older and lower-income households.

Weather protections pause shutoffs. They do not reduce what you owe, and shutoffs can resume when the protection ends.

Medical protections

A 2021 review by the National Consumer Law Center found that 46 states had some protection against shutoffs for serious illness. Usually, a doctor or other medical professional certifies that a shutoff would harm someone in the home, and the shutoff is postponed.

In most states, the person who is ill does not have to be the one whose name is on the bill. A certificate can protect your service when anyone who lives in your home, including a child, is seriously ill or relies on powered medical equipment. The rules in all four states below work this way, and so do Ohio's and Michigan's. States differ on who can sign the certificate, how long it lasts and how often it can be renewed, and many also require a payment plan or proof that you cannot pay. Check your state's rule, ask your utility for its form, and have the doctor send it before the shutoff date.

State	How long a certificate protects you
New York	30 days, renewable if you show you cannot pay
Pennsylvania	Up to 30 days, with limits on renewals for the same debt
Illinois	60 days, with a 12-installment medical payment arrangement if the certificate arrives before a shutoff
Texas	63 days for a retail electric provider's certified critical care customer on a deferred payment plan; where a non-competitive utility serves the area, 63 days for any seriously ill resident

A certificate postpones a shutoff. It does not cancel the bill. Use the time to set up a payment plan and apply for help.

If anyone in your home relies on powered medical equipment, the FDA advises telling your electric company and your local fire department. Also ask the device maker whether it can run on batteries or a generator.

Payment arrangements

There is no national rule requiring payment plans, but several states require them in certain cases.

- **New York:** an eligible customer must be offered a deferred payment agreement before a shutoff. For customers who show need, payments can be as low as $10 a month with no down payment.
- **California:** the large investor-owned utilities must offer all residential customers a 12-month payment plan before disconnection.
- **Pennsylvania:** the PUC says utilities must offer one payment arrangement and may be required to offer another.
- **Texas:** retail electric providers must offer plans in set situations, such as bills due in July, August and September.

Keep every payment on the plan, and keep paying your new bills. A broken plan can restart the shutoff process.

If a medical or payment request is refused, ask for the reason in writing. Then, before the shutoff date, contact your state utility commission, or use the appeal process of a city-owned utility or cooperative (10.8). A local legal aid office may also be able to help.

If you dispute the bill

In some states, a complaint about the bill pauses a shutoff while it is reviewed, as long as you pay the part you do not dispute.

- **New York:** no shutoff while a complaint is pending with the utility or the commission, or for 15 days after it is resolved.
- **Pennsylvania:** no termination until the dispute or complaint is resolved.
- **Texas:** no disconnection for unpaid disputed charges until the provider or the commission rules.

To dispute a charge, see 10.2 and 10.3. For where to file, see 10.8.

A safety net for notices: third-party notification

Some states let you name a relative, friend or agency to get copies of your past-due and shutoff notices. This helps if you are older, ill, often away or worried about missing mail.

- **New York** requires every utility to offer it. The third party must agree in writing, and you can keep getting the notices too.
- **Pennsylvania** requires utilities to offer it and to tell customers about it at least once a year. The third party is not responsible for your bills.

Ask your utility for its third-party notification form, even if your state does not require one.

Bills you can read

If you are blind or have low vision, or read another language more easily, ask for your bill in a format that works for you. PG&E, for example, offers monthly statements in Braille, large print or audio, and statements in Spanish or Chinese. A bill you cannot read is a notice you can miss.

SAFETY WARNING: IF YOUR POWER OR GAS IS OFF. Never use a generator, grill, camp stove or charcoal indoors or in a garage, even with doors open. CDC and CPSC say to run generators outside, more than 20 feet from windows, doors and vents. Never heat your home with a gas oven or range. Use battery-powered carbon monoxide alarms, and if one sounds, get outside and call 911. An unopened refrigerator keeps food safe for about 4 hours, and a full freezer for about 48. For warming or cooling centers, dial 2-1-1. The safety sidebar on power outages after 2.12 has the full checklist.

CASE D

The Harris household's call

The Harris household is fictional. Their utility, its programs and every figure are SAMPLE, and the outcome is illustrative. Their state is not named, and real programs and amounts vary.

Starting situation. In Chapter 10, the Harrises had a duplicate late fee removed, and the representative offered to discuss a payment arrangement and a medical certificate. The Harrises asked for a few days to think it over. Before they called back, a disconnection notice arrived. It gave them 10 days to pay the $420 past-due balance.

Household facts.

- Two retired adults in a small, paid-off house with gas heat.
- Income of about $30,000 a year from Social Security and a small pension (SAMPLE). They do not receive SNAP or Medicaid.
- One of them uses an oxygen concentrator, powered medical equipment that needs electricity.
- Their electric bill averages about $110 a month (SAMPLE).

Sample notice (SAMPLE)

Line	Amount
Past-due balance	$420.00
Current charges, not yet due	$110.00
Amount required to avoid disconnection	$420.00 within 10 days

Likely source of the problem. Not overuse. A fixed income, rising bills, and no payment plan or assistance in place, because they had never asked.

Diagnostic steps. They used Tool 12.1, the Utility Assistance Finder Checklist, to list every possible source of help, and compared their income with each limit.

Program	Their situation	Likely fit
LIHEAP crisis help	$30,000 is under the 150% limit for 2 people ($32,460); they have a shutoff notice	Strong, if their state uses the 150% limit
Weatherization	Under the 200% limit ($43,280)	Eligible, with a waiting list
Lifeline	Above the 135% limit ($29,214), and no SNAP or Medicaid	Not eligible
Medical certificate	Oxygen concentrator in the home	Strong
Utility payment arrangement	Offered in Chapter 10	Available
Utility income-qualified discount (SAMPLE)	Income under the utility's limit	Eligible

Solutions.

1. **Pay $420 within 10 days.** They could only do it by skipping other essentials. They rule it out.
2. **Medical certificate plus a payment arrangement.** The certificate postpones the shutoff. A 12-month arrangement spreads the $420 at $35 a month.
3. **Add LIHEAP crisis help.** If a crisis grant covers part of the balance, the arrangement shrinks.
4. **Add the utility's discount rate,** which lowers every future bill (SAMPLE: $15 a month).
5. **Join the weatherization waiting list** for longer-term savings, mostly on their gas heating.

Cost comparison (Estimated, monthly, for the next 12 months)

Plan	Current bill	Arrangement	Discount	Monthly total
Arrangement only	$110	$35 ($420 ÷ 12)	$0	$145
Arrangement plus a $300 LIHEAP crisis grant	$110	$10 ($120 ÷ 12)	$0	$120
All of the above, plus the discount rate	$110	$10	−$15	$105

The $216 a year they saved on trash in Chapter 9, about $18 a month, helps cover the arrangement.

The decision and why. They took every step, in this order, because the deadline mattered most:

1. **Day 1:** they called the utility, asked for the medical certificate form and said they would apply for LIHEAP. The utility agreed to hold the disconnection while the certificate was processed, and gave them a reference number for the hold. They also told their local fire department about the oxygen concentrator and asked the equipment supplier about a backup supply, as the FDA advises.
2. **Days 2 and 3:** their doctor sent the certificate, and the shutoff was postponed for 30 days (SAMPLE). They applied for LIHEAP crisis help at their community action agency, which treated it as a crisis and approved a $300 grant (Illustrative), paid to the utility.
3. **Day 5:** they set up a 12-month arrangement for the remaining $120 and enrolled in the discount rate. With the arrangement in place, the utility waived the $12.60 in late fees from Chapter 10. They also joined the weatherization waiting list and asked their gas utility about budget billing (10.5).

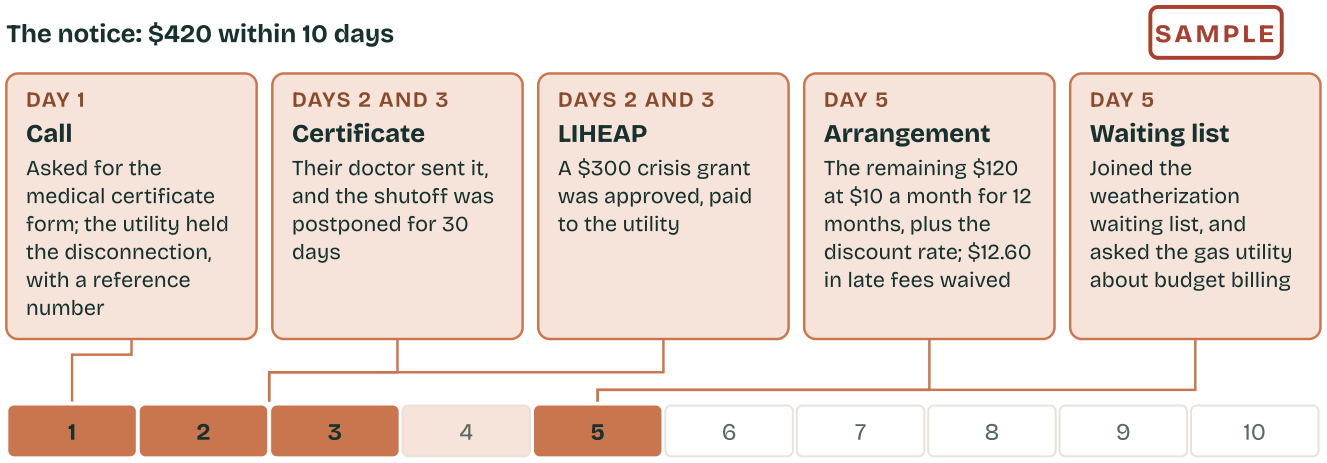


Figure 12.4. Case D: the Harris household's first five days after the shutoff notice. SAMPLE: the household, utility, programs and figures are fictional.

How they check the result. They wrote every program, date and reference number into Tool 12.1. Over the next two bills they confirmed each line: the LIHEAP payment of −$300, the $12.60 late fee reversal, the $10 arrangement installment and the −$15 discount. They noted when the medical certificate expires and when the discount must be renewed. Each line was marked **Verified** as it appeared.

The lesson: the help was there, but it all had to be asked for. Calling within a day of the notice gave them time to use it.

TOOL 12.1

The Billix Utility Assistance Finder Checklist

Use one row per program. A blank copy is in the companion worksheet pack as a fillable PDF and spreadsheet.

Your details: State: _____ Household size: ____ Yearly household income: _____ Benefits received (SNAP, SSI, TANF, Medicaid, other): _____

Part 1: Programs to check

Program	Provider or agency	Program name	Website or phone (found yourself)	Eligibility: likely yes, no or unsure	Documents needed	Deadline	Status
Energy bill help (LIHEAP)	State LIHEAP office or community action agency						
Crisis help (LIHEAP)	Same						
Weatherization	State weatherization office						
Phone or internet discount (Lifeline)	National Verifier or your provider						
Electric utility discount rate	Your electric utility						
Gas utility discount rate	Your gas utility						
Water discount or assistance	Your water utility or city						
Payment arrangement	Each utility you owe						
Arrearage or debt forgiveness program	Your utility						
Hardship fund or charity	Utility partner, 211 or a local nonprofit						
Medical or life-support protection	Your utility, with your doctor						
Senior or disability program	Your utility or state						
Free energy audit or efficiency program	Your utility						

Status options: not started, gathering documents, applied, approved, denied, waitlisted, renewal due.

Part 2: Documents ready

- ☐ Photo ID
- ☐ Social Security numbers or other ID for household members, if required
- ☐ Proof of income for everyone in the household
- ☐ Benefit award letters
- ☐ The last two bills from each utility
- ☐ Any shutoff or disconnection notice
- ☐ Lease or proof of address
- ☐ Doctor's name and phone number, for a medical certificate
- ☐ Landlord's contact details, for weatherization in a rental

Part 3: After you apply

Program	Date applied	Reference number	Decision date	Benefit or terms	Shown on bill dated	Renewal date

Reminder: real programs never charge a fee to apply. Find each website or phone number yourself, from your bill or an official .gov list.

Chapter 12 review

What you learned

- Falling behind is common. Help exists from federal programs, utilities, charities and state protections, but almost all of it must be asked for.
- LIHEAP helps with heating and cooling bills and crises. States run it and set their own limits within a federal ceiling. It never charges a fee.
- Weatherization makes homes cheaper to heat and cool for good, at no cost to eligible households, though waiting lists are common.
- Lifeline takes up to $9.25 a month off internet or bundled service, up to $5.25 off voice-only service, and up to $34.25 on qualifying Tribal lands. The Affordable Connectivity Program has ended.
- Utilities may offer discount rates, payment arrangements, debt forgiveness, hardship funds, and medical and senior protections. Availability varies by utility and state.
- Water assistance is local. The federal LIHWAP program has ended.
- Find programs through official sources, and never pay to apply.
- A shutoff notice starts a clock, not an ending. Weather, medical and dispute protections, plus payment arrangements, can buy time.

What you can do today

- If you are behind, call your utility and ask the six questions in section 12.8.
- Fill in Part 1 of Tool 12.1, the Utility Assistance Finder Checklist.
- Gather the documents in Part 2.
- If anyone in your home relies on powered medical equipment, tell your electric utility and local fire department.

What to check next month

- Follow up on every application in Part 3 of Tool 12.1.
- Confirm that each credit, discount or arrangement appears on your bill.
- Note renewal dates for discounts, certificates and Lifeline.
- Check your state's cold- or hot-weather protection dates before the season starts.

Billix Quick Check

Answer yes or no:

1. Do you know your household's yearly income and size, the two facts most programs ask for?

2. Have you asked your utility what discount rates and payment plans it offers?
3. Do you know where your state's LIHEAP program takes applications?
4. If you are on a payment plan, are you keeping up with both the plan and your new bills?
5. Does your utility know about any medical equipment in your home?
6. Do you know how to tell a real assistance site from a fake one?

Each "no" points to a section of this chapter. If you are facing a shutoff, start today with the box at the top of this chapter, then section 12.8.

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PART V · PROTECT AND DECIDE

13 Smart Upgrades and Return on Investment

Spend less than you save

Every upgrade is sold with a promise: lower bills, a fast payback, a smarter home. Some of those promises are true. Many depend on assumptions that do not fit your home, your habits or your rates. And a few products will never save as much as they cost.

This chapter is about deciding before you buy. It shows how to compare an upgrade's price with what it will really save, how long it will take to earn its cost back, and what can go wrong along the way. It covers nine common upgrades, from a $5 aerator to rooftop solar with batteries, and it is clear about which ones suit which households.

The goal is not to talk you out of every purchase. Some upgrades are among the best money a household can spend. The goal is to make sure the ones you choose pay for themselves.

The Okafor family from Chapter 2 faces a choice many homeowners meet without warning: their old electric water heater is near the end of its life. They use Tool 13.1, the "Should I Buy It?" worksheet, to decide what to replace it with.

Last Verified: Incentive rules, prices and program details in this chapter were checked in October 2026. The federal home energy tax credits ended for work completed after December 31, 2025. Check current IRS rules and your state's programs before you count any incentive.

START HERE:

- → **Holding a quote to replace something?** Go to 13.3, then Tool 13.1.
- → **Old equipment that still works?** See the rules of thumb in 13.3.
- → **A solar pitch, lease or PPA offer?** Go to 13.6 and its "Before you sign" box.
- → **Choosing which upgrade to do first?** Go to 13.4.
- → **Counting on a rebate or tax credit?** Go to 13.5.

13.1 Upfront price versus running cost

The price tag is only the first cost of anything that uses energy or water. Every month you own it, it costs something to run. Over its life, the running cost can be many times the purchase price.

When the cheap option costs more

Worked example: two light bulbs (Illustrative)

- **Assumed:** a lamp used 3 hours a day. A 60-watt incandescent bulb, or an LED that gives the same light with about 9 watts (FTC). Electricity at the rate card's $0.18 per kWh. The LED costs $2 more.
- **Incandescent:** 60 watts × 3 hours × 365 days = 65.7 kWh a year, or **$11.83.**
- **LED:** 9 watts × 3 hours × 365 days = 9.9 kWh a year, or **$1.78.**
- **The difference:** $10.05 a year. The extra $2 is earned back in about 2.4 months.
- **And it lasts longer.** DOE says LEDs last 15 to 25 times longer than incandescent bulbs, so you also buy fewer bulbs.

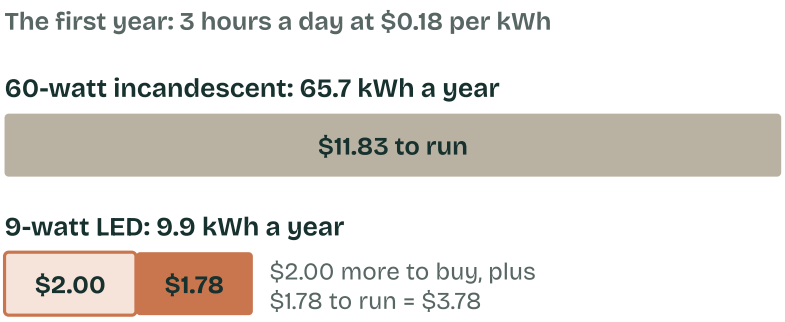


Figure 13.1. Price tag versus running cost: a 60-watt incandescent bulb and a 9-watt LED over the first year, 3 hours a day at the rate card's $0.18 per kWh. Illustrative.

When the expensive option is not a good investment

A higher price does not guarantee a better deal. What an upgrade saves depends on what it replaces and how much you use it. Section 5.4 showed this with toilets. At the rate card's prices, replacing an old 3.5-gallon toilet with an efficient one paid back in about 2.6 years. Replacing a 1.6-gallon toilet with the same model took about 18 years, because there was much less waste to remove.

Three questions before any purchase

1. **What does it replace?** The bigger the waste in the old item, the bigger the saving.
2. **How much will you use it?** A bulb in a closet saves almost nothing. A bulb in the kitchen saves the most.
3. **What is your marginal price?** Savings come only from the part of the bill that changes with use (1.8, Rule 2). Fixed charges stay.

13.2 Simple payback

Simple payback tells you how many years it takes for an upgrade's savings to equal its cost.

Net upfront cost ÷ Annual net savings = Simple payback period in years

- **Net upfront cost** is the price plus installation, minus any incentive you have qualified for.
- **Annual net savings** is the energy and water saved each year, at your marginal prices, minus any new yearly cost, such as maintenance or a subscription.

Worked example: a basic payback (Illustrative)

A $180 upgrade that produces verified annual savings of $60 has a simple payback period of $180 ÷ $60 = **3 years,** before accounting for other costs and uncertainty.

What simple payback leaves out

Simple payback is a good first test, but it ignores several things that can change the answer.

- **The equipment's life.** If the payback is longer than the expected life, the upgrade never pays for itself. Always compare the two.
- **Uncertain savings.** Estimates assume a certain use. If you use less than assumed, savings fall and payback stretches.
- **The cost of borrowing.** Interest adds to the upfront cost.

- **Changing prices.** Higher energy prices shorten payback. Lower ones, or a new rate plan, lengthen it.
- **Double counting.** A saving should be counted once (1.8, Rule 3). A new showerhead and a new water heater both save water-heating energy, but the second one saves less once the first is in.

Worked example: when the assumptions slip (Illustrative)

Scenario	Net upfront cost	Annual net savings	Simple payback
As estimated	$3,000	$300	10 years
Savings 20% lower than estimated	$3,000	$240	12.5 years
Paid with a 5-year loan at 8%	$3,649.75 total paid ($649.75 interest)	$300	About 12.2 years

If the equipment is expected to last 12 years, the first scenario earns about 2 years of net savings. Neither of the other two pays back within 12 years.

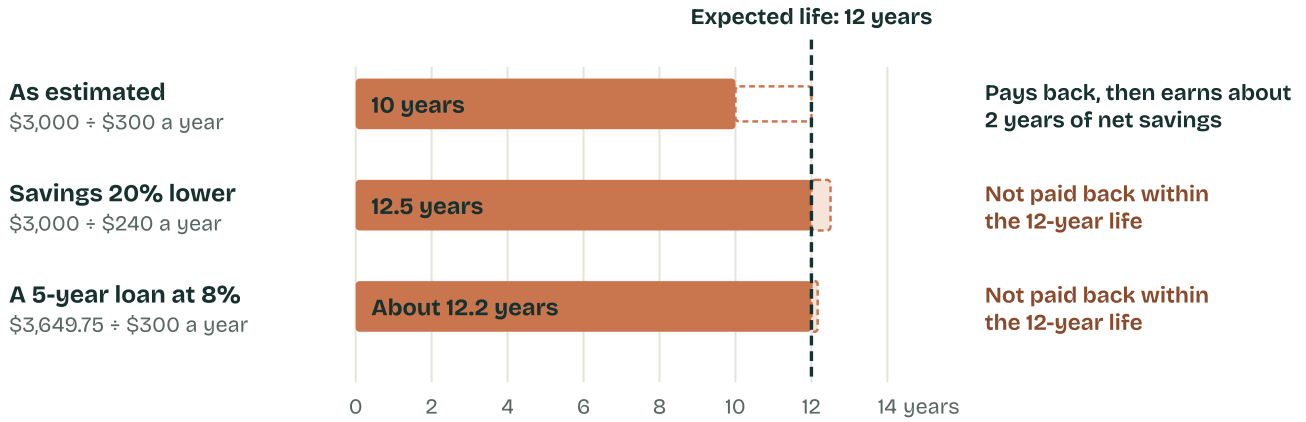


Figure 13.2. Payback against life: a $3,000 upgrade with a 12-year expected life, as estimated, with savings 20% lower, and paid with a 5-year loan at 8%. Illustrative.

What this means for you: prefer upgrades with a payback well inside their expected life. The shorter the payback, the more room there is for things to go wrong.

13.3 Total cost of ownership

Payback asks how fast an upgrade earns back its price. Total cost of ownership asks a bigger question: over the years you will own it, which choice costs the least in total? It is the better tool when you are choosing between two or more options, including keeping what you have.

What to count

Cost	What to include	Where to find it
Purchase price	The item itself	Quotes or store prices
Installation	Labor, permits, electrical or plumbing changes, removal of the old unit	Written quotes from licensed installers
Energy use	Yearly energy at your marginal price	The EnergyGuide label, the maker's rating, or your own measurement
Water use	Yearly water and sewer at your marginal price	WaterSense ratings, or your own measurement
Maintenance	Filters, servicing, cleaning	The owner's manual and the installer
Repairs	Likely repairs over its life	Warranty terms and the installer
Expected useful life	How many years it should last	DOE and ENERGY STAR guidance, the maker
Financing	Interest and fees, if you borrow	The loan's total cost of credit

Cost	What to include	Where to find it
Incentives	Only those you have qualified for	Section 13.5
Disposal	Haul-away or recycling fees for the old item	The installer or your city (9.2)

Total cost of ownership = purchase + installation + financing + disposal − confirmed incentives + (yearly energy + water + maintenance + expected repairs) × years of use

Use the same number of years for every option, usually the expected life of the new equipment.

Repair or replace?

Section 4.10 listed the factors in a repair-or-replace decision, starting with safety: then remaining life, the repair cost against the replacement cost, efficiency, comfort, incentives and your budget. A safety problem settles the question before any money math. Here is how to put money on them.

Turn each option into a cost per year: the upfront cost spread over the years it buys you, plus the yearly running cost.

Worked example: the Okafors' central air conditioner (Illustrative)

- **Repair:** $900, expected to keep their 20-year-old unit running about 3 more summers. Cooling costs about $306 a year (4.5).
- **Replace:** $7,500 installed, expected to last 15 years. Cooling would cost about $214.20 a year, assuming a 30% saving, as 4.10 does.

Option	Upfront cost per year	Running cost per year	Total per year
Repair	$900 ÷ 3 = $300	$306	**$606**
Replace	$7,500 ÷ 15 = $500	$214.20	**$714.20**

On these numbers, repairing costs less per year, as long as the old unit really lasts 3 more summers. That is the weak point. If the repair buys only one summer, it costs $900 + $306 = $1,206 for that year, and you still face the replacement. Reliability and timing matter too. A planned replacement in spring lets you compare quotes. An emergency one in August usually does not.

A note on the 15 years: it is a cautious assumption, shorter than DOE's average of about 21 years for a central air conditioner (about 18 in hot, humid areas). The answer turns on it. Over 18 years, replacing costs about $631 a year, still more than repairing; over 21 years, about $571, less. Replacing comes out ahead once the new unit lasts more than about 19 years, so try a shorter and a longer life before you decide.

Rules of thumb for the decision:

- **A large repair on equipment near the end of its life** usually points to replacement. A small repair on equipment with years left usually points to repair.
- **Still working, but in the last third of its expected life?** Plan before it fails. Get two quotes now: a like-for-like replacement, which goes in Tool 13.1 as Option A, and the efficient option. If the numbers say wait, choose the model and installer now, so a breakdown does not force a rushed like-for-like swap. DOE's average lives are about 21 years for central air (about 18 in hot, humid areas) and for a gas furnace, and 24 to 27 years for a gas boiler. For an air-source heat pump or a tank water heater (heat pump models included), the average is about 15 years, and for a refrigerator about 14. So the last third begins at about 14 years for central air or a furnace, and about 10 for a heat pump or water heater. These are averages; many units fail sooner.
- **Fix the house first.** Sealing and insulating can let a smaller, cheaper system do the job (4.5).

Call a professional: Heating, cooling and water-heating equipment involve gas, refrigerant, high voltage and venting. Get written quotes and diagnosis from licensed technicians.

13.4 Comparing nine upgrades

Not every upgrade suits every household. This table shows what each one can save, who it suits and when it usually does not pay. The savings figures come from federal sources and are national estimates. Your own result depends on your home, your use and your prices.

Cost tiers: No-cost; Low, under $50; Moderate, about $50 to $500; Major, over $500.

Upgrade	Cost tier	What federal sources say it can save	Best for	Often does not pay when	See
LED lighting	Low	Up to 90% less energy than incandescent bulbs. Replacing the five most-used bulbs or fixtures saves about $75 a year (DOE).	Nearly everyone, including renters	Bulbs are rarely used, or already LED	2.6
Faucet aerators	Low	WaterSense bathroom faucets use 1.5 gallons a minute or less (EPA)	Most homes, with landlord approval for renters	Faucets are already low-flow	5.5
Showerheads	Low	A WaterSense showerhead saves an average family about 2,700 gallons and more than 330 kWh a year (EPA)	Homes with 2.5-gallon heads, especially with electric water heating	Heads are already 2.0 gallons a minute, or showers are few	5.6
Smart thermostats	Moderate	About $100 a year for homes with high heating and cooling bills or that sit empty much of the day (ENERGY STAR)	Homes empty for long stretches, owners of compatible systems	You already set back by hand, someone is always home, or the system is not compatible	4.9
More efficient appliances	Moderate to Major	Certified refrigerators are about 9% more efficient than the federal minimum. ENERGY STAR washers use 25% less energy and 33% less water (ENERGY STAR, DOE).	Buyers who need a new appliance anyway	Replacing a working, recent appliance early	2.8, 2.9
Heat pump water heaters	Major	About 70% less electricity than a standard electric water heater (DOE)	Homes with an electric tank and a space with enough room and airflow, plus a drain nearby	You heat water with gas and would need major electrical work, or there is no suitable space	5.11, Case B
HVAC replacement	Major	Heat pumps can cut the electricity used for heating by about 50% compared with electric furnaces or baseboard heat (DOE)	Systems near the end of their life, homes with electric resistance heat	The current system is recent and working, or the house is leaky	4.10
Insulation and air sealing	Moderate to Major	10% to 20% of heating and cooling bills from sealing air leaks (DOE); about 15% on average from sealing and insulating (ENERGY STAR)	Older, leaky homes; owners	The home is already well sealed and insulated	4.5
Solar panels and batteries	Major	Depends on sun, roof, local rates and how the utility credits extra power. The 2025 median system cost $3.60 a watt before incentives (LBNL).	Owners with sunny, unshaded roofs who plan to stay	The roof is shaded or due for replacement, export credits are low, or you may move soon	13.6

How to use the table

1. **Start at the top.** The low-cost upgrades usually pay back fastest and carry the least risk.
2. **Match the upgrade to your waste.** Your own audits (Tools 2.1, 4.1 and 5.1) show where your money goes. A smart thermostat saves little in a home that is always occupied. A showerhead saves more where water heating is electric.
3. **Time the big ones.** Appliances, water heaters and HVAC systems usually make the most sense when the old one is near the end of its life.
4. **Do not stack the same saving twice.** If you seal the house and then buy a heat pump, the heat pump saves on a smaller heating load than before.
5. **Run the numbers** with Tool 13.1, "Should I Buy It?", at the end of this chapter.

13.5 Rebates and incentives

An incentive can turn a slow payback into a fast one. But it only counts if you actually qualify and the money is confirmed. It is easy to count a rebate you later find you cannot get.

Federal tax credits: ended for new work

Last Verified: October 2026. Under P.L. 119-21, signed July 4, 2025, the Energy Efficient Home Improvement Credit (25C) does not apply to property placed in service after December 31, 2025. The Residential Clean Energy Credit (25D), which covered solar, does not apply to expenditures made after that date. The IRS says an expenditure counts as made when installation is complete, so paying in 2025 for a 2026 installation does not qualify. A bill to restore 25C was introduced in 2026 but had not become law.

What this means for you:

- **Work completed in 2026** gets no federal 25C or 25D credit unless Congress changes the law.
- **Work completed in 2025** could still qualify on your 2025 return. Unused 25D credit from earlier years can be carried forward.
- **Be wary of outdated pages.** Some government and installer pages still describe a 30% credit with no end date. The IRS rules control.

State home energy rebates

Two federal programs, run by states, still pay rebates in some places.

- **HOMES** pays for whole-home upgrades based on modeled energy savings. DOE's headline figure is up to $8,000, but state amounts range from about $3,000 to $4,000 for higher incomes to $16,000 or more for low-income households. The National Association of State Energy Officials counted 8 states plus D.C. with HOMES launched as of September 30, 2026.
- **HEEHR,** the High-Efficiency Electric Home Rebates (formerly called HEAR), gives point-of-sale rebates of up to $14,000 for income-qualified households. Since May 29, 2026, it covers only replacing electric equipment with more efficient electric equipment, not switching from gas or oil. It also requires insulation and air sealing before heating and cooling upgrades, unless the home already meets the state's standard.

The National Association of State Energy Officials counted about 13 states plus D.C. with at least one program open as of September 30, 2026. "Launched" does not mean money is still available, because some programs fill up or pause. DOE tells consumers to contact their state energy office.

Utility and local rebates

Some electric and gas utilities offer their own rebates, often for smart thermostats, heat pumps, heat pump water heaters, insulation and home energy audits. Search in this order:

1. **Your utility's website,** under "rebates," "save energy" or "energy efficiency," or the number on your bill.
2. **DSIRE,** the Database of State Incentives for Renewables and Efficiency, run by N.C. State University's Clean Energy Technology Center. Search by ZIP code. Its pages carry advertising for solar quotes from a sponsor; the database is the useful part.
3. **The ENERGY STAR Rebate Finder,** by ZIP code, for specific certified products.
4. **Your state energy office,** for HOMES and HEEHR.
5. **Your local community action agency,** if your income may qualify you for free weatherization (12.3).

Count only what you qualify for

- **Read the rules before you buy.** Some programs require approval, a listed contractor or a specific model before the work starts.
- **Check income limits and location.** Many rebates are for certain incomes, utilities or counties.
- **Wait for confirmation.** Count a rebate only once it is approved or reserved for you, not when a salesperson says you will get it.
- **Subtract it from the upfront cost,** not from the yearly savings.
- **Keep every receipt and form.** Rebates often need proof of purchase, installation and model numbers.

13.6 Solar, batteries and community solar

Rooftop solar can cut electric bills for decades. It is also the largest purchase in this handbook, with the longest contracts and the most assumptions. It is not a guaranteed financial win. Whether it pays depends on your roof, your utility's rules and how long you stay.

Check the roof first

- **Sun and shade.** Panels need direct sun for most of the day. Trees, chimneys and nearby buildings cut output.
- **Age and condition.** If the roof will need replacing in the next several years, do that first. Removing and reinstalling panels later adds cost.
- **Permission.** Check HOA rules and local permits before you sign anything (FTC).

What it costs

Lawrence Berkeley National Laboratory found that the median homeowner-owned residential system installed in 2025 cost $3.60 per watt before incentives: $3.00 for known cash purchases and $4.50 for known loan-financed systems. Loan prices are higher partly because loan fees are rolled into the price. Installers' medians ranged from $2.40 to $6.30 per watt, so get at least three quotes. The median system was 7.7 kilowatts, which at $3.60 a watt is about $27,720 before incentives.

Find it on your quote: Look for the system size, in kilowatts (kW) or watts, and the total price before incentives. Divide to get the price per watt: price ÷ (kW × 1,000). For example, $31,000 for 8 kW is $31,000 ÷ 8,000 = $3.88 a watt (Illustrative). Ask for the cash price and the loan price separately, and compare each with LBNL's median for the same kind: $3.00 for cash and $4.50 for loans. **Record it in** Tool 13.1, Step 2.

Remember that the 30% federal credit (25D) ended for systems installed after December 31, 2025 (section 13.5).

How your utility credits extra power

This rule can matter as much as the price.

- **Net metering** credits power you send to the grid at the full retail rate. Maryland, for example, still nets use and production over the billing period.
- **Net billing** credits exported power at a lower rate. Since April 15, 2023, new solar customers of California's large utilities get credits based on the commission's avoided-cost values, which are usually lower than the price they pay for power. Idaho Power credits exports at seasonal and time-of-day rates set by the commission.
- **Fixed charges stay.** Solar lowers the usage part of your bill, not the customer charge (2.3).

Worked example: the same system under two rules (Illustrative)

- **Assumed:** a 7.7 kW system bought for $27,720 cash, producing 1,300 kWh per kilowatt a year, or 10,010 kWh. Half is used at home, saving the rate card's $0.18 per kWh. Half is exported.
- **Under full net metering:** all 10,010 kWh are worth $0.18. Yearly saving: **$1,801.80.** Simple payback: about **15.4 years.**
- **Under net billing at $0.06 for exports:** 5,005 kWh × $0.18 = $900.90, plus 5,005 kWh × $0.06 = $300.30. Yearly saving: **$1,201.20.** Simple payback: about **23.1 years.**

The production figure, the export rate and the share used at home are all assumptions. Ask your installer for a production estimate for your roof, ask your utility how exports are credited, and check whether those rules are guaranteed for any period.

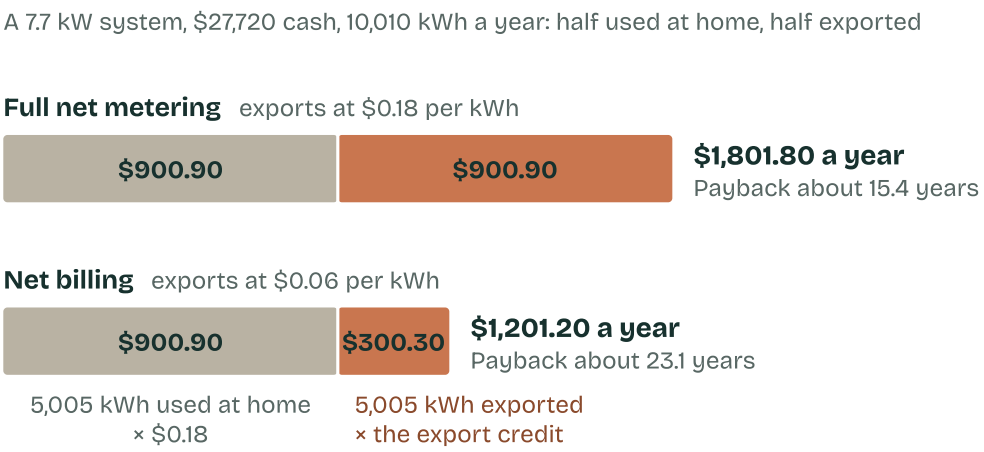


Figure 13.3. Net metering versus net billing: the same 7.7 kW system's yearly saving when exports earn $0.18 or $0.06 per kWh. Illustrative.

Owning, leasing or a power purchase agreement

The FTC explains the three ways to pay.

	Buying (cash or loan)	Lease	Power purchase agreement (PPA)
Who owns the panels	You	The company	The company
What you pay	The price, or loan payments	A monthly lease payment, often for about 20 years, which may rise	A rate per kWh the panels produce, set by the company
Who gets incentives	You, if you qualify	Usually the company	Usually the company
Maintenance	Yours, unless the contract covers it	Usually the company's	Usually the company's
Main risks	Upfront cost; loan fees	Rising payments; ending early can be hard and costly; complications when you sell	Rising rates; complications when you sell

Testing a lease or PPA

A lease or PPA pays only while your bill saving stays above the payments. Test it in three lines:

1. **Yearly bill saving** = kWh used at home × your marginal price, plus kWh exported × the export credit. Fixed charges stay, so leave them out.
2. **Yearly gain** = the bill saving − the year's lease or PPA payments. For a PPA, the payments are its rate × the kWh the panels produce.
3. **Recheck at years 10 and 20,** with the payment raised each year by the contract's increase, called the escalator.

Worked example: a lease on the same system (Illustrative)

- **Assumed:** the 7.7 kW system above, producing 10,010 kWh a year. A lease of $90 a month, or $1,080 a year, rising 3% a year. These are illustrative inputs, not typical market terms. Electricity prices stay at today's level.

Year	Lease payments	Gain under net metering ($1,801.80 saving)	Gain under net billing ($1,201.20 saving)
1	$1,080.00	$721.80	$121.20
10	$1,409.16	$392.64	-$207.96
20	$1,893.79	-$91.99	-$692.59

Under net billing, this lease costs more than it saves from year 5; under net metering, from year 19. Over 20 years the payments total about $29,020, more than the $27,720 cash price. Rising electricity prices would raise the saving, but no one can promise that. A PPA saves only while its rate is below what each kWh is worth to you: the bill saving ÷ the kWh produced, here $0.18 under net metering and $0.12 under net billing.

Batteries

A battery stores solar power for use at night or during an outage. It adds thousands of dollars to the cost. Its money value depends on your rules: a battery earns more where exports are credited at low rates or where time-of-use prices are high in the evening, and less under full net metering. For many households, its main value is backup power. Price it separately, and run its payback on its own.

Maintenance and the long run

Panels need little upkeep, but inverters and batteries may need replacing during the system's life. Rates, export rules and fixed charges can all change over 20 years or more. Treat any 25-year savings estimate as a range, not a promise.

Check the results each month, starting with the first bill:

- **Production.** Compare the kWh the panels produced, which the installer can show you how to read, with the installer's estimate for that month. Weather varies, so look for a pattern, not one low month.
- **Export credits.** On each bill, check the kWh exported, the credit rate and the credits applied. Ask what happens to leftover credits at the end of the year.

What to record: each month's kWh produced, kWh exported and export credits, in Tool 13.1, Step 7.

Community solar

If your roof does not suit panels, or you rent, community solar may let you buy a share of a local project and receive bill credits (10.7). DOE lists at least 20% savings as a best-practice target for programs, not a guarantee. It is available only where state rules allow it.

Before you sign a solar contract:

- **Test any savings claim.** Savings apply only to the part of the bill the system affects. Ask for the claim in kWh a year, and value those kWh at your marginal price and export credit. Divide by your yearly bill to see the share it could really cut. Fixed charges stay whatever you do.
- **Check the price per watt,** with cash and loan prices separately ("What it costs," above).
- **Run the lease or PPA test** for years 1, 10 and 20.
- **Read the contract:** the full length, any yearly payment increase, what happens if you sell the house, whether a lien is placed on your property, and the cancellation terms. The FTC warns that PACE financing creates a property-tax lien that can lead to foreclosure if you fall behind.
- **Do not borrow more than you save.** A loan's interest is part of the cost. If the payments are higher than the savings, the system costs you money every month until the loan is paid (13.2).
- **Do not sign under pressure.** The FTC says "free" or "no cost" solar panel offers are scams, and that the federal government does not install solar systems in homes for free. Anyone who pushes for a same-day signature or demands immediate payment is a red flag. The FTC's Cooling-Off Rule may give you three business days to cancel many home sales (10.7).

13.7 Avoiding costly mistakes

Many expensive mistakes come from deciding too fast. These are the ones to watch for.

1. **Buying before measuring.** Find where your money actually goes first, with Tools 2.1, 4.1 and 5.1. An upgrade aimed at the wrong use saves little.

2. **Buying gadgets that promise drastic savings.** Plug-in "power saver" boxes are the classic case. The National Institute of Standards and Technology found that this type of device does not save a typical household money. The FTC advises being wary of any product that promises drastic energy savings (10.7).
3. **Replacing working equipment too early.** Unless the old item is very wasteful, it usually pays to wait until it is near the end of its life (section 13.3).
4. **Upgrading equipment before the house.** Sealing and insulating first can shrink the heating and cooling load, so a smaller system will do. The state HEEHR rebates now require this order in most cases.
5. **Counting rebates you do not have** (section 13.5).
6. **Counting the same saving twice.** Two upgrades that cut the same energy use do not add up to the sum of their separate estimates (1.8, Rule 3).
7. **Skipping permits and licensed installers.** Poor installation can cut performance and create safety risks. DOE says duct sealing should be tested by a qualified professional to make sure combustion appliances vent safely.

Three more mistakes, trusting a percentage savings claim, borrowing more than you save and signing under pressure, are in the box at the end of 13.6. Those checks apply to any large upgrade, not only solar.

> **SAFETY WARNING:** Never install gas, electrical or venting equipment yourself unless you are licensed to do so. Never block vents or disable safety devices to save energy. CPSC advises carbon monoxide alarms on every level of the home and outside sleeping areas.

CASE B

The Okafors replace their water heater

The Okafors are a fictional family. Their quotes, their utility's rebate and every figure are SAMPLE or Illustrative. Real prices vary widely by area and installer.

Starting situation. During a routine check, a plumber found rust and a slow drip at the base of the Okafors' 13-year-old electric water heater. A leaking tank cannot be repaired. It would need replacing within weeks, before it failed outright.

Household facts.

- Four people in a 1962 house of 1,900 square feet (Chapter 2).
- A 50-gallon electric water heater in an unfinished basement.
- Gas service already in the house for the furnace and range, so the $15 monthly gas customer charge stays whatever they choose (Chapter 3).
- New 2.0-gallon-a-minute showerheads since Chapter 5.
- Estimated hot water use: about 60 gallons a day.
- Prices from the rate card: $0.18 per kWh and $1.30 per therm.

Sample figures: what water heating costs them now (Estimated)

- 60 gallons a day × 365 days = 21,900 gallons a year.
- At 0.186 kWh per gallon for an electric tank (5.6): about 4,073 kWh a year.
- At $0.18 per kWh: about **$733 a year.**

Your numbers: hot gallons a day (from Tool 5.1) × 365 × 0.186 kWh × your marginal price = what a standard electric tank costs you a year. Or use the yearly kWh on the EnergyGuide labels of the models you are quoted. A heat pump model could cut that by about 50% (cautious) to 70% (DOE).

Likely source of the extra cost. Water heating is one of their biggest electric uses, and a standard electric tank turns electricity into heat one for one.

Diagnostic steps.

1. They confirmed with the plumber that the tank, not a valve, was leaking, so repair was not an option.
2. They estimated hot water use from their shower and laundry habits, after the new showerheads.
3. An installer checked that the basement had enough space and airflow for a heat pump water heater, and a drain nearby.
4. They checked incentives. Their utility offers a $400 rebate on qualifying heat pump water heaters to any residential customer (SAMPLE). The federal 25C credit ended for 2026 installations. No state HEEHR program was open in their state.

Solutions. Three written quotes, all including removal of the old tank:

1. **A standard 50-gallon electric tank:** $1,600 installed (Illustrative).
2. **A 50-gallon heat pump water heater:** $3,600 installed, less the $400 rebate, for $3,200 (Illustrative). DOE says these use about 70% less electricity than a standard electric model. The Okafors also ran a cautious case at 50%.
3. **A gas tank water heater:** $2,800 installed, including a new gas line and venting (Illustrative). At 0.0097 therms per gallon (5.6), about 212 therms a year.

Cost comparison (Estimated, over 12 years, a cautious span: DOE's average life for storage water heaters, heat pump models included, is about 15 years)

Option	Net upfront cost	Yearly energy	Simple payback of the extra cost, against the standard tank	12-year total cost
Standard electric tank	$1,600	$733	Not applicable	$10,399
Heat pump water heater, 70% saving	$3,200	$220	$1,600 ÷ $513 = 3.1 years	$5,840
Heat pump water heater, cautious 50% saving	$3,200	$367	$1,600 ÷ $366 = 4.4 years	$7,599
Gas tank	$2,800	$276	$1,200 ÷ $457 = 2.6 years	$6,114

Totals use unrounded yearly costs, so they can differ by a few dollars from the rounded figures shown.

Both efficient options beat a like-for-like replacement by thousands of dollars over 12 years. Between them, the result is close. The heat pump wins if it cuts water-heating electricity by about 67% or more, close to DOE's 70%. Below that, the gas tank has the lower 12-year cost at today's prices.

The decision and why. They chose the heat pump water heater. On DOE's estimate it has the lowest total cost, and even in the cautious case it saves $2,800 over the standard tank. It needs no new gas line or venting, and it does not add another fuel-burning appliance to the house. They accepted that the gas option might cost slightly less if the heat pump underperforms.

How they check the result. They saved their utility's monthly usage data for the year before the change. At a 70% saving, their electricity use should fall by about 2,851 kWh a year, or about 238 kWh a month. At 50%, the fall would be about 2,037 kWh. They will compare use per day for the same months, before and after, and mark the saving **Verified** only after 12 months of bills. The $400 rebate counts once the check arrives.

The lesson: the time to decide on a water heater is before the old one floods the basement. A planned replacement gave them three quotes and a real choice.

TOOL 13.1

Should I Buy It?

This worksheet combines the Home Upgrade Payback Calculator with a comparison against doing nothing. A blank copy is in the companion worksheet pack; the spreadsheet does the math.

Step 1: Describe the decision

Item	Your answer
What you are thinking of buying	
What it replaces, if anything	
Why now: failure, high bills, a rebate deadline or a sales pitch	
Do you own the home, or need permission? (11.2)	

If the answer to "why now" is a sales pitch, slow down. Get at least two more quotes before you go on.

Step 2: Price each option, including doing nothing

	Option A: do nothing, repair or like-for-like	Option B	Option C
Purchase price			
Installation and permits			
Disposal of the old item			
Financing cost (total interest and fees)			
Confirmed incentives (approved or reserved only)			
Net upfront cost			
Yearly energy cost, at your marginal price			
Yearly water and sewer cost			
Yearly maintenance and expected repairs			
Yearly running cost			
Expected useful life (years)			

For a solar lease or PPA: enter $0 as the purchase price, plus any upfront fee, and add the year's lease or PPA payments to the yearly running cost. Payments rise, so fill in the column again for years 10 and 20 (13.6).

Step 3: Work out the savings

- **Yearly saving** = Option A's yearly running cost − the option's yearly running cost.
- Work out two versions: an **expected** saving, and a **cautious** one about half to two-thirds as large.
- Count each saving once. If another upgrade already cut the same use, start from the lower figure.

Step 4: Payback and total cost

	Option B	Option C
Extra upfront cost compared with Option A		
Simple payback, expected (extra cost ÷ yearly saving)		
Simple payback, cautious		
Is payback well inside the expected life?		
Total cost over the same number of years, for A, B and C		

Step 5: The questions numbers miss

- ☐ Can I pay for it without risking essential bills or debt I cannot manage?
- ☐ Will I stay long enough to collect the savings?
- ☐ Does it need a licensed installer, a permit or landlord approval?
- ☐ Does it change comfort, noise, safety or reliability?
- ☐ Are the contract terms clear, with no surprise increases or liens?
- ☐ Have I seen the rebate rules in writing, and do I qualify?

Step 6: Decide

Decision	Reason
Buy Option ___ / Wait / Do nothing	

A good decision to buy usually has three things: a cautious payback well inside the item's life, a total cost lower than doing nothing, and no answer in Step 5 that worries you.

Step 7: Verify after a year

Item	Expected	Actual	Notes
Use per day before and after, same months (1.6)			
Yearly saving at your marginal price			
Solar only: kWh produced and exported each month, and export credits (13.6)			
Rebate received			
Maintenance or repairs so far			

Mark the saving **Verified** only when 12 months of bills support it.

Chapter 13 review

What you learned

- Running costs can outweigh the price tag. A cheap product can cost more over its life, and an expensive one is not automatically a good investment.
- Simple payback = net upfront cost ÷ annual net savings. Compare it with the equipment's life, and test it with cautious savings.
- Total cost of ownership compares options, including doing nothing, over the same number of years.
- Repair or replace comes down to cost per year and how long the repair really buys you.
- Low-cost upgrades usually pay back fastest. Big upgrades make the most sense at the end of the old item's life.
- The federal 25C and 25D credits ended for work completed after December 31, 2025. Count state, local and utility incentives only once you qualify and the money is confirmed.
- Solar depends on the roof, the price, how your utility credits exports and how long you stay. It is not a guaranteed win.

What you can do today

- List your major equipment and its age: water heater, furnace or heat pump, air conditioner, refrigerator.
- Check your utility's rebate page and the ENERGY STAR Rebate Finder for your ZIP code.
- Swap the five most-used bulbs in your home for LEDs, if they are not already.

What to check next month

- For any equipment over 10 years old, get a replacement quote before it fails.
- Run Tool 13.1, "Should I Buy It?", for the next upgrade you are considering.
- If a solar salesperson has contacted you, ask your utility how it credits exported power before you talk again.

Billix Quick Check

Answer yes or no:

1. Do you know the age of your water heater and heating and cooling equipment?
2. For your last major purchase, did you compare running costs as well as prices?
3. Do you know whether your utility offers rebates?
4. Do you know that the federal home energy tax credits ended for work completed after 2025?
5. If you were offered a loan or lease for an upgrade, would you check the total cost of the contract?
6. Do you measure a saving with a year of bills before you count it as real?

Each "no" points to a section of this chapter.

Sources and further reading

Last Verified: Sources were checked between October 12 and October 23, 2026. Incentive programs and rules change often; check the official pages before you buy.

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PART VI

Your Plan

14 The Billix 30-Day and 90-Day Savings Plans

PART VI · YOUR PLAN

14 The Billix 30-Day and 90-Day Savings Plans

From reading to results

The first thirteen chapters explained how each bill works and where money leaks. This chapter turns all of that into a plan you can follow on your own, one week at a time.

The plan has two parts.

- **The 30-day plan** gets you organized and acting. Each week has one theme, a short list of tasks and a clear result.
- **The 90-day plan** checks whether your changes worked. Because bills arrive weeks after the use they measure, most results only show up in months two and three.

You do not need to do everything. A household that finishes Week 1 and makes two good changes in Week 4 is ahead of one that reads every chapter and changes nothing. Pick what fits your home, your time and your budget.

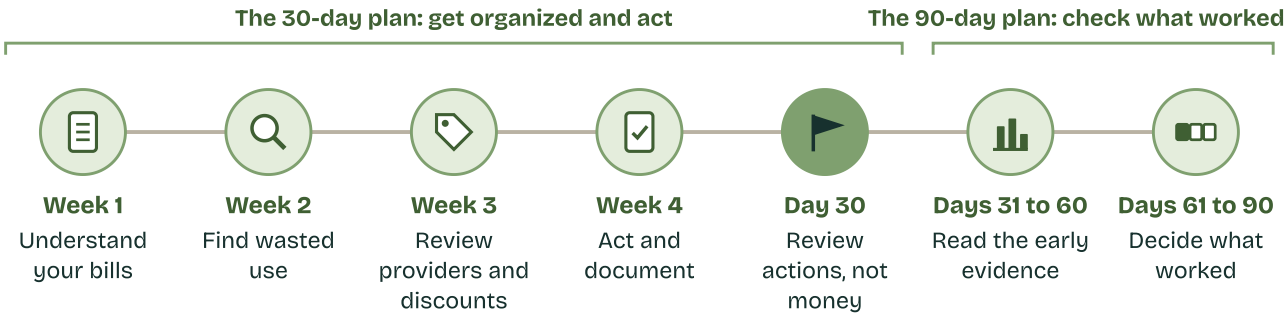


Figure 14.1. The 90-day path: one theme a week for four weeks, a review of actions at day 30, then two months of checking what worked.

Dana, the remote worker from Chapters 4 and 7, shows what a completed plan and scorecard look like.

What you need

- About 1 to 3 hours a week for four weeks, then about 30 minutes a month.
- Your last 12 months of bills, or online access to your accounts.
- A folder, paper or digital, for bills, notes and confirmations.
- The tools: the 30-Day Action Plan (Tool 14.1), the Household Savings Priority Matrix (B3), the 90-Day Household Savings Scorecard (B4) and the 12-Month Bill Calendar (Tool 14.2). All four are in the companion worksheet pack, and they work on paper with a calculator. Week 1 also uses the Spending Tracker (B1) and the 15-Minute Bill Audit record sheet (B2).

START HERE:

- → **Starting this weekend?** Go to Week 1 in 14.2, and do its five steps in order.
- → **Made changes, and the first new bills are in?** Go to "Days 31 to 60" in 14.3.
- → **At day 90, and unsure what you can claim?** Go to "Days 61 to 90" in 14.3, then 14.4.
- → **Want to see a finished scorecard?** Go to Case C.

14.1 Before you start: the three labels

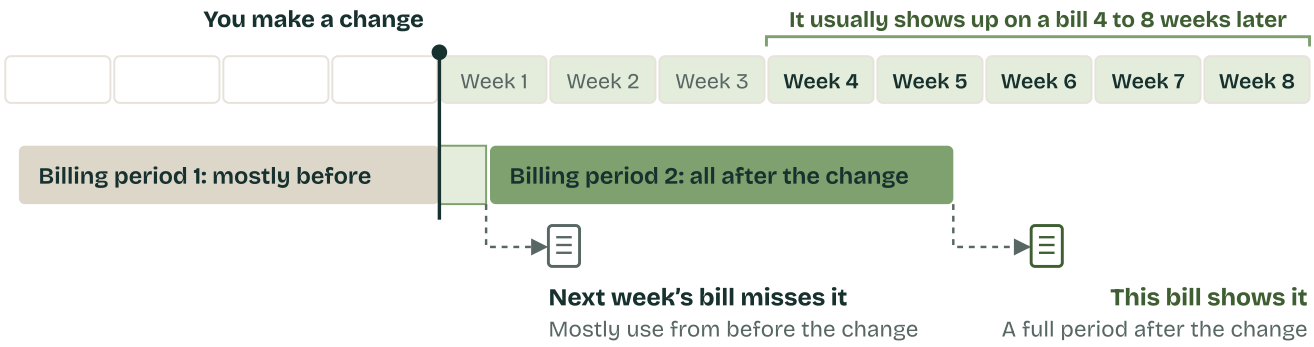
Throughout this handbook, savings carry one of three labels. Keep using them in your own plan. They are what make your results believable, to you and to anyone you share a household with.

Label	What it means	Example
Illustrative	A made-up example to show the method	"A 9-watt LED saves about $10 a year over a 60-watt bulb used 3 hours a day."
Estimated	Your own projection, based on your data, before the bills prove it	"Our new showerheads should save about 15 gallons a day."
Verified	Confirmed by your own bills over several billing periods, after adjusting for billing days, weather and price, with no other big change in the home. For a flat-price service, one full bill at the new price is enough	"Our water use per day fell 14% over two bills compared with the same months last year."

The golden rule: never add estimated savings to verified ones. Keep them in separate columns on your scorecard. An estimate becomes verified only when the bills support it.

Why results take time

A change you make today usually shows up on a bill four to eight weeks from now. The bill you receive next week mostly covers use from before your change. Water bills that arrive every two months take even longer. That is why the plan reviews actions at day 30 and money at days 60 and 90.



Water bills that arrive every two months take even longer.

Figure 14.2. Why results lag: a change made partway through a billing cycle mostly misses the next bill and shows up on the one after, usually 4 to 8 weeks later. Illustrative.

14.2 The 30-day plan

Week 1: Understand your bills

Goal: know what you pay, for what, and what "normal" looks like. **Time:** about 2 to 3 hours.

Sidebar: three terms this plan uses.

- **Baseline:** what you used and paid in the same billing months last year, before the change. Put use on a per-day basis, because billing periods differ (1.4, 1.6).
- **Marginal price:** what one more unit costs you, or what one fewer saves. It is usually not your bill divided by your use, because that average includes fixed charges. Add up every per-unit rate on your bill, or find it on your rate schedule (see 1.8, Rule 2).
- **Degree days:** a count of how far each day's average temperature falls below or rises above 65 degrees F (1.6). Compare heating and cooling per degree day, so a mild month does not pass for a saving. Some bills print them; for other sources, see "Getting your degree days" in 1.8, Rule 4.

Do these steps in order. Each one names the form its results go on.

1. **Gather your bills.** Collect 12 months of bills for every service: electricity, gas, propane or oil, water and sewer, trash, internet and TV, and phones. Download usage history from online accounts where you can. Then fill in the opening assessment in the Introduction.
2. **Enter the monthly totals.** Write each bill's total on the Spending Tracker (B1), in the month it arrived.
3. **Audit each metered service once.** Run the 15-Minute Bill Audit (B2) on the latest electricity, gas and water bills, against the same month last year. That is one audit per service, not one per bill. It gives you use per day, the price per unit and the fixed charges (1.3, 1.6). Last year's column is your baseline. For heating and cooling, note the degree days as well.
4. **Fill in B1's provider list.** For each service, record the provider, plan or rate schedule, fixed charge, marginal price (see 1.8, Rule 2), and any promotion, contract or price-lock end. Note autopay (10.6) and any deposit (10.9). Put every end date and deposit refund date in the 12-Month Bill Calendar (Tool 14.2) now.
5. **Act on errors now.** If the audit turns up one of the seven kinds of billing error (10.2), do not wait until Week 4. Follow the nine steps in 10.3 now, so the clock on any credit starts sooner.

By the end of Week 1, you have: a baseline for every bill, from the same months last year; your marginal prices; a provider list, with its key dates in your calendar; and any charges to question.

Week 2: Find wasted use

Goal: find where energy, water and money are leaking, and put a rough price on each leak. **Time:** about 2 to 3 hours, spread over the week.

Task	How to do it	See
Complete an electricity audit	Use the Home Electricity Consumption Audit to list your biggest users. Measure plug-in devices with a usage meter where you can. Check what stays on when "off."	Tool 2.1; 2.7
Check visible water leaks	Run the Water Leak Inspection Checklist: a meter test with no water running, a toilet dye test, and a look at faucets, the water heater and outdoor taps.	Tool 5.2
Review heating and cooling habits	Use the Heating and Cooling Checklist: thermostat schedule, filter, open vents and drafts.	Tool 4.1
Identify unneeded recurring services	List every subscription and add-on: streaming, device insurance, extra lines, rented equipment, unused trash add-ons and protection plans.	Chapters 7 to 11
Measure where possible	Use your meter, your utility's hourly or daily usage data, a plug-in meter and degree days. A measurement beats a guess.	1.5, 1.8, 2.5

By the end of Week 2, you have: a list of wasted use and unneeded services, each with an estimated monthly cost.

SAFETY WARNING: Audit safely. Read meters from the outside and never open, tamper with or bypass a meter, panel or gas line. Never block vents or disable safety devices. Keep your home at a safe temperature for everyone in it. If you smell gas, leave right away and call 911 or the gas utility's emergency line from well away from the building (3.7).

Week 3: Review providers and discounts

Goal: make sure every plan, rate and discount fits your household. **Time:** about 2 to 3 hours, mostly on the phone or online.

Task	How to do it	See
Compare internet offers	Work out the speed you need, read the broadband labels and compare full-term costs with the Internet Plan Comparison Worksheet. Use the Provider Negotiation Script before you switch.	7.4, 7.5; Tool 7.1; A1
Review mobile plans	Compare 24- and 36-month costs, including phones, credits, insurance and fees, with the Mobile Plan Comparison Worksheet.	8.5; Tool 8.1
Ask providers about discounts	Ask each provider which discounts you may qualify for: autopay or paperless, income-based, senior or medical programs.	10.4, 12.5
Research utility assistance	If money is tight, fill in the Utility Assistance Finder Checklist. Check LIHEAP, weatherization, Lifeline and your utility's programs.	Tool 12.1; 12.2 to 12.5
Review rate-plan suitability	Ask each utility, in writing, whether a different rate plan would cost less for your use. Consider time-of-use rates, budget billing, energy supplier offers in choice states and trash cart size.	A5; 2.10, 2.11, 9.3, 10.5

By the end of Week 3, you have: a list of better plans, discounts and programs, each with its yearly value and any switching cost.

Before you switch anything, check contract end dates and early termination fees from Week 1. Keep old service running until the new one works, and port phone numbers before cancelling (8.6).

Week 4: Act and document

Goal: make the changes worth making, and set up the records that will prove they worked. **Time:** about 2 to 4 hours, depending on what you choose.

Task	How to do it	See
Choose worthwhile changes	Score every idea from Weeks 1 to 3 on the Household Savings Priority Matrix. Start with the highest scores. Run the "Should I Buy It?" worksheet for anything with a major cost.	B3; Tool 13.1
Record upfront expenses	Enter what each change cost, including installation, fees and anything you bought, in the scorecard's "Upfront cost" column. Keep receipts.	B4
Set up tracking	Give each change its own row on the 90-Day Household Savings Scorecard. Copy in its baseline from your Week 1 audit: the same billing months last year. Note each change's start date.	B4; 14.4
Save confirmations	File every confirmation email, reference number, letter and new contract. Log each call.	A2; 10.3
Schedule follow-up measurements	Put meter readings, bill dates and the day 30, 60 and 90 reviews in your calendar. Read meters on the same day each month.	Tool 14.2

By the end of Week 4, you have: a set of changes made or scheduled, their costs recorded, and a scorecard ready to fill in.

Day 30: review actions, not money

At day 30, judge yourself on what you did, not what you saved. Most savings will not show on a bill yet, because of billing cycles. Ask:

1. Which planned changes are done, which are scheduled and which were dropped?
2. Did every promised credit, cancellation or plan change get a confirmation?
3. Is anything waiting on someone else, such as an assistance application or a repair?
4. Is the scorecard set up, with baselines and start dates?

If most planned actions are done, the first month was a success, even if no bill has changed yet.

14.3 The 90-day plan

Days 31 to 60: read the early evidence

The first bills that cover your changes arrive in this window. Compare them fairly, or you will draw the wrong conclusion.

1. **Adjust for billing days.** Compare use per day, not monthly totals. A 33-day bill and a 29-day bill are not the same month.
2. **Adjust for weather.** For heating and cooling, compare use per degree day, or compare months with similar degree days (see 1.8, Rule 4).
3. **Separate use from price.** If the rate changed, work out what the new use would have cost at the old rate. A price increase can hide a real saving, and a price cut can fake one.
4. **Note other changes.** Guests, a new appliance, a new job schedule or a long trip all change use. Write them in the scorecard's notes.
5. **Check the bill itself.** Confirm that every credit, new rate and cancellation appears as promised (10.1).

A smaller bill is not proof, and a bigger one is not failure. Adjust first. For a worked weather-adjusted heating check, see 1.8, Rule 4.

Days 61 to 90: decide what worked

By now most monthly bills have covered at least one full period since your changes. Sort every measure into one of three groups.

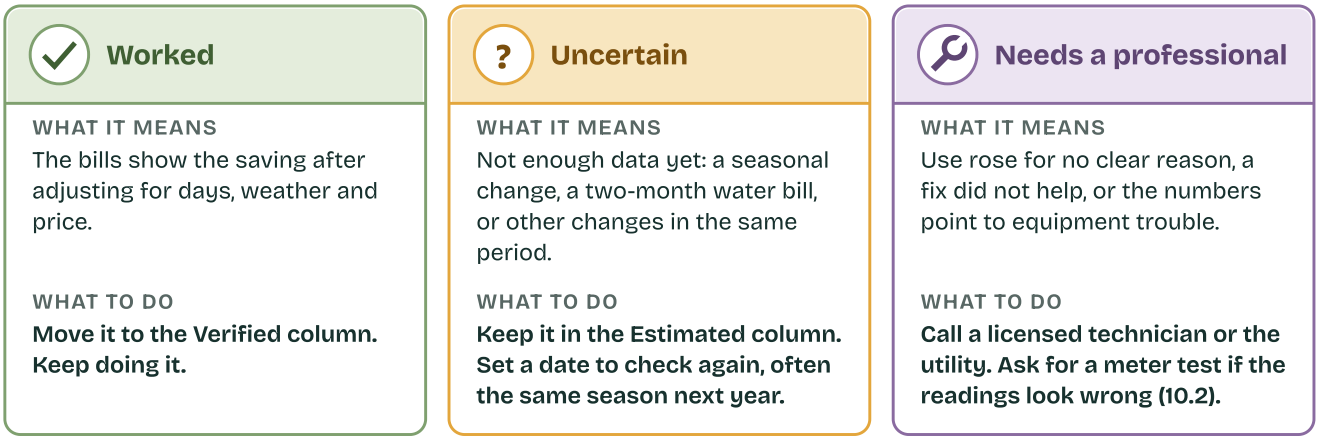


Figure 14.3. At day 90, every change is worked, uncertain or in need of a professional, and each calls for a different next step.

Group	What it means	What to do
Worked	The bills show the saving after adjusting for days, weather and price	Move it to the Verified column. Keep doing it.
Uncertain	Not enough data yet: a seasonal change, a two-month water bill, or other changes in the same period	Keep it in the Estimated column. Set a date to check again, often the same season next year.
Needs a professional	Use rose for no clear reason, a fix did not help, or the numbers point to equipment trouble	Call a licensed technician or the utility. Ask for a meter test if the readings look wrong (10.2).

What you can prove by day 90:

- **Flat-price services** (internet, phone, trash): after one full bill at the new price.
- **Other metered changes** (electricity, gas, water): after two or more bills, compared per day with the same months last year.
- **Heating and cooling:** only after a season, compared per degree day (see 1.8, Rule 4).
- **Water billed every two months:** usually still Uncertain at day 90.

Then decide, for each measure: keep it, adjust it or undo it. A thermostat setting that makes the home uncomfortable is not a saving worth keeping.

What to record: at day 90, each change's group and its verified saving so far, on the Scorecard (B4). For anything Uncertain, put the date to check again in the 12-Month Bill Calendar (Tool 14.2).

After day 90

- **Spend about 30 minutes a month** on the five-minute check for each bill (10.1) and one new scorecard line.
- **Use the 12-Month Bill Calendar** (Tool 14.2) for promotion ends, contract renewals, seasonal checks and application windows.
- **Do a full review once a year,** using the Annual Household Utility Spending Tracker (B1). Start a new 30-day plan if something has changed: a move, a new household member, new equipment or a rate increase.

14.4 Measuring and reporting your savings

The 90-Day Household Savings Scorecard (B4) is where your plan becomes a result. It uses the columns below, including one that matters most: estimated savings, kept apart from verified ones.

Column	What to enter
Change (utility)	One row per change, named with the service it affects, such as "Electricity: cooling"
Baseline usage	Use per day in the same billing months last year, before the change
Current usage	Use per day after the change
Baseline rate	Your marginal price before the change
Current rate	Your marginal price now
Baseline cost	The monthly cost in the same months last year
Current cost	The monthly cost now
Estimated savings	Your projection, before the bills prove it
Verified savings	Savings the bills confirm, after adjusting for days, weather and price
Upfront cost	What the change cost
Notes	Dates, other changes, confirmations, what to check next

How to work out verified savings

- **For metered services** (electricity, gas, water): (baseline use per day − current use per day) × days in the period × one marginal price. Use the same price for both, so a rate change does not count as your saving.
- **For heating and cooling,** use per degree day instead of use per day (see 1.8, Rule 4).
- **For flat-price services** (internet, phone, trash): baseline monthly cost − current monthly cost, once a full bill at the new price has arrived.
- **For rate or plan changes:** your actual use priced at the old rate, minus the same use priced at the new rate.

How to report your results honestly

- **Report two totals:** verified savings so far, and estimated savings still to confirm. Never add them together.
- **Annualize with care.** Multiply a verified monthly saving by 12 only if it applies all year. A cooling saving counts only for the cooling months.

- **Count each saving once.** If two changes cut the same use, record the combined effect, not the sum of two estimates.
- **Subtract upfront costs** to see what you have actually gained so far, and how long each change takes to pay back.
- **Keep the paper trail.** Bills, readings and confirmations are your evidence.

The Notes column matters more than it looks. A note such as "Guests for 10 days in March" or "Rate rose 4% on May 1" explains a result that would otherwise make no sense.

CASE C

Dana's 90 days

Dana is fictional. Providers, bills and figures are SAMPLE or Illustrative, using the Billix rate card.

Starting situation. Dana works from home five days a week in a 1,400-square-foot townhouse with a heat pump, shared with a partner who also works from home some days. Summer bills were high, and the internet bill had jumped when a promotion ended. When the high July bill arrived, Dana decided to work through the 30-day plan.

Household facts. A west-facing upstairs office and ducts in the attic (Chapter 4). Electricity at the rate card's $0.18 per kWh plus a $12 customer charge. Internet at $95 a month: $80 for 500 Mbps plus a $15 gateway rental (Chapter 7).

Sample bills (SAMPLE)

Bill	Use	Cost
April electricity, 30 days	450 kWh (15 kWh a day, almost no cooling)	
July electricity, 31 days	1,240 kWh (40.0 kWh a day), 420 cooling degree days	$235.20
Internet, monthly	500 Mbps plan	$95.00

Likely source of the extra cost. Summer cooling, worsened by a hot west office and attic ducts; and an internet plan billed at the full price after its promotion ended.

Diagnostic steps (Weeks 1 and 2). Dana started from the April and July bills. With about 465 kWh of non-cooling use in a 31-day month, July's cooling came to about 775 kWh, or 1.85 kWh per cooling degree day. A contractor had suggested replacing the heat pump, but Dana decided to try the cheap steps first (Chapter 4). Dana's household needed about 35 Mbps at its busiest hour, far below the 500 Mbps plan (Chapter 7).

Solutions (Weeks 3 and 4).

1. **Internet:** switch to Anytown Fiber at 300 Mbps for $60 all in, with a 24-month price lock and a $50 installation fee, plus about a week of overlap on the old bill ($23.75). Return the old gateway.
2. **Cooling:** in Chapter 4's order. First a daytime thermostat setting raised from 72 to 76 degrees F and a new filter, then sealing the attic hatch and duct joints, then a tune-up that found the heat pump healthy. All three were done before the August bill, for about $205.
3. **Rate plan:** with heavy daytime use at home, Dana asked the utility in writing, using the rate or plan review request (A5), whether a time-of-use plan would cost more or less.

Cost comparison: Dana's scorecard at day 90

Change (utility)	Baseline usage	Current usage	Baseline rate	Current rate	Baseline cost	Current cost	Estimated savings	Verified savings	Upfront cost	Notes
Internet	500 Mbps plan	300 Mbps plan	Flat price	Flat price	$95 a month	$60 a month	None needed	$35 a month; $70 over two full bills	$73.75 ($50 installation plus $23.75 overlap)	Price lock ends in 24 months; gateway returned
Electricity: cooling	This July, before the changes: 1.85 kWh per degree day	August: 1.40 kWh per degree day	$0.18 per kWh	$0.18 per kWh	$235.20 (July)	$201.54 (August)	About $120 for the season, including $33.66 in August	$0 so far: one weather-matched month is not yet enough (see 1.8, Rule 5)	$205	July and August both had 420 degree days; August use 1,053 kWh. An early before-and-after check; the baseline (last summer's same months) decides Verified
Electricity: rate plan	Standard plan	Under review	Standard	Not yet known	Not applicable	Not applicable	Unknown	$0	$0	Waiting for the utility's written comparison

Dana's totals at day 90:

- **Verified so far:** $70 on internet.
- **Still estimated:** about $120 on cooling for the season, including the $33.66 seen in August, and an unknown amount from the rate plan.
- **Upfront costs:** $278.75, including the $23.75 internet overlap. Dana has not yet earned that back, and that is normal at day 90. The internet costs were repaid in about two months; the cooling work should pay back in under two summers.
- **Expected yearly savings (Estimated):** about $420 on internet and about $120 on cooling, or about $540 a year.

The decision and why. Keep the internet plan, which the bills support, and the cooling changes, which August's bill supports so far. Hold the time-of-use decision until the utility's comparison arrives, and count nothing for it until then.

How Dana checks the result. The internet price-lock end goes in the 12-Month Bill Calendar (Tool 14.2). The cooling saving moves fully to Verified after the summer, once each month is compared with the same month last year, at matching degree days. In winter, Dana will check heating use per degree day the same way.

The lesson: at day 90, Dana could prove $70 and expected about $540 a year. Reporting the two numbers separately kept the plan honest.

Tools for this chapter

Tools 14.1 and 14.2 are below. The other two forms this chapter uses are printed once, in Appendix B, because the whole book uses them:

- **B3. The Household Savings Priority Matrix:** score each idea from Weeks 1 to 3 (14.2, Week 4).
- **B4. The 90-Day Household Savings Scorecard:** one row per change, with its baseline, its estimated saving and its verified saving in separate columns (14.4). Case C shows Dana's finished scorecard.

All four are in the companion worksheet pack as fillable PDFs and spreadsheets. Each works on paper with a calculator.

Tool 14.1: the 30-Day Action Plan

Purpose: track what you will do, when, and whether it is done. **How to use it:** list each task from Weeks 1 to 4, give it a date, and tick it off. Review it at day 30.

Week	Task	Target date	Done	Confirmation or result	Follow-up date
1	Gather 12 months of bills				
1	Baseline and marginal prices				
1	Providers, plans and key dates				
2	Electricity audit				
2	Water leak checks				
2	Heating and cooling review				
2	Recurring services list				
3	Internet and mobile comparisons				
3	Discounts and assistance				
3	Rate-plan review requests				
4	Chosen changes made or booked				
4	Scorecard and calendar set up				

Tool 14.2: the 12-Month Bill Calendar

Purpose: catch the dates that cost money when missed. **How to use it:** add your own dates from Week 1, then the seasonal checks below. State dates vary; check yours.

Month	Seasonal checks (examples)	Your dates
January	Compare heating use per degree day with last year; check any budget-billing balance	
February	Book a spring cooling tune-up	
March	Winter shutoff protections end in many states between mid-March and mid-April; if you are behind, call before they end (12.8). If your sewer charge uses a winter average, check that the new sewer volume matches your average water use in the averaging months (6.2).	
April	Check outdoor taps and irrigation for leaks (5.9, 5.10); replace the cooling filter	
May	Review summer rate options, such as time-of-use (2.10)	
June	Check the thermostat schedule for summer; note any hot-weather protections	
July	Compare cooling use per degree day with last year	
August	Review promotions and price locks ending in the next three months	
September	Book a heating tune-up; compare propane or oil price plans before winter (3.6); ask about budget billing	

Month	Seasonal checks (examples)	Your dates
October	Check when your state's LIHEAP applications open (12.2); test carbon monoxide and smoke alarms. If your sewer charge uses winter-average months, find out which months, and check for leaks before they begin (6.2)	
November	Winter protections begin in many states; confirm you are enrolled if you qualify	
December	Annual review with the Spending Tracker (B1) and Scorecard (B4); plan next year	

Your own dates to add: promotion and contract end dates, price-lock ends, deposit refund dates, discount and Lifeline renewals, medical certificate expiry, the budget-billing settle-up month, rebate deadlines, warranty end dates, and the days the clocks change, for new smoke and carbon monoxide alarm batteries.

Chapter 14 review

What you learned

- The 30-day plan has one theme a week: understand your bills, find wasted use, review providers and discounts, then act and document.
- Day 30 is for reviewing actions. Billing cycles mean most savings show up in months two and three.
- Compare bills fairly: use per day, use per degree day for heating and cooling, and one price for before and after.
- At day 90, every change is either worked, uncertain or in need of a professional.
- Estimated and verified savings belong in separate columns, and are never added together.
- A 12-month calendar catches the dates that cost money when missed.

What you can do today

- Start a folder and gather your last 12 months of bills.
- Fill in the first three rows of the 30-Day Action Plan (Tool 14.1) with dates.
- Put every promotion and contract end date you already know into the 12-Month Bill Calendar (Tool 14.2).

What to check next month

- At day 30, review which actions are done, scheduled or dropped.
- Make sure the scorecard (B4) has a baseline for every change.
- Confirm that every credit and plan change you asked for appears on your bills.

Billix Quick Check

Answer yes or no:

1. Do you have a baseline, in use per day, for each metered bill?
2. Have you scored your savings ideas, so you know which to do first?
3. Does each change have a start date and an upfront cost recorded?
4. Do you keep estimated and verified savings in separate columns?
5. Are your key dates in a calendar you will actually look at?

Each "no" points to a step in this chapter.

A last word

Utility bills can feel like weather: something that happens to a household, not something it controls. This handbook has tried to show the opposite. Every bill has parts you can read, choices you can make and rules that protect you.

You will not need every chapter. Some households will save most on internet and phones, others on heating, water or a single billing error. Some will find their biggest win in a program they did not know they qualified for. Start with the step that fits your home, measure what happens and keep going.

Whatever you save, save it honestly. Count what your bills prove, keep estimates separate, and come back to your plan each year. That habit will be worth more than any single tip in this book.

Sources and further reading

Last Verified: October 2026. This chapter mostly applies methods and sources from earlier chapters. The seasonal dates in the 12-Month Bill Calendar are examples; check your own state's dates.

1. NOAA Climate Prediction Center. Degree day data. https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/cdus/degree_days/
2. LIHEAP Clearinghouse. Disconnect policies and cold-weather protections by state. https://liheapch.ncat.org/node/1932
3. State winter protection examples, as cited in Chapter 12: Pennsylvania (52 Pa. Code Chapter 56), New Jersey BPU, Massachusetts (G.L. c.164 §124F), Wisconsin PSC and Michigan PSC.
4. Austin Water. Wastewater averaging (winter-average sewer billing). https://www.austintexas.gov/department/wastewater-averaging
5. Centers for Disease Control and Prevention. "About Carbon Monoxide Poisoning" (detector batteries and replacement). January 12, 2026. https://www.cdc.gov/carbon-monoxide/about/index.html
6. For the tools referenced in each week, see the sources listed in Chapters 1 to 13.

TOOLS AND REFERENCE

Appendix

How to use these pages

This appendix collects the tools you will use again and again, in a form you can copy, print or fill in. It is also the one place the book prints the blank forms that more than one chapter uses. The chapters teach each form, show it filled in where that helps, and point here. Each form below names the sections that teach it.

- **Appendix A: Scripts and letters.** What to say on a call, what to write in a dispute, and how to cancel cleanly. Each one names the sections that explain when to use it.
- **Appendix B: Core worksheets.** Four tools for tracking your bills and your progress. The Introduction shows a completed sample of B1's first part, the opening assessment, and Case C in Chapter 14 shows a filled-in B4.
- **Appendix C: Glossary.** Plain-English definitions of the terms on your bills and in this book.
- **Appendix D: Reference index.** Where to start with a common question, and where to find each topic, tool, source of help and state rule for propane and heating oil, by section.

Tool numbers. A tool printed in a chapter is numbered by chapter, such as Tool 5.2 in Chapter 5. The forms here keep their letters, A1 to A6 and B1 to B4. D2 lists them all.

The companion worksheet pack. Every fill-in tool in this book, including the ones in this appendix, is also available as a fillable PDF and a spreadsheet, so you can print fresh copies or let the spreadsheet do the math. Each tool also works on paper with a calculator. You do not need spreadsheet software.

A note on the letters. Replace every item in square brackets with your own details. Keep a copy of everything you send, and note the date. Send copies of documents, never originals.

Appendix A: Scripts and letters

A1. The Provider Negotiation Script

Purpose: to ask for a better price in one prepared call. It works for internet, TV, mobile and other competitive services, and, with the utility version below, for regulated utilities. Fill in the blanks before you dial.

Taught in: 7.7. Also used to ask a mobile carrier to match an offer (8.4) and, with the utility version, in 10.4.

Item	Your notes
Provider and account number	
Customer since	
Current all-in monthly cost	
Promotion or contract end date	
Best alternative: provider, plan or speed, two-year cost	
What I actually need: speed (7.5), data, lines or plan size	
My goal (price, plan or tier, fees removed)	

Item	Your notes
The price at which I will switch	
Number I will call (from my bill or the official website)	

Opening, with an alternative: "I've been a customer for [years]. My bill went from [old amount] to [new amount]. [Other provider] offers [plan or speed] for [amount] all in. Can you offer something close before I switch?"

Opening, without an alternative: "My bill has risen to [amount]. Can you help me lower it with a smaller plan, my own equipment, any discounts I'm missing, or an assistance plan?"

The utility version (regulated rates): "I'm not asking you to change the rate. I'd like to know whether any of these apply to my account: a better rate plan, budget billing, a payment arrangement, a discount program or a waiver of this fee."

Questions for any offer:

1. What is the total monthly price, with all fees and taxes?
2. How long does that price last, and what is the price afterward?
3. Is there a new contract, commitment or termination fee?
4. Will you send the offer in writing, with a reference number?

Call record: date, representative, offer, reference number, follow-up date. Copy it into the Provider Call Log (A2).

If the answer is no: thank them, ask whether a supervisor or the retention department can help, or call back another day. Switching is often an option for a competitive service.

A2. The Provider Call Log

Purpose: to keep a record of every contact, so promises can be checked and disputes proved.

Taught in: 10.3 (step 8). Also used to collect evidence (10.2), and to log a shutoff notice and the answers to the six questions in 12.8.

Before the call

Item	Your notes
Provider and account number	
The problem, in one sentence	
The amount in dispute	
My asks, in order	1. 2. 3.
Documents in front of me	
Number I will call (from the bill or official website)	

Each contact

Date and time	How (phone, chat, email, letter)	Representative's name or ID	What I asked	What they said or promised	Reference number	Follow-up date	Result

Reading your log: if a promised credit or change has not appeared by the follow-up date, call again with your reference number, or send the billing dispute letter (A3). A note on recording: in this book, "record" means write down. If you want to make an audio recording, check your state's law first. Some states require everyone on the call to agree.

A3. The billing dispute letter

When to use it: when a call has not fixed a billing error, or when you want a written record from the start.

Taught in: 10.2 and 10.3 (step 9). In Case F in Chapter 10, the Brooks household sends it with a receipt and meter readings.

[Your name]
[Your address]
[Account number]
[Date]

[Provider name]
[Billing dispute address or email, from your bill or the provider's official website]

Subject: Billing dispute, account [number], bill dated [date]

I am writing to dispute [describe the charge] on my bill dated [date]. The amount in dispute is [amount].

I believe this charge is wrong because [one or two sentences, for example: the meter reading was estimated, and my actual reading on [date] was [reading]].

I have paid, or will pay by the due date, the part of the bill I do not dispute: [amount].

I discussed this with your representative [name or ID] on [date]. The reference number is [number].

Please [the correction you want, for example: rebill this period using the actual reading, or remove the duplicate charge], and confirm the result to me in writing.

Copies of [list the documents] are attached.

Sincerely,
[Your name]

What to attach: copies of the evidence for your kind of error, from the table in 10.2. For a wrong reading, that is a dated meter photo and past bills; for a missing credit, your call log and any confirmation email.

How to send it: to the billing dispute address or email on your bill, by a method that shows the date you sent it, such as email. Keep a copy, and log it in the Provider Call Log (A2) with a follow-up date.

If there is no answer: if no written answer arrives by your follow-up date, or the answer is no and you still think the bill is wrong, escalate. 10.8 says which agency handles your service, and D3 how to reach it.

A4. The fee explanation request

When to use it: when a fee appears that you do not understand.

Taught in: 10.2 (charges you do not recognize) and 10.3 (step 4). Also used for an internet fee that is not on the broadband label (7.3), a trash line that does not say what it buys (9.1) and a billing company's admin fee (11.2).

Subject: Request for an explanation of a fee, account [number]

My bill dated [date] includes a charge described as [fee name] for [amount]. Please tell me in writing:

1. What the fee is for.
2. The tariff, rate schedule or contract term that allows it.
3. How the amount was calculated.
4. Whether it will appear again.
5. Whether it can be removed or waived on my account.

Thank you,
[Your name]

Sending it: attach a copy of the bill, with the fee marked. Send it by a method that shows the date, keep a copy and log it in A2 with a follow-up date. If the answer shows the fee is wrong, or no written answer arrives by then, send the billing dispute letter (A3).

A5. The rate or plan review request

When to use it: once a year for each utility, or when your household's use changes.

Taught in: 10.4 (a rate plan that fits your use better). Also used in 14.2 (Week 3); in Case C in Chapter 14, Dana sends it to ask about time-of-use.

Subject: Request for a rate or plan review, account [number]

Please review my account to see whether I am on the lowest-cost plan or rate schedule available for my use. My average use over the last 12 months is [amount and unit]. Please tell me:

1. My current rate schedule or plan.
2. Any other residential plans or programs I qualify for, such as time-of-use rates, budget billing or discount programs.
3. The estimated yearly cost of each, based on my past 12 months of use.
4. Any fees, commitments or waiting periods for switching.

Thank you,
[Your name]

Sending it: take your 12 months of use from your bills or your online usage history, and attach a copy of your latest bill. Send it by a method that shows the date, keep a copy and log it in A2 with a follow-up date. If the reply shows a rate or plan you did not choose, send the billing dispute letter (A3) and ask for a correction (10.2).

A6. The service cancellation checklist

Taught in: 7.9 (internet and TV), 9.3 (a private hauler) and 11.5 (moving). To keep a phone number, read 8.6 first.

Before you cancel

- ☐ Read your contract for early termination fees, notice periods and equipment terms.
- ☐ For a mobile number you want to keep, port it to the new carrier first. Do not cancel the old service first (8.6).
- ☐ For internet or TV, set up the new service before the old one ends, if you cannot be without it.
- ☐ For a utility, set a stop date, arrange a final reading and photograph the meter that day (11.5).

When you cancel

- ☐ Use the provider's posted cancellation process, and ask for a cancellation number.
- ☐ Log the date, time and representative's name.
- ☐ Ask what the final bill will include and when it will arrive.
- ☐ Give a forwarding address.

After you cancel

- ☐ Return rented equipment, and keep the receipt or tracking number.
- ☐ Pay the final bill, then stop autopay. Confirm no more payments are scheduled.
- ☐ Ask for any credit balance or deposit to be refunded.
- ☐ Check your next two bank or card statements for charges.

Last Verified: October 2026. The FTC's 2024 "click-to-cancel" rule was struck down by a federal appeals court on July 8, 2025, before its main requirements took effect. In March 2026 the FTC asked for comments on a possible new rule, but none was in place in October 2026. Some states have their own automatic-renewal laws. California's and Minnesota's exclude services regulated by the FCC or the state utility commission. So for phone, internet, cable and utility service, rely on your contract, the provider's posted process and, if needed, the FCC complaint route (10.8). Streaming and other online subscriptions are where these state laws most clearly apply.

Appendix B: Core worksheets

B1. The Annual Household Utility Spending Tracker

Purpose: to see a full year of household bills in one place, so you know which bills matter most, when they peak, and what to work on first.

Taught in: the opening assessment in the Introduction, which has a completed sample of Part 1 and shows how to read it. Also used in 14.2 (Week 1, steps 2 and 4) and for the yearly review (14.3).

The tracker has two parts. Part 1, the opening assessment, gives one line per service, annualized. Part 2, the month-by-month sheet, records each bill's total in the month it arrived, with a provider list. Fill in Part 1 once, then Part 2 as bills arrive through the year.

Part 1: the opening assessment

Instructions: gather up to 12 months of bills for every service. Annualize each one from actual bills: the total of the last 12 monthly bills, 6 two-month bills or 4 quarterly bills, every delivery in the last 12 months, or the yearly charge. For a service included in rent, taxes or HOA dues, use the amount the statement shows, or write "included". Then divide by 12 for the monthly cost. If you have only one or two bills, estimate the year and mark it **Estimated** in the Notes column.

Utility	Provider	Monthly cost	Annualized cost	Billing frequency	Billing type	Notes
Electricity						
Natural gas, propane, heating oil or other fuel						

Utility	Provider	Monthly cost	Annualized cost	Billing frequency	Billing type	Notes
Water						
Sewer						
Internet						
Mobile phone						
Trash or waste collection						
Other recurring household services (TV, streaming, security, service plans)						
Total						

Billing frequency: monthly, every two months, quarterly, per delivery or yearly.

Billing type: for example, metered use, flat fee, tiered rates, time-of-use, budget billing, per delivery, ratio-billed through rent, or included in rent, taxes or HOA dues.

Part 2: the month-by-month sheet

Instructions: enter each bill's total in the month it was billed. Where a bill covers two months, such as many water bills, enter it in the month it arrives and note the period. Add each column and each row.

Month	Electricity	Gas, propane or oil	Water and sewer	Trash	Internet and TV	Mobile	Other	Month total
January								
February								
March								
April								
May								
June								
July								
August								
September								
October								
November								

Month	Electricity	Gas, propane or oil	Water and sewer	Trash	Internet and TV	Mobile	Other	Month total
December								
Year total								
Monthly average								
Share of total								100%

Provider list

Service	Provider	Plan or rate schedule	Account number	Fixed monthly charge	Marginal price per unit	Promotion, contract or price-lock end
Electricity						
Gas, propane or oil						
Water and sewer						
Trash						
Internet and TV						
Mobile						

Marginal price: add up every per-unit rate on your bill; on a tiered bill, use your top tier's rate (1.8, Rule 2). Without tiers, it is close to B2's price per unit.

Calculations: monthly average = year total ÷ 12. Share of total = the service's year total ÷ the household's year total × 100.

Reading your results: start your savings work with the services that take the largest share. A month far above that service's average, without a weather or household reason, is worth a 15-Minute Bill Audit (B2). Keep each year's tracker, so you can compare one year with the next.

B2. The Billix 15-Minute Bill Audit

Purpose: to find out whether a bill changed because of your use, the price, the calendar or a mistake. You need this bill, last month's bill, the same month's bill from last year (or the usage chart), a calculator and a pen. Compare use with last year, as 1.6 explains. Last month's bill shows a new charge, a price change or a catch-up reading as soon as it starts.

Taught in: 1.3 to 1.8. The audit itself follows 1.8 in Chapter 1, with Sam's bills (1.6) as its worked check.

Instructions

Do the steps in order. A wrong reading, extra days, a new price or a new fixed charge can explain a change before you look for a cause in your home.

- ☐ **Minutes 1 to 2: confirm the account (1.7).** Check the name, service address and account number. Check that the meter number matches the number on your meter. Read meters from the outside only:

never open a cover or break a seal.

- ☐ **Minutes 3 to 4: check the reading (1.5).** Is it actual or estimated? Write down the current and previous readings. If last month's reading was estimated and this one is actual, part of this bill may be a catch-up.
- ☐ **Minutes 5 to 6: put it on a per-day basis (1.4).** Count the days. Divide the use by the days, and the total due by the days. Do the same for last year's bill.
- ☐ **Minutes 7 to 8: find the price (1.8, Rule 2).** Add up the usage charges and divide by the units used. Compare with last year. On a tiered bill, compare each tier's rate too. If none changed, a higher average price comes from more of your use falling in a higher tier, not from a price rise.
- ☐ **Minutes 9 to 10: total the fixed charges (1.3).** Add up every charge that is not per unit. Did any rise since last year, or appear for the first time since last month?
- ☐ **Minutes 11 to 12: split the change.** Put a dollar figure on the days, the price, the fixed charges and your use, with the box below.
- ☐ **Minutes 13 to 14: look for warning signs (1.7).** Late fees, a past-due balance, duplicate charges, services you do not recognize, missing credits.
- ☐ **Minute 15: read the messages.** Look for notices of rate changes, promotions ending or plan renewals.

Split the change: Put a dollar figure on each cause, using this bill and last year's. Leave taxes out. A part can be a minus number.

1. **Days:** the extra days × last year's use per day × last year's price.
2. **Price:** last year's use per day × this bill's days × the change in price per unit.
3. **Use:** the change in use per day × this bill's days × this bill's price.
4. **Fixed charges:** this bill's fixed charges minus last year's.

The four parts add up to the change in charges before taxes. If the change in the amount due differs from them by more than the taxes and a few cents of rounding, look for a credit, a late fee, a past balance or an error (1.7).

Illustrative check, Sam's bills in 1.6: days, 1 × 15.0 kWh × $0.17 = +$2.55; price, 15.0 kWh × 31 × $0.01 = +$4.65; use, −1.0 kWh × 31 × $0.18 = −$5.58; fixed, $12.00 − $11.00 = +$1.00. Together: +$2.62, the whole rise from $87.50 to $90.12. The use part matches the $5.58 that Sam's lower use saved.

Record sheet

Fill in the Last bill column for the reading type, the price, the fixed charges and any new lines. Compare use with the same month last year.

Check	This bill	Last bill	Same month last year	Notes
Meter number matches your meter				
Reading type (actual or estimated)				
Days in the period				
Use (kWh, therms, gallons or CCF)				
Use per day				
Total due				
Cost per day				

Check	This bill	Last bill	Same month last year	Notes
Price per unit (usage charges divided by use)				
Fixed charges, total				
New or changed lines				
Late fees or past-due balance				
Messages about rates or plans				

Reading your results

What you found	What it points to	Where to go next
Use per day rose 5% or more	Something in the home: weather, people, equipment or a leak	2.4 (electricity), 3.3 (gas), 5.3 (water) or 6.3 (sewer). For weather, 1.8, Rule 4
Price per unit rose	A rate change, a rider or a plan or contract change	2.3 and 2.11 (electricity), 3.2 (gas) or 5.2 (water). For how rates change, 10.4
Only the number of days changed	Nothing is wrong	No action needed
The reading was estimated	A possible catch-up bill later	Section 1.5
A charge you cannot explain	A possible error or unwanted service	10.2, then 10.3
A past-due balance or growing late fees	A payment problem, not a price problem	10.9 and 12.5. With a shutoff notice, 12.8

If more than one row applies, take them in the audit's order: reading, days, price, fixed charges, then use. The 5% line for use per day is a Billix rule of thumb, not an official figure. A smaller change is worth noting and checking again on the next bill.

Run the audit on every bill for three months. After that it goes faster, because you will know what normal looks like for your home.

B3. The Household Savings Priority Matrix

Purpose: to rank your savings ideas so you do the best ones first.

Taught in: 14.2 (Week 4, "Choose worthwhile changes").

Instructions: score each idea from 1 to 3 on four measures, using the key, then add the scores. Start with the highest totals.

Score	Cost	Effort	Potential impact	Confidence
3	No-cost	Under an hour	Over $100 a year	Measured in your home, or a confirmed price
2	Low, under $50	A few hours	$25 to $100 a year	Estimated from your own bills
1	Moderate or major	A day or more, or a professional	Under $25 a year	Based on a general claim

Idea	Cost	Effort	Impact	Confidence	Total (out of 12)	Do it?
Example: switch to a cheaper internet plan with a price lock	2	3	3	3	11	Yes, first
Example: replace a working 6-year-old refrigerator	1	1	1	1	4	No

Idea	Cost	Effort	Impact	Confidence	Total (out of 12)	Do it?

Reading your results: totals of 10 to 12 are quick wins. Totals of 7 to 9 are worth doing once the quick wins are done. Totals of 6 or less usually wait, unless something has failed or safety is involved. For any idea with a cost score of 1, run Tool 13.1 (Should I Buy It?) first.

B4. The 90-Day Household Savings Scorecard

Purpose: to record what each change saved, keeping estimates and proven results apart.

Taught in: 14.2 (Week 4, "Set up tracking"), 14.3 and 14.4. Case C in Chapter 14 shows Dana's filled-in scorecard.

Instructions: give each change its own row, named with the service it affects, such as "Electricity: cooling". Before you make the change, fill in its baseline from the same billing months last year: use per day, your marginal price and the monthly cost. Update the current and savings columns at days 60 and 90, and again after a full season for heating and cooling changes.

Change (utility)	Baseline usage	Current usage	Baseline rate	Current rate	Baseline cost	Current cost	Estimated savings	Verified savings	Upfront cost	Notes
Example: Electricity: cooling										
Totals							Report separately	Report separately		

Calculations:

- **Metered services** (electricity, gas, water): verified savings = (baseline use per day − current use per day) × days in the period × one marginal price. Use the same price for both, so a rate change does not

count as your saving.

- **Heating and cooling:** use per degree day instead of use per day. Some bills print degree days; otherwise, see "Getting your degree days" in 1.8, Rule 4.
- **Flat-price services** (internet, phone, trash): baseline monthly cost − current monthly cost, once a full bill at the new price has arrived.
- **Rate or plan changes:** your actual use priced at the old rate, minus the same use priced at the new rate.

When a saving becomes Verified: only when the bills support it. Until then, keep it in the Estimated column.

- **Flat-price services:** after one full bill at the new price.
- **Other metered changes:** after two or more bills, compared per day with the same months last year.
- **Heating and cooling:** only after a season, compared per degree day.
- **Water billed every two months:** usually still Uncertain at day 90.

At day 90, sort each change into one of three groups (14.3). **Worked:** move it to the Verified column. **Uncertain:** keep it Estimated, and set a date to check again. **Needs a professional:** use rose for no clear reason, or a fix did not help; call a licensed technician or the utility.

Reading your results: the Verified total is what your bills prove. The Estimated total is what you expect but have not yet seen. Never add them together. Compare Verified savings with Upfront cost to see how far each change is from paying for itself.

Appendix C: Glossary

The section in brackets is where each term is explained in full.

AFUE (annual fuel utilization efficiency). The share of a furnace's or boiler's fuel that becomes heat for the home over a year. An 80% AFUE furnace turns 80% of its fuel into useful heat. [3.1, 4.1]

All-in cost. An internet plan's monthly price plus equipment and every monthly provider fee. [7.3, 7.6]

All-in price. For a propane or heating oil delivery: (gallons × price per gallon + fees) ÷ gallons. For cable and satellite TV: the single price for video programming that an FCC rule requires bills and ads to show. [3.6, 7.9]

Allocated billing. See *ratio utility billing system*. [11.2]

Always-on load. The power a home draws all the time, read in kW at a quiet overnight hour from the utility's hourly data. [2.4, 2.7]

Amount required to avoid disconnection. The sum a shutoff notice says you must pay to stop the shutoff. It may be less than your full balance. [12.8]

Arrearage management plan. A utility program that forgives part of an old balance each month you pay your new bill on time. [12.5]

Automatic delivery. Propane or heating oil the dealer delivers on its own schedule, based on the weather and your past use. Also called keep-full delivery. You are less likely to run out, but unless you have a prebuy or capped plan, you pay the dealer's price on the day of delivery. [3.6]

Autopay. Automatic payment of a bill from a bank account or card on a set date. [10.6]

Auxiliary heat. A heat pump's electric backup coils, also called emergency heat, which cost far more to run than the heat pump itself. [4.1, 4.4]

Base charge. See *customer charge*. On a trash bill, see *base fee*.

Base fee. A trash fee charged whatever cart size you choose. It pays for listed services, such as recycling or bulky item pickup, so check them before you pay for those elsewhere. [9.2]

Baseline. What you used and paid in the same billing months last year, before a change, with use put on a per-day basis. A saving is measured against it. "Medical Baseline" is something else: the name of a California medical rate program (12.5). [14.2, 14.4]

Bill credit. A discount paid as monthly credits on your bill, often for a "free" phone. Also called a promotional credit. Under many offers the credits stop if you change plans or leave. [8.2, 8.5]

Billing period. The days of use a bill covers, also called the service period. [1.4]

Break-even share. (Current price − off-peak price) ÷ (peak price − off-peak price). A time-of-use plan costs less with no change in habits only when your *peak share* is below it. [2.10]

Broadband label. The standard "Broadband Facts" label that internet providers must show for each standalone plan, with price, fees and typical speeds. [7.4]

Btu (British thermal unit). A unit of heat energy, used to compare fuels. A therm is 100,000 Btu. [3.1]

Budget billing. A plan that spreads a year's energy cost into equal or near-equal monthly payments. It changes when you pay, not how much. [10.5]

CCF. One hundred cubic feet of water or gas. For water, 1 CCF is 748 gallons, and a price per CCF × 1.337 is the price per 1,000 gallons. [3.2, 5.2]

Community solar. A shared solar project whose subscribers receive credits on their electric bills, where state rules allow it. [10.7, 13.6]

Complaints. Contact the provider first, in writing (A3). If that fails, 10.8 says which agency handles your service, and D3 how to reach it. [10.8]

Contamination fee. A charge some cities bill when a recycling or organics cart contains items that do not belong there. [9.2]

Cooling degree day. A measure of how hot a day was, counted from a base of 65 degrees F and used to compare cooling use across months with different weather. For where to get them, see 1.8, Rule 4. [1.6, 1.8]

Credit months left. The bill credits still to come on a financed phone. Leaving now pays only if your monthly saving × the credit months left is more than the phone balance you would owe. [8.5]

Customer charge. A fixed monthly charge for being connected, whatever you use. Also called a base, service or basic charge. [1.3, 2.3]

Deduct meter. A second meter that measures outdoor water, so that water can be subtracted from your sewer volume. Also called a sewer credit meter. [6.4]

Deferred payment agreement. A written payment arrangement for a past-due balance, required before a shutoff in some states. [12.8]

Delivery charge. The part of an energy bill that pays for the wires or pipes that bring energy to your home. [1.2, 2.3, 3.2]

Delivery ticket. The paper left with a propane or heating oil delivery, showing the gallons, the price per gallon and each fee. [3.6]

Demand charge. A charge based on your highest rate of electricity use in a period, measured in kilowatts. Uncommon on residential bills, but some utilities use them. [2.3]

Deposit. Money a provider holds as security against unpaid bills, usually returned with interest after a period of on-time payment. Rules vary by state. [10.9, 11.5]

Deprioritization. When the mobile network is busy, your data may be slowed before other customers'. [8.2]

Device protection. Monthly phone insurance, billed as an extra. Compare it, and any deductible, with the cost of repairing or replacing the phone. [8.3, 8.4]

Disconnection notice. A written warning, also called a shutoff notice, that service will be shut off for nonpayment on or after a stated date. [12.8]

Early termination fee. A charge for ending a contract before its term is over. [2.11, 7.3, 8.6]

EER2 (energy efficiency ratio 2). A central air conditioner's or heat pump's cooling rating on the hottest days. Higher is better. New air conditioners in the Southwest must also meet an EER2 minimum. [4.10]

Energy supplier. In states with energy choice, a company other than your utility that sells the electricity or gas itself. The utility still delivers it. [2.11]

Equivalent residential unit (ERU). A unit many stormwater fees are billed in: the *impervious area* of a typical single-family lot. [6.6]

Escalator. The yearly increase in a solar lease or PPA payment, written into the contract. [13.6]

Estimated (savings label). A projected saving, based on your own data, that the bills have not yet confirmed. [1.8, 14.1]

Estimated reading. A meter reading the provider calculated instead of reading the meter. Bills often mark it E, Est. or Estimated. It can lead to a catch-up bill later. [1.5, 10.2]

Expected life. How long a type of equipment usually lasts. In this book, it is DOE's average age at which the equipment is retired; many units fail sooner. Plan a replacement once working equipment is in the last third of it. [4.10, 5.11, 13.3]

Export credit. What a utility credits you for solar power you send to the grid. [13.6]

Fixed charge. Any charge that does not change with how much you use. [1.3]

Gallons per flush (gpf) and gallons per minute (gpm). How much water a toilet uses per flush, and a faucet or showerhead uses per minute. [5.4, 5.5]

Gas cost adjustment. The price of the gas itself on a gas bill. Also called a purchased gas adjustment, gas cost recovery or gas supply charge. It can change monthly or seasonally without a rate case. [3.2]

Gateway. A modem and Wi-Fi router in one box, which internet providers often rent for a monthly fee. [7.3, 7.8]

Heat pump. Equipment that moves heat instead of making it, used to heat and cool homes and to heat water. [3.4, 4.1, 13.4]

Heating degree day. A measure of how cold a day was, counted from a base of 65 degrees F and used to compare heating use across months. For where to get them, see 1.8, Rule 4. [1.6, 1.8, 3.3]

Household hazardous waste. Leftover products that can catch fire, react, explode, or are corrosive or toxic, such as paint, cleaners, oils, batteries and pesticides. Never put them in the trash or down a drain. [9.4]

HSPF (heating seasonal performance factor). The older heat pump heating rating. See *HSPF2*, and do not compare the two. [4.10]

HSPF2 (heating seasonal performance factor 2). A heat pump's heating rating over a season, under the newer, tougher test. Higher is better. The same heat pump scores about 15% lower in HSPF2 than in the older HSPF. [4.10]

Illustrative (savings label). A made-up example that shows a method. [1.8, 14.1]

Impervious area. The hard surfaces on a lot that rain cannot soak into. Most stormwater fees are based on it. [6.6]

Investor-owned utility. A privately owned utility whose rates are set or approved by a state utility commission. [1.1, 10.4]

Kilowatt (kW) and kilowatt-hour (kWh). A kilowatt is a rate of electricity use. A kilowatt-hour is the energy used at that rate for one hour, the unit on most electric bills. [2.2]

Late fee. A charge added when a bill is paid after its due date. State rules may cap it. [10.9]

Lateral. The pipe from a house to the public sewer main. The owner is often responsible for it; rules vary by city. [6.1]

Leak adjustment. A credit some water utilities give after you repair a leak. [5.10, 10.4]

Lifeline. The federal program that discounts phone or internet service for eligible households. [12.4]

LIHEAP. The Low Income Home Energy Assistance Program, a federal block grant that states use to help with heating and cooling bills and energy crises. [12.2]

Like-for-like replacement. A new unit of the same type and efficiency as the old one. [13.3]

Manual J. The industry's standard calculation of a home's heating and cooling needs, used to size new equipment. [4.1]

Marginal price. The price of one more unit you use, or one fewer you save. Savings should be valued at this price, not at the bill's average cost per unit. To find it, add up every per-unit rate on your bill; on a tiered bill, use your top tier's rate (1.8, Rule 2). Without tiers, it is close to B2's price per unit. [1.8]

Master meter. One meter that serves a whole building, with the owner as the utility's customer. [11.2]

Matched system. A central air conditioner's or heat pump's outdoor unit together with its indoor coil or air handler. On a quote, ask for the certified rating of the whole matched system, not the outdoor unit's "up to" figure. [4.10]

Mcf. One thousand cubic feet of gas, about 10.36 therms. [3.2]

Medical certificate. A form from a doctor or other medical professional that postpones a shutoff when it would harm someone in the home. In most states it can cover anyone who lives in the home, including a child, not only the person named on the bill. [12.5, 12.8]

Mild-month use. Use per day in a month with little heating or cooling, usually in spring or fall. Use above it in winter or summer is mostly heating or cooling. [4.2]

Minimum bill. A fixed floor: the least every customer pays each month. [1.3]

Monthly provider fee. A monthly fee an internet provider adds, such as equipment rental, listed on the broadband label. It is part of the plan's price. [7.3, 7.4]

Municipal utility. A utility owned by a city or town. Its rates are usually set by a council or board, not the state commission. [1.1, 10.4]

MVNO (mobile virtual network operator). A phone company that sells service on another carrier's network. [8.1]

Net billing. A solar arrangement that credits exported power at a rate below the retail price. [13.6]

Net metering. A solar arrangement that credits exported power at the full retail rate. [13.6]

NOWData. The National Weather Service's free online weather data. Pick the station nearest your home to add up the degree days for your exact billing dates. [1.8]

Number porting. Moving your phone number to a new carrier. [8.6]

PACE financing. Financing for a home upgrade, such as solar panels, that places a property-tax lien on the home. The FTC warns that the lien can lead to foreclosure if you fall behind. [13.6]

Pass-through fee. A fee for a government program that a provider chooses to pass on to you, listed on the broadband label. [7.3, 7.4]

Pay-as-you-throw (PAYT). Trash pricing by the bag, can or weight. [9.1, 9.3]

Payment arrangement. An agreement to pay a past-due balance over time, alongside new bills. [10.4, 12.8]

Peak share. The share of your electricity used in a time-of-use plan's peak hours: peak kWh ÷ all kWh, from a month of hourly data. Compare it with the *break-even share*. [2.10]

Power purchase agreement (PPA). A solar contract in which a company owns the panels and you pay for the power they produce. [13.6]

Prebuy. A propane or heating oil plan in which you pay ahead, usually in summer or fall, for a set number of gallons at a locked price. Also called a prepaid plan. [3.6]

Prepaid and postpaid. Prepaid mobile service is paid for before each month. Postpaid service is billed after you use it, and usually needs a credit check. [8.1]

Price lock. A provider's promise to hold a price for a set period, also called a price guarantee. Some exclude fees and taxes. [7.3, 8.3]

Price per watt. A solar quote's price ÷ (system kW × 1,000), the figure used to compare quotes. [13.6]

Price to compare. The utility's own supply price per kWh, the benchmark for an electricity supplier's offer. What it includes varies by state. [2.11]

Promotional rate. A temporary lower price that rises when the promotion ends. [7.3, 7.4]

Public utility commission (PUC) or public service commission (PSC). The state agency that regulates investor-owned utilities and handles complaints about them. [1.1, 10.8]

Purchased gas adjustment. See *gas cost adjustment*. [3.2]

Rate case. The public process in which a utility asks its regulator to change its rates. [10.4]

Ratio utility billing system (RUBS). A method of splitting a building's utility bill among tenants by a formula, such as occupants or square footage, instead of measured use. Rent statements may call it a utility allocation. [11.2]

Rider. A separate charge or credit on a utility bill for a specific cost, such as fuel or storm repairs. [2.3]

SEER (seasonal energy efficiency ratio). The older cooling rating for central air conditioners and heat pumps. See *SEER2*, and do not compare the two. [4.10]

SEER2 (seasonal energy efficiency ratio 2). A central air conditioner's or heat pump's cooling rating over a season, under the newer, tougher test. Higher is better. The same system scores about 5% lower in SEER2 than in the older SEER, so do not compare an old SEER figure with a SEER2 one. [4.10]

Septic system. An on-site wastewater system that needs periodic inspection and pumping instead of a monthly sewer bill. [6.5]

Settle-up. See *true-up*. [10.5]

Shutoff date. The first day service may be cut, as printed on a shutoff notice, not the date the notice was printed. Also called the disconnection or termination date. [12.8]

Simple payback. Net upfront cost divided by annual net savings: the years an upgrade takes to earn back its cost. [13.2]

Solid waste. The usual name for the trash line on a city, water or property tax bill. Also called sanitation, garbage, refuse or environmental services. [9.1]

Stormwater charge. A fee for managing rain runoff, often based on a property's hard surfaces. [6.6]

Submeter. A meter in each unit of a building that the owner uses to bill tenants for their own use. [11.2]

Supply charge. The part of an energy bill that pays for the electricity or gas itself. [2.3, 3.2]

Tariff. A utility's official, regulator-approved schedule of rates and rules. Also called a rate schedule. [1.2, 10.4]

Therm. A unit of gas energy equal to 100,000 Btu. [3.2]

Therm factor. The number a gas bill uses to turn the volume your meter measured into therms of heat. Also called heat content or Btu factor. It is often a little above 1, but can be below 1 at high altitude, so use the one printed on your bill. [3.2]

Throttling. A cut in your mobile data speed, often a sharp one, after you use a set amount of data. [8.2]

Tiered rate. A price that rises after a set amount of use each month, so the last units cost more than the first. Value a saving at the top tier's rate. [1.8, 2.3, 5.2, 5.12]

Time-of-use rate. An electricity rate that charges more at peak hours and less at other times. Where it is part of the supply rate, buying from a supplier ends it; where it is on the delivery charge, it can stay. [2.10, 2.11]

Tipping fee. A charge for dropping trash at a landfill or transfer station. [9.1]

Total cost of ownership. Everything an item costs over its life: price, installation, energy, water, maintenance, repairs, financing and disposal, less incentives. [13.3]

True-up (also settle-up). The bill or credit that squares what you paid with what your use actually cost, as at the end of a budget-billing year. [10.5]

Unlocking. Removing a carrier's restriction so a phone can be used with another carrier. [8.6]

Usage charge. A charge that rises and falls with how much you use. [1.3]

Utility allocation. See *ratio utility billing system*. [11.2]

Verified (savings label). A saving confirmed by your own bills over several billing periods, after adjusting for billing days, weather and price, with no other big change in the home. For a flat-price service, one full bill at the new price is enough. [1.8, 14.1, 14.4]

Weatherization Assistance Program. A federal program that pays local agencies to make low-income homes more energy efficient at no cost to the household. [12.3]

Will-call delivery. Propane or heating oil you order when the tank is low, instead of on the dealer's schedule. Small or after-hours deliveries can carry fees. [3.6]

Winter averaging. A sewer billing method that sets your sewer charge from your water use in winter months. [6.2]

Winter indoor use. Gallons per person per day on a winter water bill, when outdoor use is near zero. Use above it in summer is mostly watering. [5.3]

Appendix D: Reference index

D1. Topics by chapter

If something just happened, start with the first table. To look up a topic, use the second.

If this just happened…	Start here
Your bill jumped	The 15-Minute Bill Audit (B2), including Split the change. Then 2.4 and Case B in Chapter 2 (electricity), 3.3 (gas), 5.3 and Case F in Chapter 5 (water) or 6.3 (sewer)
A shutoff notice arrived	The "Shutoff notice? Do this first" box at the start of Chapter 12, then 12.8
An offer or contract arrived	2.10 (time-of-use), 2.11 (electricity supplier), 3.6 (propane or oil price plan), 7.9 (bundle), 8.5 (a "free" phone) or 13.6 (solar). At the door, also 10.7
You are moving	11.5, Tool 11.2 and the cancellation checklist (A6)
You cannot pay a bill	12.8, then 12.2 to 12.7. For a payment arrangement, 10.4 and 12.8
A suspicious call or text	10.7, before you pay anything. Report it even if you did not pay (10.7, step 5)
You have an upgrade or repair quote	4.10 (heating and cooling), then 13.3 and Tool 13.1. For solar, 13.6

If you want to…	Go to
Understand fixed and usage charges, billing periods and estimated readings	1.3 to 1.5
Compare two bills fairly, or measure a saving honestly	1.6, 1.8 (for degree days, Rule 4)
Split a bill's change into days, price, use and fixed charges, or tell whether a change in use is real	The 15-Minute Bill Audit after 1.8: Split the change, and Reading your results (blank in B2)
Read an electric bill, or test a time-of-use plan or supplier offer against your own bill	2.3, 2.10 (The whole-bill test), 2.11 (the find-it box)
Combine a time-of-use rate with an electricity supplier	2.11 (Time-of-use and a supplier)
Find out what an appliance costs to run, or find your always-on load	2.2, 2.4, 2.5, 2.7, Tool 2.1
Judge a "power saver" or other product claim	2.12, 13.6 ("Before you sign a solar contract"), 13.7
Read a gas bill, check whether your use is normal, or compare heating fuels	3.2, 3.3, 3.4, Tool 3.1
Check a propane or oil delivery ticket, or manage deliveries and contracts	3.6 (Check your delivery ticket)
Look up state rules for propane and heating oil dealers	D4
Stay safe if you smell gas	3.7
Work out how much of a bill is heating or cooling	4.2 (mild-month use)
Cut heating and cooling costs, or set a thermostat	4.2, 4.4 to 4.9, Tool 4.1

If you want to…	Go to
Decide whether to repair or replace equipment, even if it still works	4.10 (its opening rule), Case C in Chapter 4, 13.3 (Rules of thumb)
Read the efficiency ratings on a heating or cooling quote (SEER2, EER2, HSPF2)	4.10 (Efficiency ratings on a quote)
Read a water bill or investigate a water spike	5.2, 5.3, 5.10, Case F (after 5.12)
Fix a running toilet or a leak	5.4, 5.10, Tool 5.2
Understand sewer, stormwater and septic costs, or lower a stormwater fee	6.2, 6.5, 6.6 (Lowering a stormwater fee)
Lower an internet bill or choose a speed	7.3 to 7.7, Tool 7.1
Act when a promotion ends	7.3 (the find-it box), 7.4, 7.7, Case C in Chapter 7
Question a fee you do not recognize	7.3 (internet), 9.1 (trash), 11.2 (a rent statement's admin fee), A4
Rent or buy a modem; compare TV, streaming and bundles	7.8, 7.9
Read a phone bill, price a "free" phone, or ask your carrier to match an offer	8.3 to 8.5, Tool 8.1, A1
Leave partway through phone credits	8.5 (Already on credits?), Case G in Chapter 8
Switch phone carriers safely	8.6
Check how trash is billed, or right-size a cart	9.1 to 9.3, Tool 9.1
Stop paying twice for trash	9.3 (Catch duplicate and wrong charges), Tool 9.1, Step 4
Dispose of hazardous products safely	9.4
Find and dispute a billing error, including an estimated reading	10.1 to 10.3 (Should you dispute an estimated reading? is in 10.2), A3
Negotiate, or understand what cannot be negotiated	10.4, A1
Decide on budget billing or autopay, or check a settle-up	10.5, 10.6
Spot or report a scam or high-pressure sale	10.7
File a complaint with the right agency	10.8, D3
Understand deposits, late fees and credit reports, or avoid late fees on a disputed bill	10.9; 10.3 (step 3)
Know what you can change as a renter or homeowner	11.1 to 11.3, Tool 11.1
Check a ratio-billed (RUBS) utility share	11.2 (Is your share right? Five checks), Case A in Chapter 11
Check whether a Texas landlord's water service charge is allowed	11.2 (State rules vary widely)
Split bills with housemates	11.4
Move without paying twice, including a last ratio-billed month	11.5, Tool 11.2
Get help paying an energy, phone or water bill	12.2 to 12.7 (income limits by household size: the table in 12.7), Tool 12.1
Find an internet company's own low-cost plan	12.4 (Internet companies' own low-cost plans), then 7.6 (step 6)
Stop or delay a shutoff	12.8 (Find it on your notice, and the six questions)
Find out whether a medical certificate can cover anyone in your home, including a child	12.8 (Medical protections)
Respond when a medical certificate or payment plan is refused	12.8 (Payment arrangements), then 10.8
Work out payback or total cost before buying	13.2, 13.3, Tool 13.1
Compare upgrades, rebates and solar	13.4 to 13.6
Check a solar quote, test a lease or PPA, or check what a system produces	13.6 (What it costs; Testing a lease or PPA; Maintenance and the long run)
Estimate what a heat pump water heater would save	Case B in Chapter 13 (Your numbers)
Follow a step-by-step savings plan, starting with the right form	14.2 (Week 1's five steps), 14.3
Know when a saving counts as Verified	14.3 (What you can prove by day 90), 14.4, B4

D2. Tools

Every fill-in tool is also in the companion worksheet pack.

Tool	Where it is	Taught or used in
B1: the Annual Household Utility Spending Tracker (Part 1 is the opening assessment)	Appendix B1; instructions and a completed sample of Part 1 in the Introduction	11.2, 14.2, 14.3
B2: the Billix 15-Minute Bill Audit	Chapter 1, after 1.8; blank record sheet in Appendix B2	2.4, 10.1, 14.2
Tool 2.1: the Home Electricity Consumption Audit (also the Appliance Energy Calculator)	End of Chapter 2	2.5, 14.2
Tool 3.1: the Fuel Cost Comparison	End of Chapter 3	3.4, 3.6, 4.10
Tool 4.1: the Heating and Cooling Checklist	End of Chapter 4	4.2, 4.10, 11.3, 14.2
Tool 5.1: the Household Water Audit	End of Chapter 5	5.3, 5.5, 11.3, Case B in Chapter 13
Tool 5.2: the Water Leak Inspection Checklist	End of Chapter 5	5.10, 14.2
Tool 5.3: the Water Savings Calculator	End of Chapter 5	5.2, 5.12, 6.2
Tool 7.1: the Internet Plan Comparison Worksheet	End of Chapter 7	7.3, 7.4, 7.9, 14.2
A1: the Provider Negotiation Script	Appendix A1	7.7, 8.4, 10.4, 14.2
Tool 8.1: the Mobile Plan Comparison Worksheet	End of Chapter 8	8.5, 14.2
Tool 9.1: the Waste Service Audit	Chapter 9, after 9.4, with Case D	9.1, 9.3
Tool 10.1: the Billix Electric Bill Analyzer (for any metered bill)	10.1; blank in the pack	10.1, 10.2
A2: the Provider Call Log	Appendix A2	1.7, 7.6, 10.2, 10.3, 12.8, 14.2
A3: the billing dispute letter	Appendix A3	9.3, 10.2, 10.3; Cases D and F in Chapter 10
A4: the fee explanation request	Appendix A4	7.3, 9.1, 11.2
A5: the rate or plan review request	Appendix A5	10.4, 14.2; Case C in Chapter 14
A6: the service cancellation checklist	Appendix A6	7.9, 8.6, 9.3, 11.5
Tool 11.1: the Housing Action Matrix (a reference table, not a form)	Chapter 11, after 11.5	11.1 to 11.3; the opening of Chapter 4
Tool 11.2: the Move-In and Move-Out Utility Checklist	Chapter 11, after Case A	11.5
Tool 12.1: the Billix Utility Assistance Finder Checklist	End of Chapter 12	12.7, 12.8; Case D in Chapter 12
Tool 13.1: Should I Buy It? (with the Home Upgrade Payback Calculator)	End of Chapter 13	11.3, 13.3, 13.4, 13.6, 14.2
Tool 14.1: the 30-Day Action Plan	End of Chapter 14	14.2
B3: the Household Savings Priority Matrix	Appendix B3	14.2 (Week 4)
B4: the 90-Day Household Savings Scorecard	Appendix B4; Dana's filled-in copy in Case C, Chapter 14	1.8, 2.10, 2.11, 14.2 to 14.4
Tool 14.2: the 12-Month Bill Calendar	End of Chapter 14	14.2, 14.3

D3. Where to get help

Contact the provider first, in writing (A3), and give it a fair chance to answer. Go to an agency when that has failed (10.8). Find phone numbers and websites yourself, from your bill or the official sources below. Never use a number from an unexpected call, text or ad.

For	Contact	Chapter
A dispute with an investor-owned electric, gas or water utility	Your state utility commission; NARUC's directory lists every commission	10.8
Phone, internet or TV complaints	FCC: fcc.gov/complaints or 1-888-225-5322	10.8

For	Contact	Chapter
Scams and fake utility callers	FTC: ReportFraud.ftc.gov; also your utility and state attorney general	10.7
Deceptive sales	Your state attorney general (NAAG's "Find my AG") or state consumer office (USA.gov)	10.8
Credit report errors and debt collection	The credit bureau first, then the CFPB: consumerfinance.gov/complaint or (855) 411-2372	10.9
Free credit reports	AnnualCreditReport.com; NCTUE (utility and telecom data): 866-349-5185	10.9
Identity theft	IdentityTheft.gov	10.8
Energy bill help (LIHEAP)	EnergyHelp.us, or the National Energy Assistance Referral line: 1-866-674-6327	12.2
LIHEAP fee requests or fraud	HHS fraud hotline: 1-800-447-8477	12.2
Weatherization	Your state weatherization office (DOE's map)	12.3
Phone and internet discounts (Lifeline)	lifelinesupport.org, or 1-800-234-9473	12.4
Local help with any bill	Dial 2-1-1, or visit 211.org	12.7
Federal guide to energy bill help	USA.gov, or 1-844-872-4681	12.7
Rebates	Your utility; DSIRE; the ENERGY STAR Rebate Finder; your state energy office	13.5
Hazardous waste drop-off	Your local solid waste agency, or Earth911's search	9.4

Contact details were checked in October 2026. Agencies change numbers and web addresses; confirm them before you rely on them.

D4. State rules for propane and heating-oil deliveries (examples)

These state examples go with 3.6, which explains tank ownership, delivery plans, price plans and fees.

State	Rule
Vermont	No termination or tank-removal fee once a tank has been on site 12 months. Leftover propane refunded within 20 days, and tank removed within 20 days (30 if underground), with penalties of $250 plus $75 a day for late refunds or removals. A dealer may not refuse to fill a tank you own if you show proof and the system passes a safety check.
New Hampshire	From August 31, 2026: no termination fees once an aboveground tank has been on site 24 months (60 if underground). Removal and refunds within 30 days, 60 days' notice of fee changes, and no rental fee on a tank owned by someone else.
Maine and New Hampshire	Prepaid fuel contracts must be secured: by a commitment for 75% of the fuel, a 50% bond or a 100% letter of credit. New Hampshire allows prepaid contracts to be marketed only from May 1 to October 31.
Connecticut	Fees on customer-ordered (will-call) deliveries only in limited cases, such as small deliveries, off-hours or extra labor; no fees on dealer-scheduled deliveries; prepaid contracts must be secured.
New York	From November 1 to April 15, automatic-delivery customers must get written notice before deliveries stop.
Maine, Minnesota, Pennsylvania	Only the tank's owner, or someone the owner authorizes, may fill a propane tank. Pennsylvania requires removal within 30 days of the property owner's notice.

Last Verified: October 2026. State propane and heating oil rules change. Check your state attorney general or consumer protection office for current rules. The statutes and guides behind each row are listed in Chapter 3's sources (14 to 20).

About Billix

Billix helps households understand and lower the bills they pay every month. It starts with the ones nearly every home has, such as electricity, heating fuel, water and sewer, trash, internet and phone. It also covers the wider costs of running a household, from housing and insurance to subscriptions and local fees.

Billix explains bills in plain English. It shows the math, cites its sources, says where rules vary by state, and labels every example as illustrative, estimated or verified. That approach runs through this handbook.

Independence

[PLACEHOLDER: independence statement, with Billix's legal status (for example, nonprofit or company) and whether Billix receives payment, referral fees or advertising from any provider, product or program mentioned in this book]

The providers named "Anytown" in this book are fictional and marked SAMPLE. Real companies, agencies and programs are named only as examples, with sources, and not as endorsements.

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- **Website:** billixapp.com
- **Video channel:** [PLACEHOLDER: video channel name and address]
- **Corrections and feedback:** [PLACEHOLDER: contact address for corrections and feedback]

About this edition

This is the 2026 edition of The Billix Household Utility Savings Handbook. Facts, rates and programs were last verified in October 2026. Prices, rules and programs change, so check the official sources cited in each chapter before you act. This handbook is general education, not legal, tax or financial advice for your situation.